

Relative entropy of mixtures of excited states in CFT

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ABSTRACT: We compute the universal terms in the relative entropy of mixtures of energy eigenstates in 2d Conformal Field Theory (CFT), in both the small and large interval limits. Numerical reduced-state calculations for the compact free boson and Ising CFTs test the predicted powers and amplitudes, with finite-lattice and fitting-window deviations reported explicitly. We illustrate an application to approximate quantum error correction in conformal field theory.

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1 Introduction

The study of entanglement entropy in conformal field theory (CFT) has shed light on many aspects of universal strongly coupled phases and dynamics of many-body quantum systems. However, entanglement entropy of relativistic theories suffers from ultraviolet divergences, which is

Quantum relative entropy is a measure of distinguishability of quantum states, with an asymptotic interpretation in terms of hypothesis testing.

Consider a 2d CFT, and two energy eigenstates of $|\Psi\rangle$ and $|\Phi\rangle$ corresponding to Virasoro primaries Φ and Ψ . Denote by ρ_Φ and ρ_Ψ , the reduced density matrices of these states to a region A , and consider the mixture state $\rho_y = y\rho_\Phi + (1 - y)\rho_\Psi$ for some $0 \leq y \leq 1$. Motivated by connections to quantum error correction codes [1], in this note, we study the relative entropies $S(\rho_\Phi || \rho_y)$ in the small and large $|A|$ limits. We first review and extend the pure-state calculation because it fixes the replica conventions and the two- and three-point building blocks used throughout the analysis of mixtures.

Universal terms in classical relative entropy To argue for $\log y$ term in the mixture.

2 Relative entropy of excited states

2.1 Setup and Replica Trick

Consider a conformal field theory (CFT) in $1+1$ dimensions on a cylinder $\mathcal{C}_1$ of circumference R . We focus on a spatial subsystem A defined by the interval $(-l/2, l/2)$ at a fixed time slice. We simplify our notation by taking $A = (-\theta, \theta)$ on a cylinder of circumference 2π such that $\theta = \pi x$, where $x = l/R$ is the interval fraction.

Our states of interest are the reduced density matrices corresponding to energy eigenstates corresponding to the insertion of Virasoro primary fields at the center of radial quantization. These states are generated by the action of operators Ψ and Φ with conformal weights $(h_\Psi, \bar{h}_\Psi)$ and $(h_\Phi, \bar{h}_\Phi)$ on the vacuum at past infinity in imaginary time on the Euclidean cylinder $\mathcal{C}_1$:

$$|\Psi\rangle = \lim_{u \rightarrow i\infty} \frac{\Psi(u)|\Omega\rangle}{\langle \Psi^*(-u)\Psi(u) \rangle^{1/2}} \quad |\Phi\rangle = \lim_{u \rightarrow i\infty} \frac{\Phi(u)|\Omega\rangle}{\langle \Phi^*(-u)\Phi(u) \rangle^{1/2}} \quad (2.1)$$

While considering the special case of massless free boson theory, we will replace these operators with Vertex operators. The reduced density matrix of the state $|\Psi\rangle$ on the subsystem A is

denoted by ρ_Ψ . The replica trick that compute the relative entropy of excited states is

$$\begin{aligned}
S(\rho_\Psi \| \rho_\Phi) &= \left(\partial_n \log \left[\frac{\text{tr}(\rho_\Psi^n)}{\text{tr}(\rho_\Psi \rho_\Phi^{n-1})} \right] \right)_{n \rightarrow 1} \\
&= \left(\partial_n \log \left[\frac{\langle \prod_{i=1}^n \Psi(u_{i,n}^+, \bar{u}_{i,n}^+) \Psi^*(u_{i,n}^-, \bar{u}_{i,n}^-) \rangle_{\mathcal{C}_n} \langle \Phi(u^+, \bar{u}^+) \Phi^*(u^-, \bar{u}^-) \rangle_{\mathcal{C}_1}^{n-1}}{\langle \Psi(u_{n,n}^+, \bar{u}_{n,n}^+) \Psi^*(u_{n,n}^-, \bar{u}_{n,n}^-) \mathcal{O}_\Phi^{(n-1)} \rangle_{\mathcal{C}_n} \langle \Psi(u^+, \bar{u}^+) \Psi^*(u^-, \bar{u}^-) \rangle_{\mathcal{C}_1}^{n-1}} \right] \right)_{n=1}
\end{aligned} \tag{2.2} \quad \{\text{relEntropyRep}\}$$

$$\mathcal{O}_\Phi^{(n-1)} = \Phi(u_{1,n}^+, \bar{u}_{1,n}^+) \Phi^*(u_{1,n}^-, \bar{u}_{1,n}^-) \cdots \Phi(u_{n-1,n}^+, \bar{u}_{n-1,n}^+) \Phi^*(u_{n-1,n}^-, \bar{u}_{n-1,n}^-). \tag{2.3} \quad \{\text{corrFncyl}\}$$

The conformal map [2]

$$z = \left(\frac{\sin((u - \theta)/2)}{\sin((u + \theta)/2)} \right)^{1/n} \tag{2.4}$$

sends the interval $A = (-\theta, \theta)$ to $A = (0, \infty)$ uniformizing $\mathcal{C}_n$ to the complex plane. Under this transformation Virasoro primaries transform as

$$\begin{aligned}
\mathcal{O}(u, \bar{u}) &= \left(\frac{\partial z}{\partial u} \right)^{h_\mathcal{O}} \left(\frac{\partial \bar{z}}{\partial \bar{u}} \right)^{\bar{h}_\mathcal{O}} \mathcal{O}(z, \bar{z}) \\
\Psi(u_{k,n}, \bar{u}_{k,n}) &= \left(\frac{z_{k,n} \Lambda}{n} \right)^{h_\Psi} \left(\frac{\bar{z}_{k,n} \Lambda}{n} \right)^{\bar{h}_\Psi} \Psi(z_{k,n}, \bar{z}_{k,n}),
\end{aligned} \tag{2.5}$$

where in our example

$$z_{k,n} = e^{\frac{2\pi i k}{n}}, \quad z_{k,n}^\pm = e^{\frac{i\pi}{n}(2k \pm x)}, \quad \Lambda = \frac{\sin(\theta)}{\cos(\theta) - \cos(u)} \tag{2.6}$$

We define

$$\begin{aligned}
\mathcal{J}_{n,\Psi} &= \prod_{k=1}^{n-1} \left(\frac{\partial z_{k,n}^+}{\partial u_{k,n}^+} \right)^{h_\Psi} \left(\frac{\partial \bar{z}_{k,n}^+}{\partial \bar{u}_{k,n}^+} \right)^{\bar{h}_\Psi} \left(\frac{\partial z_{k,n}^-}{\partial u_{k,n}^-} \right)^{h_\Psi} \left(\frac{\partial \bar{z}_{k,n}^-}{\partial \bar{u}_{k,n}^-} \right)^{\bar{h}_\Psi} \\
&= \prod_{k=1}^{n-1} \left(\frac{z_{k,n}^+ z_{k,n}^- \Lambda^2}{n^2} \right)^{h_\Psi} \left(\frac{\bar{z}_{k,n}^+ \bar{z}_{k,n}^- \Lambda^2}{n^2} \right)^{\bar{h}_\Psi} \\
&= \prod_{k=1}^{n-1} (\Lambda/n)^{2(h_\Psi + \bar{h}_\Psi)} e^{4\pi i(h_\Psi - \bar{h}_\Psi)k/n} \\
&= (\Lambda/n)^{2(n-1)\Delta_\Psi} e^{2\pi i(n-1)s_\Psi}
\end{aligned} \tag{2.7}$$

The overall conformal factor that captures the transformation of the correlator in (2.3) from the cylinder to the complex plane is

$$\begin{aligned}
\mathcal{J} &= \frac{\mathcal{J}_{n,\Psi}}{\mathcal{J}_{n,\Phi}} \left(z_{1,1}^+ z_{1,1}^- \Lambda^2 \right)^{(n-1)(h_\Phi - h_\Psi)} \left(\bar{z}_{1,1}^+ \bar{z}_{1,1}^- \Lambda^2 \right)^{(n-1)(\bar{h}_\Phi - \bar{h}_\Psi)} \\
&= n^{-2(n-1)(\Delta_\Psi - \Delta_\Phi)} e^{2\pi i(n-1)(s_\Psi - s_\Phi)}
\end{aligned} \tag{2.8}$$

Therefore, mapping the correlator to the complex plane, we have

$$S(\rho_\Psi \parallel \rho_\Phi) = (\partial_n \log \mathcal{F}_n) |_{n=1}$$

$$\mathcal{F}_n = \left[\frac{\langle \prod_{i=1}^n \Psi(z_{i,n}^+, \bar{z}_{i,n}^+) \Psi^*(z_{i,n}^-, \bar{z}_{i,n}^-) \rangle \langle \Phi(z^+, \bar{z}^+) \Phi^*(z^-, \bar{z}^-) \rangle^{n-1}}{\langle \Psi(z_{n,n}^+, \bar{z}_{n,n}^+) \Psi^*(z_{n,n}^-, \bar{z}_{n,n}^-) \tilde{\mathcal{O}}_\Phi^{(n-1)} \rangle \langle \Psi(z^+, \bar{z}^+) \Psi^*(z^-, \bar{z}^-) \rangle^{n-1}} \right] \left(\frac{e^{2\pi i(s_\Psi - s_\Phi)}}{n^{2(\Delta_\Psi - \Delta_\Phi)}} \right)^{n-1} \quad (2.9) \quad \{\text{fncorrspin}\}$$

$$\tilde{\mathcal{O}}_\Phi^{(n-1)} = \prod_{k=1}^{n-1} \Phi(z_{k,n}^+, \bar{z}_{k,n}^+) \Phi^*(z_{k,n}^-, \bar{z}_{k,n}^-) \quad (2.10) \quad \{\text{fncorrspin}\}$$

Since $\Psi\Psi^*$ and $\Phi\Phi^*$ OPEs are invariant under a rotation of π about its midpoint [3], the total conformal spin should be an even integer. This requires $2s_\Psi, 2s_\Phi \in 2\mathbb{Z}$. Thus we have, $s_\Psi, s_\Phi \in \mathbb{Z}$ [3]. Since $s_\Psi - s_\Phi$ is an integer, we drop the $e^{2\pi i(s_\Psi - s_\Phi)}$ term. [Is this statement correct?](#)

2.2 Small interval limit $x \rightarrow 0^+$

Label by $\mathcal{O}_L$, $\mathcal{O}_{L''}$, and $\mathcal{O}_{L'}$ the lightest Virasoro primaries that appear, respectively, in the OPE of $\Psi\Psi^*$, $\Phi\Phi^*$, and $\Psi\Phi^*$ ¹. To simplify our notation, we define $C_\Psi^L = C_{\Psi\Psi^*}^L$, $C_\Phi^{L''} = C_{\Phi\Phi^*}^{L''}$ and $C_{\Psi\Phi^*}^{L'} = C_{\Psi\Phi^*}^{L'} = C_{\Psi^*\Phi}^{L'}$ ². In the small x limit, we can expand in the OPE channel

$$\frac{\Psi(z_k^+) \Psi^*(z_k^-)}{\langle \Psi(z^+) \Psi^*(z^-) \rangle} = e^{-4\pi i k s_\Psi / n} \left(\frac{\sin(\pi x)}{\sin(\pi x/n)} \right)^{2\Delta_\Psi} \times$$

$$\left(1 + C_\Psi^L (2 \sin(\pi x/n))^{\Delta_L} e^{\pi i s_L (2k/n + 1/2)} \mathcal{O}_L(e^{2\pi i k/n}) + \dots \right) \quad (2.11)$$

$$\frac{\Phi(z_k^+) \Phi^*(z_k^-)}{\langle \Phi(z^+) \Phi^*(z^-) \rangle} = e^{-4\pi i k s_\Phi / n} \left(\frac{\sin(\pi x)}{\sin(\pi x/n)} \right)^{2\Delta_\Phi} \times$$

$$\left(1 + C_\Phi^{L''} (2 \sin(\pi x/n))^{\Delta_{L''}} e^{\pi i s_{L''} (2k/n + 1/2)} \mathcal{O}_{L''}(e^{2\pi i k/n}) + \dots \right) \quad (2.12)$$

In our notation, the Virasoro primary $\mathcal{O}_L$ and $\mathcal{O}_{L''}$ have dimensions $(h_L, \bar{h}_L)$ and $(h_{L''}, \bar{h}_{L''})$, weights $\Delta_L = h_L + \bar{h}_L < 2$ and $\Delta_{L''} = h_{L''} + \bar{h}_{L''} < 2$, and spins $s_L = h_L - \bar{h}_L$ and $s_{L''} = h_{L''} - \bar{h}_{L''}$ respectively. Depending on the theory, there might be several operators contributing at the lowest order. In this case, we will have to sum over all possible operators. We simplify our notation by keeping the sum over all possible operators implicit. Note that the factors coming from spins of Ψ and Φ becomes

$$\prod_{k=1}^{n-1} e^{-4\pi i k (s_\Psi - s_\Phi) / n} = e^{-2\pi i (n-1)(s_\Psi - s_\Phi)} \quad (2.13)$$

¹Otherwise the lowest order term would come from stress tensor.

²Note that $C_{\Psi\Phi^*}^{L'} = (C_{\Psi^*\Phi}^{L'})^*$, but in a unitary CFT, the OPE coefficients are real.

which precisely cancels out the that of equation (2.9). Therefore, the analytic correlator we need to compute to find the relative entropy is

$$\mathcal{F}_n = \frac{(1 + (2 \sin(\pi x/n))^{2\Delta_L} f(n) + \dots)}{(D_n(x)^{n-1} [1 + (2 \sin(\pi x/n))^{2\Delta_L} g(n) + \dots])} \quad (2.14) \quad \{\text{eq:F_n}\}$$

$$D_n(x) = \left(\frac{n \sin(\pi x/n)}{\sin(\pi x)} \right)^{2(\Delta_\Psi - \Delta_\Phi)}, \quad D_1(x) = 1, \quad (2.15) \quad \{\text{Dn}\}$$

where the leading correction $f(n)$ and $g(n)$ are given by the sums in (2.18) and (2.23), respectively, over OPE coefficients and light operator correlators on the replica manifold. First, observe that $D_n(x)^{n-1}$ term does not contribute to relative entropy:

$$-\partial_n ((n-1) \log D_n(x))_{n \rightarrow 1} = 0, \quad (2.16)$$

Hence, we drop it from the denominator, and

$$\begin{aligned} S(\rho_\Psi \| \rho_\Phi) &= \partial_n \left[\log \frac{1 + \xi^{2\Delta_L} f(n) + \dots}{1 + \xi^{2\Delta_L} g(n) + \dots} \right]_{n=1} = \xi^{2\Delta_L} \partial_n [f(n) - g(n)]_{n=1} + \dots \\ \xi &= \frac{2\pi x}{n} \ll 1. \end{aligned} \quad (2.17)$$

where

$$\begin{aligned} f(n) &= e^{i\pi s_L} \frac{(C_\Psi^L)^2}{2} \sum_{k \neq k'=0}^{n-1} e^{2\pi i s_L (k+k')/n} \langle \mathcal{O}_L(e^{2\pi i k/n}) \mathcal{O}_L(e^{2\pi i k'/n}) \rangle \\ &= \frac{e^{i\pi s_L} (C_\Psi^L)^2}{2} \sum_{k \neq k'=0}^{n-1} e^{-i\pi s_L (2i \sin(\pi(k-k')/n))^{-2h_L} (-2i \sin(\pi(k-k')/n))^{-2\bar{h}_L}} \\ &= \frac{(C_\Psi^L)^2}{2} \sum_{k \neq k'=0}^{n-1} \frac{1}{(2 \sin(\pi|k-k'|/n))^{2\Delta_L}}. \end{aligned} \quad (2.18) \quad \{\text{fnexpression1}\}$$

The analytic continuation

$$S_2(n, \Delta_L) = \frac{n}{2} \sum_{k=1}^{n-1} (2 \sin(\pi k/n))^{-2\Delta_L}, \quad S_2(1, \Delta_L) = 0, \quad (2.19)$$

was discussed in [3]³

$$S'_2(1, \Delta) = \frac{2^{-2\Delta} \sqrt{\pi} \Gamma(\Delta+1)}{4\Gamma(\Delta+3/2)}. \quad (2.20)$$

Therefore,

$$f'(1) = (C_\Psi^L)^2 \frac{2^{-2\Delta} \sqrt{\pi} \Gamma(\Delta+1)}{4\Gamma(\Delta+3/2)} \quad (2.21)$$

³Note that our notation has an extra factor $2^{-2\Delta_L}$.

In a general CFT, there could be several inequivalent choices of the lightest primary operator $\mathcal{O}_L$ that contribute at the same order. In this case, the leading correction $f(n)$ would be given by the sum over all such operators, that following [3] we denote by

$$f(n) = (C_\Psi^L)^2 S_2(n, \Delta) \quad (2.22)$$

The denominator in $\mathcal{F}_n$ depends on how Δ_L and $\Delta_{L''}$ compare. We first consider the case, $\Delta_L = \Delta_{L''}$. Then,

$$\begin{aligned} g(n) &= C_\Psi^L C_\Phi^L \sum_{k=1}^{n-1} \frac{1}{(2 \sin(\pi k/n))^{2\Delta_L}} + \frac{(C_\Phi^L)^2}{2} \sum_{k \neq k'=1}^{n-1} \frac{1}{(2 \sin(\pi |k - k'|/n))^{2\Delta_L}}, \\ &= (C_\Psi^L - C_\Phi^L) C_\Phi^L \sum_{k=1}^{n-1} \frac{1}{(2 \sin(\pi k/n))^{2\Delta_L}} + (C_\Phi^L)^2 S_2(n, \Delta_L) \end{aligned} \quad (2.23) \quad \{\text{gnw1}\}$$

Therefore, expanding in small x we find

$$\log \mathcal{F}_n(x) = (2\pi x/n)^{2\Delta_L} \left(((C_\Psi^L)^2 - (C_\Phi^L)^2) S_2(n) - (C_\Psi^L - C_\Phi^L) C_\Phi^L \frac{2S_2(n, \Delta_L)}{n} \right) + \dots \quad (2.24)$$

Since $S_2(1, \Delta) = 0$ we find

$$\begin{aligned} S(\rho_\Psi \| \rho_\Phi) &= (C_\Psi^L - C_\Phi^L)^2 S_2'(1, \Delta_L) (2\pi x)^{2\Delta_L} + \dots \\ &= (C_\Psi^L - C_\Phi^L)^2 \frac{\sqrt{\pi} \Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} (\pi x)^{2\Delta_L} + \dots \end{aligned} \quad (2.25)$$

Next, consider the case $\Delta_{L''} < \Delta_L$. In this case, only the denominator of $\mathcal{F}_n$ contributes to the lowest order, and we have

$$\begin{aligned} g(n) &= \frac{(C_\Phi^{L''})^2}{2} \sum_{k \neq k'=1}^{n-1} \frac{1}{(2 \sin(\pi |k - k'|/n))^{2\Delta_{L''}}}, \\ &= - (C_\Phi^{L''})^2 \sum_{k=1}^{n-1} \frac{1}{(2 \sin(\pi k/n))^{2\Delta_{L''}}} + (C_\Phi^{L''})^2 S_2(n, \Delta_{L''}) \end{aligned} \quad (2.26) \quad \{\text{gnw1p}\}$$

and since we assumed $\Delta_{L''}$

$$\begin{aligned} \log \mathcal{F}_n(x) &= (2\pi x/n)^{2\Delta_{L''}} \left(- (C_\Phi^{L''})^2 S_2(n, \Delta_{L''}) + (C_\Phi^{L''})^2 \frac{2S_2(n, \Delta_{L''})}{n} \right) + \dots \\ \implies S(\rho_\Psi \| \rho_\Phi) &= (C_\Phi^{L''})^2 S_2'(1, \Delta_{L''}) (2\pi x)^{2\Delta_{L''}} + \dots \\ &= (C_\Phi^{L''})^2 \frac{\sqrt{\pi} \Gamma(\Delta_{L''} + 1)}{4\Gamma(\Delta_{L''} + 3/2)} (\pi x)^{2\Delta_{L''}} + \dots \end{aligned} \quad (2.27)$$

Finally, consider the case $\Delta_{L''} > \Delta_L$, then only the numerator contributes to the lowest order to $\mathcal{F}_n$, and we have

$$\begin{aligned} S(\rho_\Psi \| \rho_\Phi) &= (C_\Psi^L)^2 S_2'(1, \Delta_L) (2\pi x)^{2\Delta_L} + \dots \\ &= (C_\Psi^L)^2 \frac{\sqrt{\pi} \Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} (\pi x)^{2\Delta_L} + \dots \end{aligned} \quad (2.28)$$

The lesson to learn here is that when $\Delta_{L''} < \Delta_L$, to the lowest order the calculation is the same as $S(\rho_0 \| \rho_\Phi)$, and when $\Delta_{L''} > \Delta_L$, to the lowest order, it is the same as $S(\rho_\Psi \| \rho_0)$, with ρ_0 the vacuum reduced state. As a result, in the remainder of this draft, with no loss of generality, we focus on the case $\Delta_L = \Delta_{L''}$ to simplify our notation.

Stress tensors T and $\bar{T}$ contribute to the OPE expansion with weights $(2, 0)$ and $(0, 2)$. In theories with conserved currents J and $\bar{J}$ (e.g. Kac-Moody algebras), these currents have weights $(1, 0)$ and $(0, 1)$. Therefore, to the lowest order in small x expansion, in any 2d CFT with a relevant deformation $\Delta_L < 1$, we can ignore these corrections.

In the small interval limit, $x \rightarrow 0^+$, the relative entropy of primary states of dimension $(h_\Psi, \bar{h}_\Psi)$ and $(h_\Phi, \bar{h}_\Phi)$ is universal and given by [2.34] where Δ_L is the scaling dimension of the lightest spinless primary in the spectrum that appears in the OPE.

If we consider the case where $\Delta_L > 2$, then to the lowest order we will have the contribution of T and $\bar{T}$ in the OPE that has scaling dimension 2. So we need the following OPEs,

$$\begin{aligned} \frac{\Psi(z_k^+) \Psi^*(z_k^-)}{\langle \Psi(z^+) \Psi^*(z^-) \rangle} &= e^{-4\pi i k s_\Psi / n} \left(\frac{\sin(\pi x)}{\sin(\pi x/n)} \right)^{2\Delta_\Psi} \\ &\times \left(1 + \frac{2h_\Psi}{c} (2\sin(\pi x/n))^2 e^{2\pi i(2k/n+1/2)} T(e^{2\pi i k/n}) \right. \\ &\quad \left. + \frac{2\bar{h}_\Psi}{c} (2\sin(\pi x/n))^2 e^{-2\pi i(2k/n+1/2)} \bar{T}(e^{2\pi i k/n}) + \dots \right), \end{aligned} \quad (2.29)$$

$$\begin{aligned} \frac{\Phi(z_k^+) \Phi^*(z_k^-)}{\langle \Phi(z^+) \Phi^*(z^-) \rangle} &= e^{-4\pi i k s_\Phi / n} \left(\frac{\sin(\pi x)}{\sin(\pi x/n)} \right)^{2\Delta_\Phi} \\ &\times \left(1 + \frac{2h_\Phi}{c} (2\sin(\pi x/n))^2 e^{2\pi i(2k/n+1/2)} T(e^{2\pi i k/n}) \right. \\ &\quad \left. + \frac{2\bar{h}_\Phi}{c} (2\sin(\pi x/n))^2 e^{-2\pi i(2k/n+1/2)} \bar{T}(e^{2\pi i k/n}) + \dots \right), \end{aligned} \quad (2.30)$$

where the $\dots$ refer to the descendants of $T, \bar{T}$, the primary $\mathcal{O}_L$ and its descendants. The two-point function of stress tensor is given by,

$$\langle T(z_{k'}) T(z_k) \rangle = e^{-4\pi i((k'+k)/n+1/2)} \frac{c}{2(2\sin(\pi(k'-k)/n))^4} \quad (2.31)$$

$$\langle \bar{T}(\bar{z}_{k'}) \bar{T}(\bar{z}_k) \rangle = e^{4\pi i((k'+k)/n+1/2)} \frac{c}{2(2\sin(\pi(k'-k)/n))^4} \quad (2.32)$$

It is clear that the phase term from the OPE gets cancelled from the phase in the two-point function. We can now just substitute $C_\Psi = \frac{2h_\Psi}{c}$ and $C_\Phi = \frac{2h_\Phi}{c}$ in the final answer, with a factor of $c/2$ coming from the two-point function of stress tensor. By adding the contribution from both the holomorphic and anti-holomorphic part, we get,

$$\begin{aligned} S(\rho_\Psi \|\rho_\Phi) &= (\pi x)^4 \frac{c\sqrt{\pi}\Gamma(3)}{8\Gamma(7/2)} \left[\left(\frac{2h_\Psi}{c} - \frac{2h_\Phi}{c} \right)^2 + \left(\frac{2\bar{h}_\Psi}{c} - \frac{2\bar{h}_\Phi}{c} \right)^2 \right] + \dots \\ &= (\pi x)^4 \frac{8}{15c} ((h_\Psi - h_\Phi)^2 + (\bar{h}_\Psi - \bar{h}_\Phi)^2) + \dots \end{aligned} \quad (2.33)$$

In the small interval limit, $x \rightarrow 0^+$, the relative entropy of primary states of dimension $(h_\Psi, \bar{h}_\Psi)$ and $(h_\Phi, \bar{h}_\Phi)$ is universal and given by

$$S(\rho_\Psi \|\rho_\Phi) = \begin{cases} \sum_L \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + \frac{3}{2})} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L} + \dots & \text{if } \Delta_L < 2, \\ \frac{8}{15} \left[\frac{(h_\Psi - h_\Phi)^2}{c} + \frac{(\bar{h}_\Psi - \bar{h}_\Phi)^2}{\bar{c}} \right] (\pi x)^4 + \dots & \text{if } \Delta_L > 2. \end{cases} \quad (2.34) \quad \{\text{Ssmallx}\}$$

Note that we have included a sum over L because there can be multiple scalar primaries of the same lowest dimension Δ_L .

Additivity of relative entropy: Assume we have a theory that is the tensor product of left and right movers, so that any Ψ can be decomposed into $\Psi_L \otimes \Psi_R$ with dimensions, respectively, $(h_\Psi, 0)$ and $(0, \bar{h}_\Psi)$. Then, for the second case $\Delta_L > 2$, the relative entropy is additive, meaning that at least to the lowest order, we have $\rho_\Psi \simeq \rho_{\Psi_L} \otimes \rho_{\Psi_R}$, and similarly for Φ . In the case $\Delta_L < 2$, we continue to have the symmetry if the chiral and anti-chiral copies of the theory are isomorphic.

Light Primaries: We consider light primaries, such that the leading order contribution comes from the two and three-point function of the primaries. In other words, we focus on $\Delta_L < 4/3$ so that the three-point function of the primary contributes at a smaller order than the two-point function of stress tensor.

It turns out that that the three-point function $\langle \mathcal{O}_L \mathcal{O}_L \mathcal{O}_L \rangle$ it is also independent of spin. The three-point function contribution from the numerator is given by,

$$(C_\Psi^L)^3 C_{LLL} (\pi x)^{3\Delta_L} \mathcal{T}_{n,LLL}, \quad (2.35)$$

where

$$\mathcal{T}_{n,LLL} \equiv \frac{1}{6} \sum_{k_1 \neq k_2 \neq k_3=0}^{n-1} \frac{1}{(\sin(\pi(k_3 - k_2)/n) \sin(\pi(k_2 - k_1)/n) \sin(\pi(k_3 - k_1)/n))^{\Delta_L}}. \quad (2.36)$$

The contribution of it to the relative entropy is given by $\partial_n \mathcal{T}_{n,LLL}|_{n=1}$. The analytic continuation of the three-point function has already been computed in [4] using modular flow, and

is given by,

$$\partial_n \mathcal{T}_{n,LLL}|_{n=1} = -\frac{2^{3\Delta_L} \Gamma\left(\frac{1+\Delta_L}{2}\right)^3}{12\pi \Gamma\left(\frac{3+3\Delta_L}{2}\right)} \quad (2.37)$$

This analytic continuation is the special case of the three-point function of three different primaries. We have performed its analytic continuation in Appendix G. The total contribution from the denominator consists of two classes of three-point functions. The first arises from inserting the lightest primary in each of the three $\Phi\Phi^*$ OPEs, and therefore is proportional to C_Φ^3 .

$$\begin{aligned} \sum_{k_1 \neq k_2 \neq k_3=1}^{n-1} C_{LLL} (C_\Phi^L)^3 (\pi x)^{3\Delta_L} e^{i\pi s_L(2(k_1+k_2+k_3)/n+3/2)} \langle \mathcal{O}_L(z_{k_3}) \mathcal{O}_L(z_{k_2}) \mathcal{O}_L(z_{k_1}) \rangle \\ = C_{LLL} (C_\Phi^L)^3 \frac{(\pi x)^{3\Delta_L}}{6} \sum_{k_1 \neq k_2 \neq k_3=1}^{n-1} \frac{1}{(\sin(\pi(k_3 - k_2)/n) \sin(\pi(k_2 - k_1)/n) \sin(\pi(k_3 - k_1)/n))^{\Delta_L}} \end{aligned} \quad (2.38)$$

$$(2.39)$$

Another contribution arises from inserting the lightest primary in two $\Phi\Phi^*$ OPEs and one from $\Psi\Psi^*$ OPE, which is proportional to $C_\Psi C_\Phi^2$.

$$\begin{aligned} \sum_{k_1 \neq k_2=1}^{n-1} C_{LLL} C_\Psi^L (C_\Phi^L)^2 (\pi x)^{3\Delta_L} e^{i\pi s_L(2(k_1+k_2)/n+3/2)} \langle \mathcal{O}_L(z_{k_2}) \mathcal{O}_L(z_{k_1}) \mathcal{O}_L(z_0) \rangle \\ = C_{LLL} C_\Psi^L (C_\Phi^L)^2 \frac{(\pi x)^{3\Delta_L}}{2} \sum_{k_1 \neq k_2=1}^{n-1} \frac{1}{(\sin(\pi(k_2 - k_1)/n) \sin(\pi k_2/n) \sin(\pi k_1/n))^{\Delta_L}} \end{aligned} \quad (2.40)$$

Using translation invariance, this sum can be written in terms of $\mathcal{T}_n$,

$$\frac{1}{2} \sum_{k_1 \neq k_2=1}^{n-1} \frac{1}{(\sin(\pi(k_2 - k_1)/n) \sin(\pi k_2/n) \sin(\pi k_1/n))^{\Delta_L}} = \frac{3}{n} \mathcal{T}_{n,LLL}. \quad (2.41)$$

Then the contributions to the denominator would be of the form,

$$C_{LLL} \left((3/n) C_\Psi^L (C_\Phi^L)^2 + \left(1 - \frac{3}{n}\right) (C_\Phi^L)^3 \right) \mathcal{T}_{n,LLL} (\pi x)^{3\Delta_L} \quad (2.42)$$

The net contribution to relative entropy from the three-point function would be,

$$\begin{aligned}
S(\rho_\Psi \parallel \rho_\Phi) &= \sum_L \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L} \\
&\quad + \sum_L C_{LLL} \partial_n \left[\left((C_\Psi^L)^3 - (C_\Phi^L)^3 - 3/n (C_\Psi^L (C_\Phi^L)^2 - (C_\Phi^L)^3) \right) \mathcal{T}_{n,LLL} \right] \Big|_{n=1} (\pi x)^{3\Delta_L} + \dots
\end{aligned} \tag{2.43}$$

$$\begin{aligned}
&= \sum_L \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L} \\
&\quad + \sum_L C_{LLL} (C_\Psi^L - C_\Phi^L)^2 (C_\Psi^L + 2C_\Phi^L) \partial_n \mathcal{T}_{n,LLL} |_{n=1} (\pi x)^{3\Delta_L} + \dots
\end{aligned} \tag{2.44}$$

$$\begin{aligned}
&= \sum_L \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L} \\
&\quad + \sum_L \frac{2^{3\Delta_L} \Gamma\left(\frac{1+\Delta_L}{2}\right)^3}{12\pi\Gamma\left(\frac{3+3\Delta_L}{2}\right)} C_{LLL} (C_\Psi^L - C_\Phi^L)^2 (C_\Psi^L + 2C_\Phi^L) (\pi x)^{3\Delta_L} + \dots
\end{aligned} \tag{2.45}$$

where in the last step, we analytic continued $\partial_n \mathcal{T}_{n,LLL}|_{n=1}$ (see [4] and Appendix G for details). We can generalize this further for unequal dimensions of the primaries, i.e. $\Delta_L \neq \Delta_M \neq \Delta_N$. Again, we focus on light primaries, i.e. $\Delta_L < 2$ and $\Delta_L + \Delta_M + \Delta_N < 4$, thus ignoring the contribution from Stress tensor. We obtain,

$$\begin{aligned}
\log \mathcal{F}|_n &= \sum_L (C_\Psi^L - C_\Phi^L)^2 S_2(n, \Delta_L) (2\pi x)^{2\Delta_L} \\
&\quad + \sum_{LMN} \mathcal{T}_{n,LMN} (\pi x)^{\Delta_L + \Delta_M + \Delta_N} C_{LMN} \left(C_\Psi^L C_\Psi^M C_\Psi^N \right. \\
&\quad \left. - \frac{1}{n} (C_\Psi^L C_\Phi^M C_\Phi^N + C_\Phi^L C_\Psi^M C_\Phi^N + C_\Phi^L C_\Phi^M C_\Psi^N) \right. \\
&\quad \left. - \left(1 - \frac{3}{n} \right) C_\Phi^L C_\Phi^M C_\Phi^N \right) + \dots
\end{aligned} \tag{2.46}$$

$$\begin{aligned}
\Rightarrow S(\rho_\Psi \parallel \rho_\Phi) &= \sum_L \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L} \\
&\quad + \sum_{LMN} \partial_n \mathcal{T}_{n,LMN} |_{n=1} (\pi x)^{\Delta_L + \Delta_M + \Delta_N} C_{LMN} \left[C_\Psi^L C_\Psi^M C_\Psi^N \right. \\
&\quad \left. - (C_\Psi^L C_\Phi^M C_\Phi^N + C_\Phi^L C_\Psi^M C_\Phi^N + C_\Phi^L C_\Phi^M C_\Psi^N) \right. \\
&\quad \left. + 2C_\Phi^L C_\Phi^M C_\Phi^N \right] + \dots
\end{aligned} \tag{2.47}$$

where C_{LMN} is the three-point OPE coefficient. We have performed the analytic continuation of the three-point function of three different primaries in Appendix G. For simplicity of

notation, we define the following,

$$f_{LMN} \equiv -\frac{\sqrt{\pi}}{12} \frac{\Gamma(\Delta_L + 1)\Gamma(\Delta_M + 1)\Gamma(\Delta_N + 1)}{\Gamma\left(1 + \frac{\Delta_L + \Delta_M - \Delta_N}{2}\right)\Gamma\left(1 + \frac{\Delta_M + \Delta_N - \Delta_L}{2}\right)\Gamma\left(1 + \frac{\Delta_N + \Delta_L - \Delta_M}{2}\right)\Gamma\left(\frac{\Delta_L + \Delta_M + \Delta_N + 3}{2}\right)}, \quad (2.48)$$

$$f_{LL} \equiv \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma\left(\Delta_L + \frac{3}{2}\right)} = -3f_{LL0}, \quad (2.49)$$

$$\mathcal{B}_{LMN}^{\Psi|\Phi} = C_{\Psi}^L C_{\Psi}^M C_{\Psi}^N - C_{\Psi}^L C_{\Phi}^M C_{\Phi}^N - C_{\Phi}^L C_{\Psi}^M C_{\Phi}^N - C_{\Phi}^L C_{\Phi}^M C_{\Psi}^N + 2C_{\Phi}^L C_{\Phi}^M C_{\Phi}^N. \quad (2.50)$$

In terms of the newly defined variables, we obtain,

$$S(\rho_{\Psi}||\rho_{\Phi}) = \sum_L f_{LL}(C_{\Psi}^L - C_{\Phi}^L)^2(\pi x)^{2\Delta_L} + \sum_{LMN} f_{LMN} C_{LMN} \mathcal{B}_{LMN}^{\Psi|\Phi}(\pi x)^{\Delta_L + \Delta_M + \Delta_N} + \dots \quad (2.51)$$

Heavy primaries: Consider the case where $\Delta_L > 3$, which would correspond to primaries whose two-point function will contribute at a higher order than the three-point function or two-point function of descendants of stress tensor. Then the leading contribution only from the two-point, three-point and descendants of stress tensors. We thus need to consider the following OPE (we drop the phases coming from spin for simplicity, as they don't contribute to the relative entropy),

$$\begin{aligned} \frac{\Psi(z_k^+)\Psi(z_k^-)}{\langle\Psi(z^+)\Psi(z^-)\rangle} &= \left(\frac{\sin(\pi x)}{\sin(\pi x/n)}\right)^{2\Delta_{\Psi}} \left(1 + \frac{2h_{\Psi}}{c}(2\sin(\pi x/n))^2 T(z_k)\right. \\ &\quad \left.+ (2\sin(\pi x/n))^4 \left[\frac{h_{\Psi}}{20c}\partial^2 T(z_k) + \frac{2h_{\Psi}(5h_{\Psi} + 1)}{c(5c + 22)}\left(:TT:(z_k) - \frac{3}{10}\partial^2 T(z_k)\right)\right] + \dots\right) \end{aligned} \quad (2.52)$$

where $\dots$ include the anti-holomorphic and the higher order terms. This is obtained by using the end-point OPEs given in [5],[6], and then shifting the operators to write the mid-point OPEs. If we denote the level-4 operator by

$$\Lambda \equiv :TT: - \frac{3}{10}\partial^2 T, \quad (2.53)$$

then the two-point function $\langle T\Lambda \rangle$ vanishes as they are linearly independent [7]. Note that, we ignored the level-6 insertions, since the contribution of it to relative entropy would vanish. It vanishes precisely because of the $(n-1)$ factor sitting in front of it after taking the log of $\mathcal{F}_n$. Thus, at the order x^6 , we are left with two-point function of $\langle T\partial^2 T \rangle$ and three-point function of $\langle TTT \rangle$, with their anti-holomorphic components.

Naively if we expand the two-point function contribution, which goes as $\sin(\pi x)^4$, to obtain the $O(x^6)$ contribution, we obtain,

$$S(\rho_{\Psi}||\rho_{\Phi})\Big|_{\langle TT \rangle} = \frac{8}{15c}(h_{\Psi} - h_{\Phi})^2(\pi x)^4 - \frac{16}{45c}(h_{\Psi} - h_{\Phi})^2(\pi x)^6 + O(x^8). \quad (2.54)$$

Since, $\langle T(z)T(w) \rangle = \frac{c}{2(z-w)^4}$, then we have,

$$\langle T(z)\partial^2 T(w) \rangle = \partial_w^2 \langle T(z)T(w) \rangle \quad (2.55)$$

$$= \partial_w^2 \frac{c}{2(z-w)^4} \quad (2.56)$$

$$= \frac{10c}{(z-w)^6}. \quad (2.57)$$

There are two contributions from this two-point function, precisely because we have $T(z)\partial^2 T(w)$ and $T(w)\partial^2 T(z)$. Then we have,

$$S(\rho_\Psi \| \rho_\Phi) \Big|_{\langle T\partial^2 T \rangle} = 2 \left(\frac{2h_\Psi}{c} - \frac{2h_\Phi}{c} \right) (10c) \left(\frac{h_\Psi}{20c} - \frac{h_\Phi}{20c} \right) (2 \sin(\pi x))^6 S'_2(1, 3) + O(x^8) \quad (2.58)$$

$$= \frac{16}{35c} (h_\Psi - h_\Phi)^2 (\pi x)^6 + O(x^8). \quad (2.59)$$

where in last step we used $S'_2(1, 3) = 1/280$.

And finally, we have the contribution from the three-point function $\langle TTT \rangle$, given by,

$$S(\rho_\Psi \| \rho_\Phi) \Big|_{\langle TTT \rangle} = -\frac{2^6 \Gamma(\frac{3}{2})^3}{12\pi \Gamma(\frac{9}{2})} c \left(\frac{2h_\Psi}{c} - \frac{2h_\Phi}{c} \right)^2 \left(\frac{2h_\Psi}{c} + 2\frac{2h_\Phi}{c} \right) (\pi x)^6 + O(x^8) \quad (2.60)$$

$$= -\frac{256}{315} \left[\frac{(h_\Psi - h_\Phi)^2 (h_\Psi + 2h_\Phi)}{c^2} \right] (\pi x)^6 + O(x^8) \quad (2.61)$$

where we used $C_{TTT} = c$, $C_{\bar{T}\bar{T}\bar{T}} = \bar{c}$, and $\langle TTT \rangle = c / (z_{12} z_{23} z_{31})^2$, $\langle \bar{T}\bar{T}\bar{T} \rangle = \bar{c} / (\bar{z}_{12} \bar{z}_{23} \bar{z}_{31})^2$. Thus, the full contribution up to the order x^6 is given by,

$$\begin{aligned} S(\rho_\Psi \| \rho_\Phi) &= \frac{8}{15} \left[\frac{(h_\Psi - h_\Phi)^2}{c} + \frac{(\bar{h}_\Psi - \bar{h}_\Phi)^2}{\bar{c}} \right] (\pi x)^4 \\ &+ \frac{32}{315} \left[\frac{(h_\Psi - h_\Phi)^2}{c} \left(1 - \frac{8(h_\Psi + 2h_\Phi)}{c} \right) + \frac{(\bar{h}_\Psi - \bar{h}_\Phi)^2}{\bar{c}} \left(1 - \frac{8(\bar{h}_\Psi + 2\bar{h}_\Phi)}{\bar{c}} \right) \right] (\pi x)^6 + \dots \end{aligned} \quad (2.62)$$

Comments on Table 1: Note that, we did not include the contribution from the first level descendant, since they vanish explicitly in our mid-point OPEs. Also, if we consider the case $\Delta_L = 1$, for example coherent states, one has to include the four-point function $\langle \mathcal{O}_L \mathcal{O}_L \mathcal{O}_L \mathcal{O}_L \rangle$ into account if one wants to go beyond three-point-function. In other words, one has to include $\langle TT \rangle$, $\langle T \mathcal{O}_L \mathcal{O}_L \rangle$, $[\mathcal{O}_L \mathcal{O}_L]_{\text{level } 2}$ and $\langle \mathcal{O}_L \mathcal{O}_L \mathcal{O}_L \mathcal{O}_L \rangle$, all of which have the same dimension. The level 2 descendants ($[\mathcal{O}_L \mathcal{O}_L]_{\text{level } 2}$) include terms like $\langle L_{-1}^2 \mathcal{O}_L \mathcal{O}_L \rangle$ and $\langle L_{-2} \mathcal{O}_L \mathcal{O}_L \rangle$, where L_{-n} are Virasoro generators. Note that, there is no contribution from level 3 descendant since it vanishes from the midpoint OPEs. Similarly, it has been shown in the Appendix J that all the odd level descendants vanish. If we consider level 4 descendant, for example, $\langle L_{-4} \mathcal{O}_L \mathcal{O}_L \rangle$, which has dimension $2\Delta_L + 4$. That will contribute only if $2\Delta_L + 4 < 4\Delta_L$,

Range of Δ_L	$\mathcal{O}_L \mathcal{O}_L$	$\mathcal{O}_L \mathcal{O}_L \mathcal{O}_L$	TT	$[\mathcal{O}_L \mathcal{O}_L]_{\text{sub.}}, [\mathcal{O}_L \mathcal{O}_L]_{\text{level 2}}, T\mathcal{O}_L \mathcal{O}_L$	$[TT]_{\text{sub.}}, T\partial^2 T, TTT$
$0 < \Delta_L \leq 1$	✓	✓	×	×	×
$1 < \Delta_L \leq \frac{3}{2}$	✓	✓	✓	✓	×
$\frac{3}{2} < \Delta_L < \frac{8}{3}$	✓	✓	✓	✓	✓
$\frac{8}{3} \leq \Delta_L < 3$	✓	×	✓	✓	✓
$3 \leq \Delta_L < 4$	✓	×	✓	×	✓
$4 \leq \Delta_L$	×	×	✓	×	✓

Table 1. Contributions retained in the small-interval expansion up to the first order at which an un-computed four-point contribution enters. A cross marked entry does not imply that the corresponding correlator vanishes, but only that it contributes beyond the order of four point function of stress tensor or the primary. Antiholomorphic counterparts are implicit.

{tab:small-x-co

which says $\Delta_L > 2$. However, the four point function of stress tensor is already of dimension 8. Comparing it with the four point function of stress tensor, $2\Delta_L + 4 < 8$, gives $\Delta_L < 2$. But then, it will have dimension larger than $\langle \mathcal{O}_L \mathcal{O}_L \mathcal{O}_L \mathcal{O}_L \rangle$. Then, we can again ignore the contribution of level 4 descendant, since, we are restricting our computation till the 4-point function of primaries or the stress tensor. Same argument holds for operators of higher dimensions, such as $\langle L_{-2} \mathcal{O}_L \mathcal{O}_L \mathcal{O}_L \rangle$, etc.

We now collect the contributions to the short-interval expansion that are needed before four-point data enter. In addition to the leading two-point term, there are contributions from three-point functions, from the first subleading term in the two-point conformal family, and from the vacuum Virasoro family. It is useful to organize these terms according to their scaling with the interval size.

Primary contributions: The leading contribution associated with a non-vacuum primary $\mathcal{O}_L$ is

$$\mathcal{S}_{LL}^{[2\Delta_L]} \equiv f_{LL} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L}, \quad (2.63)$$

At the next orders, three distinct primaries can contribute through their three-point function. We denote this contribution by

$$\mathcal{S}_{LMN}^{[\Delta_L + \Delta_M + \Delta_N]} \equiv f_{LMN} C_{LMN} \mathcal{B}_{LMN}^{\Psi|\Phi} (\pi x)^{\Delta_L + \Delta_M + \Delta_N}. \quad (2.64)$$

First conformal-family correction: At order $x^{2\Delta_L+2}$, several contributions associated with the same primary family occur simultaneously. They arise from the first subleading term in the two-point contribution, the level-two descendants of $\mathcal{O}_L$, and the mixed three-point functions $T\mathcal{O}_L \mathcal{O}_L$ and $\bar{T}\mathcal{O}_L \mathcal{O}_L$. Since these terms contribute at the same order, it is natural to combine them.

$$\mathcal{S}_{LL,\text{sub}}^{[2\Delta_L+2]} \equiv -\frac{\Delta_L}{3} f_{LL} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L+2} = \Delta_L f_{LL0} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L+2}. \quad (2.65)$$

$$\mathcal{S}_{LL,\text{desc}}^{[2\Delta_L+2]} \equiv \frac{\Delta_L(\Delta_L+1)}{2\Delta_L+3} f_{LL} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L+2}. \quad (2.66)$$

The two-point and level-two descendant pieces combine to

$$\begin{aligned} \mathcal{S}_{LL,\text{fam}}^{[2\Delta_L+2]} &\equiv \mathcal{S}_{LL,\text{sub}}^{[2\Delta_L+2]} + \mathcal{S}_{LL,\text{desc}}^{[2\Delta_L+2]} \\ &= \frac{\Delta_L^2}{3(2\Delta_L+3)} f_{LL} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L+2}. \end{aligned} \quad (2.67)$$

The remaining contributions at this order arise from $T\mathcal{O}_L\mathcal{O}_L$ and $\bar{T}\mathcal{O}_L\mathcal{O}_L$. The corresponding three-point coefficients are fixed by the conformal Ward identities, $C_{TLL} = h_L$ and $C_{\bar{T}LL} = \bar{h}_L$ [8]. Thus,

$$\mathcal{S}_{TLL}^{[2\Delta_L+2]} \equiv 3f_{TLL} h_L \mathcal{B}_{TLL}^{\Psi|\Phi} (\pi x)^{2\Delta_L+2}, \quad (2.68)$$

$$\mathcal{S}_{\bar{T}LL}^{[2\Delta_L+2]} \equiv 3f_{\bar{T}LL} \bar{h}_L \mathcal{B}_{\bar{T}LL}^{\Psi|\Phi} (\pi x)^{2\Delta_L+2}. \quad (2.69)$$

where

$$f_{TLL} = f_{\bar{T}LL} = -\frac{\sqrt{\pi}\Delta_L\Gamma(\Delta_L+1)}{6\Gamma(\Delta_L+\frac{5}{2})} = -\frac{4\Delta_L}{3(2\Delta_L+3)} f_{LL}. \quad (2.70)$$

We therefore define the complete contribution of the $\mathcal{O}_L$ family at this order by

$$\mathcal{S}_L^{[2\Delta_L+2]} \equiv \mathcal{S}_{LL,\text{fam}}^{[2\Delta_L+2]} + \mathcal{S}_{TLL}^{[2\Delta_L+2]} + \mathcal{S}_{\bar{T}LL}^{[2\Delta_L+2]}. \quad (2.71)$$

The above equation simplifies to,

$$\begin{aligned} \mathcal{S}_L^{[2\Delta_L+2]} &= \frac{\Delta_L f_{LL}}{2\Delta_L+3} (C_\Psi^L - C_\Phi^L) \left[\frac{\Delta_L}{3} (C_\Psi^L - C_\Phi^L) - \frac{8h_L}{c} (h_\Psi(C_\Psi^L + C_\Phi^L) - 2h_\Phi C_\Phi^L) \right. \\ &\quad \left. - \frac{8\bar{h}_L}{\bar{c}} (\bar{h}_\Psi(C_\Psi^L + C_\Phi^L) - 2\bar{h}_\Phi C_\Phi^L) \right] (\pi x)^{2\Delta_L+2} \\ &= \frac{\Delta_L f_{LL}}{2\Delta_L+3} \left[\frac{\Delta_L}{3} (C_\Psi^L - C_\Phi^L)^2 - 4h_L \mathcal{B}_{TLL}^{\Psi|\Phi} - 4\bar{h}_L \mathcal{B}_{\bar{T}LL}^{\Psi|\Phi} \right] (\pi x)^{2\Delta_L+2} \end{aligned} \quad (2.72)$$

where

$$\mathcal{B}_{TLL}^{\Psi|\Phi} = \frac{2}{c} (C_\Psi^L - C_\Phi^L) [h_\Psi(C_\Psi^L + C_\Phi^L) - 2h_\Phi C_\Phi^L] \quad (2.73)$$

$$\mathcal{B}_{\bar{T}LL}^{\Psi|\Phi} = \frac{2}{\bar{c}} (C_\Psi^L - C_\Phi^L) [\bar{h}_\Psi(C_\Psi^L + C_\Phi^L) - 2\bar{h}_\Phi C_\Phi^L] \quad (2.74)$$

Vacuum-family contributions: The first vacuum-family contribution arises from the stress-tensor two-point functions,

$$\mathcal{S}_{\text{vac}}^{[4]} \equiv \frac{8}{15} \left[\frac{(h_\Psi - h_\Phi)^2}{c} + \frac{(\bar{h}_\Psi - \bar{h}_\Phi)^2}{\bar{c}} \right] (\pi x)^4. \quad (2.75)$$

At order x^6 , the corresponding vacuum-family terms combine to

$$\mathcal{S}_{\text{vac}}^{[6]} \equiv \frac{32}{315} \left[\frac{(h_\Psi - h_\Phi)^2}{c} \left(1 - \frac{8(h_\Psi + 2h_\Phi)}{c} \right) + \frac{(\bar{h}_\Psi - \bar{h}_\Phi)^2}{\bar{c}} \left(1 - \frac{8(\bar{h}_\Psi + 2\bar{h}_\Phi)}{\bar{c}} \right) \right] (\pi x)^6. \quad (2.76)$$

Combining all of the above contribution, we finally obtain,

$$\begin{aligned} S(\rho_\Psi \| \rho_\Phi) = & \sum_{\substack{L \\ 2\Delta_L < \gamma}} \mathcal{S}_{LL}^{[2\Delta_L]} + \sum_{\substack{\text{ord } L, M, N \\ \Delta_L + \Delta_M + \Delta_N < \gamma}} \mathcal{S}_{LMN}^{[\Delta_L + \Delta_M + \Delta_N]} + \Theta(\gamma - 4) \mathcal{S}_{\text{vac}}^{[4]} \\ & + \sum_{\substack{L \\ 2\Delta_L + 2 < \gamma}} \mathcal{S}_L^{[2\Delta_L + 2]} + \Theta(\gamma - 6) \mathcal{S}_{\text{vac}}^{[6]} + O(x^\gamma). \\ & \gamma = \min(4\Delta_*, 8) \\ & \Delta_* = \min\{\Delta_L\} \end{aligned}$$

where Θ is the Heaviside theta function. Here Δ_* denotes the smallest scaling dimension among the non-vacuum primaries appearing in the relevant diagonal OPEs. It need not be relevant operator; if $\Delta_* > 2$, the truncation is set by the vacuum-family contribution at the order x^8 . This pure-state expansion above will provide the template for the generalization to mixtures considered below.

Compute $S(\rho_\Psi \| \rho_\Phi) - S(\rho_\Phi \| \rho_\Psi) = \Delta \hat{K}_\Phi$ this is proportional to $(C_\Phi - C_\Psi)^3$ and due to the $\Delta \hat{K}_\Phi$.

$$S(\rho_\Psi \| \rho_\Phi) - S(\rho_\Phi \| \rho_\Psi) = \sum_L \frac{2^{3\Delta_L} \Gamma\left(\frac{1+\Delta_L}{2}\right)^3}{12\pi \Gamma\left(\frac{3+3\Delta_L}{2}\right)} C_{LLL} (C_\Psi^L - C_\Phi^L)^3 (\pi x)^{3\Delta_L} + \dots \quad (2.77)$$

$$= (\Delta K_\Psi - (S_\Psi - S_\Phi)) - (\Delta K_\Phi - (S_\Phi - S_\Psi)) \quad (2.78)$$

$$= (\Delta K_\Psi - \Delta K_\Phi) - 2(S_\Psi - S_\Phi) \quad (2.79)$$

Consider the generalization of the calculation above in the Heavy limit $\Delta_\Phi, \Delta_\Psi \gg 1$ with $\Delta_\Phi/c \sim \Delta_\Psi/c = O(1)$. In this limit, the expectation is that the correlation function is dominated by the Virasoro identity block, and is often said to be holographic. In this limit, we can compute the relative entropy above non-perturbatively using Zamolodchikov's exponentiation of conformal blocks. However, this is surprising because even though ΔS can be calculated holographically, there is no known way of computing $\Delta \hat{K}_\Phi$ holographically. What happens to the assumption of sparse spectrum?

Another very exciting observation about the 2d Ising CFT is the following: since $C_{\epsilon\epsilon}^\epsilon = C_{\epsilon\epsilon}^\sigma = 0$, the OPE of $[\epsilon][\epsilon] \sim [1]$, which means the correlation functions pick up contribution only from the Virasoro identity block. Does this mean that $S(\rho_I \| \rho_\epsilon)$ can be computed

holographically? For example, in holography, if we used gravity to compute the 2n-point functions associated with $S(\rho_I||\rho_\Phi)$ for any holographic state ρ_Φ , up to a bunch of Φ -dependent constants we can figure out the x dependence should exactly match that of the Ising CFT $S(\rho_I||\rho_\epsilon)$.

If there were two Virasoro primaries $[\epsilon_1]$ and $[\epsilon_2]$ that each squared to identity, then we would have had a similar conclusion for relative entropy $S(\rho_1||\rho_2)$, but it turns out this never happens in minimal models.

In fact, every time we have primaries with the property

2.2.1 Some examples

The short-interval expansion derived above can be checked against several known exact results in the free boson and Ising CFT. We collect these checks here, emphasizing only the terms that are controlled by the two- and three-point structures retained in our expansion.

Chiral vertex operators: In massless free boson theory, chiral coherent states V_α and $V_{i\partial\phi}$ are eigenstates of the Hamiltonian with eigenvalues $(\alpha^2/2, 0)$ and $(\beta^2/2, 0)$ respectively. The lowest-dimension primary field that appears in the OPE of a pair of V_α is the holomorphic and anti-holomorphic current. Focussing on the holomorphic sector, the current $J = i(\partial\varphi)$ has dimension $(1, 0)$, and $C_{\alpha\alpha}^{i\partial\varphi} = -\alpha$. Therefore,

$$S(\rho_\alpha||\rho_\beta) = \frac{(\alpha - \beta)^2(\pi x)^2}{3} + O(x^4) \quad (2.80)$$

which matches the expansion of

$$S(\rho_\alpha||\rho_\beta) = (\alpha - \beta)^2(1 - \pi x \cot(\pi x)) = \frac{(\alpha - \beta)^2(\pi x)^2}{3} + O(x^4) \quad (2.81)$$

Non-chiral vertex operators: This is precisely an example of additivity of relative entropy, which was discussed earlier. The algebra of free massless boson is a tensor product of the $U(1)$ algebra $U(1) \times U(1)$, with generators $(J, \bar{J})$ and conformal weights $(h_\alpha, \bar{h}_\alpha)$. A general vertex operator is given by

$$V_{\alpha, \bar{\alpha}}(z, \bar{z}) = (e^{i\alpha\varphi(z)} e^{i\bar{\alpha}\bar{\varphi}(\bar{z})}) \quad (2.82)$$

with dimesnion $(\alpha^2/2, \bar{\alpha}^2/2)$. The n -point function of these correlators factor

$$\langle V_{\alpha_1, \bar{\alpha}_1} \cdots V_{\alpha_n, \bar{\alpha}_n} \rangle = \langle V_{\alpha_1} \cdots V_{\alpha_n} \rangle \langle V_{\bar{\alpha}_1} \cdots V_{\bar{\alpha}_n} \rangle \quad (2.83)$$

and as a result, their relative entropies factor

$$S(\rho_{\alpha, \bar{\alpha}}||\rho_{\beta, \bar{\beta}}) = S(\rho_\alpha||\rho_\beta) + S(\rho_{\bar{\alpha}}||\rho_{\bar{\beta}}) \quad (2.84)$$

Therefore, in the small interval expansion, we have

$$S(\rho_{\alpha, \bar{\alpha}}||\rho_{\beta, \bar{\beta}}) = ((\alpha - \beta)^2 + (\bar{\alpha} - \bar{\beta})^2)(1 - \pi x \cot(\pi x)) = \frac{((\alpha - \beta)^2 + (\bar{\alpha} - \bar{\beta})^2)(\pi x)^2}{3} + O(x^4) \quad (2.85) \quad \{\text{smallxCoherent}\}$$

Spinless vertex operators of a massless boson have the same charges in holomorphic and anti-holomorphic sector ($\alpha = \bar{\alpha}$), and dimension $\Delta = \alpha^2$. Therefore,

$$S(\rho_\alpha \| \rho_\beta) = 2(\alpha - \beta)^2(1 - \pi x \cot(\pi x)) \quad (2.86)$$

$$= 2(\alpha - \beta)^2 \left(\frac{(\pi x)^2}{3} + \frac{(\pi x)^4}{45} + \frac{2(\pi x)^6}{945} + O(x^8) \right) \quad (2.87)$$

Current operators: The relative entropy of the primary state created by $J = i\partial\varphi$ with respect to a general chiral vertex operator is [9–11]

$$S(\rho_{i\partial\varphi} \| \rho_\beta) = S(\rho_{i\partial\varphi} \| \rho_0) + S(\rho_0 \| \rho_\beta) \quad (2.88)$$

At small x we have

$$S(\rho_0 \| \rho_\beta) = \beta^2 \left(\frac{(\pi x)^2}{3} + \frac{(\pi x)^4}{45} + \frac{2(\pi x)^6}{945} + O(x^8) \right) \quad (2.89)$$

and

$$\begin{aligned} S(\rho_{i\partial\varphi} \| \rho_0) &= 2(1 - \pi x \cot(\pi x)) + 2(\log(2 \sin(\pi x)) + \psi_0(\csc(\pi x)/2) + \sin(\pi x)) \\ &\simeq_{x \rightarrow 0^+} \frac{8}{15}(\pi x)^4 - \frac{32}{45}(\pi x)^6 + O(x^8) \end{aligned} \quad (2.90)$$

where we used the large z expansion of ψ_0 ,

$$\psi_0(z) = \log z - \frac{1}{2z} - \frac{1}{12z^2} + \frac{1}{120z^4} + \cdots, \quad z = \frac{1}{2 \sin(\pi x)} \quad (2.91)$$

Using $(h_\Psi, \bar{h}_\Psi) = (1, 0)$, $(h_\Phi, \bar{h}_\Phi) = (0, 0)$ and $c = \bar{c} = 1$ out earlier result on vacuum contribution, reproduces the above expansion in small x . Adding the above two gives,

$$S(\rho_{i\partial\varphi} \| \rho_\beta) \simeq_{x \rightarrow 0^+} \frac{\beta^2}{3}(\pi x)^2 + \left(\frac{8}{15} + \frac{\beta^2}{45} \right) (\pi x)^4 + O(x^6) \quad (2.92)$$

To derive the fourth order term using our OPEs, we need to consider corrections in the denominator from $\langle JJJJ \rangle, \langle TT \rangle, \langle TJJ \rangle, \langle J\partial^2 J \rangle, \langle (\partial J)^2 \rangle, \langle J^4 \rangle$ terms. At higher orders additional four-point and descendant data enter, which are not contained in the universal two- and three-point structures retained above.

Ising CFT: We have two non-identity primaries, namely, $\sigma \equiv (\frac{1}{16}, \frac{1}{16})$ and $\varepsilon \equiv (\frac{1}{2}, \frac{1}{2})$. The relative entropies of the primaries with respect to vacuum have been calculated exactly by

bosonization [11],

$$S(\rho_\sigma \| \rho_0) = S(\rho_0 \| \rho_\sigma) \quad (2.93)$$

$$= \frac{1}{4}(1 - \pi x \cot(\pi x)) \quad (2.94)$$

$$\underset{x \rightarrow 0^+}{\simeq} \frac{(\pi x)^2}{12} + \frac{(\pi x)^4}{180} + \frac{(\pi x)^6}{1890} + O(x^8) \quad (2.95)$$

$$S(\rho_\varepsilon \| \rho_0) = S(\rho_{i\partial\phi} \| \rho_0) \quad (2.96)$$

$$= 2(1 - \pi x \cot(\pi x)) + 2(\log(2 \sin(\pi x)) + \psi_0(\csc(\pi x)/2) + \sin(\pi x)) \quad (2.97)$$

$$\underset{x \rightarrow 0^+}{\simeq} \frac{8}{15}(\pi x)^4 - \frac{32}{45}(\pi x)^6 + O(x^8) \quad (2.98)$$

The above equation agrees with our result of the vacuum contribution up to the order x^6 , when we substitute $h_\Psi = \bar{h}_\Psi = 1/2$, $h_\Phi = \bar{h}_\Phi = 0$ and $c = \bar{c} = 1/2$. They also showed,

$$S(\rho_\varepsilon \| \rho_\sigma) = S(\rho_\varepsilon \| \rho_0) + S(\rho_0 \| \rho_\sigma) \quad (2.99) \quad \{\text{Sepsigma}\}$$

$$= \frac{9}{4}(1 - \pi x \cot(\pi x)) + 2(\log(2 \sin(\pi x)) + \psi_0(\csc(\pi x)/2) + \sin(\pi x)) \quad (2.100) \quad \{\text{Sepsigma}\}$$

$$\underset{x \rightarrow 0^+}{\simeq} \frac{(\pi x)^2}{12} + \frac{97}{180}(\pi x)^4 + O(x^6) \quad (2.101)$$

The above results of Ising CFT are a special case for the calculation of relative entropy of mixtures. These results do agree with our OPE calculations, up to the lowest-order contribution. Naively, one would have expected that relative entropy in (2.99) to start at order $x^{2\Delta_L}$ where the lightest primary is σ with $\Delta_L = 1/16 + 1/16 = 1/8$; however, this term is absent due to the fact that in the Ising model $C_{\varepsilon\varepsilon}^\sigma = C_{\sigma\sigma}^\sigma = 0$. So, the term proportional to x^2 arises from the next primary ε , which has $\Delta_\varepsilon = 1/2 + 1/2 = 1$. Note that $C_{\sigma\sigma}^\varepsilon = 1/2$ and $C_{\varepsilon\varepsilon}^\sigma = 0$, which upon using in our result, does reproduce the same expansion of the non-perturbative result, up to the lowest order.

Now, if you expand in the $x \rightarrow 1^-$ limit you will have the OPE of $\sigma\varepsilon$, which has σ in it, so the expectation is that besides the divergent term there will be a correction at order $\epsilon^{1/4}$. Check explicitly!

2.3 Large interval, $x \rightarrow 1^-$ limit

Before we start the replica trick, we note that for any pair of reduced states

$$\begin{aligned} S(\psi_A \| \chi_A) &= \text{tr}((\psi_A - \chi_A)K_{\chi_A}) - \Delta S(A) \\ S(\psi_{\bar{A}} \| \chi_{\bar{A}}) &= \text{tr}((\psi_{\bar{A}} - \chi_{\bar{A}})K_{\chi_{\bar{A}}}) - \Delta S(\bar{A}) \\ \Delta S(A) &= S(\psi_A) - S(\chi_A), \quad K_\rho \equiv -\log \rho. \end{aligned} \quad (2.102)$$

Since for pure global states we have $S(A) = S(\bar{A})$ we define the global modular Hamiltonian $\hat{K}_A = K_A - K_{\bar{A}}$ and find

$$S(\psi_{\bar{A}} \| \chi_{\bar{A}}) - S(\psi_A \| \chi_A) = -\langle \Psi | \hat{K}_{\chi, A} | \Psi \rangle. \quad (2.103) \quad \{\text{Khatformula}\}$$

Therefore, in the small $1 - x$ limit, we should expect a term that is order $\epsilon^{2\Delta_L}$ coming from $S(\rho_\Psi \|\rho_\Phi)$, and then a correction term due to the modular Hamiltonian $\hat{K}_{\chi,A}$. As we see below, the key observation is that the $\hat{K}$ term diverges as $x \rightarrow 1$ with corrections starting at order $1/\epsilon$. Note that further corrections to $\hat{K}$ can appear at lower order than $\epsilon^{2\Delta_L}$, as well.

In the large interval limit, we expand the correlator in (2.2) in the limit of small $\epsilon \equiv 1 - x$. In this limit, we have to expand in a different OPE channel (at z_k^+ and z_{k+1}^-) where operators from sheet k are close to operators on sheet $k+1$. The correlators with $n = 1$ (normalizations of density matrices) are symmetric under $x \rightarrow 1 - x$.

$$\frac{\Psi(z_k^+) \Phi^*(z_{k+1}^-)}{\sqrt{\langle \Psi(z^+) \Psi^*(z^-) \rangle \langle \Phi(z^+) \Phi^*(z^-) \rangle}} = \sum_p C_{\Psi\Phi^*}^p \zeta_1^{\Delta_p} \left(\frac{\zeta_n}{\zeta_1} \right)^{\Delta_p - \Delta_\Psi - \Delta_\Phi} e^{i\pi \Omega_k^{(p)}} \mathcal{O}_p(e^{i\pi(2k+1)/n})$$

$$\zeta_n = 2 \sin(\pi\epsilon/n), \quad \Omega_k^{(p)} = s_p(\theta_k - 1/2) - (s_\Psi + s_\Phi)(\theta_k - 1), \quad \theta_k = \frac{2k+1}{n} \quad (2.104)$$

For the n -dependent correlators, at small $1 - x$ the OPE expansion of the numerator remains the same as small x with $x \rightarrow 1 - x$. However, the denominator changes because for $k = 0$ and $k = n - 1$ we now have the OPE of Ψ and Φ^* and Φ and Ψ^* respectively. For $h_\Psi \neq h_\Phi$, there cannot be an identity contribution in the OPE of Ψ and Φ . Therefore, in the denominator, to the lowest order, we need the following OPEs

$$\frac{\Phi^*(z_{k+1}^-) \Phi(z_k^+)}{\langle \Phi(z^+) \Phi^*(z^-) \rangle} = \left(\frac{\zeta_1}{\zeta_n} \right)^{2\Delta_\Phi} e^{-2i\pi s_\Phi(\theta_k)} \left(1 + \sum_j C_\Phi^j \zeta_n^{\Delta_j} e^{i\pi s_j(\theta_k+1/2)} \mathcal{O}_j(e^{i\pi\theta_k}) \dots \right)$$

$$\frac{\Phi^*(z_1^-) \Psi(z_0^+)}{\sqrt{\langle \Psi(z^+) \Psi^*(z^-) \rangle \langle \Phi(z^+) \Phi^*(z^-) \rangle}} = \left(\frac{\zeta_1}{\zeta_n} \right)^{\Delta_\Psi + \Delta_\Phi} C_{\Phi\Psi}^{(0)}(e^{-2\pi i s_\Phi/n} 1 +$$

$$\sum_j C_{\Phi\Psi}^j \zeta_n^{\Delta'_j} e^{i\pi(s'_j - s_\Psi - s_\Phi)(1/n+1/2)} e^{i\pi(s_\Psi + s_\Phi)/2} \mathcal{O}_j(e^{i\pi/n}) + \dots)$$

$$\frac{\Phi(z_{n-1}^+) \Psi^*(z_0^-)}{\sqrt{\langle \Psi(z^+) \Psi^*(z^-) \rangle \langle \Phi(z^+) \Phi^*(z^-) \rangle}} = \left(\frac{\zeta_1}{\zeta_n} \right)^{\Delta_\Psi + \Delta_\Phi} C_{\Phi\Psi}^{(0)}(e^{-2\pi i s_\Phi(-1/n+2)} 1 +$$

$$\sum_j C_{\Psi\Phi}^j \zeta_n^{\Delta'_j} e^{i\pi(s'_j - s_\Psi - s_\Phi)(-1/n+5/2)} e^{i\pi(s_\Psi + s_\Phi)/2} \mathcal{O}_j(e^{i(2n-1)\pi/n}) + \dots)$$

$$(2.105) \quad \{\text{eq: general OPE}\}$$

where $\mathcal{O}_j$ is the j^{th} lowest dimension operator in the OPE $\Psi\Psi^*$ and $\Phi\Phi^*$,⁴ and $\mathcal{O}'_j$ is the j^{th} lowest dimension primary in the OPE $\Psi\Phi^*$ (and similarly $(\mathcal{O}'_j)^*$ is the j^{th} lowest dimension primary in the OPE of $\Psi^*\Phi$) which need not be the same as $\mathcal{O}_L$ which is the lowest dimension operator in the OPE of Φ and Φ^* . To match the notation from previous section, we use $\Delta_L \equiv \Delta_1$, $C_\Phi = C_\Phi^{(1)}$, $C_{\Psi\Phi} = C_{\Psi\Phi}^{(1)}$, and $C_{\Psi\Phi^*} \equiv C_{\Psi\Phi^*}^{L'}$.

During computation of the contribution to $\mathcal{F}_n$, we want to keep only those terms which are invariant under a rotation of π about its midpoint. The numerator would contribute only

⁴As before, we continue to assume, without loss of generality, that the primaries in the OPEs of $\Psi\Psi^*$ and $\Phi\Phi^*$ have the same dimension.

if $s_\Psi \in \mathbb{Z}$. This implies that the denominator will contribute only if $s_\Phi \in \mathbb{Z}$, since we would require $s_\Psi + s_\Phi \in \mathbb{Z}$, and hence $s_\Phi \in \mathbb{Z}$.

In the numerator of $\mathcal{F}_n$, the phases due to the identity piece is given by:

$$\prod_{k=0}^{n-1} e^{-2\pi i s_\Psi(\theta_k)} = e^{-2\pi i n s_\Psi} \quad (2.106)$$

and to the lowest order $\zeta_n^{2\Delta_L}$ we have

$$\begin{aligned} & \left(\frac{\zeta_1}{\zeta_n} \right)^{2n\Delta_\Psi} \left(1 + \zeta_n^{2\Delta_L} \frac{(C_\Psi^L)^2}{2} \sum_{k \neq k'} e^{i\pi s_L(\theta_k + \theta_{k'} + 1)} \langle \mathcal{O}_L(e^{i\pi\theta_k}) \mathcal{O}_L(e^{i\pi\theta_{k'}}) \rangle + \dots \right) \\ &= \left(\frac{\zeta_1}{\zeta_n} \right)^{2n\Delta_\Psi} \left(1 + \zeta_n^{2\Delta_L} \frac{(C_\Psi^L)^2}{2} \sum_{k \neq k'} \frac{1}{(2 \sin(\pi|k - k'|/n))^{2\Delta_L}} + \dots \right) \\ &= \left(\frac{\zeta_1}{\zeta_n} \right)^{2n\Delta_\Psi} \left(1 + \zeta_n^{2\Delta_L} (C_\Psi^L)^2 S_2(n) + \dots \right) \end{aligned} \quad (2.107)$$

To the lowest order, in the denominator, we have $n - 2$ copies of the OPE of Φ and Φ^* and the OPEs of $\Psi\Phi^*$ and $\Psi^*\Phi$. The phase from the identity piece in the denominator is given by,

$$\prod_{k=1}^{n-2} e^{-2\pi i s_\Phi \theta_k} = e^{-2\pi i s_\Phi (n-2)} \quad (2.108)$$

and to the lowest order we have

$$\begin{aligned} & \left(\frac{\zeta_1}{\zeta_n} \right)^{2((n-1)\Delta_\Phi + \Delta_\Psi)} \zeta_n^{2\Delta_{L'}} (C_{\Psi\Phi^*}^{L'})^2 e^{-2\pi i(n-2)s_\Phi} e^{i\pi s_{L'}} e^{2\pi i(s_{L'} - s_\Psi - s_\Phi)} \langle \mathcal{O}_{L'}(1) \mathcal{O}_{L'}(e^{\pi/n}) \rangle \\ &= \left(\frac{\zeta_1}{\zeta_n} \right)^{2((n-1)\Delta_\Phi + \Delta_\Psi)} (C_{\Psi\Phi^*}^{L'})^2 e^{-2\pi i s_\Phi (n-2)} e^{-2\pi i(s_\Psi + s_\Phi)} \zeta_n^{2\Delta_{L'}} \\ \xi_n &= \frac{\sin(\pi\epsilon/n)}{\sin(\pi/n)} \end{aligned} \quad (2.109)$$

The $e^{2\pi i s_{L'}}$ gets cancelled from the phase of the two-point function.

The denominator takes the following form,

$$\begin{aligned} & C^2 \left[1 + \zeta_n^{2\Delta_L} \left((C_\Phi^L)^2 \sum_{\substack{1 \leq k, k' \leq n-2 \\ k \neq k'}} \left[\frac{1}{2 \sin\left(\frac{\pi(k-k')}{n}\right)} \right]^{2\Delta_L} \right. \right. \\ & \quad \left. \left. + (C_{\Psi\Phi^*}^L)^2 \left[\frac{1}{2 \sin(\pi/n)} \right]^{2\Delta_L} \right. \right. \\ & \quad \left. \left. + 2C_{\Psi\Phi^*}^L C_\Phi^L \left[\frac{2S_2(n, \Delta_L)}{n} - \left(\frac{1}{2 \sin(\pi/n)} \right)^{2\Delta_L} \right] \right) \right]. \end{aligned} \quad (2.110)$$

Using the sum

$$\sum_{\substack{1 \leq k, k' \leq n-2 \\ k \neq k'}} \left[\frac{1}{2 \sin\left(\frac{\pi(k-k')}{n}\right)} \right]^{2\Delta_L} = S_2(n, \Delta_L) - \frac{4}{n} S_2(n, \Delta_L) + \left[\frac{1}{\sin(\pi/n)} \right]^{2\Delta_L} \quad (2.111)$$

we can simplify the contribution coming from the two-point functions to the following,

$$(C_\Phi^L)^2 \left(S_2(n, \Delta_L) - \frac{4S_2(n, \Delta_L)}{n} \right) + 2C_{\Psi\Phi^*}^L C_\Phi^L \left(\frac{2S_2(n, \Delta_L)}{n} \right) + (C_\Phi^L - C_{\Psi\Phi^*}^L)^2 \left(\frac{1}{2 \sin(\pi/n)} \right)^{2\Delta_L}. \quad (2.112)$$

Notice, we have a divergent piece $(1/(2 \sin(\pi/n)))^{2\Delta_L}$ which clearly blows up. Our aim was to use the OPE Coefficient C in the mixing channel as a regulator, to control the divergence of relative entropy between two non-orthogonal states. However, the result still blows up. We will consider the relative entropy of mixtures in the following section, where we will see that the classical probability of the mixture will regulate the divergence in the small $1 - x$ limit.

At this point, the leading term in the OPE of $\Psi\Phi^*$ results in a divergent relative entropy because $\partial_n \xi_n|_{n \rightarrow 1} = \infty$. The origin of this divergence can be traced back to the fact that the two-point function of $\mathcal{O}_{L'}\mathcal{O}_{L'}$ is evaluated at distance $(2 \sin(\pi/n))$ which vanishes at $n \rightarrow 1$. In fact, the limit $\epsilon \rightarrow 0$ and $n \rightarrow 1$ do not commute here. Note that the corrections to the denominator can come from summing over L' , namely the 2-point function of $\mathcal{O}_{L'}\mathcal{O}_{L'}$ coming from two pairs of OPEs of $\Phi\Psi^*$, and the three-point function coming from $\mathcal{O}_{L'}\mathcal{O}_{L''}\mathcal{O}_L$, where $\mathcal{O}_{L'}$ and $\mathcal{O}_{L''}$ come from the OPEs of $\Phi\Psi^*$, and $\mathcal{O}_L$ comes from the OPE of $\Phi\Phi$. There could also be higher point functions. Each correction from the two-point function also carries the same divergence! Going back to the same x expansion of relative entropy in (2.18), we note that if we only pick one of the terms with fixed $k \neq k'$, the analytic continuation gives the same divergence.

Let us explain this point more explicitly. The difference between the numerator and the denominator of $\mathcal{F}_n$ is the switching of one copy $\Phi\Phi^*$ to a copy of $\Psi\Phi^*$ in the $2n$ -point function. Define the formal primary operator $(\Psi\Phi^{-1})$ with the property:

$$\lim_{\delta \rightarrow 0} \delta^{\Delta(\Psi\Phi^{-1})} (\Psi\Phi^{-1})(\delta)\Phi(0) = \Psi(0) \quad (2.113)$$

so that in the denominator, we have the correlator

$$\lim_{\delta \rightarrow 0} \langle (\Psi\Phi^{-1})(z_0^+ - \delta)(\Psi\Phi^{-1})^*(z_0^- + \delta) \prod_{k=0}^{n-1} \Phi(z_k^+) \Phi^*(z_k^-) \rangle \quad (2.114)$$

Note that in general, such a primary might not exist. The intuition originates from the example of coherent states of $U(1)$ current algebra, the primary $: e^{i(\alpha-\beta)\phi} :$ plays the role of $(e^{i\alpha\phi} e^{-i\beta\psi})$. We can expand the normalized version of this correlator by taking the OPE of

$(\Psi\Phi^{-1})(\Psi\Phi^{-1})^*$ with distance $(2\sin(\pi(1-x)/n))$, and taking pair-wise OPEs of $\Phi\Phi^*$ with distance $(2\sin(\pi\epsilon/n))$ so that we have

$$\left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)}\right)^{-2\Delta_{(\Psi\Phi^{-1})}} \left(1 + \sum_p C_{(\Psi\Phi^{-1})(\Psi\Phi^{-1})}^p C_{\Phi\Phi}^p \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)}\right)^{-2\Delta_p} \times \sum_{k=0}^n \frac{1}{(2\sin(\pi(k+1/2)/n)^{2\Delta_p}} + \dots\right) \quad (2.115)$$

Note that at small ϵ and integer n this is a sensible expansion, and higher order terms are suppressed; however, the small ϵ does not commute with the small $n-1$. If we first expand in ϵ at integer n we find

$$-2\Delta_{(\Psi\Phi^{-1})} \partial_n \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)}\right)_{n \rightarrow 1} = 2\Delta_{(\Psi\Phi^{-1})} \cot(\pi\epsilon) \quad (2.116)$$

which diverges as $\epsilon \rightarrow 0$. However, keeping ϵ small and fixed, and taking the $n \rightarrow 1$ limit we find that

$$\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} = 1 - \pi \cot(\pi\epsilon)(n-1) + \dots \quad (2.117)$$

which means that we are not allowed to truncate the series expansion, as higher order Δ_p also contribute at the same $1/\epsilon$ order. To better understand the relative entropy in the limit $x \rightarrow 1^-$ we consider special cases.

Can we use any of this to compute the expansion of relative entropy of excited states in the Ising model, because there is σ in the OPE of ϵ and σ , which makes the calculation highly non-trivial in the case $\lim_{x \rightarrow 1^-} S(\rho_\epsilon \| \rho_\sigma)$.

2.3.1 Special cases:

Relative entropy $S(\rho_\Phi \| \rho_0)$ at $x \rightarrow 1^-$: The replica trick for this relative entropy in the $x \rightarrow 1^-$ limit is

$$\begin{aligned} \mathcal{F}_n &= \frac{1 + (C_\Phi^L)^2 (2\sin(\pi\epsilon/n))^{2\Delta_L} f(n) + \dots}{\left(\frac{n\sin(\pi\epsilon/n)}{\sin\pi\epsilon}\right)^{2\Delta_\Phi(n-1)} \left(\frac{e^{2\pi i n s_\Phi} e^{i\pi s_\Phi}}{e^{2\pi i(n-1)s_\Phi}} (2\sin(\pi\epsilon/n))^{2\Delta_\Phi} \langle \Phi(z_{0,n}^+) \Phi^*(z_{0,n}^-) \rangle + \dots\right)} \\ &= \frac{1 + (C_\Phi^L)^2 (2\sin(\pi\epsilon/n))^{2\Delta_L} f(n) + \dots}{D_n(\epsilon)^{n-1} \left(\left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)}\right)^{2\Delta_\Phi} + \dots\right)} \end{aligned} \quad (2.118)$$

where the $\dots$ refer to higher order corrections. Therefore,

$$\begin{aligned} \log \mathcal{F}_n &= (2\pi\epsilon/n)^{2\Delta_L} (C_\Phi^L)^2 f(n) - (n-1)2\Delta_\Phi \log \left(\frac{n\sin(\pi\epsilon/n)}{\sin(\pi\epsilon)}\right) - 2\Delta_\Phi \log \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)}\right) + \dots \\ &= (2\pi\epsilon/n)^{2\Delta_L} (C_\Phi^L)^2 f(n) - 2n\Delta_\Phi \log \left(\frac{n\sin(\pi\epsilon/n)}{\sin(\pi\epsilon)}\right) - 2\Delta_\Phi \log \left(\frac{\sin(\pi\epsilon)}{n\sin(\pi(1-\epsilon)/n)}\right) + \dots \end{aligned} \quad (2.119)$$

We then obtain,

$$\begin{aligned}
S(\rho_\Phi \| \rho_0) &= \sum_L (\pi\epsilon)^{2\Delta_L} (C_\Phi^L)^2 \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} + 2\Delta_\Phi \pi \cot(\pi\epsilon) + \dots \\
&\simeq \sum_L (\pi\epsilon)^{2\Delta_L} (C_\Phi^L)^2 \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} + 2\Delta_\Phi \left(\frac{1}{\epsilon} - \frac{\pi^2\epsilon}{3} - \frac{\pi^4\epsilon^3}{45} - \frac{2\pi^6\epsilon^5}{945} + \dots \right)
\end{aligned} \tag{2.120}$$

We would like to highlight the appearance of the term $2\Delta_\Phi \pi \cot(\pi\epsilon)$. This is the expectation value of the full modular Hamiltonian of the vacuum in the state created by $\mathcal{O}_L$:

$$- \langle \mathcal{O}_\Phi | \hat{K}_0 | \mathcal{O}_\Phi \rangle = 2\Delta_\Phi \pi \cot(\pi\epsilon) . \tag{2.121}$$

So the above result of relative entropy in large interval does satisfy,

$$S(\rho_{\Phi, \bar{A}} \| \rho_{0, \bar{A}}) - S(\rho_{\Phi, A} \| \rho_{0, A}) = - \langle \Phi | \hat{K}_0 | \Phi \rangle \tag{2.122}$$

where A corresponds to small interval, and $\bar{A}$ is the complement of A , i.e. large interval.

Relative Entropy $S(\rho_0 \| \rho_\Phi)$ at $x \rightarrow 1^-$: We would like to study the $x \rightarrow 1^-$ limit of relative entropy $S(\rho_0 \| \rho_\Phi)$. In this limit, the replica trick requires us to compute the $2(n-1)$ -point function of Φ in the OPE channel where $\Phi(z_k^+)$ approaches $\Phi^*(z_{k+1}^-)$. In this limit, the lowest order term in $\mathcal{F}_n$ comes from the two-point function at $z_{n-1, n}^+ = \exp(\frac{2i\pi}{n}(n-1+x/2))$ and $z_{1, n}^- = \exp(\frac{2i\pi}{n}(1-x/2))$ and its OPE with $\mathcal{O}_L$ inserted at $z_3 = \exp(i\pi 2k/n)$ for $k = 1, 2, \dots, n-2$:

$$\begin{aligned}
\log \mathcal{F}_n &= -\log \langle \tilde{\mathcal{O}}_\Phi^{(n-1)} \rangle \\
\langle \tilde{\mathcal{O}}_\Phi^{(n-1)} \rangle &= \left(\frac{n \sin(\pi\epsilon/n)}{\sin(\pi\epsilon)} \right)^{-4(n-1)h_\Phi} (2 \sin(\pi\epsilon/n))^{4h_\Phi} e^{2\pi i(n-1)s_\Phi} e^{-2\pi i(n-2)s_\Phi} e^{i\pi s_\Phi} \times \\
&\left(\langle \Phi(z_{n-1, n}^+) \Phi^*(z_{1, n}^-) \rangle + (C_\Phi^L)^2 \sum_{k=1}^{n-2} \left(2 \sin\left(\frac{\pi\epsilon}{n}\right) \right)^{\Delta_L} e^{i\pi s_L((2k+1)/n+1/2)} \langle \Phi(z_{n-1, n}^+) \Phi^*(z_{1, n}^-) \mathcal{O}_L(e^{i\pi(2k+1)/n}) \rangle + \dots \right)
\end{aligned} \tag{2.123}$$

If we naively truncate the denominator at the lowest order, and analytically continue in n we find that this term results in a large negative relative entropy! This clearly means that our OPE expansion is invalid. What has gone wrong is that operators $\Phi(z_{n-1, n}^+, n) \Phi^*(z_{1, n}^-)$ are separated by angle $2\pi(1+x)/n$ which in the limit $n \rightarrow 1$ is larger than 2π !

2.3.2 Some examples

The goal of this subsection is to reproduce these divergences from the replica trick. For the massless free boson theory, we consider two types of conformal primaries, the chiral vertex operator $V_{\alpha, \alpha}$, and the primaries $i\partial\phi$ with $(h, \bar{h}) = (1, 0)$ and $i\bar{\partial}\bar{\phi}$ with $(h, \bar{h}) = (0, 1)$. The

relative entropies in these cases are computed in [12]. They diverge in large subsystem limit $\epsilon = (1-x) \rightarrow 0^+$ as

$$S(\rho_\alpha \| \rho_\beta) = (\alpha - \beta)^2 (1 - \pi x \cot(\pi x)) \simeq_{x \rightarrow 1^-} (\alpha - \beta)^2 \left(\frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} + \frac{\pi^2 \epsilon^2}{3} - \frac{\pi^4 \epsilon^3}{45} + \frac{\pi^4 \epsilon^4}{45} - \frac{2\pi^6}{945} \epsilon^5 + \frac{2\pi^6}{945} \epsilon^6 + O(\epsilon^7) \right) \quad (2.124) \quad \{\text{eq: free mass}\}$$

The above result up to the order ϵ^2 using replica trick has been derived in the Appendix C.

$$S(\rho_{i\partial\phi} \| \rho_0) = 2(1 - \pi x \cot(\pi x)) + 2(\log(2 \sin(\pi x)) + \psi_0(\csc(\pi x)/2) + \sin(\pi x)) \quad (2.125)$$

$$\simeq_{x \rightarrow 1^-} \frac{2}{\epsilon} - \frac{2\pi^2}{3} \epsilon - \frac{2\pi^4}{45} \epsilon^3 + \frac{8\pi^4}{15} \epsilon^4 - \frac{4\pi^6}{945} \epsilon^5 - \frac{32\pi^6}{45} \epsilon^6 + O(\epsilon^7) \quad (2.126)$$

The above result is also obtained using our OPE up to ϵ^4 . The numerator of $\mathcal{F}_n$ will give the contribution from $\langle TT \rangle$ term proportional to x^4 . While the terms up to ϵ^3 comes from the denominator, which is just the two-point function of current. Then the small ϵ expansion of $S(\rho_{i\partial\phi} \| \rho_\beta)$ is given by,

$$\begin{aligned} S(\rho_{i\partial\phi} \| \rho_\beta) &= S(\rho_{i\partial\phi} \| \rho_0) + S(\rho_0 \| \rho_\beta) \simeq_{x \rightarrow 1^-} \frac{2 + \beta^2}{\epsilon} - \frac{(2 + \beta^2)\pi^2}{3} + \frac{\beta^2 \pi^2}{3} \epsilon^2 - \frac{(2 + \beta^2)\pi^4}{45} \epsilon^3 \\ &\quad + \left(\frac{8\pi^4}{15} + \frac{\beta^2 \pi^4}{45} \right) \epsilon^4 + O(\epsilon^5) \quad (2.127) \\ &= \left[\frac{2}{\epsilon} - \frac{2\pi^2}{3} \epsilon - \frac{2\pi^4}{45} \epsilon^3 + \frac{8\pi^4}{15} \epsilon^4 \right] \\ &\quad + \beta^2 \left[\frac{1}{\epsilon} - \frac{\pi^2}{3} \epsilon + \frac{\pi^2}{3} \epsilon^2 - \frac{\pi^4}{45} \epsilon^3 + \frac{\pi^4}{45} \epsilon^4 \right] + O(\epsilon^5) \quad (2.128) \end{aligned}$$

The numerator of $\mathcal{F}_n$ will give just the term $\frac{8\pi^4}{15} \epsilon^4$ from the two-point function of stress tensor, i.e. $\langle TT \rangle$. While the rest of the terms comes from the denominator, which consist of mixing OPEs of J and $V_{\pm\beta}$ at two sites, and non-mixing Vertex operator OPEs at the remaining sites. The contributions up to the order of ϵ^4 requires us to compute several three-point and four point functions in the denominator.

This term is precisely the divergent term we get from the contribution from just the two-point function of the mixing OPE. There should be a factor in the denominator of $\mathcal{F}_n$ from the contribution of higher order terms, that will exactly cancel the numerator, and thus reproducing the massless free boson result (once we replace the general operators with Vertex operators).

3 Relative entropy of mixed excited states

We aim to compute the relative entropy $S(\rho \| \sigma_y)$ where $\rho = \rho_\Psi$ and the reference state is the mixture $\sigma_y = y\rho_\Psi + (1-y)\rho_\Phi$ for $0 \leq y \leq 1$. The standard replica trick for relative

entropy relates this quantity to the analytic continuation of trace functionals that break replica symmetry. Specifically, for the mixed state case, we utilize the following identity derived from the generalized replica trick for normalized density matrices:

$$\begin{aligned} S\left(\rho_\Psi \parallel y\rho_\Psi + (1-y)\rho_\Phi\right) &= \left(\partial_n \log \left[\frac{\text{tr}(\rho_\Psi^n)}{\text{tr}(\rho_\Psi(y\rho_\Psi + (1-y)\rho_\Phi)^{n-1})} \right]\right)_{n \rightarrow 1} \\ &= (\partial_n \log \mathcal{F}_n(\Psi|\Phi; y))_{n \rightarrow 1} \end{aligned} \quad (3.1)$$

where

$$\begin{aligned} \mathcal{F}_n(\Psi|\Phi; y) &\equiv \frac{\langle \prod_{k=1}^n \tilde{\Psi}(z_{k,n}^+) \tilde{\Psi}^*(z_{k,n}^-) \rangle}{\langle \tilde{\Psi}(z_{n,n}^+) \tilde{\Psi}^*(z_{n,n}^-) \tilde{\mathcal{O}}_{\Psi|\Phi}^{(n-1)} \rangle} \\ \tilde{\mathcal{O}}_{\Psi|\Phi; y}^{(n-1)} &= \sum_{p=0}^{n-1} y^{n-1-p} (1-y)^p n^{4p(h_\Psi - h_\Phi)} \sum_{\substack{w \in \{\Psi, \Phi\}^{n-1} \\ N_\Phi(w)=p}} \prod_{k=1}^{n-1} \frac{\mathcal{O}_{w_k}(z_{k,n}^+) \mathcal{O}_{w_k^*}(z_{k,n}^-)}{\langle \mathcal{O}_{w_k}(z_{1,1}^+) \mathcal{O}_{w_k^*}(z_{1,1}^-) \rangle} \end{aligned} \quad (3.2)$$

Here, $w = (w_1, \dots, w_{n-1})$ is an ordered word and $N_\Phi(w)$ counts how many letters equal Φ , and the operators are already mapped to the complex plane, as before.

For simplicity, we often suppress the notation $(\Psi|\Phi; y)$ in $\mathcal{F}_n$. The analytic continuation of the numerator in $\mathcal{F}_n$ after the n -derivative and the limit $n \rightarrow 1$ contributes as the vacuum subtracted entropy

$$[\partial_n \log(\text{tr}(\rho_\Psi^n)) - \partial_n \log(\text{tr}(\rho_0^n))]_{n \rightarrow 1} = S(\rho_0) - S(\rho_\Psi) \quad (3.3)$$

whereas the analytic continuation of the denominator of $\mathcal{F}_n$ contributes as

$$\begin{aligned} -[\partial_n \log(\text{tr}(\rho_\Psi \sigma_y^{n-1})) - \log(\text{tr}(\rho_0^n))]_{n \rightarrow 1} &= \text{tr}(\rho_\Psi \log \sigma_y) + S(\rho_0) \\ &= \text{tr}(\rho_\Psi (\log \sigma_y - \log \rho_0)) - \text{tr}((\rho_0 - \rho_\Psi) \log \rho_0) \\ &= \langle V_\alpha | (H_0 - H_{\sigma_y}) | V_\alpha \rangle - \text{tr}((\rho_\Psi - \rho_0) H_0) \end{aligned} \quad (3.4)$$

$$\begin{aligned} [\partial_n \log(\text{tr}(\rho_\Psi \sigma_y^{n-1})) - \partial_n \log(\text{tr}(\rho_0^n))]_{n \rightarrow 1} &= \text{tr}(\rho_\Psi \log \sigma_y) + S(\rho_0) \\ &= \text{tr}(\rho_\Psi (\log \sigma_y - \log \rho_0)) - \text{tr}((\rho_0 - \rho_\Psi) \log \rho_0) \\ &= \langle \Psi | (H_0 - H_{\sigma_y}) | \Psi \rangle - \text{tr}((\rho_\Psi - \rho_0) H_0) \end{aligned} \quad (3.5)$$

where $\sigma_y = y\rho_\Psi + (1-y)\rho_\Phi$, and we have the modular Hamiltonians

$$H_{\sigma_y} := -\log \sigma_y; \quad H_0 = -\log \rho_0. \quad (3.6)$$

The reason we split the terms this way is that in vacuum 2d CFTs H_0 has a universal form that is a local integral over stress tensor. The modular Hamiltonian of the vacuum on a cylinder of circumference 2π in the interval $(-x, x)$ is given by

$$H_0 = \frac{(2\pi)^2}{2} \int_{-x}^x \frac{\cos(\pi y) - \cos(\pi x)}{\sin(\pi x)} T_{00}(x). \quad (3.7)$$

In a state with uniform energy density, such as an exact energy eigenstate with dimension h_Φ on a cylinder of circumference 2π we have $\Delta T_{00}(y) = h/\pi$, and as a result

$$\Delta H_\sigma \equiv \text{tr}((\rho_\Phi - \rho_0)H_0) = 4h_\Psi(1 - \pi x \cot(\pi x)) \quad (3.8)$$

This means that in computing the contribution of the exact energy eigenkets $|V_\alpha\rangle$ we should always expect a term of the form

$$-4h_\Psi(1 - \pi x \cot(\pi x)). \quad (3.9)$$

In other words,

$$S(\rho_\Psi \| \sigma_y) = S(0) - S(\rho_\Psi) - \langle \Psi | (H_0 - H_{\sigma_y}) | \Psi \rangle + 4h_\Psi(1 - \pi x \cot(\pi x)). \quad (3.10)$$

This way of splitting the relative entropy has the advantage that it makes the analysis of regions that are larger than half, i.e. $x > 1/2$, clearer. Note that the ΔS term is symmetric under $x \rightarrow 1 - x$. The term proportional to h_α at $x \rightarrow 0$ starts at order x^2 , whereas at $x \rightarrow 1^-$ it has a simple pole $(1 - x)^{-1}$. The remainder term is the vacuum-subtracted modular Hamiltonian whose $x \rightarrow 1 - x$ behavior is to be studied.

As we will see, in free massless boson, the coherent states of charge α have the property that $\rho_\alpha = U_\alpha \rho_0 U_\alpha^\dagger$ for some unitary operator U_α . In this case, the relative entropy simplifies to

$$S(\rho_\alpha \| y\rho_\alpha + (1 - y)\rho_\beta) = S(\rho_0 \| y\rho_0 + (1 - y)\rho_{\beta-\alpha}). \quad (3.11)$$

Another special case is ρ_0 the vacuum reduced density matrix. In this case, the numerator drops out, and the denominator becomes

$$S(\rho_0 \| \sigma_y) = \text{tr}(\rho_0(H_{\sigma_y} - H_0)) = \langle H_{\sigma_y} - H_0 \rangle. \quad (3.12)$$

3.1 Small x limit

The replica representation is given by,

$$\mathcal{F}_n = \frac{1 + (2 \sin(\pi x/n))^{2\Delta_L} f(n) + \dots}{\sum_{p=0}^{n-1} y^{n-1-p} ((1-y)D_n(x))^p \sum_{\substack{w \in \{\Psi, \Phi\}^{n-1} \\ N_\Phi(w)=p}} [1 + (2 \sin(\pi x/n))^{2\Delta_L} g_w(n) + \dots]} \quad (3.13) \quad \{\text{eq:F_n_mixed}\}$$

$$D_n(x) = \left(\frac{n \sin(\pi x/n)}{\sin(\pi x)} \right)^{2(\Delta_\Psi - \Delta_\Phi)} \simeq 1 + \frac{(\Delta_\Psi - \Delta_\Phi)(\pi x)^2}{3} \left(1 - \frac{1}{n^2} \right) + \dots, \quad D_1(x) = 1, \quad (3.14) \quad \{\text{Dn_mixed}\}$$

where the leading correction $f(n)$ and $g_w(n)$ are given by the sums in (??) and (3.18), respectively, over OPE coefficients and light operator correlators on the replica manifold. It can be shown that the phase terms also get cancelled exactly, one from the conformal factor

and the other from the OPE. First observation is that if $\Delta_L > 1$, the dominant contribution to $\mathcal{F}_n$ comes from the $D_n(x)$ term and the denominator becomes

$$\begin{aligned} & \sum_{p=0}^{n-1} y^{n-1-p} (1-y)^p \binom{n-1}{p} \left(1 + \frac{2p(1-1/n^2)}{3} (h_\Psi - h_\Phi) (\pi x)^2 + \dots \right) \\ &= 1 + \frac{2(n-1)^2(n+1)(h_\Psi - h_\Phi)(\pi x)^2(1-y)}{3n^2} + \dots \end{aligned} \quad (3.15)$$

where we have used

$$\sum_{p=0}^{n-1} y^{n-1-p} (1-y)^p p = (n-1)(1-y) \quad (3.16)$$

Taking the logarithm and the n -derivative at $n = 1$ we find that the x^2 term does not contribute to the relative entropy

$$\partial_n \log \mathcal{F}_n(x) = -\frac{2}{3} (h_\Psi - h_\Phi) (1-y) (\pi x)^2 \partial_n \left(\frac{(n-1)^2(n+1)}{n^2} \right)_{n \rightarrow 1} + O(x^{2\Delta_L}) = O(x^{2\Delta_L}) \quad (3.17)$$

Therefore, we can ignore the contribution of the $D_n(x)$ term in $\mathcal{F}_n$. For future convenience, we keep $D_n(x)$ in the calculation for now, because it does not play an important role in the analytic continuation discussed below, and it will help us generalize to general x later.

The expression for $f(n)$ is given in (2.18). Similarly, we have

$$g_w(n) = C_\Psi^L \sum_{k=1}^{n-1} \frac{C_{w_k}^L}{(2 \sin(\pi k/n))^{2\Delta_L}} + \frac{1}{2} \sum_{k \neq k'=1}^{n-1} \frac{C_{w_k}^L C_{w_{k'}}^L}{(2 \sin(\pi |k - k'|/n))^{2\Delta_L}}, \quad (3.18) \quad \{\text{gnw}\}$$

We can simplify the denominator of $\mathcal{F}_n$ by noting that

$$\begin{aligned} & \sum_{\substack{w \in \{\alpha, \beta\}^{n-1} \\ N_\beta(w) = p}} [1 + (2 \sin(\pi x/n))^{2\Delta_L} g_w(n) + \dots] = \binom{n-1}{p} + (2 \sin(\pi x/n))^{2\Delta_L} \bar{g}_p(n) + \dots \\ \bar{g}_p(n) &= \sum_{\substack{w \in \{\Psi, \Phi\}^{n-1} \\ N_\Phi(w) = p}} g_w(n) = C_\Psi^L \sum_{k=1}^{n-1} \frac{A_k}{(2 \sin(\pi k/n))^{2\Delta_L}} + \frac{1}{2} \sum_{k \neq k'=1}^{n-1} \frac{B_{kk'}}{(2 \sin(\pi |k - k'|/n))^{2\Delta_L}}, \end{aligned} \quad (3.19)$$

where we have exchanged the order of the sum over words and the sum over replica indices and defined

$$\begin{aligned} A_k(p, n) &= \sum_{\substack{w \in \{\Psi, \Phi\}^{n-1} \\ N_\Phi(w) = p}} C_{w_k}^L = \binom{n-2}{p} C_\Psi^L + \binom{n-2}{p-1} C_\Phi^L \equiv A(p, n) \\ B_{kk'}(p, n) &= \sum_{\substack{w \in \{\Psi, \Phi\}^{n-1} \\ N_\Phi(w) = p}} C_{w_k}^L C_{w_{k'}}^L = \binom{n-3}{p} (C_\Psi^L)^2 + 2 \binom{n-3}{p-1} C_\Psi^L C_\Phi^L + \binom{n-3}{p-2} (C_\Phi^L)^2 \equiv B(p, n), \end{aligned} \quad (3.20)$$

where we performed the sums over words for fixed $k \neq k'$ and found that the result is independent of k and k' . Therefore, we obtain

$$\begin{aligned}
g_p(n) &= C_\Psi^L A(p, n) \sum_{k=1}^{n-1} \frac{1}{(2 \sin(\pi k/n))^{2\Delta_L}} + \frac{B(p, n)}{2} \sum_{k \neq k'=1}^{n-1} \frac{1}{(2 \sin(\pi |k - k'|/n))^{2\Delta_L}} \\
&= (C_\Psi^L A(p, n) - B(p, n)) \sum_{k=1}^{n-1} \frac{1}{(2 \sin(\pi k/n))^{2\Delta_L}} + \frac{B(p, n)}{2} \sum_{k \neq k'=0}^{n-1} \frac{1}{(2 \sin(\pi |k - k'|/n))^{2\Delta_L}} \\
&= \frac{2S_2(n)}{n} (C_\Psi^L A(p, n) - B(p, n)) + B(p, n) S_2(n)
\end{aligned} \tag{3.21}$$

and as a result we can write

$$[1 + (2 \sin(\pi x/n))^{2\Delta_L} g_w(n) + \dots] = \binom{n-1}{p} + (2 \sin(\pi x/n))^{2\Delta_L} g_p(n) + \dots \tag{3.22}$$

For notational convenience, we suppress the argument x below in $D_n(x)$, writing simply D_n . We note the following identities,

$$Z(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p} ((1-y)D_n)^p \binom{n-1}{p} = (y + (1-y)D_n)^{n-1} \tag{3.23}$$

$$A_\Psi(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p} ((1-y)D_n)^p \binom{n-2}{p} = y(y + (1-y)D_n)^{n-2} \tag{3.24}$$

$$A_\Phi(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p} ((1-y)D_n)^p \binom{n-2}{p-1} = (1-y)D_n(y + (1-y)D_n)^{n-2} \tag{3.25}$$

$$B_{\Psi\Psi}(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p} ((1-y)D_n)^p \binom{n-3}{p} = y^2(y + (1-y)D_n)^{n-3} \tag{3.26}$$

$$B_{\Psi\Phi}(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p} ((1-y)D_n)^p \binom{n-3}{p-1} = y(1-y)D_n(y + (1-y)D_n)^{n-3} \tag{3.27}$$

$$B_{\Phi\Phi}(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p} ((1-y)D_n)^p \binom{n-3}{p-2} = (1-y)^2 D_n^2 (y + (1-y)D_n)^{n-3}, \tag{3.28}$$

For simplicity, we define

$$\begin{aligned}
X(n) &= (y + (1-y)D_n), \quad X(1) = 1 \\
Y_{\Psi\Phi}(n) &= (yC_\Psi^L + (1-y)D_n C_\Phi^L)
\end{aligned} \tag{3.29}$$

so that

$$\begin{aligned}
B(p, n) &= (C_\Psi^L)^2 B_{\Psi\Psi}(n) + 2C_\Psi^L C_\Phi^L B_{\Psi\Phi}(n) + (C_\Phi^L)^2 B_{\Phi\Phi}(n) \\
&= Y_{\Psi\Phi}(n)^2 X(n)^{n-3} \\
A(p, n) &= C_\Psi^L A_\Psi(n) + C_\Phi^L A_\Phi(n) = Y_{\Psi\Phi}(n) X(n)^{n-2}
\end{aligned} \tag{3.30}$$

Then, in the denominator we have

$$X(n)^{n-1} \left(1 + \left(\frac{2\pi x}{n} \right)^{2\Delta_L} \Delta(n) + \dots \right) \quad (3.31)$$

where

$$\begin{aligned} \Delta(n) &\equiv \frac{2S_2(n)}{n} \frac{(C_\Psi^L A(p, n) - B(p, n))}{X(n)^{n-1}} + \frac{B(p, n)}{X(n)^{n-1}} S_2(n) \\ &= \frac{2S_2(n)}{n} \left(\frac{C_\Psi^L Y_{\Psi\Phi}(n)}{X(n)} - \frac{Y_{\Psi\Phi}(n)^2}{X(n)^2} \right) + S_2(n) \left(\frac{Y_{\Psi\Phi}(n)^2}{X(n)^2} \right) \end{aligned} \quad (3.32)$$

$$= S_2(n) \left(\frac{Y_{\Psi\Phi}(n)}{X(n)} \left(\frac{2C_\Psi^L}{n} \right) + \frac{Y_{\Psi\Phi}^2(n)}{X(n)^2} \left(1 - \frac{2}{n} \right) \right) \quad (3.33)$$

Putting this back into (3.13) we find

$$\log \mathcal{F}_n = (1 - n) \log X(n) + \log \left[\frac{1 + \left(\frac{2\pi x}{n} \right)^{2\Delta_L} S_2(n) \left((C_\Psi^L)^2 + \dots \right)}{1 + \left(\frac{2\pi x}{n} \right)^{2\Delta_L} \Delta(n) + \dots} \right] \quad (3.34)$$

We first note that taking the n -derivative of the first factor in $\mathcal{F}_n$ and sending $n \rightarrow 1$ vanishes

$$\partial_n [(1 - n) \log(y + (1 - y)D_n)] \Big|_{n=1} = 0 \quad (3.35)$$

and we are left with

$$S(\rho_\Psi \| y\rho_\Psi + (1 - y)\rho_\Phi) = \partial_n \left[\left(\frac{2\pi x}{n} \right)^{2\Delta_L} \left(S_2(n)(C_\Psi^L)^2 - \Delta(n) \right) + \dots \right]_{n=1} \quad (3.36)$$

Note that this is consistent with our previous observation that $D_n(x)$ does not contribute to the relative entropy at order x^2 . However, the analytic continuation we achieved shows that the higher order terms of $D_n(x)$ do not contribute either.

Since $S_2(1) = 0$ and $X(1) = 1$, we have

$$\begin{aligned} S(\rho_\Psi \| y\rho_\Psi + (1 - y)\rho_\Phi) &= (2\pi x)^{2\Delta_L} S_2'(1) \left((C_\Psi^L)^2 - 2C_\Psi^L Y_{\Psi\Phi}(1) + Y_{\Psi\Phi}(1)^2 \right) + \dots \\ &= (\pi x)^{2\Delta_L} \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} (1 - y)^2 (C_\Psi^L - C_\Phi^L)^2 + \dots \end{aligned} \quad (3.37)$$

In the special case, $\Psi = \Phi$ or $y \rightarrow 1$ the term vanishes as expected. In the case of vertex operators in free massless bosons the equations of motion are $\partial_z \partial_{\bar{z}} \phi(z, \bar{z}) = 0$, therefore the field splits into a chiral and anti-chiral mode $\phi(z, \bar{z}) = \varphi(z) + \bar{\varphi}(\bar{z})$. In the case $y \rightarrow 0$ we have

$$S(\rho_\alpha \| \rho_\beta) = (\alpha - \beta)^2 \frac{\pi^2 x^2}{3} + \dots \quad (3.38)$$

which agrees with the result of [13, 14]. Similarly, repeating our calculation for the case where $\Delta_L > 2$, in the lowest order we get the contribution from the two-point function of stress tensors. In full generality, we finally obtain,

$$S(\rho_\Psi \| y\rho_\Psi + (1-y)\rho_\Phi) = \begin{cases} \sum_L \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + \frac{3}{2})} (C_\Psi^L - C_\Phi^L)^2 (1-y)^2 (\pi x)^{2\Delta_L} + \dots & \text{if } \Delta_L < 2, \\ \frac{8}{15} \left[\frac{(h_\Psi - h_\Phi)^2}{c} + \frac{(\bar{h}_\Psi - \bar{h}_\Phi)^2}{\bar{c}} \right] (1-y)^2 (\pi x)^4 + \dots & \text{if } \Delta_L > 2. \end{cases}$$

(3.39) {smallxMixture}

where the sum over L is over all operator of lowest dimension Δ_L in the OPE of $\Psi\Psi^*$ and $\Phi\Phi^*$. An interesting feature of this equation is that it is symmetric under switching Ψ and Φ :

$$S(\rho_\Psi \| y\rho_\Psi + (1-y)\rho_\Phi) \simeq S(\rho_\Phi \| y\rho_\Phi + (1-y)\rho_\Psi) + \dots \quad (3.40)$$

which is in accordance to our prediction that the above quantity in leading order should be quadratic, as it is the quantum Fisher information metric.

Once again assuming that h_Φ and h_Ψ are very heavy but $h_\Phi/c \sim h_\Psi/c = O(1)$, can we resum this using the identity block and is the answer holographic in any sense?

3.1.1 Relative entropy of two mixtures at $x \rightarrow 0^+$

This can be generalized further by computing the relative entropy of a mixture with respect to another mixture, namely $S(\bar{\rho}_{y_1} \| \bar{\rho}_{y_2})$, where $\bar{\rho}_{y_1} = y_1\rho_\Psi + (1-y_1)\rho_\Phi$ and $\bar{\rho}_{y_2} = y_2\rho_\Psi + (1-y_2)\rho_\Phi$. We want to compute it using replica trick, by computing the following,

$$S(\bar{\rho}_{y_1} \| \bar{\rho}_{y_2}) = \partial_n \log \left[\frac{\text{tr}(\bar{\rho}_{y_1}^n)}{\text{tr}(\bar{\rho}_{y_1} \bar{\rho}_{y_2}^{n-1})} \right] \quad (3.41)$$

For the numerator, the sum over words gives

$$B_{kk'}(p, n) = \sum_{\substack{w \in \{\Psi, \Phi\}^n \\ N_\Phi(w) = p}} (C_\Psi^L)^2 \binom{n-2}{p} + 2C_\Psi^L C_\Phi^L \binom{n-2}{p-1} + (C_\Phi^L)^2 \binom{n-2}{p-2} \quad (3.42)$$

The term other than the identity piece in the numerator evaluates to the following,

$$\sum_{p=0}^n y_1^{n-p} ((1-y_1)D_n)^p \left((C_\Psi^L)^2 \binom{n-2}{p} + 2C_\Psi^L C_\Phi^L \binom{n-2}{p-1} + (C_\Phi^L)^2 \binom{n-2}{p-2} \right) \quad (3.43)$$

$$= (y_1 + (1-y_1)D_n)^{n-2} \left((C_\Psi^L)^2 y_1^2 + 2C_\Psi^L C_\Phi^L y_1(1-y_1)D_n + (C_\Phi^L)^2 ((1-y_1)D_n)^2 \right) \quad (3.44)$$

$$= X_1(n)^{n-2} (C_\Psi^L y_1 + C_\Phi^L (1-y_1)D_n)^2 \quad (3.45)$$

$$= X_1(n)^{n-2} Y_1(n)^2 \quad (3.46)$$

where we defined

$$X_1(n) = y_1 + (1 - y_1)D_n \quad (3.47)$$

$$Y_1(n) = C_\Psi^L y_1 + C_\Phi^L (1 - y_1)D_n \quad (3.48)$$

The numerator is then given by,

$$\text{tr}(\bar{\rho}_{y_1}^n) = X_1^n \left(1 + \left(\frac{2\pi x}{n} \right)^{2\Delta_L} S_2(n) \frac{Y_1(n)^2}{X_1(n)^2} + \dots \right) \quad (3.49)$$

Now for the denominator, we want to evaluate the following,

$$\text{tr}(\bar{\rho}_{y_1} \bar{\rho}_{y_2}^{n-1}) \quad (3.50)$$

Again, we can define the following for the denominator,

$$X_2(n) = y_2 + (1 - y_2)D_n \quad (3.51)$$

$$Y_2(n) = C_\Psi^L y_2 + C_\Phi^L (1 - y_2)D_n \quad (3.52)$$

The identity piece comes from precisely from $X_1 X_2^{n-1}$. The piece coming from the two-point function evaluates to (following the same method we used earlier),

$$X_2(n)^{n-2} \left(Y_1(n) Y_2(n) \frac{2S_2(n)}{n} + \frac{1}{2} Y_2(n)^2 \left(S_2(n) - \frac{2S_2(n)}{n} \right) \right) \quad (3.53)$$

$$(3.54)$$

For simplicity, we use the notation,

$$X_1(n) \equiv X_1, \quad Y_1(n) \equiv Y_1, \quad (3.55)$$

$$X_2(n) \equiv X_2, \quad Y_2(n) \equiv Y_2. \quad (3.56)$$

The denominator is then given by,

$$\text{tr}(\bar{\rho}_{y_1} \bar{\rho}_{y_2}^{n-1}) = X_1 X_2^{n-1} \left[1 + \left(\frac{2\pi x}{n} \right)^{2\Delta_L} \left(\frac{Y_1 Y_2}{X_1 X_2} \left(\frac{2S_2(n)}{n} \right) + \left(\frac{Y_2}{X_2} \right)^2 \left(S_2(n) - \frac{2S_2(n)}{n} \right) \right) + \dots \right] \quad (3.57)$$

Combining numerator and denominator gives

$$S(\bar{\rho}_{y_1} \parallel \bar{\rho}_{y_2}) = \partial_n \log \left[\frac{X_1^n \left(1 + \left(\frac{2\pi x}{n} \right)^{2\Delta_L} S_2(n) \frac{Y_1^2}{X_1^2} + \dots \right)}{X_1 X_2^{n-1} \left[1 + \left(\frac{2\pi x}{n} \right)^{2\Delta_L} \left(\frac{Y_1 Y_2}{X_1 X_2} \left(\frac{2S_2(n)}{n} \right) + \left(\frac{Y_2}{X_2} \right)^2 \left(S_2(n) - \frac{2S_2(n)}{n} \right) \right) + \dots \right]} \right] \quad (3.58)$$

$$= \partial_n \log \left(\frac{X_1}{X_2} \right)^{n-1} \left(1 + \left(\frac{2\pi x}{n} \right)^{2\Delta_L} S_2'(1) (Y_1^2 - 2Y_1 Y_2 + Y_2^2) \right) \quad (3.59)$$

where we used $X_1(1) = X_2(1) = 1$. Using the definitions of Y_1, Y_2 , we finally obtain the relative entropy of mixtures of two excited states,

$$S(\bar{\rho}_{y_1} \parallel \bar{\rho}_{y_2}) = \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} \sum_L (C_\Psi^L - C_\Phi^L)^2 (y_1 - y_2)^2 (\pi x)^{2\Delta_L} + \dots \quad \text{if } \Delta_L \leq 2. \quad (3.60)$$

Generalizing our results for $\Delta_L \geq 2$, we thus have,

$$S(\bar{\rho}_{y_1} \parallel \bar{\rho}_{y_2}) = \begin{cases} \sum_L \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + \frac{3}{2})} (C_\Psi^L - C_\Phi^L)^2 (y_1 - y_2)^2 (\pi x)^{2\Delta_L} + \dots & \text{if } \Delta_L < 2, \\ \frac{8}{15} \left[\frac{(h_\Psi - h_\Phi)^2}{c} + \frac{(\bar{h}_\Psi - \bar{h}_\Phi)^2}{\bar{c}} \right] (y_1 - y_2)^2 (\pi x)^4 + \dots & \text{if } \Delta_L > 2. \end{cases} \quad (3.61) \quad \{\text{smallxMixture}\}$$

Understand why this is symmetric under $(y_1, y_2) \rightarrow (y_1 + c, y_2 + c)$. The fact that it is proportional to $(y_1 - y_2)^2$ is expected from the Fisher information form of it, but in principle it could be proportional to $(y_1 - y_2)^2 f(y_1)$.

3.2 Relative entropy of a M -mixture with respect to a N -mixture at $x \rightarrow 0^+$

We further generalize our results by considering two mixtures by a set of M $\{\Psi_a\}$ and N $\{\Phi_b\}$. In particular, we consider,

$$\bar{\rho}_{\mathbf{p}} = \sum_{a=1}^M p_a \rho_{\Psi_a}, \quad \sum_{a=1}^M p_a = 1, \quad (3.62)$$

$$\bar{\rho}_{\mathbf{q}} = \sum_{b=1}^N q_b \rho_{\Phi_b}, \quad \sum_{b=1}^N q_b = 1. \quad (3.63)$$

Again, for simplicity, we assume that the same lightest operator $\mathcal{O}_L$ with dimension Δ_L appears in all relevant diagonal OPEs. We can write down the OPE of a diagonal channel now,

$$\frac{\Psi_a(z_k^+) \Psi_a^*(z_k^-)}{\langle \Psi_a(z^+) \Psi_a^*(z^-) \rangle} = D_{a,n}(x) (1 + \zeta_n^{\Delta_L} C_a^L \mathcal{O}_L(z_k) + \dots), \quad (3.64)$$

$$\frac{\Phi_b(z_k^+) \Phi_b^*(z_k^-)}{\langle \Phi_b(z^+) \Phi_b^*(z^-) \rangle} = \tilde{D}_{b,n}(x) (1 + \zeta_n^{\Delta_L} C_b^L \mathcal{O}_L(z_k) + \dots) \quad (3.65)$$

where $D_{a,n}(x), \tilde{D}_{b,n}(x)$ are the normalization and conformal factors corresponding to the diagonal OPEs. For simplicity, we used the notation $C_a^L \equiv C_{\Psi_a \Psi_a^*}^L, C_b^L \equiv C_{\Phi_b \Phi_b^*}^L$. Then we can generalize our previous notation by defining,

$$X_{\mathbf{p}}(n) = \sum_{a=1}^M p_a D_{a,n}(x), \quad Y_{\mathbf{p}} = \sum_{a=1}^M p_a D_{a,n} C_a^L, \quad (3.66)$$

$$X_{\mathbf{q}}(n) = \sum_{b=1}^N q_b \tilde{D}_{b,n}(x), \quad Y_{\mathbf{q}} = \sum_{b=1}^N q_b \tilde{D}_{b,n} C_b^L. \quad (3.67)$$

We consider the numerator now, i.e. $\text{tr}(\rho_p^n)$. We have,

$$\bar{\rho}_{\mathbf{p}}^n = \sum_{a_0, \dots, a_{n-1}=1}^M \left(\prod_{j=0}^{n-1} p_{a_j} \right) \rho_{\Psi_{a_0}} \rho_{\Psi_{a_1}} \cdots \rho_{\Psi_{a_{n-1}}}. \quad (3.68)$$

Then the identity contribution will pick a factor of D_a from every diagonal OPE, and we get,

$$\sum_{\{a_j\}} \prod_{j=0}^{n-1} p_{a_j} D_{a_j, n} = \prod_{j=0}^{n-1} \left(\sum_{a=1}^M p_a D_{a, n} \right) = X_{\mathbf{p}}(n)^n. \quad (3.69)$$

Next we want to evaluate the contribution from the two-point functions, which would take the following form,

$$\sum_{\{a_j\}} \left(\prod_{j=0}^{n-1} p_{a_j} D_{a_j, n} \right) C_{a_k}^L C_{a_{k'}}^L = \left[\prod_{j \neq \{k, k'\}} \sum_{a_j=1}^M p_{a_j} D_{a_j, n} \right] \left[\sum_{a_k=1}^M p_{a_k} D_{a_k, n} C_{a_k}^L \right] \left[\sum_{a_{k'}=1}^M p_{a_{k'}} D_{a_{k'}, n} C_{a_{k'}}^L \right] \quad (3.70)$$

$$= X_{\mathbf{p}}^{n-2} Y_{\mathbf{p}}^2 \quad (3.71)$$

Then, we have the following simplification,

$$\text{tr}(\bar{\rho}_{\mathbf{p}}^n) = X_{\mathbf{p}}^n \left(1 + \zeta_n^{2\Delta_L} S_2(n, \Delta_L) \left(\frac{Y_{\mathbf{p}}}{X_{\mathbf{p}}} \right)^2 + \cdots \right). \quad (3.72)$$

Notice, this is the generalization of the previous calculation. Hence, we can directly write down the denominator,

$$\text{tr}(\bar{\rho}_{\mathbf{p}} \bar{\rho}_{\mathbf{q}}^{n-1}) = X_{\mathbf{p}} X_{\mathbf{q}}^{n-1} \left(1 + \zeta_n^{2\Delta_L} \left(\frac{Y_{\mathbf{p}} Y_{\mathbf{q}}}{X_{\mathbf{p}} X_{\mathbf{q}}} \left(\frac{2S_2(n, \Delta_L)}{n} \right) + \left(\frac{Y_{\mathbf{q}}}{X_{\mathbf{q}}} \right)^2 \left(S_2(n, \Delta_L) - \frac{2S_2(n, \Delta_L)}{n} \right) \right) + \cdots \right) \quad (3.73)$$

Finally, we define the following averaged OPE coefficients,

$$\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} = \sum_{a=1}^M p_a C_{\Psi_a \Psi_a^*}^L, \quad \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} = \sum_{b=1}^N q_b C_{\Phi_b \Phi_b^*}^L, \quad (3.74)$$

$$\langle h \rangle_{\mathbf{p}} = \sum_{a=1}^M p_a h_{\Psi_a}, \quad \langle \bar{h} \rangle_{\mathbf{p}} = \sum_{a=1}^M p_a \bar{h}_{\Psi_a} \quad (3.75)$$

$$\langle h \rangle_{\mathbf{q}} = \sum_{b=1}^N q_b h_{\Phi_b}, \quad \langle \bar{h} \rangle_{\mathbf{q}} = \sum_{b=1}^N q_b \bar{h}_{\Phi_b} \quad (3.76)$$

to obtain the relative entropy,

$$S(\bar{\rho}_{\mathbf{p}}\|\bar{\rho}_{\mathbf{q}}) = \begin{cases} \sum_L \frac{\sqrt{\pi}\Gamma(\Delta_L+1)}{4\Gamma(\Delta_L+\frac{3}{2})} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right)^2 (\pi x)^{2\Delta_L} + \dots & \text{if } \Delta_L < 2, \\ \frac{8}{15} \left[\frac{(\langle h \rangle_{\mathbf{p}} - \langle h \rangle_{\mathbf{q}})^2}{c} + \frac{(\langle \bar{h} \rangle_{\mathbf{p}} - \langle \bar{h} \rangle_{\mathbf{q}})^2}{\bar{c}} \right] (\pi x)^4 + \dots & \text{if } \Delta_L > 2. \end{cases} \quad (3.77) \quad \{\text{small } x \text{ general}\}$$

The above Eq. (3.77) is the main result of the short-interval analysis. At leading order, the individual components of the mixtures enter only through probability-averaged local CFT data. For a light exchanged primary, the relative entropy is controlled by the difference of the averaged diagonal OPE coefficients, $\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}}$. If no primary with $\Delta_L < 2$ contributes, the stress tensor gives the leading term and the corresponding role is played by the averaged conformal weights. Notice, the leading contribution is quadratic in the difference between the two sets of averaged CFT data. Consequently it is symmetric under interchange of the two density matrices at this order,

$$S(\bar{\rho}_{\mathbf{p}}\|\bar{\rho}_{\mathbf{q}}) = S(\bar{\rho}_{\mathbf{q}}\|\bar{\rho}_{\mathbf{p}}) + \dots, \quad (3.78) \quad \{\text{eq:leading-sm}\}$$

where $\dots$ denotes the higher order correction. Although relative entropy is not symmetric in general, the symmetry in (3.78) is natural from the information-geometric viewpoint. The mixtures $\bar{\rho}_{\mathbf{p}}$ and $\bar{\rho}_{\mathbf{q}}$ need not be close in mixture-parameter space. However, in the short-interval limit $x \rightarrow 0$ their reduced density matrices approach the same local reference state, so that their difference is perturbatively small in x . Then, for two nearby density matrices, the linear variation of the relative entropy vanishes by the First law of Entanglement Entropy [15], and the first non-vanishing contribution is quadratic in their difference. This defines the quantum Fisher Information metric associated with Relative entropy. Exchanging two density matrices reverses the sign of their difference but leaves its quadratic norm invariant, explaining the symmetry above.

The subleading short-interval expansion has the same conformal-family structure as in the pure-state case. The difference is that the state-dependent diagonal OPE data associated with each replica sheet are replaced, after summing over replica words, by their probability averages.

Primary contributions: The leading two-point contribution is

$$\mathcal{S}_{LL,\mathbf{p}|\mathbf{q}}^{[2\Delta_L]} \equiv f_{LL} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right)^2 (\pi x)^{2\Delta_L}. \quad (3.79)$$

For the three-point contribution it is also convenient to define

$$\begin{aligned} \mathcal{B}_{LMN}^{\Psi|\Phi;\mathbf{p}|\mathbf{q}} &\equiv \langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} \langle C_{\Psi\Psi^*}^M \rangle_{\mathbf{p}} \langle C_{\Psi\Psi^*}^N \rangle_{\mathbf{p}} - \langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} \langle C_{\Phi\Phi^*}^M \rangle_{\mathbf{q}} \langle C_{\Phi\Phi^*}^N \rangle_{\mathbf{q}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \langle C_{\Psi\Psi^*}^M \rangle_{\mathbf{p}} \langle C_{\Phi\Phi^*}^N \rangle_{\mathbf{q}} \\ &\quad - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \langle C_{\Phi\Phi^*}^M \rangle_{\mathbf{q}} \langle C_{\Psi\Psi^*}^N \rangle_{\mathbf{p}} + 2 \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \langle C_{\Phi\Phi^*}^M \rangle_{\mathbf{q}} \langle C_{\Phi\Phi^*}^N \rangle_{\mathbf{q}}. \end{aligned} \quad (3.80)$$

Three-point functions of non-vacuum primaries give

$$\mathcal{S}_{LMN,\mathbf{p}|\mathbf{q}}^{[\Delta_L+\Delta_M+\Delta_N]} \equiv f_{LMN} C_{LMN} \mathcal{B}_{LMN}^{\Psi|\Phi;\mathbf{p}|\mathbf{q}} (\pi x)^{\Delta_L+\Delta_M+\Delta_N}. \quad (3.81)$$

First conformal-family correction: At order $x^{2\Delta_L+2}$, the first subleading term in the two-point contribution and the level-two descendants combine exactly as in the pure-state calculation. Replacing the single-state OPE coefficients by their ensemble averages gives

$$\begin{aligned} \mathcal{S}_{LL,\text{sub},\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} &\equiv -\frac{\Delta_L}{3} f_{LL} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right)^2 (\pi x)^{2\Delta_L+2} \\ &= \Delta_L f_{LL0} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right)^2 (\pi x)^{2\Delta_L+2}. \end{aligned} \quad (3.82)$$

$$\mathcal{S}_{LL,\text{desc},\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} \equiv \frac{\Delta_L(\Delta_L+1)}{2\Delta_L+3} f_{LL} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right)^2 (\pi x)^{2\Delta_L+2}. \quad (3.83)$$

Combining the above two contributions,

$$\begin{aligned} \mathcal{S}_{LL,\text{fam},\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} &\equiv \mathcal{S}_{LL,\text{sub},\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} + \mathcal{S}_{LL,\text{desc},\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} \\ &= \frac{\Delta_L^2}{3(2\Delta_L+3)} f_{LL} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right)^2 (\pi x)^{2\Delta_L+2}. \end{aligned} \quad (3.84)$$

The remaining terms at this order arise from $T\mathcal{O}_L\mathcal{O}_L$ and $\bar{T}\mathcal{O}_L\mathcal{O}_L$. Using the conformal Ward identities $C_{TLL} = h_L$ and $C_{\bar{T}LL} = \bar{h}_L$ [8], together with the coefficients derived in the pure-state calculation, we obtain

$$\mathcal{S}_{TLL,\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} \equiv 3f_{TLL} h_L \mathcal{B}_{TLL}^{\Psi|\Phi;\mathbf{p}|\mathbf{q}} (\pi x)^{2\Delta_L+2}, \quad (3.85)$$

$$\mathcal{S}_{\bar{T}LL,\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} \equiv 3f_{\bar{T}LL} \bar{h}_L \mathcal{B}_{\bar{T}LL}^{\Psi|\Phi;\mathbf{p}|\mathbf{q}} (\pi x)^{2\Delta_L+2}. \quad (3.86)$$

where

$$\mathcal{B}_{TLL}^{\mathbf{p}|\mathbf{q}} = \frac{2}{c} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) \left(\langle h_{\Psi} \rangle_{\mathbf{p}} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} + \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) - 2 \langle h_{\Phi} \rangle_{\mathbf{q}} \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) \quad (3.87)$$

$$\mathcal{B}_{\bar{T}LL}^{\mathbf{p}|\mathbf{q}} = \frac{2}{\bar{c}} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) \left(\langle \bar{h}_{\Psi} \rangle_{\mathbf{p}} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} + \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) - 2 \langle \bar{h}_{\Phi} \rangle_{\mathbf{q}} \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right). \quad (3.88)$$

Combining all contributions at this order, we define

$$\mathcal{S}_{L,\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} \equiv \mathcal{S}_{LL,\text{fam},\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} + \mathcal{S}_{TLL,\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} + \mathcal{S}_{\bar{T}LL,\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]}. \quad (3.89)$$

The above equation simplifies to,

$$\begin{aligned}
\mathcal{S}_{L,\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} &= \frac{\Delta_L f_{LL}}{2\Delta_L+3} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) \left[\frac{\Delta_L}{3} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) \right. \\
&\quad - \frac{8h_L}{c} \left(\langle h_{\Psi} \rangle_{\mathbf{p}} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} + \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) - 2 \langle h_{\Phi} \rangle_{\mathbf{q}} \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) \\
&\quad \left. - \frac{8\bar{h}_L}{\bar{c}} \left(\langle \bar{h}_{\Psi} \rangle_{\mathbf{p}} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} + \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) - 2 \langle \bar{h}_{\Phi} \rangle_{\mathbf{q}} \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right) \right] (\pi x)^{2\Delta_L+2} \\
&= \frac{\Delta_L f_{LL}}{2\Delta_L+3} \left[\frac{\Delta_L}{3} \left(\langle C_{\Psi\Psi^*}^L \rangle_{\mathbf{p}} - \langle C_{\Phi\Phi^*}^L \rangle_{\mathbf{q}} \right)^2 - 4h_L \mathcal{B}_{TLL}^{\Psi|\Phi;\mathbf{p}|\mathbf{q}} - 4\bar{h}_L \mathcal{B}_{\bar{T}LL}^{\Psi|\Phi;\mathbf{p}|\mathbf{q}} \right] (\pi x)^{2\Delta_L+2}
\end{aligned} \tag{3.90}$$

Vacuum-family contributions: The leading vacuum-family contribution is

$$\mathcal{S}_{\text{vac},\mathbf{p}|\mathbf{q}}^{[4]} \equiv \frac{8}{15} \left[\frac{\left(\langle h_{\Psi} \rangle_{\mathbf{p}} - \langle h_{\Phi} \rangle_{\mathbf{q}} \right)^2}{c} + \frac{\left(\langle \bar{h}_{\Psi} \rangle_{\mathbf{p}} - \langle \bar{h}_{\Phi} \rangle_{\mathbf{q}} \right)^2}{\bar{c}} \right] (\pi x)^4. \tag{3.91}$$

At order x^6 , the corresponding vacuum-family terms combine to

$$\begin{aligned}
\mathcal{S}_{\text{vac},\mathbf{p}|\mathbf{q}}^{[6]} &\equiv \frac{32}{315} \left[\frac{\left(\langle h_{\Psi} \rangle_{\mathbf{p}} - \langle h_{\Phi} \rangle_{\mathbf{q}} \right)^2}{c} \left(1 - \frac{8 \left(\langle h_{\Psi} \rangle_{\mathbf{p}} + 2 \langle h_{\Phi} \rangle_{\mathbf{q}} \right)}{c} \right) \right. \\
&\quad \left. + \frac{\left(\langle \bar{h}_{\Psi} \rangle_{\mathbf{p}} - \langle \bar{h}_{\Phi} \rangle_{\mathbf{q}} \right)^2}{\bar{c}} \left(1 - \frac{8 \left(\langle \bar{h}_{\Psi} \rangle_{\mathbf{p}} + 2 \langle \bar{h}_{\Phi} \rangle_{\mathbf{q}} \right)}{\bar{c}} \right) \right] (\pi x)^6.
\end{aligned} \tag{3.92}$$

Collecting the non-vacuum primary, conformal-family, and vacuum-family contributions, we obtain

$$\begin{aligned}
S(\bar{\rho}_{\mathbf{p}} || \bar{\rho}_{\mathbf{q}}) &= \sum_{\substack{L \\ 2\Delta_L < \gamma}} \mathcal{S}_{LL,\mathbf{p}|\mathbf{q}}^{[2\Delta_L]} + \sum_{\substack{\text{ord } L,M,N \\ \Delta_L + \Delta_M + \Delta_N < \gamma}} \mathcal{S}_{LMN,\mathbf{p}|\mathbf{q}}^{[\Delta_L + \Delta_M + \Delta_N]} + \Theta(\gamma - 4) \mathcal{S}_{\text{vac},\mathbf{p}|\mathbf{q}}^{[4]} \\
&\quad + \sum_{\substack{L \\ 2\Delta_L + 2 < \gamma}} \mathcal{S}_{L,\mathbf{p}|\mathbf{q}}^{[2\Delta_L+2]} + \Theta(\gamma - 6) \mathcal{S}_{\text{vac},\mathbf{p}|\mathbf{q}}^{[6]} + O(x^\gamma). \\
\gamma &= \min(4\Delta_*, 8), \\
\Delta_* &= \min\{\Delta_L\}.
\end{aligned}$$

3.3 Small $1 - x$ limit

Next, we would like to compute the relative entropy of mixtures in the limit $x \rightarrow 1^-$. In this limit, we have to expand (3.93) in a different OPE channel (at z_k^+ and z_{k+1}^-) where operators

from sheet k are close to operators on sheet $k+1$. The correlators with $n=1$ (normalizations of density matrices) are symmetric under $x \rightarrow 1-x$. For the n -dependent correlators, at small $1-x$ the OPE expansion of the numerator remains the same as small x with $x \rightarrow 1-x$. However, the denominator changes significantly because for $p > 0$ there are terms that involve the OPE of operators Ψ and Φ^* of weights h_Ψ and h_Φ at z_k^+ and z_{k+1}^- . It can be shown that the phase terms also get cancelled exactly, one from the conformal factor and the other from ζ_n from each OPE. So, there is no dependency on spin, which showed up from these phases. The key difference compared to the previous case is that previously, all OPEs involved operators of the same weight, which starts with the identity term, whereas now, for $h_\Phi \neq h_\Psi$, there is no identity term in the OPE, and the fact that the lowest dimension operator is $\mathcal{O}_{L'}$ comes with a new OPE coefficient that we denote $C_{\Psi\Phi^*}$. Since the overall factors $(2\sin(\pi(1-x)/n))$ come with powers that are additive in the operator dimensions, the new expression for $\mathcal{F}_n$ takes the form

$$\begin{aligned}\mathcal{F}_n &= \frac{1 + (2\pi(1-x)/n)^{2\Delta_L} f(n) + \dots}{y^{n-1} \left(1 + (2\pi(1-x)/n)^{2\Delta_L} f(n) + \sum_{L'} (2\pi(1-x)/n)^{2\Delta_{L'}} (C_{\Psi\Phi}^{L'})^2 \mu(n, x, y; \Delta_{L'}) + \dots \right)} \\ &= y^{1-n} \left(1 - \sum_{L'} \left(\frac{2\pi(1-x)}{n} \right)^{2\Delta_{L'}} (C_{\Psi\Phi}^{L'})^2 \mu(n, x, y; \Delta_{L'}) + \dots \right) \\ \mu(n, x, y; \Delta_{L'}) &= \sum_{p=1}^{n-1} \frac{(n-p)((y^{-1}-1)D_n(1-x))^p}{(2\sin(\pi p/n))^{2\Delta_{L'}}}\end{aligned}\tag{3.93} \quad \{\text{eq: } \mu(n, x, y; \Delta_{L'})\}$$

In the first line, the $p=0$ in the denominator becomes

$$y^{n-1}(1 + (2\pi(1-x)/n)^{2\Delta_L} f(n))\tag{3.94}$$

whereas for non-zero p we get

$$\frac{1}{2} \sum_{k \neq k'} \frac{\delta_{p|k-k'|}}{(2\sin(\pi|k-k'|/n))^{2\Delta_{L'}}} = \frac{1}{2} \frac{2(n-p)}{(2\sin(\pi p/n))^{2\Delta_{L'}}}\tag{3.95}$$

and as a result we pick up the contribution

$$y^{n-1-p}((1-y)D_n(1-x))^p \frac{(n-p)(C_{\Psi\Phi}^{L'})^2}{(2\sin(\pi p/n))^{2\Delta_{L'}}} (2\pi(1-x)/n)^{2\Delta_{L'}}\tag{3.96}$$

Note that the $p=0$ term in the denominator corresponds to the same correlation function as the one appearing in the numerator with an extra factor of y^{n-1} , therefore we only need to keep track of $p > 0$ terms. Plugging these back in $\mathcal{F}_n$ and expanding in small $1-x$, gives us the relative entropy,

$$S(\rho_\Psi \| y\rho_\Psi + (1-y)\rho_\Phi) = -\log y - (C_{\Psi\Phi}^{L'})^2 (2\pi(1-x))^{2\Delta_{L'}} \partial_n \mu(n, x, y; \Delta_{L'})|_{n=1} + \dots\tag{3.97}$$

The result separates naturally into a classical and a quantum piece. The leading term $-\log y$ depends only on the weight assigned by the mixture to the component ρ_Ψ . The first correction probes the mixing OPE between Ψ and Φ^* and is controlled by the lightest operator that can interpolate between the two excited states.

The reader should also note that, in our previous calculation of relative entropy $S(\rho_\Psi \parallel \rho_\Phi)$, we were essentially computing the $y \rightarrow 0$ limit of this relative entropy of mixture calculation. The leading divergent piece came from the fact that we had just $p = n-2$ $\Phi\Phi^*$ OPE insertions, and the mixing OPEs of $\Psi\Phi^*$ and $\Phi\Psi^*$. That gave us a term in the denominator of $\mathcal{F}_n$,

$$\frac{1}{(2 \sin(\pi/n))^{2\Delta_{L'}}} \quad (3.98)$$

which is clearly divergent when we compute the relative entropy. However, in the mixture, we have a summation over p in (3.93), where p is the number of pairs of $\Phi\Phi^*$ insertions. We expect this summation to regulate our final answer, as we saw previously in the case where

$$S_2(n, \Delta) = \sum_{k=1}^{n-1} \frac{1}{(2 \sin(\pi k/n))^{2\Delta}} \quad (3.99)$$

in the limit $n \rightarrow 1$ vanishes, however, the individual terms diverge.

For simplicity, assume $y = 1/2$ so that we can expand in small $\epsilon = 1 - x$ as

$$\mu(n, \epsilon, \Delta_{L'}) = \sum_{p=1}^{n-1} \frac{(n-p)(1 + 2p\pi^2\epsilon^2(n^2-1)(h_\Psi - h_\Phi)/(3n^2) + O(\epsilon^4))}{(2 \sin(\pi p/n))^{2\Delta_{L'}}} = \mu_0(n, \Delta_{L'}) + \mu_1(n, \Delta_{L'})\epsilon^2 + \dots \quad (3.100)$$

We note that since the denominator of the summand is symmetric under $p \rightarrow n-p$ we have

$$\begin{aligned} \mu_0(n, \Delta_L) &= \sum_{p=1}^{n-1} \frac{n-p}{(2 \sin(\pi p/n))^{2\Delta_L}} = \frac{1}{2} \sum_{p=1}^{n-1} \frac{p + (n-p)}{(2 \sin(\pi p/n))^{2\Delta_L}} \\ &= \frac{n}{2} \sum_{p=1}^{n-1} \frac{1}{(2 \sin(\pi p/n))^{2\Delta_L}} = S_2(n, \Delta_L) . \end{aligned} \quad (3.101)$$

We find

$$S(\rho_\Psi \parallel (\rho_\Psi + \rho_\Phi)/2) = \log 2 - \sum_{L'} (2\pi\epsilon)^{2\Delta_{L'}} (C_{\Psi\Phi}^{L'})^2 (\mu'_0(1) + \mu'_1(1)\epsilon^2 + \dots) \quad (3.102)$$

Therefore,

$$\boxed{S(\rho_\Psi \parallel (\rho_\Psi + \rho_\Phi)/2) = \log 2 - \sum_{L'} (\pi\epsilon)^{2\Delta_{L'}} (C_{\Psi\Phi}^{L'})^2 \frac{\sqrt{\pi}\Gamma(\Delta_{L'} + 1)}{4\Gamma(\Delta_{L'} + 3/2)} + \dots} \quad (3.103)$$

For the next order term, we find

$$\mu_1(n) = \sum_{p=1}^{n-1} 2\pi^2 (n - 1/n) (h_\Psi - h_\Phi) \frac{p}{(2 \sin(\pi p/n))^{2\Delta_{L'}}} - \sum_{p=1}^{n-1} 2\pi^2 (1 - 1/n^2) (h_\Psi - h_\Phi) \frac{p^2}{(2 \sin(\pi p/n))^{2\Delta_{L'}}} \quad (3.104)$$

The first sum can be carried out again using the symmetry $\sin(x) = \sin(\pi - x)$, and that would give $S_2(n)$. Here the derivative at $n = 1$ from that term would contribute $S'_2(1)$. The analytic continuation of the second series is given in the Appendix B,

$$\begin{aligned} \mu'_1(n)|_{n=1} &= 2^{-2\Delta_{L'}+1} \pi^2 (1 - 1/n^2) (h_\Psi - h_\Phi) \left(\frac{\sqrt{\pi} \Gamma(\Delta_{L'} + 1)}{4 \Gamma(\Delta_{L'} + 3/2)} \right. \\ &\quad \left. - \frac{B(\Delta_{L'}, -2\Delta_{L'}) \cos(\pi \Delta_{L'})}{4} \left[-\frac{12}{\Delta_{L'}} + \frac{2\Delta_{L'}}{2\Delta_{L'} + 1} \left(\pi^2 - 6\psi_1(\Delta_{L'}) + \frac{6}{\Delta_{L'}^2} \right) \right] \right) \Big|_{n=1} \end{aligned} \quad (3.105)$$

$$= 0. \quad (3.106)$$

Special case of averaged mixture: As a special case, we consider the following,

$$S(\rho_0 \| (\rho_0 + \rho_\Phi)/2) = \partial_n \log 2^{n-1} \Big|_{n=1} - \partial_n \log \left(1 + \sum_{L'} (2\pi(1-x)/n)^{2\Delta_L} (C_{\Psi\Phi}^{L'})^2 \mu(n, x, 1/2; \Delta_{L'}) + \dots \right) \Big|_{n=1} \quad (3.107)$$

$$\begin{aligned} S(\rho_\Phi \| (\rho_0 + \rho_\Phi)/2) &= \partial_n \log 2^{n-1} \Big|_{n=1} + \partial_n \log \left[1 + (2\pi(1-x)/n)^{2\Delta_L} f(n) + \dots \right] \Big|_{n=1} \\ &\quad - \partial_n \log \left(1 + (2\pi(1-x)/n)^{2\Delta_L} f(n) \right. \\ &\quad \left. + \sum_{L'} (2\pi(1-x)/n)^{2\Delta_L} (C_{\Psi\Phi}^{L'})^2 \mu(n, x, 1/2; \Delta_{L'}) + \dots \right) \Big|_{n=1} \end{aligned} \quad (3.108)$$

In the second line, the contribution from the numerator gets cancelled exactly from the $p = n - 1$ contribution from the denominator (terms of the order $\epsilon^{2\Delta_L}$), and we will be left with $\mu(n, x, 1/2; \Delta_{L'})$. Hence, we have,

$$S(\rho_0 \| (\rho_0 + \rho_\Phi)/2) \simeq S(\rho_\Phi \| (\rho_0 + \rho_\Phi)/2) + \dots \quad (3.109)$$

up to the leading order contribution coming from the two-point function of the primaries from the mixing OPE. Similarly, this can be extended to show,

$$S(\rho_\Psi \| (\rho_\Psi + \rho_\Phi)/2) \simeq S(\rho_\Phi \| (\rho_\Psi + \rho_\Phi)/2) + \dots \quad (3.110)$$

Even further, it can be shown that the relative entropy is symmetric under the exchange of $\Psi \leftrightarrow \Phi$ for general binary mixture,

$$S(\rho_\Psi \| y\rho_\Psi + (1-y)\rho_\Phi) \simeq S(\rho_\Phi \| y\rho_\Phi + (1-y)\rho_\Psi) \quad (3.111)$$

$$= -\log y - (C_{\Psi\Phi}^{L'})^2 (2\pi(1-x))^{2\Delta_{L'}} \partial_n \mu(n, x, y; \Delta_{L'})|_{n=1} + \dots \quad (3.112)$$

Similarly, we obtain,

$$\begin{aligned} S(\rho_\Psi \| y\rho_\Phi + (1-y)\rho_\Psi) &\simeq S(\rho_\Phi \| y\rho_\Psi + (1-y)\rho_\Phi) \\ &= -\log(1-y) - (C_{\Psi\Phi}^{L'})^2 (2\pi(1-x))^{2\Delta_{L'}} \partial_n \mu(n, x, 1-y; \Delta_{L'})|_{n=1} + \dots \end{aligned} \quad (3.113)$$

We have analytically continued $\partial_n \mu(n, x, y; \Delta_{L'})$ as $n \rightarrow 1$ (see Appendix F for details). Here we quote the result,

$$\partial_n \mu(n, x, y; \Delta)|_{n=1} = \frac{\sqrt{\frac{1-y}{y}}}{\Gamma(2\Delta+2)} \left| \Gamma\left(\Delta+1 + \frac{i \log \frac{1-y}{y}}{2\pi}\right) \right|^2 \left[\frac{1}{2} + \frac{1}{\pi} \Im \psi\left(\Delta+1 + \frac{i \log \frac{1-y}{y}}{2\pi}\right) \right] \quad (3.114) \quad \{\text{eq:general-y-r}$$

Notice, $y = 1/2$ gives us the old perturbative result back,

$$\partial_n \mu(n, x, 1/2; \Delta_{L'})|_{n=1} = \frac{\Gamma(\Delta_{L'}+1)^2}{2\Gamma(2\Delta_{L'}+2)} = 2^{-2\Delta_{L'}} \frac{\sqrt{\pi}\Gamma(\Delta_{L'}+1)}{4\Gamma(\Delta_{L'}+3/2)} \quad (3.115)$$

An important feature of the above equation is that the result is independent of $\epsilon = 1-x$. Thus, at the leading two-point order, all dependence on the large-interval parameter is contained in the explicit prefactor $(2\pi\epsilon)^{2\Delta_{L'}}$ in the relative entropy; no additional ϵ dependence arises from the analytically continued replica sum. This is consistent with the small- ϵ expansion $\mu = \mu_0 + \epsilon^2 \mu_1 + \dots$, for which we found $\partial_n \mu_1|_{n=1} = 0$. The learning is that $D_n(\epsilon)$ does not contribute to the relative entropy.

3.4 Three-point function

We compute the contribution of three-point function which we expect to be suppressed compared to the two-point function. For $p = 0$, the contribution from both numerator and denominator,

$$f_{3pt}(n) = (C_\Psi^L)^3 \zeta_n^{3\Delta_L} \mathcal{T}_{n,LLL} \quad (3.116)$$

which is suppressed by a factor of $\zeta_n^{3\Delta_L}$ coming from the OPEs of $\Psi\Psi^*$. Note that, for $p = 0$ case, the three-point function contribution from the numerator and the denominator of $\mathcal{F}_n$ are identical, so they cancel each other.

The numerator of $\mathcal{F}_n$, including the corrections from the three-point function would be,

$$1 + \zeta_n^{2\Delta_L} f(n) + \zeta_n^{3\Delta_L} f_{3pt}(n) + \dots \quad (3.117)$$

The denominator of $\mathcal{F}_n$ including the corrections from the three-point functions for $p > 0$ $\langle \mathcal{O}_{L'} \mathcal{O}_{L'} \mathcal{O}_L \rangle$ for $p > 0$, would be,

$$y^{n-1} \left(1 + \zeta_n^{2\Delta_L} f(n) + \zeta_n^{3\Delta_L} f_{3pt}(n) + \sum_{L'} \zeta_n^{2\Delta_{L'}} (C_{\Psi\Phi}^{L'})^2 \mu(n, x, y; \Delta_{L'}) \right. \\ \left. + \sum_{L, L'} \zeta_n^{2\Delta_{L'} + \Delta_L} (C_{\Psi\Phi}^{L'})^2 \mu_G^{\Psi|\Phi}(n, x, y; \Delta_{L'}, \Delta_L) + \dots \right) \quad (3.118)$$

where we define

$$q = y^{-1} - 1 \quad (3.119)$$

$$G_p(n) = \sum_{m=1}^{p-1} \frac{1}{(\sin(\pi m/n) \sin(\pi(p-m)/n))^{\Delta_L}} \quad (3.120)$$

$$\mu_G^{\Psi|\Phi}(n, x, y; \Delta_{L'}, \Delta_L) = \sum_{p=1}^{n-1} \frac{q^p (n-p) D_n(\epsilon)^p}{(\sin(\pi p/n))^{2\Delta_{L'} - \Delta_L}} (C_{\Phi}^L G_p(n) + C_{\Psi}^L G_{n-p}(n)) \quad (3.121)$$

We performed the analytic continuation of the three-point function in Appendix H, and is given by,

$$\partial_n \mu_G^{\Psi|\Phi}(n, x, y; \Delta_{L'}, \Delta_L)|_{n=1} = \frac{3i^{\Delta_L+1} 2^{3+\Delta_L+2\Delta_{L'}} e^{\pi\nu}}{128\pi} \left[C_{\Phi} \left(-\mathcal{H}_{\Delta_L, \Delta_{L'}}(\nu) + \frac{\mathcal{H}_{\Delta_L, \Delta_{L'}}''(\nu)}{\pi^2} \right) \right. \\ \left. - C_{\Psi} \left(\mathcal{H}_{\Delta_L, \Delta_{L'}}(-\nu) + \frac{2}{\pi} \mathcal{H}_{\Delta_L, \Delta_{L'}}'(-\nu) + \frac{\mathcal{H}_{\Delta_L, \Delta_{L'}}''(-\nu)}{\pi^2} \right) \right] \quad (3.122) \quad \{\text{eq: analytic}\}$$

where $\mathcal{H}_{\Delta_L, \Delta_{L'}}(\nu)$ is defined in (H.46). Apriori we don't see any reason why the three-point function would be equal upon swapping $\Psi \leftrightarrow \Phi$.

[Do the analytic continuation of the above sum using the modular flow method](#)

3.5 Some examples

Chiral vertex operators: We have computed the relative entropy of a pure coherent state with respect to averaged binary mixture of coherent state upto the sub-leading order in Appendix I. It is given by,

$$S \left(\rho_{\alpha} \left\| \frac{\rho_{\alpha} + \rho_{\beta}}{2} \right\| \right) = S \left(\rho_{\beta} \left\| \frac{\rho_{\alpha} + \rho_{\beta}}{2} \right\| \right) + \dots \quad (3.123)$$

$$= \log 2 - \frac{\sqrt{\pi} \Gamma \left(1 + \frac{\delta^2}{2} \right)}{4\Gamma \left(\frac{3}{2} + \frac{\delta^2}{2} \right)} (\pi\epsilon)^{\delta^2} + \frac{3\delta^2}{2\pi} \frac{\sqrt{\pi} \Gamma \left(1 + \frac{\delta^2}{2} \right)}{4\Gamma \left(\frac{3}{2} + \frac{\delta^2}{2} \right)} (\pi\epsilon)^{\delta^2+1} + \dots \quad (3.124)$$

where $\delta^2 = (\alpha - \beta)^2$. Similarly, one can extend this result to non-chiral coherent states, where we consider excited states $\rho_{\alpha, \bar{\alpha}}$ and $\rho_{\beta, \bar{\beta}}$. Then we have $\delta^2 = (\alpha - \beta)^2 + (\bar{\alpha} - \bar{\beta})^2$. Notice that the above result is symmetric under $\alpha \leftrightarrow \beta$. This is an artifact of $U(1)$ current algebra.

3.6 Some limits

Limit of $y \rightarrow 1$: This is the trivial limit. Indeed, from (3.93), all terms containing at least one ρ_Φ in the denominator are multiplied by a positive power of $(1 - y)$ and vanish in this limit. Hence, the denominator of $\mathcal{F}_n$ becomes identical to the numerator,

$$\mathcal{F}_n \Big|_{y=1} = 1, \quad (3.125)$$

and therefore

$$S \left(\rho_\Psi \parallel y \rho_\Psi + (1 - y) \rho_\Phi \right) \Big|_{y=1} = S(\rho_\Psi \parallel \rho_\Psi) = 0. \quad (3.126)$$

This is also evident from the classical term $(\log y)$ and the two-point correction term of the relative entropy vanishes identically as $y \rightarrow 1$.

Limit of $y \rightarrow 0$: It is useful to examine the $y \rightarrow 0$ limit of the large-interval result. The leading two-point contribution can be written as

$$S \left(\rho_\Psi \parallel y \rho_\Psi + (1 - y) \rho_\Phi \right) = -\log y - (C_{\Psi\Phi}^{L'})^2 (2\pi\epsilon)^{2\Delta_{L'}} \partial_n \mu(n, x, y; \Delta) \Big|_{n=1} + \dots \quad (3.127)$$

Using the large-imaginary-argument asymptotics of the Gamma and digamma functions, we find

$$\partial_n \mu(n, x, y; \Delta) \Big|_{n=1} = \frac{\left(\log \frac{1-y}{y} \right)^{2\Delta+1}}{(2\pi)^{2\Delta} \Gamma(2\Delta+2)} \left[1 + O \left(\frac{1}{\log \frac{1-y}{y}} \right) \right], \quad y \rightarrow 0. \quad (3.128)$$

Therefore,

$$S \left(\rho_\Psi \parallel y \rho_\Psi + (1 - y) \rho_\Phi \right) = -\log y - \frac{(C_{\Psi\Phi}^{L'})^2}{\Gamma(2\Delta_{L'}+2)} \epsilon^{2\Delta_{L'}} \left(\log \frac{1-y}{y} \right)^{2\Delta_{L'}+1} + \dots \quad (3.129)$$

Since $\log \frac{1-y}{y} = -\log y + O(y)$, the relative correction to the leading $-\log y$ term is controlled by

$$\left(\epsilon \log \frac{1}{y} \right)^{2\Delta_{L'}}. \quad (3.130)$$

Thus, the large-interval expansion obtained at fixed y is not uniform in the limit $y \rightarrow 0$. In particular, when $\log(1/y) \sim O(1/\epsilon)$, higher orders in the large-interval expansion can become equally important and must be resummed.

4 Examples of $x \rightarrow 0^+$ and $x \rightarrow 1^-$ limits

We use Ising and compact free boson CFT to test our analytical results, first at the equal mixing probability $y = 1/2$ and then near $y = 1/2$ in Sec. 4.4. For an ordered pair (Ψ, Φ) , we use the reduced states and averaged mixture of the analytical discussion,

$$\bar{\rho}_C = \frac{\rho_C^\Psi + \rho_C^\Phi}{2}, \quad C = A, B. \quad (4.1)$$

in which A and B represent the small and large interval, separately.

For the spin chains contain N sites with unit lattice spacing, the circumference is $R = N$. For a short interval A of length l , the plotted quantity is $S(\rho_A^\Psi \| \bar{\rho}_A)$ and $x = l/R$. For its large complement $B = \bar{A}$ of length $R - l$, the interval fraction is $x = |B|/R = 1 - l/R$; we plot $\log 2 - S(\rho_B^\Psi \| \bar{\rho}_B)$ against $\epsilon = 1 - x = l/R$.

The equal- and unequal-mixture calculations use the complete grid of system sizes $N \in 4\mathbb{N}$ through $N = 8192$ for Ising and $N = 16384$ for the compact boson, with $l = 6, \dots, 10$. Every geometry satisfying $x \leq 0.1$ or $\epsilon \leq 0.05$ is evaluated in the corresponding regime. We compare three fitting models: a free power law, a one-term fit M_1 with the theoretical leading power, and a two-term fit M_2 with fixed leading and next powers. All displayed mixture fits use the same windows for both CFTs: $0 < x \leq 0.02$ and $0 < \epsilon \leq 0.005$. These are comparison windows; the leading amplitudes and free powers do not uniformly attain one-percent agreement with CFT. Sampling is uniform in N , not in x or ϵ . We denote a fitted power by $\hat{r}$ and the theoretical CFT leading and next leading exponent by r_0 and r_1 , respectively. The small-interval leading power is $2\Delta_L$ for a contributing light primary or current, and 4 for the stress-tensor contribution; the large-interval power is $2\Delta_{L'}$. Appendix L gives the constructions of the primary states on spin chains, entropy algorithms, sampling grid, fitting procedure, and numerical precision checks.

4.1 Compact free boson

With $s = 1/\sqrt{2}$, $|00\rangle$ denotes the ground state, while $|ss\rangle$ denotes the primary state that corresponds to the vertex operator $V(z, \bar{z}) = \exp\left(\frac{i}{\sqrt{2}}\phi(z) + \frac{i}{\sqrt{2}}\bar{\phi}(\bar{z})\right)$. We also consider another primary $|3s, s\rangle$ that corresponds to the vertex operator $V(z, \bar{z}) = \exp\left(i\frac{3}{\sqrt{2}}\phi(z) + \frac{i}{\sqrt{2}}\bar{\phi}(\bar{z})\right)$. Consider the mixture state with $y = 1/2$, $\bar{\rho}_A^{(\chi)} = \frac{1}{2}(\rho_A^{00} + \rho_A^\chi)$.

For $x \rightarrow 0^+$, the current contribution has $\Delta_L = 1$. The vertex charges are $(\alpha, \bar{\alpha}) = (s, s)$ for $|ss\rangle$ and $(\alpha, \bar{\alpha}) = (3s, s)$ for $|3s, s\rangle$. Hence,

$$S(\rho_A^{00} \| \bar{\rho}_A^{(\chi)}) = \begin{cases} \frac{\pi^2}{12}x^2 + \dots & \chi = |ss\rangle \\ \frac{5\pi^2}{12}x^2 + \dots & \chi = |3s, s\rangle \end{cases} \quad (4.2)$$

For $x \rightarrow 1^-$ limit, we have to consider the $\Delta_{L'}$ for the mixing OPE. For $|ss\rangle$, $\Delta_{L'} = \frac{1}{2} \left((1/\sqrt{2} - 0)^2 + (1/\sqrt{2} - 0)^2 \right) = 1/2$. The mixing OPE coefficient $C_{\alpha,I}^\alpha = \bar{C}_{\bar{\alpha},I}^{\bar{\alpha}} = 1$. Hence,

$$\delta_\beta^{(\chi)} = \log 2 - \frac{1}{2} \left(S(\rho_B^{00} \|\bar{\rho}_B^{(\chi)}) + S(\rho_B^\chi \|\bar{\rho}_B^{(\chi)}) \right) \quad (4.3)$$

$$= \log 2 - \frac{1}{2} \left(\log 2 - (\pi\epsilon) \frac{\sqrt{\pi}\Gamma(3/2)}{4\Gamma(2)} + O(\epsilon^2) \right) - \frac{1}{2} \left(\log 2 - (\pi\epsilon) \frac{\sqrt{\pi}\Gamma(3/2)}{4\Gamma(2)} + O(\epsilon^2) \right) \quad (4.4)$$

$$= \frac{\pi^2\epsilon}{8} + O(\epsilon^2) \quad (4.5)$$

$$= 1.234\epsilon + O(\epsilon^2) \quad (4.6)$$

Similarly, for $|3s, s\rangle$, $\Delta_{L'} = \frac{1}{2} \left((3/\sqrt{2} - 0)^2 + (1/\sqrt{2} - 0)^2 \right) = 5/2$. And the mixing OPE coefficients are $C_{\alpha,I}^\alpha = 1$ and $\bar{C}_{\bar{\alpha},I}^{\bar{\alpha}} = 1$. Hence,

$$\delta_\beta^{(\chi)} = \log 2 - \frac{1}{2} \left(\log 2 - (\pi\epsilon)^5 \frac{\sqrt{\pi}\Gamma(7/2)}{4\Gamma(4)} + O(\epsilon^6) \right) - \frac{1}{2} \left(\log 2 - (\pi\epsilon)^5 \frac{\sqrt{\pi}\Gamma(7/2)}{4\Gamma(4)} + O(\epsilon^6) \right) \quad (4.7)$$

$$= (\pi\epsilon)^5 \frac{\sqrt{\pi}\Gamma(7/2)}{4\Gamma(4)} + O(\epsilon^6) \quad (4.8)$$

$$= 75.108\epsilon^5 + O(\epsilon^6). \quad (4.9)$$

In the above two cases, notice, $S(\rho_B^{00} \|\bar{\rho}_B^{(\chi)}) = S(\rho_B^\chi \|\bar{\rho}_B^{(\chi)}) + \dots$.

For the states $|s, s\rangle$ and $|3s, s\rangle$, we have $\delta^2 = 2$, and therefore

$$\begin{aligned} S\left(\rho_{s,s} \left\| \frac{\rho_{s,s} + \rho_{3s,s}}{2} \right.\right) &= S\left(\rho_{3s,s} \left\| \frac{\rho_{s,s} + \rho_{3s,s}}{2} \right.\right) + O(\epsilon^4) \\ &= \log 2 - \frac{\pi^2}{3}\epsilon^2 + \pi^2\epsilon^3 + O(\epsilon^4) \\ &= \log 2 - 3.289868\epsilon^2 + 9.869604\epsilon^3 + O(\epsilon^4). \end{aligned} \quad (4.10)$$

For the states $|00\rangle$ and $|3s, s\rangle$, the total squared charge difference is $\delta^2 = 5$, and we obtain

$$\begin{aligned} S\left(\rho_{00} \left\| \frac{\rho_{00} + \rho_{3s,s}}{2} \right.\right) &= S\left(\rho_{3s,s} \left\| \frac{\rho_{00} + \rho_{3s,s}}{2} \right.\right) + O(\epsilon^7) \\ &= \log 2 - \frac{5\pi^6}{64}\epsilon^5 + \frac{75\pi^6}{128}\epsilon^6 + O(\epsilon^7) \\ &= \log 2 - 75.10853075\epsilon^5 + 563.31398061\epsilon^6 + O(\epsilon^7). \end{aligned} \quad (4.11)$$

4.1.1 Numerical verification at small intervals

{sec:num-boson-}

The current contribution predicts $r_0 = 2\Delta_L = 2$ in all six directions. The leading amplitudes for $00 \leftrightarrow ss$, $00 \leftrightarrow 3s, s$, and $ss \leftrightarrow 3s, s$ are $\pi^2/12$, $5\pi^2/12$, and $\pi^2/6$, respectively. Figure 1

and Table 2 use all 19 984 geometries per direction in $0 < x \leq 0.02$. The free exponents range from 1.99873 to 1.99984, within 0.064% of the quadratic prediction. The two-term fits use powers two and four. Their leading-amplitude errors are approximately -5.023% , -3.410% , and -0.990% for $00 \leftrightarrow ss$, $00 \leftrightarrow 3s, s$, and $ss \leftrightarrow 3s, s$, respectively. Each full-sample M_2 amplitude is closer to CFT than its M_1 counterpart.

The persistent offsets are visible in Fig. 2. Increasing N makes x small while keeping the subsystem resolution fixed at six to ten sites; it does not establish a continuum extrapolation. For $00 \leftrightarrow ss$, the improvement of c_0 is only about 0.015 percentage points and reverses for 11 of the 4022 fits that omit one complete chain size. These deletions test coefficient sensitivity, not out-of-sample prediction or statistical uncertainty. The fourth-order fitted amplitudes have no complete analytical coefficient for comparison and are therefore numerical estimates.

(Ψ, Φ)	$\hat{r}$	C_{free}	c_0^{CFT}	$c_0(M_1)$	$c_0(M_2)$	c_1^{CFT}	$c_1(M_2)$	$\eta_0(M_1)$	$\eta_0(M_2)$	$\eta_1(M_2)$
$(00, ss)$	1.99984	0.780482	0.822467	0.781025	0.781152	—	-0.523334	-5.03871	-5.02334	—
$(ss, 00)$	1.99984	0.780488	0.822467	0.781026	0.781152	—	-0.518795	-5.03859	-5.02336	—
$(00, (3s, s))$	1.99873	3.94643	4.11234	3.96781	3.97209	—	-17.7358	-3.51439	-3.41026	—
$((3s, s), 00)$	1.99876	3.94717	4.11234	3.96793	3.97209	—	-17.2311	-3.51152	-3.41035	—
$(ss, (3s, s))$	1.99958	1.62517	1.64493	1.62806	1.62864	—	-2.41934	-1.02581	-0.990298	—
$((3s, s), ss)$	1.99964	1.62564	1.64493	1.62814	1.62864	—	-2.09719	-1.02115	-0.990366	—

Table 2. Compact-boson small-interval fits on $0 < x \leq 0.02$, using all 19 984 geometries per direction from 4022 chain sizes on the complete step-4 grid, with $l = 6, \dots, 10$. The free fit is $C_{\text{free}}x^{\hat{r}}$; M_1, M_2 use the powers in Table 7. All amplitudes include their powers of π . The η_j columns give signed percentage discrepancies from CFT, Eq. (L.22), computed before rounding. A dash denotes an unavailable complete next coefficient.

{tab:num-boson-

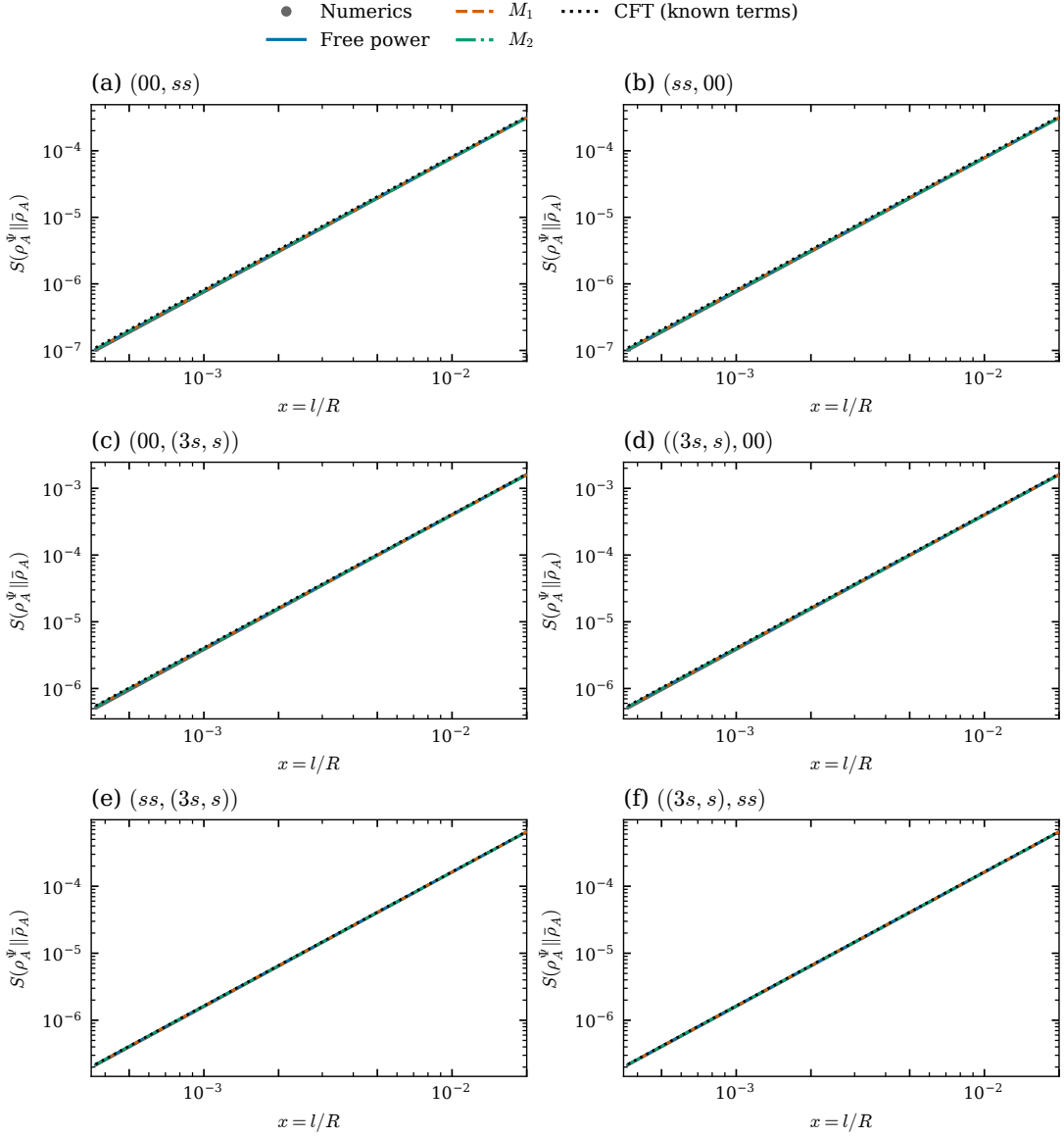


Figure 1. Compact-boson small-interval $S(\rho_A^\Psi || \bar{\rho}_A)$ for all six ordered pairs (Ψ, Φ) . Every panel contains all 19984 observations in $0 < x \leq 0.02$ from 4022 step-4 chain sizes, with interval lengths $l = 6, \dots, 10$. Gray circles show the data; colored curves are free, M_1 , and M_2 fits. The black CFT curve includes the complete known next term where available and the leading term otherwise. Both axes are logarithmic.

{fig:num-boson-

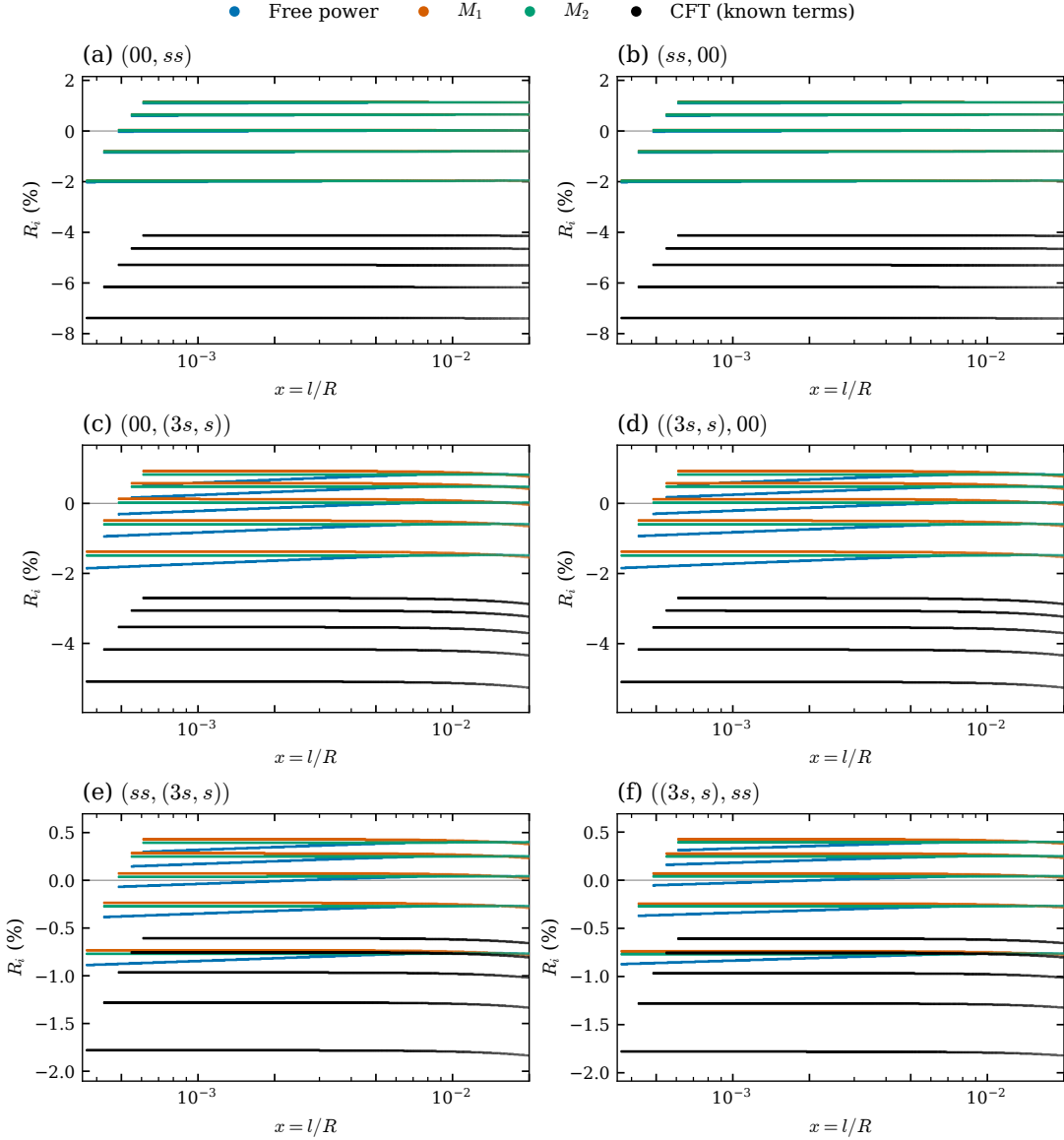


Figure 2. Signed percentage residuals $R_i = 100[v_i - M(z_i)]/v_i$ [Eq. (L.23)] for Fig. 1, using the same 19984 observations per direction and window $0 < x \leq 0.02$. Colors identify the three fitted models and the CFT curve with the same analytical terms as in the value figure. All lengths share circular markers. The residual ordinate is linear and the fraction axis logarithmic.

{fig:num-boson-}

4.1.2 Numerical verification at large intervals

{sec:num-boson-}

The deficit resolves the approach to $S(\rho_B^\Psi \| \bar{\rho}_B) = \log 2$ when B is not the entire spin chain. For the same three unordered pairs, theory predicts

$$(2\Delta_{L'}, c_0^{\text{CFT}}) = (1, \pi^2/8), \quad (5, 5\pi^6/64), \quad (2, \pi^2/3), \quad (4.12) \quad \text{{eq:num-boson-}}$$

with the same leading term in both directions. Each plotted deficit uses one source state. Fig. 3 and Table 3 show free exponents within 0.671% of the predicted values on $0 < \epsilon \leq 0.005$. The fifth-power channel gives $\hat{r} = 4.96690, 4.96646$; its M_2 leading amplitudes are 75.8559, 75.8548, with signed errors +0.995%, +0.994% relative to $5\pi^6/64$.

The added term improves c_0 in four of the six directions. For $00 \leftrightarrow ss$, the M_1 error of approximately +2.233% instead becomes +2.354% in M_2 ; for $ss \leftrightarrow 3s, s$, the M_2 leading errors are about -1.009%. Thus the window does not give one-percent leading agreement in every channel. Figure 4 compares the residuals of all fitted models with the available CFT series.

The complete next-order deficit coefficients are $c_1^{\text{CFT}} = -75\pi^6/128$ for $00 \leftrightarrow 3s, s$ and $c_1^{\text{CFT}} = -\pi^2$ for $ss \leftrightarrow 3s, s$; their signs are opposite to those in the expansion of S . The signed discrepancies are +6.475% and +7.851% for $00 \leftrightarrow 3s, s$, and -6.616% and -4.262% for $ss \leftrightarrow 3s, s$. All four known next coefficients therefore agree within 7.851% on the complete system-size grid. The CFT curves in the figures include these terms. For $00 \leftrightarrow ss$, the four-point (broken-word) contribution also enters at $O(\epsilon^2)$, so the three-point result alone does not supply a complete c_1^{CFT} (Appendix I).

(Ψ, Φ)	$\hat{r}$	C_{free}	c_0^{CFT}	$c_0(M_1)$	$c_0(M_2)$	c_1^{CFT}	$c_1(M_2)$	$\eta_0(M_1)$	$\eta_0(M_2)$	$\eta_1(M_2)$
$(00, ss)$	0.99898	1.2535	1.2337	1.26125	1.26274	—	-0.540093	2.2332	2.35376	—
$(ss, 00)$	0.998976	1.25347	1.2337	1.26124	1.26274	—	-0.542542	2.23266	2.35377	—
$(00, (3s, s))$	4.9669	61.1646	75.1085	73.1534	75.8559	-563.314	-599.791	-2.60301	0.995014	6.47545
$((3s, s), 00)$	4.96646	60.9885	75.1085	73.1175	75.8548	-563.314	-607.538	-2.65093	0.993565	7.85072
$(ss, (3s, s))$	1.99177	3.07628	3.28987	3.22204	3.25668	-9.8696	-9.21666	-2.06178	-1.00885	-6.61574
$((3s, s), ss)$	1.99157	3.07181	3.28987	3.22115	3.25666	-9.8696	-9.44894	-2.08874	-1.00928	-4.26223

Table 3. Compact-boson large-interval fits on $0 < \epsilon \leq 0.005$, using all 18 485 geometries per direction from 3797 chain sizes on the complete step-4 grid, with $l = 6, \dots, 10$. The free fit is $C_{\text{free}}\epsilon^{\hat{r}}$; M_1, M_2 use the powers in Table 7. All amplitudes include their powers of π . The η_j columns give signed percentage discrepancies from CFT, Eq. (L.22), computed before rounding. A dash denotes an unavailable complete next coefficient.

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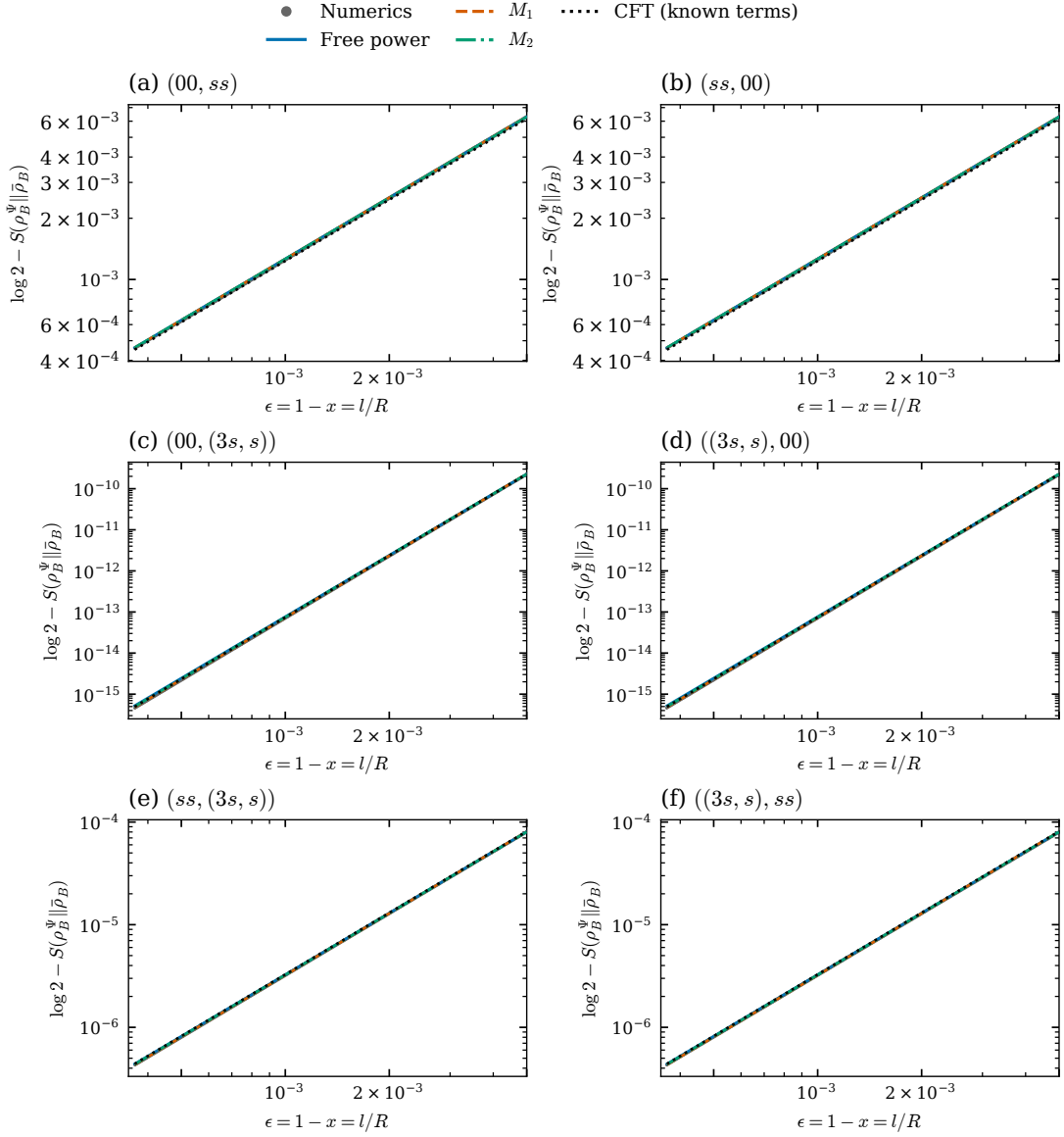


Figure 3. Compact-boson large-interval $\log 2 - S(\rho_B^\Psi || \bar{\rho}_B)$ for all six ordered pairs (Ψ, Φ) . Every panel contains all 18 485 observations in $0 < \epsilon \leq 0.005$ from 3797 step-4 chain sizes, with short-complement lengths $l = 6, \dots, 10$. Gray circles show the data; colored curves are free, M_1 , and M_2 fits. The black CFT curve includes the complete known next term where available and the leading term otherwise. Both axes are logarithmic.

{fig:num-boson-

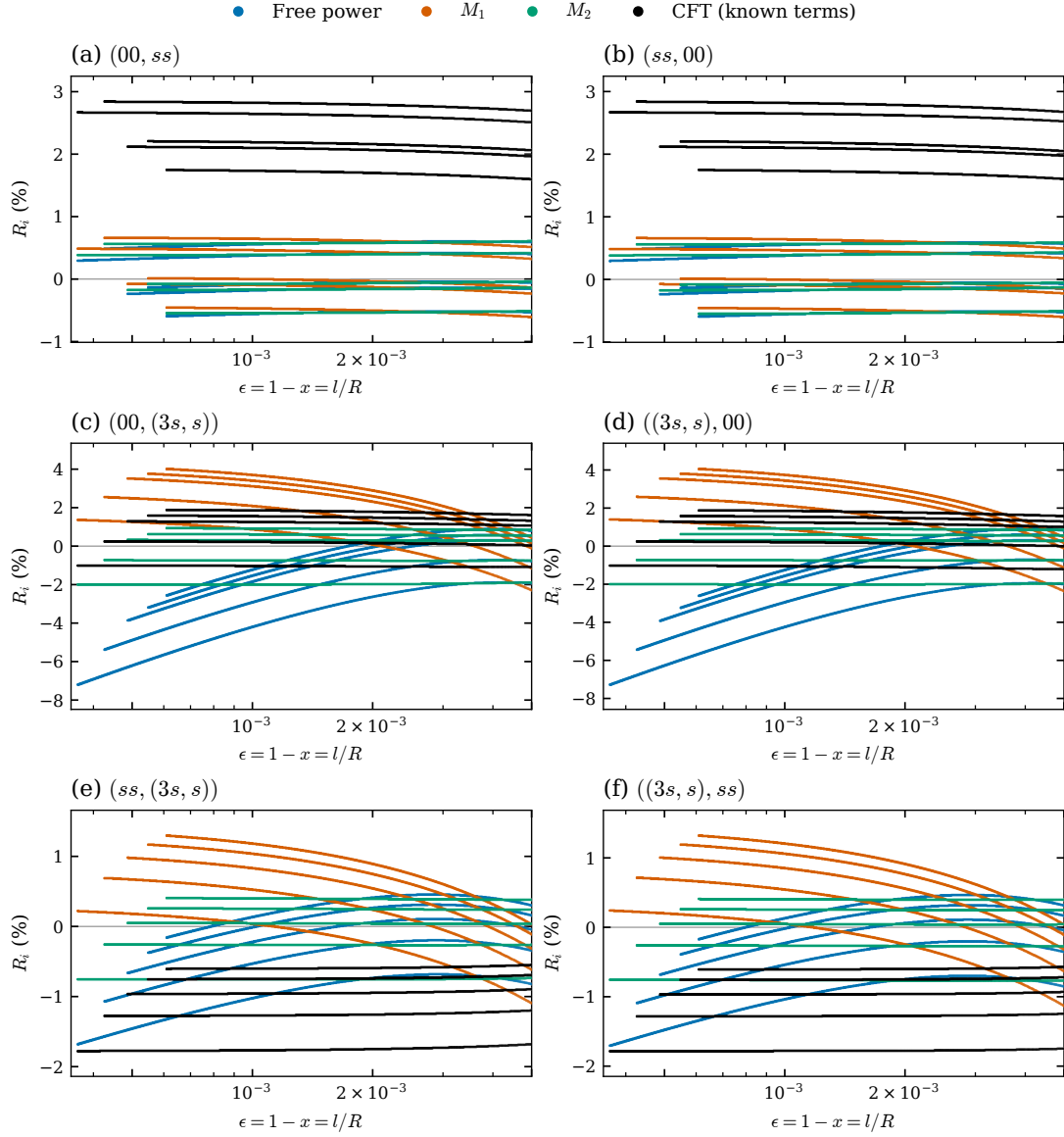


Figure 4. Signed percentage residuals $R_i = 100[v_i - M(z_i)]/v_i$ [Eq. (L.23)] for Fig. 3, using the same 18 485 observations per direction and window $0 < \epsilon \leq 0.005$. Colors identify the three fitted models and the CFT curve with the same analytical terms as in the value figure. All lengths share circular markers. The residual ordinate is linear and the fraction axis logarithmic.

{fig:num-boson-

4.2 Ising CFT

We have two non-identity primaries, namely, $\sigma \equiv (\frac{1}{16}, \frac{1}{16})$ and $\varepsilon \equiv (\frac{1}{2}, \frac{1}{2})$. We have the following OPEs,

$$\sigma(z_1) \times \sigma(z_2) = \frac{1}{|z_{12}|^{1/4}} \left(1 + \frac{1}{2}|z_{12}|\varepsilon(z_2) + \dots \right) \quad (4.13)$$

$$\varepsilon(z_1) \times \varepsilon(z_2) = \frac{1}{|z_{12}|^2} \left(1 + 2|z_{12}|^2 T(z_2) + 2|\bar{z}_{12}|^2 \bar{T}(\bar{z}_2) + \dots \right) \quad (4.14)$$

$$\varepsilon(z_1) \times \sigma(z_2) = \frac{1}{2|z_{12}|} \left(\sigma(z_2) + 4|z_{12}|\partial\sigma(z_2) + 4|\bar{z}_{12}|\bar{\partial}\sigma(z_2) + \dots \right) \quad (4.15)$$

$$\sigma(z_1) \times \varepsilon(z_2) = \frac{1}{2|z_{12}|} \left(\sigma(z_2) - 3|z_{12}|\partial\sigma(z_2) - 3|\bar{z}_{12}|\bar{\partial}\sigma(z_2) + \dots \right) \quad (4.16)$$

So, for small interval, we have the following results,

$$S(\rho_0 \| y\rho_0 + (1-y)\rho_\sigma) = \frac{1}{12}(1-y)^2(\pi x)^2 + \underbrace{O(x^4)}_{\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle, \langle T\varepsilon\varepsilon\rangle, \langle\bar{T}\varepsilon\varepsilon\rangle, \langle TT\rangle, \langle\bar{T}\bar{T}\rangle} \quad (4.17)$$

$$S(\rho_\sigma \| y\rho_0 + (1-y)\rho_\sigma) = \frac{1}{12}y^2(\pi x)^2 + \underbrace{O(x^4)}_{\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle, \langle T\varepsilon\varepsilon\rangle, \langle\bar{T}\varepsilon\varepsilon\rangle, \langle TT\rangle, \langle\bar{T}\bar{T}\rangle} \quad (4.18)$$

$$S(\rho_0 \| y\rho_0 + (1-y)\rho_\varepsilon) = \frac{8}{15}(1-y)^2(\pi x)^4 + \frac{32}{315}(1-y)^2(16y-15)(\pi x)^6 + \underbrace{O(x^8)}_{\langle TTTT\rangle, \langle\bar{T}\bar{T}\bar{T}\bar{T}\rangle, \text{desc.}} \quad (4.19)$$

$$S(\rho_\varepsilon \| y\rho_0 + (1-y)\rho_\varepsilon) = \frac{8}{15}y^2(\pi x)^4 + \frac{32}{315}y^2(16y-23)(\pi x)^6 + \underbrace{O(x^8)}_{\langle TTTT\rangle, \langle\bar{T}\bar{T}\bar{T}\bar{T}\rangle, \text{desc.}} \quad (4.20)$$

$$S(\rho_\sigma \| y\rho_\sigma + (1-y)\rho_\varepsilon) = \frac{1}{12}(1-y)^2(\pi x)^2 + \underbrace{O(x^4)}_{\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle, \langle T\varepsilon\varepsilon\rangle, \langle\bar{T}\varepsilon\varepsilon\rangle, \langle TT\rangle, \langle\bar{T}\bar{T}\rangle} \quad (4.21)$$

$$S(\rho_\varepsilon \| y\rho_\sigma + (1-y)\rho_\varepsilon) = \frac{1}{12}y^2(\pi x)^2 + \underbrace{O(x^4)}_{\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle, \langle T\varepsilon\varepsilon\rangle, \langle\bar{T}\varepsilon\varepsilon\rangle, \langle TT\rangle, \langle\bar{T}\bar{T}\rangle} \quad (4.22)$$

where we used the two-point function, three-point and descendant of stress tensor in the third and fourth line, which comes from the OPEs of a pair of ε . For the remaining equations, the $O(x^4)$ will involve the computation of 4 point function $\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle$ which is not included in our computation, since the 4 point function for a general CFT is non-universal.

Similarly, for the large interval limit, we have the following,

$$S(\rho_0 \| y\rho_0 + (1-y)\rho_\sigma) = S(\rho_\sigma \| y\rho_\sigma + (1-y)\rho_0) + \dots \quad (4.23)$$

$$= -\log y - (2\pi\epsilon)^{1/4} \mu'(1, x, y; 1/8) + \underbrace{O(\epsilon^{1/2})}_{\langle \sigma\sigma\sigma\sigma \rangle} + \underbrace{O(\epsilon^{5/4})}_{\langle \sigma\sigma\epsilon \rangle} + \dots \quad (4.24)$$

$$= -\log y - \left[(2\pi\epsilon)^{1/4} \frac{\sqrt{\frac{1-y}{y}}}{\Gamma(\frac{9}{4})} \left| \Gamma\left(\frac{9}{8} + \frac{i}{2\pi} \log \frac{1-y}{y}\right) \right|^2 \right. \\ \left. \times \left(\frac{1}{2} + \frac{1}{\pi} \Im \psi\left(\frac{9}{8} + \frac{i}{2\pi} \log \frac{1-y}{y}\right) \right) \right] + \underbrace{O(\epsilon^{1/2})}_{\langle \sigma\sigma\sigma\sigma \rangle} + \underbrace{O(\epsilon^{5/4})}_{\langle \sigma\sigma\epsilon \rangle} + \dots \quad (4.25)$$

$$S(\rho_0 \| y\rho_0 + (1-y)\rho_\epsilon) = S(\rho_\epsilon \| y\rho_\epsilon + (1-y)\rho_0) + \dots \quad (4.26)$$

$$= -\log y - (2\pi\epsilon)^2 \mu'(1, x, y; 1) + \underbrace{O(\epsilon^4)}_{\substack{\langle \epsilon\epsilon\epsilon\epsilon \rangle, \langle T\epsilon\epsilon \rangle, \langle \bar{T}\epsilon\epsilon \rangle, \\ \langle TT \rangle, \langle \bar{T}\bar{T} \rangle}} + \dots \quad (4.27)$$

$$= -\log y - \left[\frac{(2\pi\epsilon)^2}{6} \sqrt{\frac{1-y}{y}} \left| \Gamma\left(2 + \frac{i}{2\pi} \log \frac{1-y}{y}\right) \right|^2 \right. \\ \left. \times \left(\frac{1}{2} + \frac{1}{\pi} \Im \psi\left(2 + \frac{i}{2\pi} \log \frac{1-y}{y}\right) \right) \right] + \underbrace{O(\epsilon^4)}_{\substack{\langle \epsilon\epsilon\epsilon\epsilon \rangle, \langle T\epsilon\epsilon \rangle, \langle \bar{T}\epsilon\epsilon \rangle, \\ \langle TT \rangle, \langle \bar{T}\bar{T} \rangle}} + \dots \quad (4.28)$$

$$S(\rho_\sigma \| y\rho_\sigma + (1-y)\rho_\epsilon) = S(\rho_\epsilon \| y\rho_\epsilon + (1-y)\rho_\sigma) + \dots \quad (4.29)$$

$$= -\log y - \frac{1}{4} (2\pi\epsilon)^{1/4} \mu'(1, x, y; 1/8) + \underbrace{O(\epsilon^{1/2})}_{\langle \sigma\sigma\sigma\sigma \rangle} + \underbrace{O(\epsilon^{5/4})}_{\langle \sigma\sigma\epsilon \rangle} + \dots \quad (4.30)$$

$$= -\log y - \left[\frac{1}{4} (2\pi\epsilon)^{1/4} \frac{\sqrt{\frac{1-y}{y}}}{\Gamma(\frac{9}{4})} \left| \Gamma\left(\frac{9}{8} + \frac{i}{2\pi} \log \frac{1-y}{y}\right) \right|^2 \right. \\ \left. \times \left(\frac{1}{2} + \frac{1}{\pi} \Im \psi\left(\frac{9}{8} + \frac{i}{2\pi} \log \frac{1-y}{y}\right) \right) \right] + \underbrace{O(\epsilon^{1/2})}_{\langle \sigma\sigma\sigma\sigma \rangle} + \underbrace{O(\epsilon^{5/4})}_{\langle \sigma\sigma\epsilon \rangle} + \dots \quad (4.31)$$

where the 1/4 in the third case comes from the square of the OPE coefficient $C_{\sigma\epsilon}^\sigma = 1/2$.

Particular to the averaged mixture ($y = 1/2$), we have the following result,

$$S\left(\rho_0 \left\| \frac{\rho_0 + \rho_\sigma}{2} \right\| \right) = S\left(\rho_\sigma \left\| \frac{\rho_0 + \rho_\sigma}{2} \right\| \right) + \dots \quad (4.32)$$

$$= \log 2 - \frac{\sqrt{\pi}\Gamma(9/8)}{4\Gamma(13/8)}(\pi\epsilon)^{1/4} + \underbrace{O(\epsilon^{1/2})}_{\langle\sigma\sigma\sigma\sigma\rangle} + \underbrace{O(\epsilon^{5/4})}_{\langle\sigma\sigma\epsilon\rangle} + \dots \quad (4.33)$$

$$S\left(\rho_0 \left\| \frac{\rho_0 + \rho_\epsilon}{2} \right\| \right) = S\left(\rho_\epsilon \left\| \frac{\rho_0 + \rho_\epsilon}{2} \right\| \right) + \dots \quad (4.34)$$

$$= \log 2 - \frac{(\pi\epsilon)^2}{3} + \underbrace{O(\epsilon^4)}_{\langle\epsilon\epsilon\epsilon\epsilon\rangle, \langle T\epsilon\epsilon\rangle, \langle\bar{T}\epsilon\epsilon\rangle, \langle TT\rangle, \langle\bar{T}\bar{T}\rangle} + \dots \quad (4.35)$$

$$S\left(\rho_\sigma \left\| \frac{\rho_\sigma + \rho_\epsilon}{2} \right\| \right) = S\left(\rho_\epsilon \left\| \frac{\rho_\sigma + \rho_\epsilon}{2} \right\| \right) + \dots \quad (4.36)$$

$$= \log 2 - \frac{1}{4} \frac{\sqrt{\pi}\Gamma(9/8)}{4\Gamma(13/8)}(\pi\epsilon)^{1/4} + \underbrace{O(\epsilon^{1/2})}_{\langle\sigma\sigma\sigma\sigma\rangle} + \underbrace{O(\epsilon^{5/4})}_{\langle\sigma\sigma\epsilon\rangle} + \dots \quad (4.37)$$

4.2.1 Numerical verification at small intervals

At $y = 1/2$, the analytical results above predict $r_0 = 2$ and $c_0^{\text{CFT}} = \pi^2/48$ for the four directions involving σ , and $r_0 = 4$ and $c_0^{\text{CFT}} = 2\pi^4/15$ for $0 \leftrightarrow \epsilon$. Fig. 5 and Table. 4 test these two distinct powers. On $0 < x \leq 0.02$, the free exponents are 2.00015, 2.00012 for $0 \leftrightarrow \sigma$, 3.99357, 3.98652 for $0 \leftrightarrow \epsilon$, and 2.01778, 2.01316 for $\sigma \leftrightarrow \epsilon$. The largest relative power discrepancy is 0.889%.

The two-term fits use the exponents $r_0, r_0 + 2$ and improve the leading amplitude in all six directions. Their absolute leading errors are below 0.809% for $0 \leftrightarrow \epsilon$ and below 0.042% for the other directions. The two complete sixth-order coefficients provide a further test:

$$\begin{aligned} c_1^{\text{CFT}}(0, \epsilon) &= -8\pi^6/45, & \hat{c}_1 &= -170.284, \\ c_1^{\text{CFT}}(\epsilon, 0) &= -8\pi^6/21, & \hat{c}_1 &= -355.101. \end{aligned} \quad (4.38)$$

The signed discrepancies are -0.369% and -3.042% . These results resolve the directional asymmetry of the sixth-order term as well as the common leading power, with a measurable discrepancy in the reverse direction. Figure 6 shows the residuals; the other fitted next coefficients have no complete analytical value for comparison.

(Ψ, Φ)	$\hat{r}$	C_{free}	c_0^{CFT}	$c_0(M_1)$	$c_0(M_2)$	c_1^{CFT}	$c_1(M_2)$	$\eta_0(M_1)$	$\eta_0(M_2)$	$\eta_1(M_2)$
$(0, \sigma)$	2.00015	0.205857	0.205617	0.205727	0.205702	—	0.103855	0.0537552	0.0415586	—
$(\sigma, 0)$	2.00012	0.205827	0.205617	0.20572	0.205699	—	0.0868425	0.0502305	0.0400317	—
$(0, \varepsilon)$	3.99357	12.5007	12.9879	12.8306	12.8839	-170.914	-170.284	-1.21105	-0.800709	-0.368568
$(\varepsilon, 0)$	3.98652	12.0927	12.9879	12.7718	12.8829	-366.244	-355.101	-1.66383	-0.808121	-3.04231
(σ, ε)	2.01778	0.225042	0.205617	0.208697	0.205657	—	12.5928	1.49828	0.0193938	—
(ε, σ)	2.01316	0.21985	0.205617	0.207913	0.205665	—	9.30932	1.11671	0.0234328	—

Table 4. Ising small-interval fits on $0 < x \leq 0.02$, using all 9 744 geometries per direction from 1974 chain sizes on the complete step-4 grid, with $l = 6, \dots, 10$. The free fit is $C_{\text{free}} x^{\hat{r}}$; M_1, M_2 use the powers in Table 7. All amplitudes include their powers of π . The η_j columns give signed percentage discrepancies from CFT, Eq. (L.22), computed before rounding. A dash denotes an unavailable complete next coefficient.

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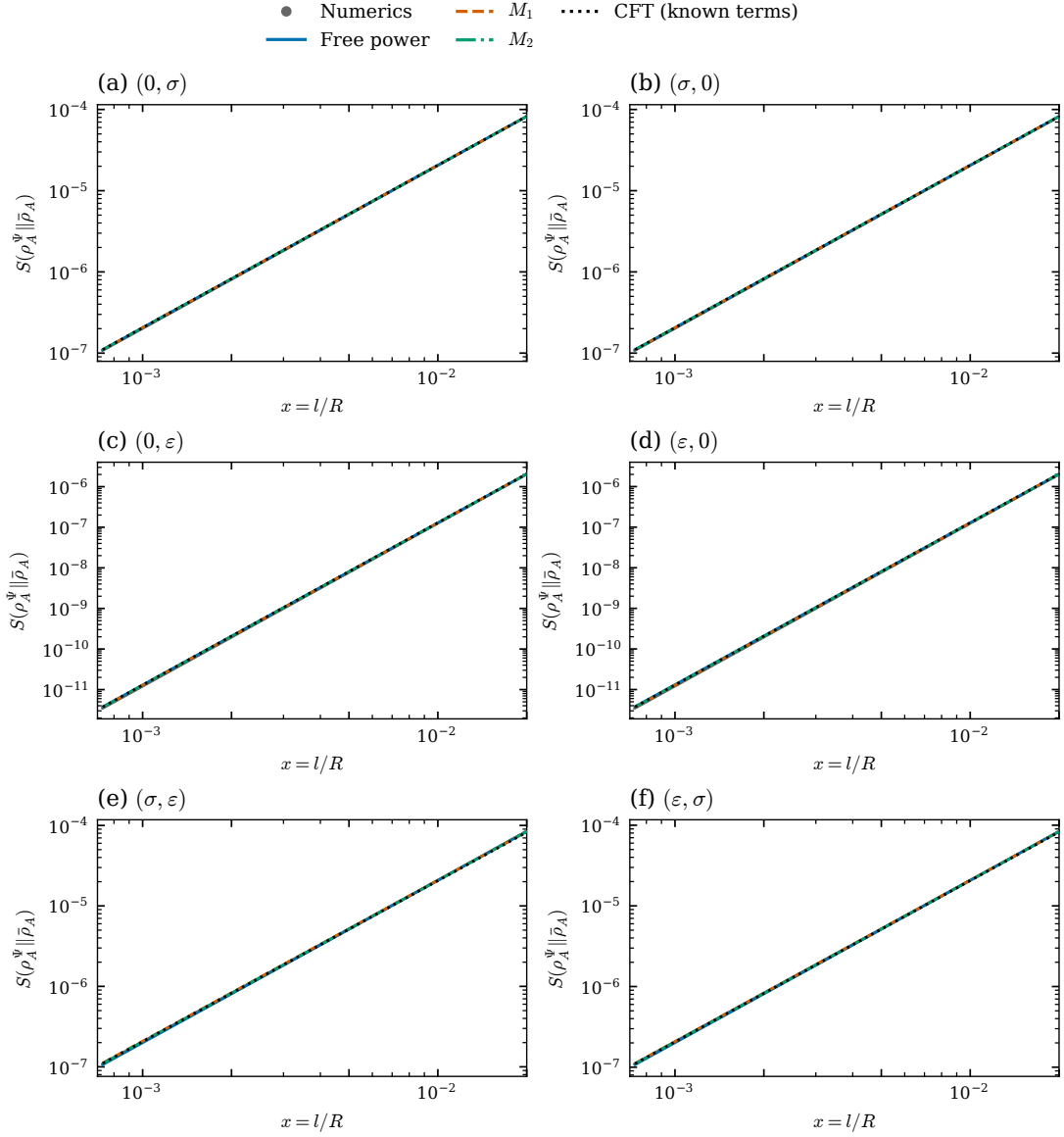


Figure 5. Ising small-interval $S(\rho_A^\Psi || \bar{\rho}_A)$ for all six ordered pairs (Ψ, Φ) . Every panel contains all 9744 observations in $0 < x \leq 0.02$ from 1974 step-4 chain sizes, with interval lengths $l = 6, \dots, 10$. Gray circles show the data; colored curves are free, M_1 , and M_2 fits. The black CFT curve includes the complete known next term where available and the leading term otherwise. Both axes are logarithmic.

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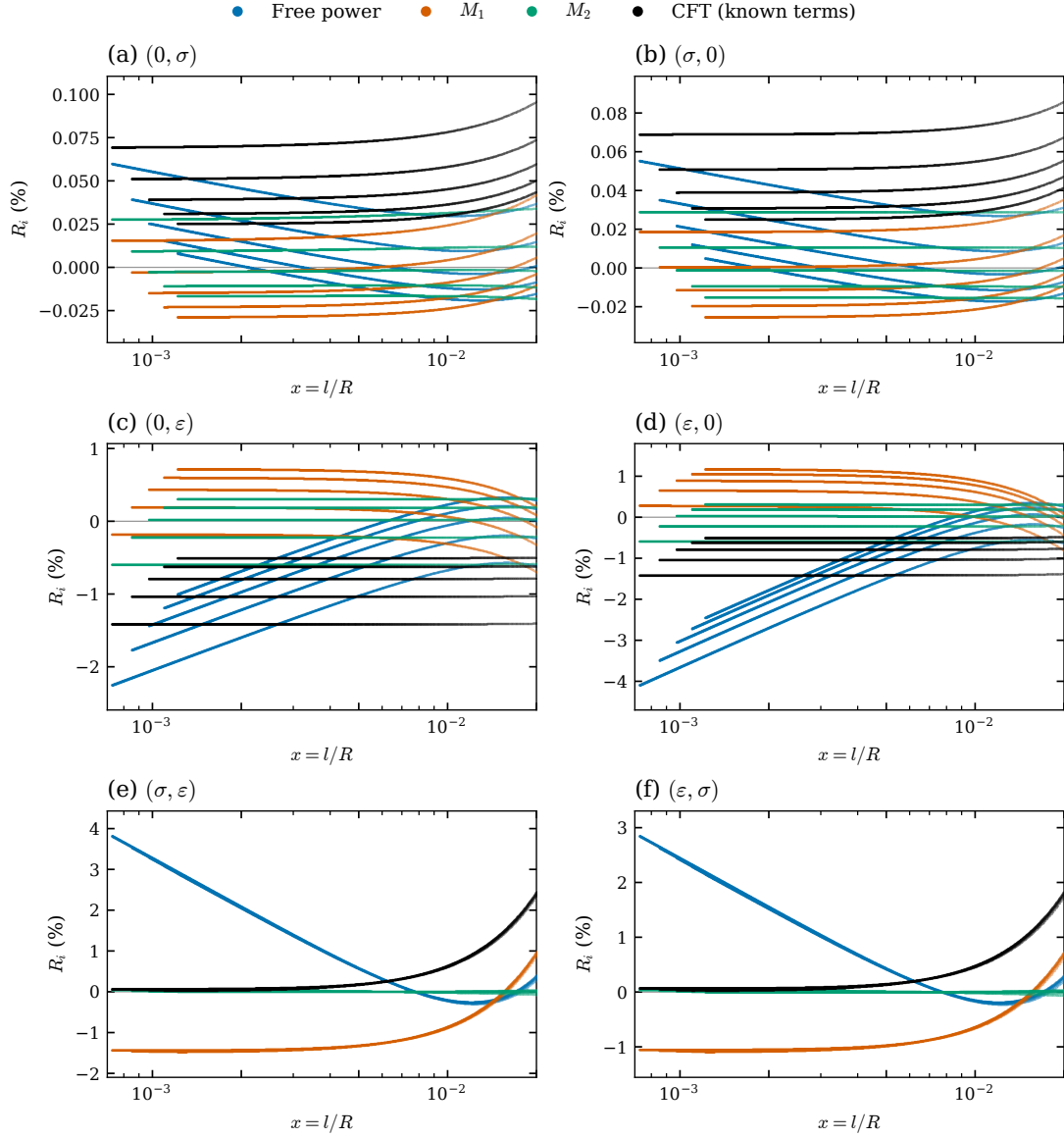


Figure 6. Signed percentage residuals $R_i = 100[v_i - M(z_i)]/v_i$ [Eq. (L.23)] for Fig. 5, using the same 9744 observations per direction and window $0 < x \leq 0.02$. Colors identify the three fitted models and the CFT curve with the same analytical terms as in the value figure. All lengths share circular markers. The residual ordinate is linear and the fraction axis logarithmic.

{fig:num-ising}

4.2.2 Numerical verification at large intervals

{sec:num-ising-

The different mixing OPEs predict deficit powers $2\Delta_{L'} = 1/4, 2, 1/4$ for $0 \leftrightarrow \sigma$, $0 \leftrightarrow \varepsilon$, and $\sigma \leftrightarrow \varepsilon$, respectively. Their leading amplitudes are

$$\begin{aligned} c_0^{\text{CFT}}(0, \sigma) &= \frac{\pi^{3/4} \Gamma(9/8)}{4\Gamma(13/8)} \simeq 0.61965, \\ c_0^{\text{CFT}}(0, \varepsilon) &= \frac{\pi^2}{3}, \quad c_0^{\text{CFT}}(\sigma, \varepsilon) = \frac{1}{4} c_0^{\text{CFT}}(0, \sigma). \end{aligned} \quad (4.39) \quad \{\text{eq:num-ising-}$$

Fig. 7 and Table. 5 compare these predictions with the independently evaluated deficits.

On $0 < \epsilon \leq 0.005$, $0 \leftrightarrow \sigma$ has effective free powers 0.266806, 0.266809, or 6.722%, 6.724% above $1/4$. Including the $\epsilon^{1/2}$ term reduces the leading-amplitude error from about +6.131% in M_1 to -0.970% , -0.971% in M_2 . For (σ, ε) and (ε, σ) , the free powers are 0.249116 and 0.257340, with relative errors -0.353% and $+2.936\%$. Their M_2 leading errors are $+1.245\%$ and -1.278% . For $0 \leftrightarrow \varepsilon$, the free powers 1.99705, 2.00278 remain close to two, and the M_2 amplitudes differ from $\pi^2/3$ by -0.115% , $+0.196\%$.

Five of the six directions improve in leading-amplitude accuracy. The exception is (σ, ε) , whose absolute leading error increases from 0.870% to 1.245%. Figure 8 displays the remaining deviations. In particular, the vacuum-spin free exponent does not verify the quarter-power prediction to one percent on this window. The fixed powers of M_2 are inputs to that fit, and its next coefficients remain numerical estimates without a complete analytical comparison.

(Ψ, Φ)	$\hat{r}$	C_{free}	c_0^{CFT}	$c_0(M_1)$	$c_0(M_2)$	c_1^{CFT}	$c_1(M_2)$	$\eta_0(M_1)$	$\eta_0(M_2)$	$\eta_1(M_2)$
$(0, \sigma)$	0.266806	0.730252	0.619653	0.657641	0.613643	—	0.207521	6.13062	-0.969902	—
$(\sigma, 0)$	0.266809	0.730268	0.619653	0.657644	0.613637	—	0.207559	6.13101	-0.970817	—
$(0, \varepsilon)$	1.99705	3.22532	3.28987	3.27922	3.28607	—	-451.714	-0.32368	-0.115459	—
$(\varepsilon, 0)$	2.00278	3.35484	3.28987	3.30282	3.29633	—	428.379	0.393833	0.196368	—
(σ, ε)	0.249116	0.155403	0.154913	0.156261	0.156842	—	-0.00273668	0.870241	1.24479	—
(ε, σ)	0.25734	0.164933	0.154913	0.157558	0.152934	—	0.0218096	1.70704	-1.2779	—

Table 5. Ising large-interval fits on $0 < \epsilon \leq 0.005$, using all 8 245 geometries per direction from 1749 chain sizes on the complete step-4 grid, with $l = 6, \dots, 10$. The free fit is $C_{\text{free}} \hat{r}$; M_1, M_2 use the powers in Table 7. All amplitudes include their powers of π . The η_j columns give signed percentage discrepancies from CFT, Eq. (L.22), computed before rounding. A dash denotes an unavailable complete next coefficient.

{tab:num-ising-

4.3 Four-point function

We want to know if the four point function contribution of $\langle \mathcal{O}_{L'} \mathcal{O}_{L'} \mathcal{O}_{L'} \mathcal{O}_{L'} \rangle$, in the case of $S(\rho_\Psi \| y \rho_\Psi + (1-y) \rho_\Phi)$ and exchanging $\Psi \leftrightarrow \Phi$, i.e. $S(\rho_\Phi \| y \rho_\Phi + (1-y) \rho_\Psi)$ the same.

Consider the case, when $\Delta_L > 2\Delta_{L'}$. This corresponds to the case where the four-point function $\langle \mathcal{O}_{L'} \mathcal{O}_{L'} \mathcal{O}_{L'} \mathcal{O}_{L'} \rangle$ is of lower order than the three-point function $\langle \mathcal{O}_L \mathcal{O}_{L'} \mathcal{O}_{L'} \rangle$. Using

Replica trick, we want to evaluate the following,

$$\mathcal{F}_n^{\Psi|\Phi} \equiv \frac{\text{tr} \rho_\Psi^n}{\text{tr} \left(\rho_\Psi (y \rho_\Psi + (1-y) \rho_\Phi)^{n-1} \right)} \quad (4.40)$$

$$= \frac{1 + \zeta_n^{2\Delta_L} f(n) + \dots}{y^{n-1} \left[1 + \zeta_n^{2\Delta_L} f(n) + (C_{\Psi\Phi}^{L'})^2 \zeta_n^{2\Delta_{L'}} \sum_{p=0}^{n-1} \frac{\left(\frac{1-y}{y} D_n^{\Psi|\Phi}(\epsilon) \right)^p}{(2 \sin(\pi p/n))^{2\Delta_{L'}}} + (C_{\Psi\Phi}^{L'})^4 \zeta_n^{4\Delta_{L'}} \mu_4^{\Psi|\Phi}(n, x, y; \Delta_{L'}) + \dots \right]} \quad (4.41)$$

$$\mu_4^{\Psi|\Phi}(n, x, y; \Delta_{L'}) = \sum_{\substack{a, b, c, d \in \mathbb{Z}_{>0} \\ a+b+c+d=n}} a \left(\frac{1-y}{y} D_n(\epsilon)^{\Psi|\Phi} \right)^{b+d} \mathcal{G}_{4,n}^{L'}(b, c, d), \quad (4.42)$$

$$\mathcal{G}_{4,n}^{L'}(b, c, d) \equiv \langle \mathcal{O}_{L'}(u_0) \mathcal{O}_{L'}(u_b) \mathcal{O}_{L'}(u_{b+c}) \mathcal{O}_{L'}(u_{b+c+d}) \rangle. \quad (4.43)$$

Note that, the uniform word contribution ($p = 0$) in the denominator gets cancelled exactly from the numerator. So, we are left with the four point function contribution from $p > 1$ cases. Similarly, for the swapped case, one can write its denominator as

$$\text{tr} \left(\rho_\Phi (y \rho_\Phi + (1-y) \rho_\Psi)^{n-1} \right) = y^{n-1} \text{tr} \left[1 + (C_{\Psi\Phi}^{L'})^2 \zeta_n^{2\Delta_{L'}} \mu(n, x, y; \Delta_{L'}) + (C_{\Psi\Phi}^{L'})^4 \zeta_n^{4\Delta_{L'}} \mu_4^{\Phi|\Psi} + \dots \right], \quad (4.44)$$

$$\mu_4^{\Phi|\Psi}(n, x, y; \Delta_{L'}) = \sum_{\substack{a, b, c, d \in \mathbb{Z}_{>0} \\ a+b+c+d=n}} a \left(\frac{1-y}{y} D_n^{\Psi|\Phi}(\epsilon)^{-1} \right)^{b+d} \mathcal{G}_{4,n}^{L'}(b, c, d). \quad (4.45)$$

Here we used $D_n^{\Psi|\Phi}(\epsilon) = D_n^{\Phi|\Psi}(\epsilon)^{-1}$. Let us define the following quantities,

$$q \equiv \frac{1-y}{y}, \quad \lambda \equiv q D_n^{\Psi|\Phi}(\epsilon), \quad (4.46)$$

$$\mathcal{M}_4(n, \lambda) \equiv \mu_4^{\Psi|\Phi}(n, x, y; \Delta_{L'}) = \sum_{\substack{a+b+c+d \geq 1 \\ a+b+c+d=n}} a \lambda^{b+d} \mathcal{G}_{4,n}^{L'}(b, c, d). \quad (4.47)$$

Then, we take the derivative of the contribution of the four-point function with respect to n ,

$$\partial_n \mu_4^{\Psi|\Phi}(n, x, y; \Delta_{L'}) = \partial_n \mathcal{M}_4(n, \lambda) + q D_n^{\Psi|\Phi}(\epsilon)' \partial_\lambda \mathcal{M}_4(n, \lambda), \quad (4.48)$$

$$\implies \partial_n \mu_4^{\Psi|\Phi}(n, x, y; \Delta_{L'}) \Big|_{n=1, \lambda=q} = \partial_n \mathcal{M}_4(n, \lambda) \Big|_{n=1} + q D_n^{\Psi|\Phi}(\epsilon)' \partial_\lambda \mathcal{M}_4(n, \lambda) \Big|_{n=1, \lambda=q}. \quad (4.49) \quad \{\text{eq: 4 point}\}$$

However, we have

$$\mathcal{M}_4(1, \lambda) = 0, \quad \forall \lambda > 0, \quad (4.50)$$

$$\implies \partial_\lambda \mathcal{M}_4(1, \lambda) = 0. \quad (4.51)$$

Thus the second term in (4.49) vanishes. We are left with,

$$\partial_n \mu_4^{\Psi|\Phi}(n, x, y; \Delta_{L'})|_{n=1} = \partial_n \mathcal{M}_4(n, \lambda)|_{n=1, \lambda=q}. \quad (4.52)$$

Similarly, we can evaluate the derivative of the swapped case,

$$\partial_n \mu_4^{\Phi|\Psi}(n, x, y; \Delta_{L'})|_{n=1} = \partial_n \mathcal{M}_4(n, \lambda)|_{n=1} - q D_n^{\Psi|\Phi}(\epsilon)' \partial_\lambda \mathcal{M}_4(n, \lambda)|_{n=1, \lambda=q} \quad (4.53)$$

$$= \partial_n \mathcal{M}_4(n, \lambda)|_{n=1, \lambda=q} \quad (4.54)$$

We finally obtain that the four point function contribution to the relative entropy remain invariant under swapping $\Psi \leftrightarrow \Phi$. In other words,

$$\partial_n \mu_4^{\Psi|\Phi}(n, x, y; \Delta_{L'}) = \partial_n \mu_4^{\Phi|\Psi}(n, x, y; \Delta_{L'}) \quad (4.55)$$

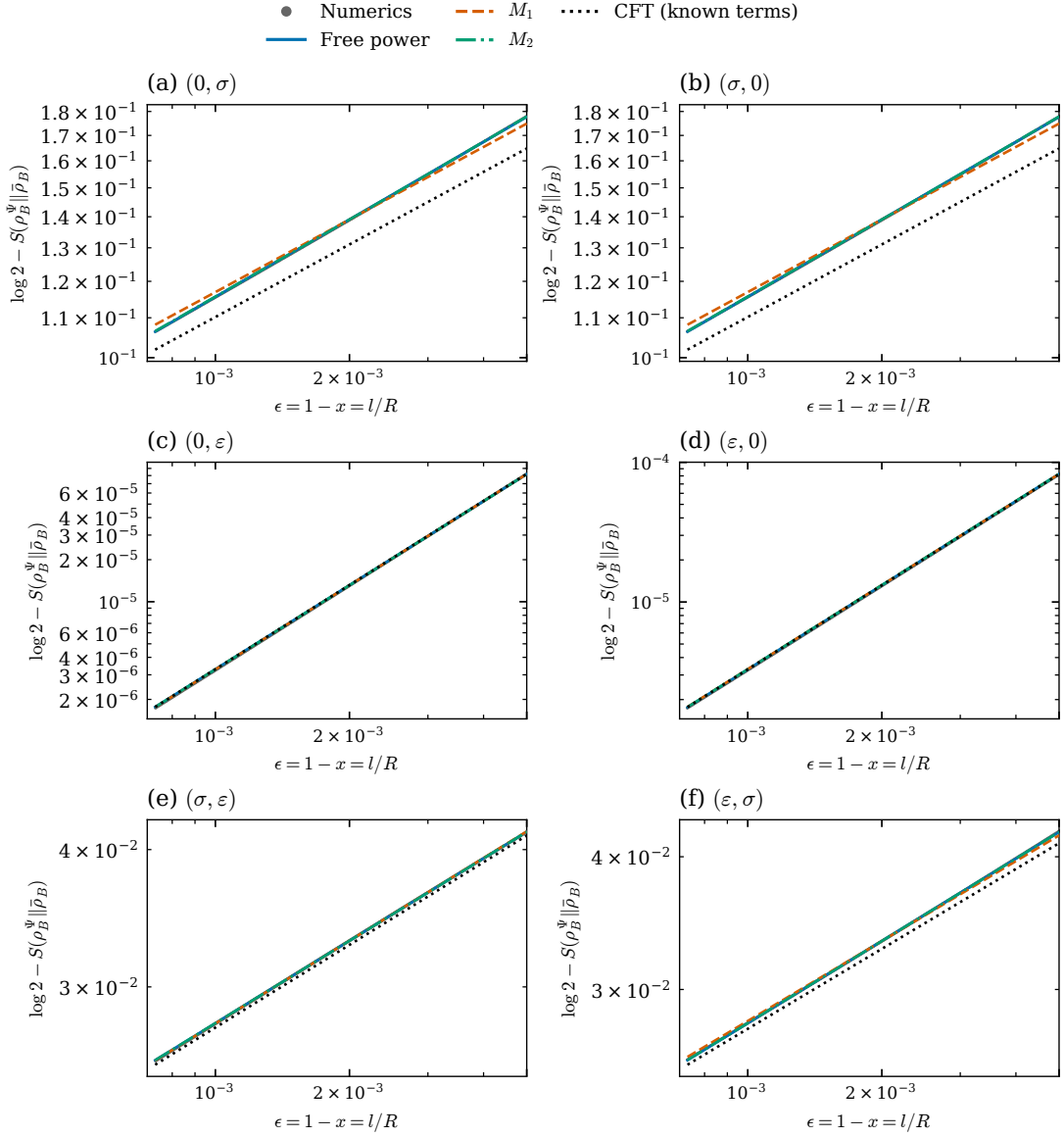


Figure 7. Ising large-interval $\log 2 - S(\rho_B^\Psi \| \bar{\rho}_B)$ for all six ordered pairs (Ψ, Φ) . Every panel contains all 8245 observations in $0 < \epsilon \leq 0.005$ from 1749 step-4 chain sizes, with short-complement lengths $l = 6, \dots, 10$. Gray circles show the data; colored curves are free, M_1 , and M_2 fits. The black CFT curve includes the complete known next term where available and the leading term otherwise. Both axes are logarithmic.

{fig:num-ising-

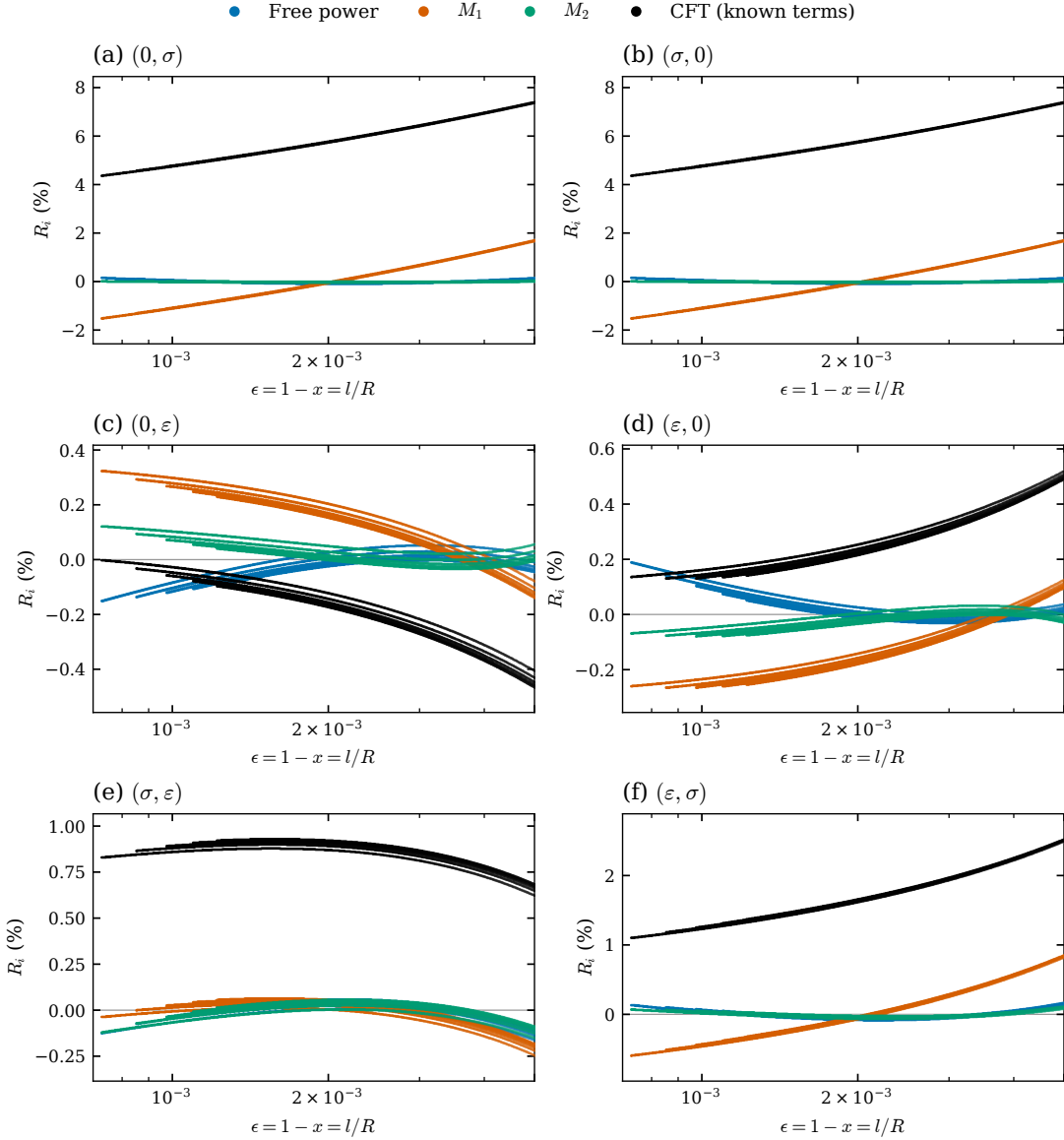


Figure 8. Signed percentage residuals $R_i = 100[v_i - M(z_i)]/v_i$ [Eq. (L.23)] for Fig. 7, using the same 8245 observations per direction and window $0 < \epsilon \leq 0.005$. Colors identify the three fitted models and the CFT curve with the same analytical terms as in the value figure. All lengths share circular markers. The residual ordinate is linear and the fraction axis logarithmic.

{fig:num-ising}

4.4 Mixing-probability dependence near $y = 1/2$

{sec:num-gener}

To test the mixing-probability dependence, we use $\rho_{C,y} = y\rho_C^\Psi + (1-y)\rho_C^\Phi$, with $C = A, B$, and write $\delta = y - 1/2$. Thus y is always the weight of the source state in $S(\rho_C^\Psi \| \rho_{C,y})$; exchanging the ordered pair exchanges the source and its weight together. The reverse relative entropy to the same fixed mixture is instead obtained by replacing y with $1 - y$.

For small intervals, Eq. (3.39) gives the exact mixing-probability dependence of the leading coefficient:

$$c_0^{\text{CFT}}(y) = 4(1-y)^2 c_0^{\text{CFT}}(1/2) = c_0^{\text{CFT}}(1/2)(1-4\delta+4\delta^2), \quad S(\rho_A^\Psi \| \rho_{A,y}) = c_0^{\text{CFT}}(y)x^{r_0} + \dots \quad (4.56)$$

For the reverse relative entropy to a fixed mixture, the corresponding factor is $4y^2 = 1+4\delta+4\delta^2$.

For large intervals, set $\Delta = \Delta_{L'}$ and $m_\Delta(y) = \partial_n \mu(n, x, y; \Delta)|_{n=1}$ in Eq. (3.114). Its expansion follows from $\log[(1-y)/y] = -4\delta + O(\delta^3)$ and the derivatives of the Gamma and digamma functions:

$$m_\Delta(y) = \frac{\Gamma(\Delta+1)^2}{2\Gamma(2\Delta+2)} [1 - A_\Delta \delta + A_\Delta \delta^2 + O(\delta^3)],$$

$$A_\Delta = 2 + \frac{4}{\pi^2} \psi_1(\Delta+1), \quad \psi_1(z) = \frac{d\psi(z)}{dz}. \quad (4.57)$$

Consequently, the leading coefficient of the deficit is

$$c_0^{\text{CFT}}(y) = (C_{\Psi\Phi}^{L'})^2 (2\pi)^{2\Delta} m_\Delta(y) = c_0^{\text{CFT}}(1/2) [1 - A_\Delta \delta + A_\Delta \delta^2 + O(\delta^3)],$$

$$c_0^{\text{CFT}}(1/2) = (C_{\Psi\Phi}^{L'})^2 \pi^{2\Delta} \frac{\sqrt{\pi} \Gamma(\Delta+1)}{4\Gamma(\Delta+3/2)}, \quad -\log y - S(\rho_B^\Psi \| \rho_{B,y}) = c_0^{\text{CFT}}(y) \epsilon^{2\Delta} + \dots \quad (4.58)$$

Including $-\log y = \log 2 - 2\delta + 2\delta^2 + O(\delta^3)$ gives

$$S(\rho_B^\Psi \| \rho_{B,y}) = \log 2 - 2\delta + 2\delta^2 - c_0^{\text{CFT}}(1/2) \epsilon^{2\Delta} (1 - A_\Delta \delta + A_\Delta \delta^2) + O(\delta^3) + o(\epsilon^{2\Delta}). \quad (4.59)$$

The last remainder denotes terms beyond the leading large-interval order. The values of A_Δ for $\Delta = 1/8, 1/2, 1$, and $5/2$ are approximately 2.562589, 2.378861, 2.261382, and 2.133889, respectively.

Figures 9–12 compare the leading CFT coefficients with $\bar{c}_0(y)$, the coefficient c_0 fitted independently at each y using M_2 in Eq. (L.20); the bar denotes a fitted value, not an average over probabilities or directions. The lower panels use the signed percentage discrepancy η_0 of Eq. (L.22), with $c_0 = \bar{c}_0(y)$ and $c_0^{\text{CFT}} = c_0^{\text{CFT}}(y)$. The state construction and fitting procedure are those of Appendix L. The probability-dependent study now uses the same complete step-4 grid as the equal-mixture results: $l = 6, \dots, 10$, $N \in 4\mathbb{N}$ through 8192 for Ising and 16384 for the compact boson. Each ordered pair is evaluated at the forty source weights $y_j = 0.4 + j/195$, $j = 0, \dots, 39$. The production windows $x \leq 0.1$ and $\epsilon \leq 0.05$ contain 14 605 920 unequal-weight observations. The displayed fits use $0 < x \leq 0.02$ and $0 < \epsilon \leq 0.005$ for both CFTs; their exact grids are specified in the captions. The forty-point probability grid excludes $y = 1/2$, whose complete 365 148 observations are retained separately as a reference.

The small-interval M_2 leading amplitudes agree with CFT within 0.811% for Ising, whereas the largest compact-boson discrepancy is 5.024%. The corresponding large-interval

bounds are 1.662% and 2.411%. To distinguish these absolute offsets from the probability dependence, we also compare $\bar{c}_0(y)/\bar{c}_0(1/2)$ with $c_0^{\text{CFT}}(y)/c_0^{\text{CFT}}(1/2)$, using the same fitting window at every weight. The absolute percentage differences stay below 0.00205% and 0.000288% for small Ising and boson intervals, and below 0.412% and 0.147% for their large complements. This agreement of the normalized dependence does not remove the absolute amplitude offsets; the figures display the unnormalized coefficients. The known small-interval Ising $0 \leftrightarrow \varepsilon$ sixth-order coefficients also agree with the analytical expressions within 3.408% at all forty weights, after expressing each direction in terms of its source weight y . The unrestricted small-interval powers remain within 0.938% of CFT for all weights and directions. The largest large-Ising free-power error is approximately 6.92%, so this window does not give uniform one-percent verification of the leading powers.

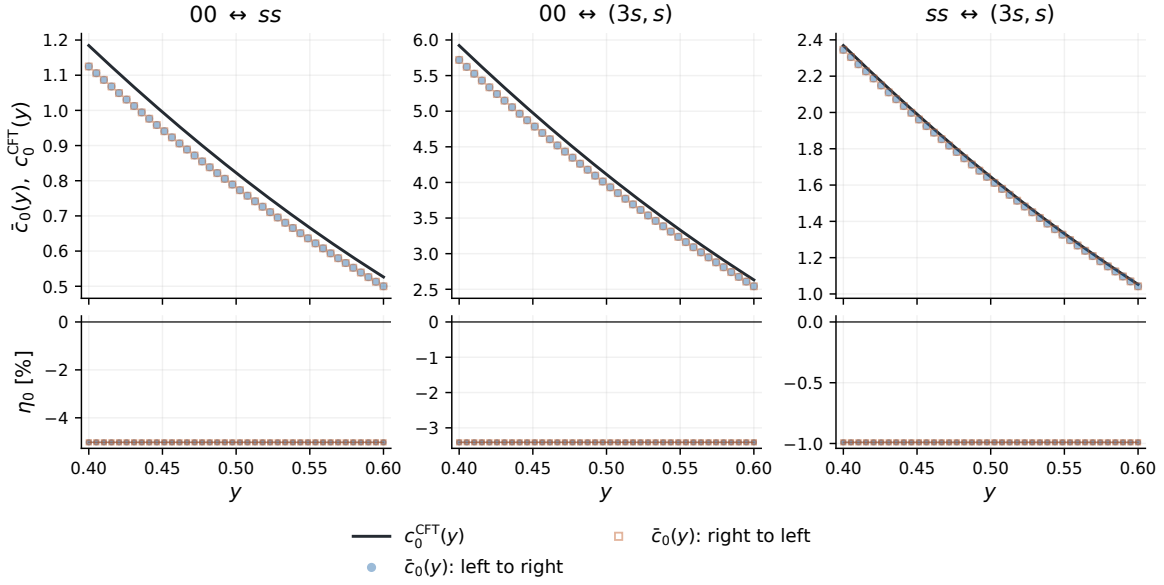


Figure 9. Compact-boson small-interval leading coefficients versus the source weight y . The upper panels show the fitted $\bar{c}_0(y)$ from $S(\rho_A^\Psi \| \rho_{A,y})$ and the solid theoretical curves $c_0^{\text{CFT}}(y) = K\pi^2(1-y)^2/3$, with $K = 1, 5, 2$ from left to right [Eq. (4.56)]; $s = 1/\sqrt{2}$ and $r_0 = 2$. Filled blue circles take the source as the left state in each panel title, and open orange squares take the right state; y is the source weight in both cases. Each direction contains exactly 40 equally spaced probabilities $y_j = 0.4 + j/195$, $j = 0, \dots, 39$. Each M_2 fit uses 19984 geometries from 4022 chain sizes $N = 300, 304, \dots, 16384$, with $l = 6, \dots, 10$ and $0 < x = l/N \leq 0.02$. At each N , only lengths satisfying this cutoff enter the fit. The powers are those in Table 7. The lower panels show $\eta_0[\%] = 100[\bar{c}_0(y)/c_0^{\text{CFT}}(y) - 1]$; zero indicates agreement, and the largest absolute discrepancy is 5.023%. All axes are linear; the coefficients include their powers of π .

{fig:num-boson-

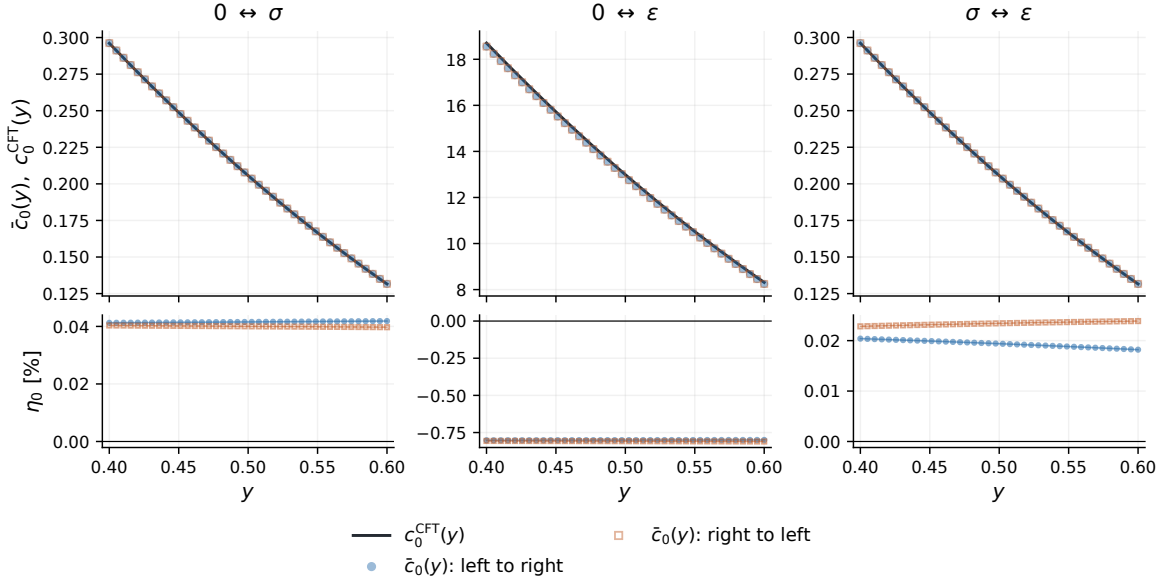


Figure 10. Ising small-interval leading coefficients versus the source weight y . The upper panels compare the independently fitted $\bar{c}_0(y)$ from $S(\rho_A^\Psi \|\rho_{A,y})$ with the solid CFT curves [Eq. (4.56)]. For $0 \leftrightarrow \sigma$, $0 \leftrightarrow \varepsilon$, and $\sigma \leftrightarrow \varepsilon$, respectively, the predictions are $c_0^{\text{CFT}}(y) = \pi^2(1-y)^2/12$, $8\pi^4(1-y)^2/15$, and $\pi^2(1-y)^2/12$, with $r_0 = 2, 4, 2$. Filled blue circles use the left state in each panel title as source; open orange squares use the right state. Each direction has 40 equally spaced $y_j = 0.4 + j/195$, $j = 0, \dots, 39$, with y always the source weight. Each M_2 fit uses 9744 geometries from 1974 chain sizes $N = 300, 304, \dots, 8192$, with $l = 6, \dots, 10$ and $0 < x = l/N \leq 0.02$. Only lengths satisfying the cutoff at each N enter the fit. The powers are those in Table 7. The lower panels give $\eta_0[\%] = 100[\bar{c}_0(y)/c_0^{\text{CFT}}(y) - 1]$; the largest absolute discrepancy is 0.810%. All axes are linear.

{fig:num-ising}

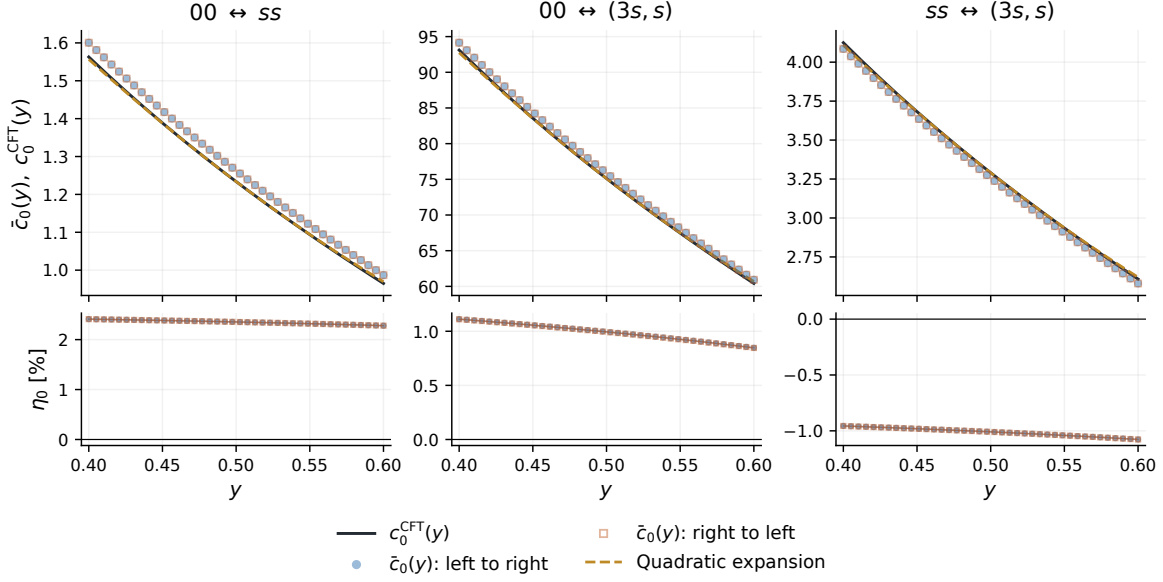


Figure 11. Compact-boson large-interval leading coefficients versus the source weight y . The upper panels show $\bar{c}_0(y)$ fitted to the deficit $-\log y - S(\rho_B^\Psi \| \rho_{B,y})$ and the solid CFT curves from Eq. (4.58). From left to right, $\Delta_{L'} = 1/2, 5/2, 1$ and $(C_{\Psi\Phi}^{L'})^2 = 1$; the half-weight coefficients are $\pi^2/8$, $5\pi^6/64$, and $\pi^2/3$. Dashed gold curves show the quadratic expansion in $\delta = y - 1/2$. Filled blue circles use the left state in each panel title as source; open orange squares use the right state, with y the source weight in both cases. Each direction contains 40 equally spaced $y_j = 0.4 + j/195$, $j = 0, \dots, 39$. Each M_2 fit uses 18 485 geometries from 3797 chain sizes $N = 1200, 1204, \dots, 16384$, with $l = 6, \dots, 10$ and $0 < \epsilon = l/N \leq 0.005$. Only eligible lengths at each N enter the fit. The powers are those in Table 7. The lower panels give $\eta_0[\%] = 100[\bar{c}_0(y)/c_0^{\text{CFT}}(y) - 1]$ relative to the full theoretical coefficient, not its quadratic approximation; the largest absolute discrepancy is 2.410%. All axes are linear.

{fig:num-boson-

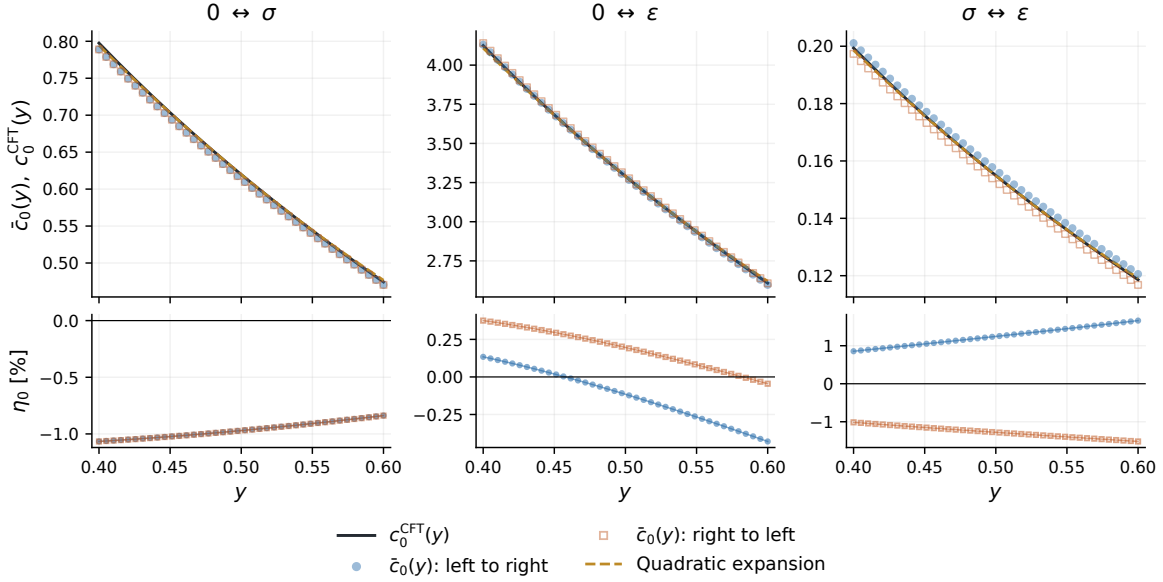


Figure 12. Ising large-interval leading coefficients versus the source weight y . The upper panels compare $\bar{c}_0(y)$ fitted to $-\log y - S(\rho_B^\Psi \| \rho_{B,y})$ with the solid CFT curves from Eq. (4.58). From left to right, $(\Delta_{L'}, (C_{\Psi\Phi}^{L'})^2) = (1/8, 1), (1, 1), (1/8, 1/4)$; the half-weight coefficients are given in Eq. (4.39). Dashed gold curves show the quadratic expansion in $\delta = y - 1/2$. Filled blue circles use the left state in each panel title as source, and open orange squares use the right state. Each direction contains 40 equally spaced $y_j = 0.4 + j/195$, $j = 0, \dots, 39$, with y always the source weight. Each M_2 fit uses 8245 geometries from 1749 chain sizes $N = 1200, 1204, \dots, 8192$, with $l = 6, \dots, 10$ and $0 < \epsilon = l/N \leq 0.005$. Only lengths satisfying the cutoff at each N enter the fit. The powers are those in Table 7. The lower panels show $\eta_0[\%] = 100[\bar{c}_0(y)/c_0^{\text{CFT}}(y) - 1]$ relative to the full theoretical coefficient, not its quadratic approximation; the largest absolute discrepancy is 1.661%. All axes are linear.

{fig:num-ising}

5 An application to Quantum Error Correcting Codes

{sec:QEC}

The relative entropies studied in this work have a direct application to approximate quantum error correction in conformal field theories. In particular, Ref. [1], considers code subspaces spanned by a finite collection of CFT primary states and characterizes the information lost under erasure of a spatial region in terms of the coherent-information loss. The latter can be written entirely in terms of relative entropies between the reduced code states and their incoherent mixture. We briefly explain how the results of the previous sections determine the corresponding error correction exponents.

Consider a D -dimensional code subspace

$$\mathbb{V} = \text{span}\{|\Psi_1\rangle, \dots, |\Psi_D\rangle\}, \quad (5.1)$$

and let

$$\bar{\rho} = \frac{1}{D} \sum_{a=1}^D |\Psi_a\rangle \langle \Psi_a| \quad (5.2)$$

be the maximally mixed state on the code subspace. We denote by ρ_A^a and $\bar{\rho}_A$ the corresponding reduced density matrices on an interval A , and by $\bar{A}$ its complement. The coherent-information loss (BS) Add the definition of Δ . under erasure of A can be expressed as [1]

$$\Delta(A; \mathbb{V}) = \log D + \frac{1}{D} \sum_{a=1}^D [S(\rho_A^a \| \bar{\rho}_A) - S(\rho_{\bar{A}}^a \| \bar{\rho}_{\bar{A}})] . \quad (5.3) \quad \{\text{eq:coherent-def}\}$$

Thus the error-correcting properties of the code are controlled by precisely the relative entropies of mixtures computed in this paper.

Let $x = |A|/R \ll 1$. Following Ref. [1], introduce the exponents α and β by

$$\frac{1}{D} \sum_a S(\rho_A^a \| \bar{\rho}_A) \sim x^\alpha , \quad (5.4) \quad \{\text{eq:alpha-def}\}$$

$$\log D - \frac{1}{D} \sum_a S(\rho_{\bar{A}}^a \| \bar{\rho}_{\bar{A}}) \sim x^\beta . \quad (5.5) \quad \{\text{eq:beta-def}\}$$

Equation (5.3) immediately gives

$$\Delta(A; \mathcal{V}) \sim c_\alpha x^\alpha + c_\beta x^\beta \sim x^{\min\{\alpha, \beta\}} , \quad (5.6) \quad \{\text{eq:QEC-loss}\}$$

for non-vanishing positive coefficients c_α, c_β . The two exponents have rather different CFT origins, which our calculations make explicit.

Small interval and the exponent α . Our general result in Eq. (3.77) can be applied by taking one of the two mixtures to be a single code state and the other to be the uniform code mixture. Define

$$\bar{C}^L \equiv \frac{1}{D} \sum_{b=1}^D C_{\Psi_b \Psi_b^*}^L . \quad (5.7)$$

For a light primary O_L of dimension $\Delta_L < 2$, one obtains

$$\frac{1}{D} \sum_{a=1}^D S(\rho_A^a \| \bar{\rho}_A) = \sum_L f_{LL} \left[\frac{1}{D} \sum_{a=1}^D \left(C_{\Psi_a \Psi_a^*}^L - \bar{C}^L \right)^2 \right] (\pi x)^{2\Delta_L} + \dots , \quad (5.8) \quad \{\text{eq:qec-alpha-}\}$$

where

$$f_{LL} = \frac{\sqrt{\pi} \Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} . \quad (5.9) \quad \{\text{eq:qec-alpha-}\}$$

The leading exponent α is therefore set by the lowest-dimension local operator whose diagonal OPE coefficient is not identical for all code states. In other words, the erased interval learns about the logical state through the lowest-dimension local CFT datum that varies across the code subspace.

If no such primary with $\Delta_L < 2$ contributes, the leading distinguishability is due to the stress tensor. In this case,

$$\frac{1}{D} \sum_a S(\rho_A^a \| \bar{\rho}_A) = \frac{8}{15} \left[\frac{\text{Var}_{\mathbb{V}}(h)}{c} + \frac{\text{Var}_{\mathbb{V}}(\bar{h})}{\bar{c}} \right] (\pi x)^4 + \dots , \quad (5.10) \quad \{\text{eq:qec-alpha-}\}$$

so that $\alpha = 4$ whenever the conformal dimensions distinguish the code states and no lighter primary does.

Large interval and the exponent β . The second term in Eq. (5.3) has a qualitatively different origin. As $x \rightarrow 0$, the complement $\bar{A}$ approaches the entire system. The different global code states become perfectly distinguishable, and hence

$$S(\rho_A^a \| \bar{\rho}_{\bar{A}}) \longrightarrow \log D. \quad (5.11)$$

The quantity relevant to error correction is the deficit from this classical value.

For a binary code,

$$\mathbb{V} = \text{span}\{|\Psi\rangle, |\Phi\rangle\}, \quad \bar{\rho} = \frac{1}{2}(\rho_\Psi + \rho_\Phi), \quad (5.12)$$

our large-interval result gives

$$\log 2 - S\left(\rho_{\bar{A}}^\Psi \left\| \frac{\rho_{\bar{A}}^\Psi + \rho_{\bar{A}}^\Phi}{2}\right.\right) = \sum_{L'} f_{L'L'} \left(C_{\Psi\Phi^*}^{L'}\right)^2 (\pi x)^{2\Delta_{L'}} + \dots, \quad (5.13) \quad \{\text{eq:qec-beta}\}$$

and, to the same order, the expression is invariant under $\Psi \leftrightarrow \Phi$. Here $O_{L'}$ is the lightest primary appearing in the *mixing* OPE $\Psi\Phi^*$. Consequently,

$$\beta = 2\Delta_{L'} \quad (5.14) \quad \{\text{eq:qec-beta}\}$$

for the lightest mixing operator with non-zero OPE coefficient. Whereas α is controlled by differences of diagonal OPE data, β probes operators that interpolate between distinct codewords.

This distinction also explains why either exponent can dominate the error-correction problem. The small erased interval can distinguish the code states directly through local one-point data, producing the α contribution, while its large complement fails to distinguish the codewords perfectly because of the mixing channel, producing the β contribution. The coherent-information loss is controlled by the slower of these two mechanisms.

Examples. The examples studied in Sec. 4 reproduce the exponents used in Ref. [1]. For the compact boson one finds

$\mathbb{V}$	α	β
$\text{span}\{ 00\rangle, ss\rangle\}$	2	1
$\text{span}\{ 00\rangle, 3s, s\rangle\}$	2	5

(5.15)

where $\alpha = 2$ follows from the $U(1)$ current, while β is twice the dimension of the vertex operator in the corresponding mixing OPE. Similarly, for the Ising CFT,

$\mathbb{V}$	α	β
$\text{span}\{ 0\rangle, \sigma\rangle\}$	2	1/4
$\text{span}\{ 0\rangle, \epsilon\rangle\}$	4	2
$\text{span}\{ \sigma\rangle, \epsilon\rangle\}$	2	1/4

(5.16)

These values agree with the power laws extracted numerically in Ref. [1].

Finally, the application in Ref. [1] goes beyond the dimensionally reduced CFT code. For a two-dimensional chiral edge code, an erasure localized near one physical edge has coherent-information loss $\Delta_{2d} \sim x^\gamma$. Monotonicity of relative entropy implies

$$\gamma \geq \alpha \geq \min\{\alpha, \beta\}. \quad (5.17)$$

Thus the CFT quantities calculated here provide a lower bound on the power governing local information leakage from the chiral edge code.

(BS) (1) I think we have obtained a few other values. We can summarize these into another example. (2) Let me think if the sub-leading power law has any physical implication. (3) Double check if the results here varifies all the claims made in the previous paper about this work.

(YS) With current relative entropy results, we may extend to quasi-local Petz recovery for single-interval erasure on 1D CFT code, since we have both α and β .

6 Discussions

Numerical verification and its limits. The equal-mixture calculations in Sec. 4 use every step-4 chain size on the stated finite grid. The common small window $x \leq 0.02$ gives free powers within 0.889% of CFT and tests the two Ising sixth-order coefficients to 0.369% and 3.042%. Small-boson leading amplitudes still differ by up to 5.024%, and their slight improvement in $00 \leftrightarrow ss$ is sensitive to omitted chain sizes. On the common large window $\epsilon \leq 0.005$, the four complete boson next coefficients agree within 7.851%, but the boson $00 \leftrightarrow ss$ leading error remains about 2.354% and the Ising vacuum–spin free power is about 6.72% above $1/4$. Adding the next term improves every small-interval leading amplitude, five of six large-Ising amplitudes, and four of six large-boson amplitudes. It does not provide uniform one-percent verification of the leading laws. The forty-weight study in Sec. 4.4 now uses the same complete grid and fitting windows. Its normalized leading-amplitude ratios follow the predicted probability dependence within 0.00205% for small intervals and 0.412% for large complements, while absolute leading-amplitude errors remain as large as 5.024% and 2.411%, respectively. Agreement of the probability dependence therefore coexists with finite-resolution offsets in the coefficients. Neither dense sampling in N nor the numerical precision checks constitute a continuum extrapolation in the short length l .

Divergences in relative entropy of the free boson theory To argue for a universal $1/\epsilon$ term in the $\epsilon \rightarrow 1^-$ limit.

The basic idea is that for a mixture of states, the divergences in the $x \rightarrow 1^-$ is controlled by the worst divergence among any pair of pure states. In the free boson theory, we consider the pure states (coherent states) created by V_α , the state $J = i\partial\phi$. The relative entropy of $S(\rho_\alpha\|\rho_0)$ diverges as $1/\epsilon$ because $\langle\alpha|\hat{K}_\Omega\alpha\rangle \sim 1/\epsilon$. Similarly, since $S(\rho_\alpha\|\rho_\beta) = S(\rho_{\alpha-\beta}\|\rho_0)$, the divergences are again $1/\epsilon$. Finally, the divergences of $S(\rho_J\|\rho_\beta)$ are controlled by the divergences in $\hat{K}_\beta$ which should be connected to $\hat{K}_\Omega$ by unitaries. But another formula that makes this explicit is $S(\rho_J\|\rho_\beta) = S(\rho_J\|\rho_0) + S(\rho_0\|\rho_\beta)$, therefore the divergences in this relative entropy is also $1/\epsilon$.

A Analytic continuation of the replica sum with a phase (Irrelevant since it already exists in literature)

{app:analytic-}

In this appendix we derive the analytic continuation in the replica index of the sum

$$\mu(n) = \sum_{p=1}^{n-1} \frac{(n-p)e^{-2\pi i \kappa_n p}}{\sin^{2\Delta}\left(\frac{\pi p}{n}\right)}$$

$$e^{-2\pi i \kappa_n} := r_n(x) := (y^{-1} - 1)D_n(1-x) = (y^{-1} - 1) \left(\frac{n \sin(\pi(1-x)/n)}{\sin(\pi(1-x))} \right)^{4(h_\alpha - h_\beta)}, \quad (\text{A.1})$$

and compute its derivative at $n = 1$. First note that for all positive integer n the function $D_n(1-x) \geq 1$, which means that for $h_\beta \geq h_\alpha$ and $0 \leq y \leq 1/2$ the function $r_n(x)$ is real, and bounded above by one:

$$\forall 0 \leq y \leq 1/2, h_\alpha \leq h_\beta : \quad 0 \leq r_n(x) \leq 1, \quad (\text{A.2})$$

We are going to focus on this case. Redefining $p \rightarrow n - p$ we have

$$\mu(n) = r_n(x)^n \sum_{p=1}^{n-1} \frac{p e^{2\pi i \kappa_n p}}{\sin^{2\Delta}\left(\frac{\pi p}{n}\right)}. \quad (\text{A.3})$$

Our derivation of the analytic continuation $\mu(1)$ and $\mu'(1)$ in this limit follows closely the method used by Cardy, Tonni and collaborators[3], based on Fourier analysis and Carlson's theorem.

A.1 Fourier coefficients of the sine kernel

Define the periodic kernel

$$f(x) = \sin^{-2\Delta}(\pi x). \quad (\text{A.4})$$

Its Fourier coefficients are

$$f_\Delta(m) = \int_0^1 dx \frac{e^{-2\pi i m x}}{\sin^{2\Delta}(\pi x)}, \quad m \in \mathbb{Z}. \quad (\text{A.5})$$

Using the standard beta-function integral one finds

$$f_\Delta(m) = \frac{2^{2\Delta} \Gamma(1-2\Delta) e^{-i\pi m}}{\Gamma(1-\Delta+m) \Gamma(1-\Delta-m)}. \quad (\text{A.6})$$

For analytic continuation it is convenient to introduce a holomorphic interpolation of the positive Fourier coefficients to complex m ,

$$g_\Delta(m) = c_\Delta \frac{\Gamma(m+\Delta)}{\Gamma(m+1-\Delta)}, \quad c_\Delta = \frac{2^{2\Delta} \Gamma(1-2\Delta) \sin(\pi\Delta)}{\pi}. \quad (\text{A.7}) \quad \{\text{eq: Fourier}$$

This function reproduces the discrete data $g_\Delta(m) = f_\Delta(m)$ for integer $m \geq 0$ and satisfies the growth conditions required for the application of Carlson's theorem.

A.2 Analytic continuation of the replica sum

Following the Poisson-resummation strategy used in [3], the analytic continuation of the replica sum can be written as

$$S_n(k_0, \Delta) = \frac{1}{2} \sum_{m \geq 1} [n g(nm - \kappa_n) - g(m - \kappa_n)] + \sum_{m \geq 0} [n g(-nm - \kappa_n) - g(-m - \kappa_n)], \quad (\text{A.8}) \quad \{\text{eq: Fourier}\}$$

where

$$\kappa_n = -\frac{nk_0}{2\pi}. \quad (\text{A.9})$$

This expression reproduces the original discrete sum for integer n and provides its analytic continuation.

A.3 Derivative at $n = 1$

To compute the derivative at $n = 1$ we differentiate term by term. In the strip $0 < \Re \Delta < 1/2$ the series converge absolutely, allowing the interchange of differentiation and summation. One obtains

$$\begin{aligned} \partial_n S_n|_{n=1} &= \frac{1}{2} \sum_{m=1}^{\infty} [g(m - \kappa) + mg'(m - \kappa)] + \sum_{m=0}^{\infty} [g(-m - \kappa) - mg'(-m - \kappa)], \\ &= \frac{1}{2} \sum_{m=1}^{\infty} [g(m - \kappa) + mg'(m - \kappa)] + \sum_{m=0}^{\infty} [g(m - \kappa) + mg'(m - \kappa)]. \end{aligned} \quad (\text{A.10})$$

where $\kappa = \kappa_{n=1} = -k_0/(2\pi)$.

A.4 Integral representation

To evaluate these sums we use the beta-function identity

$$\frac{\Gamma(z + \Delta)}{\Gamma(z + 1 - \Delta)} = \frac{1}{\Gamma(1 - 2\Delta)} \int_0^1 dt t^{z+\Delta-1} (1-t)^{-2\Delta}, \quad (\text{A.11})$$

valid for $0 < \Re \Delta < 1/2$.

This gives

$$g(z) = A_\Delta \int_0^1 dt t^{z+\Delta-1} (1-t)^{-2\Delta}, \quad A_\Delta = \frac{2^{2\Delta} \sin(\pi \Delta)}{\pi}. \quad (\text{A.12})$$

Differentiating,

$$g'(z) = A_\Delta \int_0^1 dt t^{z+\Delta-1} (1-t)^{-2\Delta} \ln t. \quad (\text{A.13})$$

Substituting these expressions into the derivative formula and interchanging the order of summation and integration we obtain

$$\partial_n S_n|_{n=1} = \frac{A_\Delta}{2} \int_0^1 dt (1-t)^{-2\Delta} \left[\sum_{m=1}^{\infty} t^{m-\kappa+\Delta-1} (1+m \ln t) + \sum_{m=0}^{\infty} t^{m-\kappa+\Delta-1} (1+m \ln t) \right]. \quad (\text{A.14})$$

A.5 Performing the sums

Using the geometric-series identities

$$\sum_{m=1}^{\infty} t^m = \frac{t}{1-t}, \quad (\text{A.15})$$

$$\sum_{m=1}^{\infty} m t^m = \frac{t}{(1-t)^2}, \quad (\text{A.16})$$

$$\sum_{m=0}^{\infty} t^m = \frac{1}{1-t}, \quad (\text{A.17})$$

the sums can be evaluated explicitly. This yields

$$\partial_n S_n|_{n=1} = \frac{A_\Delta}{2} \int_0^1 dt \left[\frac{t^{\Delta-\kappa} + t^{\Delta-\kappa-1}}{(1-t)^{2\Delta+1}} + \frac{2t^{\Delta-\kappa} \ln t}{(1-t)^{2\Delta+2}} \right]. \quad (\text{A.18})$$

A.6 Final evaluation

The remaining integrals can be evaluated using

$$\int_0^1 dt t^{a-1} (1-t)^{b-1} = B(a, b), \quad (\text{A.19})$$

$$\int_0^1 dt t^{a-1} (1-t)^{b-1} \ln t = B(a, b) [\psi(a) - \psi(a+b)]. \quad (\text{A.20})$$

After straightforward algebra we obtain

$$\begin{aligned} \partial_n S_n(k_0, \Delta)|_{n=1} = & \frac{2^{2\Delta-1} \sin(\pi\Delta)}{\pi} \left[B(1+\Delta-\kappa, -2\Delta) + B(\Delta-\kappa, -2\Delta) \right. \\ & \left. + 2B(1+\Delta-\kappa, -1-2\Delta)(\psi(1+\Delta-\kappa) - \psi(-\Delta-\kappa)) \right], \end{aligned} \quad (\text{A.21})$$

with

$$\kappa = -\frac{k_0}{2\pi}. \quad (\text{A.22})$$

This expression gives the analytically continued derivative of the replica sum at $n = 1$.

Now if we set $\kappa = 0$, we get the desired result. This can be shown by using the identities

of Gamma and digamma functions.

$$\begin{aligned}
\mu'_0(1) = \partial_n S_n(k_0 = 0, \Delta)|_{n=1} &= \frac{2^{2\Delta-1} \sin(\pi\Delta)}{\pi} \left[\frac{\Gamma(\Delta+1)\Gamma(-2\Delta)}{\Gamma(-\Delta+1)} + \frac{\Gamma(\Delta)\Gamma(-2\Delta)}{\Gamma(-\Delta)} \right. \\
&\quad \left. + 2B(\Delta+1, -2\Delta-1)(\psi(1+\Delta) - \psi(-\Delta)) \right] \\
&= \frac{2^{2\Delta-1} \sin(\pi\Delta)}{\pi} \left[\frac{\Delta\Gamma(\Delta)\Gamma(-2\Delta)}{-\Delta\Gamma(-\Delta)} + \frac{\Gamma(\Delta)\Gamma(-2\Delta)}{\Gamma(-\Delta)} \right. \\
&\quad \left. + \frac{2\Gamma(\Delta+1)\Gamma(-2\Delta-1)}{\Gamma(-\Delta)} (-\pi \cot(\pi\Delta)) \right] \\
&= 2^{2\Delta} \cos(\pi\Delta) \left[\frac{\Gamma(\Delta+1)\Gamma(-2\Delta-1)}{\frac{\pi}{\sin(\pi\Delta)\Gamma(1+\Delta)}} \right] \\
&= \frac{2^{2\Delta-1}}{\pi} \sin(2\pi\Delta) (\Gamma(1+\Delta))^2 \Gamma(-2\Delta-1) \\
&= 2^{2\Delta-1} \frac{(\Gamma(1+\Delta))^2}{\Gamma(2+2\Delta)} \\
&= \frac{2^{2\Delta-1} (\Gamma(1+\Delta))^2}{\left(\frac{\Gamma(1+\Delta)\Gamma(\Delta+3/2)}{\sqrt{\pi} 2^{1-2(1+\Delta)}} \right)} \\
&= \frac{\sqrt{\pi} \Gamma(1+\Delta)}{4\Gamma(\Delta+3/2)}. \tag{A.23}
\end{aligned}$$

where we used the following identities,

$$\begin{aligned}
\psi(1+\Delta) - \psi(-\Delta) &= -\pi \cot(\pi\Delta) \\
\Gamma(1+z)\Gamma(-z) &= -\frac{\pi}{\sin(\pi z)} \\
\Gamma(z)\Gamma(z+1/2) &= 2^{1-2z} \sqrt{\pi} \Gamma(2z)
\end{aligned}$$

B Analytic continuation of $\mu_1(n)$ (Irrelevant since we have the general result now)

Cardy, Tonni and collaborators analytically continued the sum $S_2(n)$ and its derivative at $n = 1$ using Carlson's theorem. They analytically continued

{Analytic cont.

$$S_2(n) = \frac{1}{2} \sum_{j=1}^{n-1} \frac{1}{\left[\sin^2 \left(\frac{2\pi j}{2n} \right) \right]^\alpha} \equiv \frac{1}{2} \sum_{k=-\infty}^{\infty} [ng(nk) - g(k)]$$

where $g(k)$ is the analytic continuation of the Fourier modes $f(k)$ of $(\sin^2(u/2))^{-\alpha}$. However, for $\mu_1(n)$, we want the analytic continuation of the sum involving,

$$\mu_1(n) = \frac{1}{2} \sum_{p=1}^{n-1} \frac{p^2}{\left(\sin^2 \left(\frac{\pi p}{n} \right) \right)^\Delta} \tag{B.1}$$

For this higher moment, we will consider the Fourier modes of the generating function,

$$\frac{e^{-2i\kappa u}}{(\sin^2(u))^\Delta} \quad (\text{B.2})$$

We can take two derivatives with respect to κ to obtain the desired sum, and set $\kappa = 0$ in the end. By adding this extra source term, we replace m with $m + \kappa$ in eq. [A.7], and obtain,

$$g(m, \kappa) = c_\Delta \frac{\Gamma(1 - 2\Delta)\Gamma(m + \kappa + \Delta)}{\Gamma(m + \kappa + 1 - \Delta)} \quad (\text{B.3})$$

Then the sum is given by,

$$\begin{aligned} \mu_1(n) &= -\frac{1}{4} \frac{1}{2} \partial_\kappa^2 \left[\sum_{m \geq 1} (ng(n(m + \kappa)) - g(m + \kappa)) + \sum_{m \geq 0} (ng(-nm + n\kappa) - g(-m - \kappa)) \right] \\ &= -\frac{1}{8} \left[\sum_{m \geq 1} (n^3 g''(n(m + \kappa)) - g''(m + \kappa)) + \sum_{\geq 0} (n^3 g''(-nm + n\kappa) - g''(-m - \kappa)) \right]. \end{aligned} \quad (\text{B.4})$$

Then, we take the derivative with respect to n , and take the limit $n = 1$ and $\kappa = 0$, along with using the reflection $g(-m) = g(m)$ for integer m , to obtain,

$$\partial_n \mu_1|_{n=1, \kappa=0} = -\frac{1}{8} \left[\sum_{m \geq 1} (3g''(m) + mg'''(m)) + \sum_{m \geq 0} (3g''(m) + mg'''(m)) \right] \quad (\text{B.5})$$

Making use of the following integral representation of the analytically continued Fourier modes,

$$g(m + \kappa) = A_\Delta \int_0^1 dt t^{m+\kappa+\Delta-1} (1-t)^{-2\Delta} \quad (\text{B.6})$$

$$g'(m + \kappa) = A_\Delta \int_0^1 dt t^{m+\kappa+\Delta-1} (1-t)^{-2\Delta} \ln t \quad (\text{B.7})$$

$$g''(m + \kappa) = A_\Delta \int_0^1 dt t^{m+\kappa+\Delta-1} (1-t)^{-2\Delta} (\ln t)^2 \quad (\text{B.8})$$

$$g'''(m + \kappa) = A_\Delta \int_0^1 dt t^{m+\kappa+\Delta-1} (1-t)^{-2\Delta} (\ln t)^3 \quad (\text{B.9})$$

$$(\text{B.10})$$

We obtain,

$$\partial_n \mu_1(n)|_{n=1} = -\frac{A_\Delta}{8} \left(\sum_{m \geq 1} 3 \int_0^1 dt t^{m+\kappa+\Delta-1} (1-t)^{-2\Delta} (\ln t)^2 + \sum_{m \geq 1} m \int_0^1 dt t^{m+\kappa+\Delta-1} (1-t)^{-2\Delta} (\ln t)^3 \right) \quad (\text{B.11})$$

$$\sum_{m \geq 0} 3 \int_0^1 dt t^{m+\kappa+\Delta-1} (1-t)^{-2\Delta} (\ln t)^2 + \sum_{m \geq 1} m \int_0^1 dt t^{m+\kappa+\Delta-1} (1-t)^{-2\Delta} (\ln t)^3 \quad (\text{B.12})$$

Doing the sum over m , we get,

$$\partial_n \mu_1(n)|_{n=1} = -\frac{A_\Delta}{8} \left(3 \int_0^1 dt \frac{t^{\kappa+\Delta} (\ln t)^2}{(1-t)^{2\Delta+1}} + 2 \int_0^1 dt \frac{t^{\kappa+\Delta} (\ln t)^3}{(1-t)^{2\Delta+2}} + 3 \int_0^1 dt \frac{t^{\kappa+\Delta-1} (\ln t)^2}{(1-t)^{2\Delta+1}} \right) \quad (\text{B.13})$$

We use the following integrals,

$$\int_0^1 dt t^{a-1} (1-t)^{b-1} (\ln t)^2 = B(a, b) \left[(\psi(a) - \psi(a+b))^2 + (\psi_1(a) - \psi_1(a+b)) \right] \quad (\text{B.14})$$

$$\begin{aligned} \int_0^1 dt t^{a-1} (1-t)^{b-1} (\ln t)^3 = & B(a, b) (\psi(a) - \psi(a+b)) \left[(\psi(a) - \psi(a+b))^2 + (\psi_1(a) - \psi_1(a+b)) \right] \\ & + B(a, b) [2 (\psi(a) - \psi(a+b)) (\psi_1(a) - \psi_1(a+b)) + (\psi_2(a) - \psi_2(a+b))] \end{aligned} \quad (\text{B.15})$$

to obtain,

$$\begin{aligned} \partial_n \mu_1(n)|_{n=1, \kappa=0} = & -\frac{A_\Delta}{8} \left[3B(1+\Delta, -2\Delta) \left((\psi(\Delta+1) - \psi(-\Delta+1))^2 + (\psi_1(\Delta+1) - \psi_1(-\Delta+1)) \right) \right] \\ & - \frac{A_\Delta}{8} \left[3B(\Delta, -2\Delta) \left((\psi(\Delta) - \psi(-\Delta))^2 + (\psi_1(\Delta) - \psi_1(-\Delta)) \right) \right] \\ & - \frac{A_\Delta}{8} \left[2B(\Delta+1, -2\Delta-1) \left[(\psi(\Delta+1) - \psi(-\Delta)) \times \right. \right. \\ & \quad \left. \left((\psi(\Delta+1) - \psi(-\Delta))^2 + (\psi_1(\Delta+1) - \psi_1(-\Delta)) \right) \right. \\ & \quad \left. \left. + 2 (\psi(\Delta+1) - \psi(-\Delta)) (\psi_1(\Delta+1) - \psi_1(-\Delta)) + (\psi_2(\Delta+1) - \psi_2(-\Delta)) \right] \right] \end{aligned} \quad (\text{B.16}) \quad \{\text{eq: analytic}\}$$

We use the following identities of Beta function, digamma, trigamma and poly-gamma function,

$$B(1 + \Delta, -2\Delta) = -B(\Delta, -2\Delta) \quad (\text{B.17})$$

$$B(\Delta + 1, -2\Delta - 1) = -\frac{\Delta}{2\Delta + 1} B(\Delta, -2\Delta) \quad (\text{B.18})$$

$$\psi(\Delta) - \psi(-\Delta) = -\pi \cot(\pi\Delta) - \frac{1}{\Delta} \quad (\text{B.19})$$

$$\psi(1 + \Delta) - \psi(1 - \Delta) = -\pi \cot(\pi\Delta) + \frac{1}{\Delta} \quad (\text{B.20})$$

$$\psi(1 + \Delta) - \psi(-\Delta) = -\pi \cot(\pi\Delta) \quad (\text{B.21})$$

$$\psi_1(\Delta) - \psi_1(-\Delta) = 2\psi_1(\Delta) - \pi^2 \csc^2(\pi\Delta) - \frac{1}{\Delta^2} \quad (\text{B.22})$$

$$\psi_1(1 + \Delta) - \psi_1(1 - \Delta) = 2\psi_1(\Delta) - \pi^2 \csc^2(\pi\Delta) - \frac{1}{\Delta^2} \quad (\text{B.23})$$

$$\psi_1(1 + \Delta) - \psi_1(-\Delta) = 2\psi_1(\Delta) - \pi^2 \csc^2(\pi\Delta) - \frac{2}{\Delta^2} \quad (\text{B.24})$$

$$\psi_2(\Delta) - \psi(-\Delta) = -2\pi^3 \csc^2(\pi\Delta \cot(\pi\Delta)) - \frac{2}{\Delta^3} \quad (\text{B.25})$$

$$\psi_2(1 + \Delta) - \psi_2(1 - \Delta) = -2\pi^3 \csc^2(\pi\Delta) \cot(\pi\Delta) + \frac{2}{\Delta^3} \quad (\text{B.26})$$

$$\psi_2(1 + \Delta) - \psi_2(-\Delta) = -2\pi^3 \csc^2(\pi\Delta) \cot(\pi\Delta) \quad (\text{B.27})$$

Plugging these in (B.16) to obtain,

$$\begin{aligned} \partial_n \mu_1|_{n=1} &= -\frac{A_\Delta}{8} B(\Delta, -2\Delta) \left(3 \left[\left(-\pi \cot(\pi\Delta) - \frac{1}{\Delta} \right)^2 - \left(-\pi \cot(\pi\Delta) + \frac{1}{\Delta} \right)^2 \right] \right) \\ &\quad + \frac{A_\Delta}{4} \frac{\Delta}{2\Delta + 1} B(\Delta, -2\Delta) \left(C(C^2 + D) + 2CD + F \right) \end{aligned} \quad (\text{B.28})$$

$$\equiv T_1 + T_2 \quad (\text{B.29})$$

where

$$T_1 = -\frac{A_\Delta}{8} B(\Delta, -2\Delta) \left(3 \left[\left(-\pi \cot(\pi\Delta) - \frac{1}{\Delta} \right)^2 - \left(-\pi \cot(\pi\Delta) + \frac{1}{\Delta} \right)^2 \right] \right) \quad (\text{B.30})$$

$$T_2 = \frac{A_\Delta}{4} \frac{\Delta}{2\Delta + 1} B(\Delta, -2\Delta) \left(C(C^2 + D) + 2CD + F \right). \quad (\text{B.31})$$

Define,

$$P = \pi \cot(\pi\Delta) \quad (\text{B.32})$$

$$Q = \pi^2 \csc^2(\pi\Delta) \quad (\text{B.33})$$

Then,

$$C = \psi(1 + \Delta) - \psi(-\Delta) = -P \quad (\text{B.34})$$

$$D = \psi_1(1 + \Delta) - \psi_1(-\Delta) = 2\psi_1(\Delta) - Q - \frac{2}{\Delta^2} \quad (\text{B.35})$$

$$F = \psi_2(1 + \Delta) - \psi_2(-\Delta) = -2PQ \quad (\text{B.36})$$

Then the second term (T_2) in these new variables is given by,

$$\begin{aligned} T_2 &= -P(P^2 + D) - 2PD - 2PQ \\ &= -P^3 - 3PD - 2PQ \\ &= -P^3 - 3P(2\psi_1(\Delta) - Q - \frac{2}{\Delta^2}) - 2PQ \\ &= -P^3 - 6P\psi_1(\Delta) + PQ + \frac{6P}{\Delta^2} \\ &= P \left[\pi^2 - 6\psi_1(\Delta) + \frac{6}{\Delta^2} \right] \end{aligned} \quad (\text{B.37})$$

where in the last step we used $Q^2 = \pi^2 + P^2$. Finally we get,

$$\begin{aligned} \partial_n \mu_1|_{n=1, \kappa=0} &= -\frac{3A_\Delta B(\Delta, -2\Delta)\pi \cot(\pi\Delta)}{2\Delta} + \frac{A_\Delta B(\Delta, -2\Delta)\Delta\pi \cot(\pi\Delta)}{4(2\Delta + 1)} \left(\pi^2 - 6\psi_1(\Delta) + \frac{6}{\Delta^2} \right) \\ &= \frac{A_\Delta B(\Delta, -2\Delta)\pi \cot(\pi\Delta)}{8} \left[-\frac{12}{\Delta} + \frac{2\Delta}{2\Delta + 1} \left(\pi^2 - 6\psi_1(\Delta) + \frac{6}{\Delta^2} \right) \right] \end{aligned} \quad (\text{B.38})$$

B.1 $\Delta = 1$ case

We can also check that it does not blow at $\Delta = 1$, which is of interest as it corresponds to massless free boson theory. We use the following identity,

$$B(\Delta, -2\Delta)\Big|_{\Delta \rightarrow N} = (-1)^N \frac{(N-1)!(N)!}{2(2N)!}. \quad (\text{B.39})$$

to get $B(1, -2) = -1/4$. Hence,

$$\partial_n \mu_1(\Delta = 1)|_{n=1, \kappa=0} = \frac{1}{8} \left[-12 + \frac{2}{3} (\pi^2 - 6\psi_1(1) + 6) \right] \quad (\text{B.40})$$

We use the following series expansion of poly-gamma function,

$$\psi_m(z) = (-1)^{m+1} (m!) \sum_{k=0}^{\infty} \frac{1}{(z+k)^{m+1}}, \quad m \geq 1 \quad (\text{B.41})$$

to get,

$$\partial_n \mu_1|_{n=1, \kappa=0}(\Delta = 1) = -1 \quad (\text{B.42})$$

Similarly, we can check $\partial_n \mu_1|_{n=1, \kappa=0}(\Delta = 2) = 0$.

C Vertex operators in large interval limit

{Vertex operators}

We consider chiral vertex operators, and a simple case for the calculation of relative entropy. We want to calculate $S(\rho_0\|\rho_\beta)$, where V_β is a chiral vertex operator with conformal weights $(h_\beta, \bar{h}_\beta) = (\beta^2/2, 0)$. In our previous calculation, we had,

$$\mathcal{F}_n = \frac{1}{D(n)^{n-1} \zeta_n^{2\Delta_\beta} \langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) \rangle + \dots}. \quad (\text{C.1})$$

This gave us negative divergence, precisely because the two vertex operators are $\sin(\pi(1+\epsilon)/n)$ separated, instead of $\sin(\pi(1-\epsilon)/n)$ which was the case when we were computing $S(\rho_\beta\|\rho_0)$. We indeed got the result for the later case that matched with the exact analytical expression. However, in the former case, i.e. $S(\rho_0\|\rho_\beta)$, we want to show that the other terms (in fact, all the terms) have a non vanishing contribution in the limit $x \rightarrow 1$ limit. This happens precisely because we will have higher point functions such as $\langle V_{-\beta} V_\beta J^m \rangle$ will be of the order of $(\epsilon/\sin(\pi/n))^m \sim (\epsilon/(n-1))^m$. Hence, in the limit $n \rightarrow 1$, we need to take all the higher point functions into account, in order to get the correct expression by eliminating the $\sin(\pi(1+\epsilon)/n)$ which gives the negative divergence. In other words, this is equivalent to the case if we can show that the $2n$ -point function of Vertex operators would eventually get rid of the $\sin(\pi(1+\epsilon)/n)$ and give the correct result.

C.1 Exact analytical calculation (Irrelevant)

In the denominator, we have to compute $\text{tr}(\rho_0\|\rho_\beta^{n-1})$. There are $(n-2)$ sites with $V_\beta V_{-\beta}$ OPEs. And we also have $V_\beta(z_{n-1,n}^+)$ and $V_{-\beta}(z_{1,n}^-)$. Using BCH, we have the OPEs,

$$V_\beta(z) V_{-\beta}(w) =: e^{i\beta\phi(z)} :: e^{-i\beta\phi(w)} : \quad (\text{C.2})$$

$$= e^{\langle i\beta\phi(z)(-i\beta\phi(w)) \rangle} : e^{i\beta(\phi(z)-\phi(w))} : \quad (\text{C.3})$$

In terms of current, $J(z) = i\partial\phi(z)$, we thus have,

$$V_\beta(z) V_{-\beta}(w) = (z-w)^{-\beta^2} : e^{\beta \int_w^z du J(u)} : \quad (\text{C.4}) \quad \{\text{eq: non mixing}\}$$

Then the denominator of $\mathcal{F}_n$,

$$\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) \prod_{k=1}^{n-2} : e^{Q_k} : \rangle \quad (\text{C.5})$$

up to the OPE factors which we will add in the end. Here we defined,

$$Q_k = \beta \int_{z_{k+1,n}^-}^{z_{k,n}^+} du J(u). \quad (\text{C.6})$$

Since the theory is Gaussian, using Wick's theorem, we have,

$$\frac{\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) \prod_{k=1}^{n-2} : e^{Q_k} : \rangle}{\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) \rangle} = e^{\sum_k \langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) Q_k \rangle} e^{\sum_{k<l} \langle Q_k Q_l \rangle_c} \quad (\text{C.7})$$

We obtain this by contracting every pair of currents in the OPEs. This gives the second term of right hand side which is the connected component part ($\langle Q_k Q_l \rangle_c$). The remaining current part would give the first term on the right hand side. We now want to evaluate the right hand side.

Thus, we have,

$$e^{\sum_k \langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) Q_k \rangle} = \prod_{k=1}^{n-2} \left[\frac{(z_{k+1,n}^- - z_{1,n}^-)(z_{k,n}^+ - z_{n-1,n}^+)}{(z_{k,n}^+ - z_{1,n}^-)(z_{k+1,n}^- - z_{n-1,n}^+)} \right]^{\beta^2} \quad (\text{C.8})$$

$$= \prod_{k=1}^{n-2} \left[\frac{\sin(\pi k/n) \sin(\pi(k+1)/n)}{\sin(\pi(k-\epsilon)/n) \sin(\pi(k+1+\epsilon)/n)} \right]^{\beta^2} \quad (\text{C.9})$$

Using the following identities,

$$\prod_{k=1}^{n-1} \sin(\pi k/n) = \frac{n}{2^{n-1}} \quad (\text{C.10})$$

$$\prod_{m=0}^{n-1} \sin(\pi(m+x)/n) = \frac{\sin(\pi x)}{2^{n-1}}, \quad (\text{C.11})$$

we finally have,

$$e^{\sum_k \langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) Q_k \rangle} = \left[\frac{n \sin(\pi \epsilon/n) \sin(\pi(1+\epsilon)/n)}{\sin(\pi/n) \sin(\pi \epsilon)} \right]^{2\beta^2} \quad (\text{C.12})$$

Clearly, we can see that there is some cancellation of $\sin(\pi(1+\epsilon)/n)$ coming from this contribution. Now, let's write down the contribution from the connected component of the current,

$$e^{\sum_{k<l} \langle Q_k Q_l \rangle_c} = \prod_{1 \leq k < l \leq n-2} \left[\frac{(z_{k,n}^+ - z_{l,n}^+)(z_{k+1,n}^- - z_{l+1,n}^-)}{(z_{k,n}^+ - z_{l+1,n}^-)(z_{k+1,n}^- - z_{l,n}^+)} \right]^{\beta^2} \quad (\text{C.13})$$

$$= \prod_{p=1}^{n-3} \left[\frac{\sin^2(\pi p/n)}{\sin(\pi(p+\epsilon)/n) \sin(\pi(p-\epsilon)/n)} \right]^{\beta^2(n-2-p)} \quad (\text{C.14})$$

We can use the following to simplify the above expression,

$$\prod_{p=1}^{n-3} \sin(\pi p/n)^{n-2-p} = \prod_{r=1}^{n-3} \prod_{p=1}^r \sin(\pi p/n). \quad (\text{C.15})$$

We thus have,

$$e^{\sum_{k<l} \langle Q_k Q_l \rangle_c} = \prod_{r=1}^{n-3} \left[\frac{\left(\prod_{p=1}^r \sin(\pi p/n) \right)^2}{\left(\prod_{p=1}^r \sin(\pi(p-\epsilon)/n) \right) \left(\prod_{p=1}^r \sin(\pi(p+\epsilon)/n) \right)} \right]^{\beta^2} \quad (\text{C.16})$$

$$= \left[n^{n-4} \left(\frac{\sin(\pi \epsilon/n)}{\sin(\pi \epsilon)} \right)^{n-4} \frac{\sin^2(\pi/n)}{\sin(\pi(1-\epsilon)/n) \sin(\pi(1+\epsilon)/n)} \right]^{\beta^2} \quad (\text{C.17})$$

Now, clearly by considering its contribution, we get rid off the $\sin(\pi(1+\epsilon)/n)$ factor completely.

We now write down the contribution coming from the two-point function of the Vertex operators sitting at $z_{1,n}^-$ and $z_{n-1,n}^+$, the OPE factors, normalization factors and the conformal transformation factor,

$$(\sin(\pi(1+\epsilon)/n))^{-\beta^2} (\sin(\pi\epsilon/n))^{-(n-2)\beta^2} (\sin(\pi(1-\epsilon)/n))^{(n-1)\beta^2} n^{-\beta^2(n-1)}. \quad (\text{C.18})$$

The final result is obtained by multiplying this with the contribution from the $n-2$ OPEs, we obtain,

$$\left(\frac{\sin(\pi x)}{n \sin(\pi x/n)} \right)^{\beta^2}. \quad (\text{C.19})$$

Thus,

$$\partial_n \mathcal{F}_n|_{n=1} = \beta^2(1 - \pi x \cot(\pi x)). \quad (\text{C.20})$$

If we write the contribution of both of them together,

$$\left[\frac{\prod_{1 \leq k < l \leq n-1} (z_{k,n}^+ - z_{l,n}^+)(z_{k+1,n}^- - z_{l+1,n}^-)}{\prod_{k,l=1}^{n-1} (z_{k,n}^+ - z_{l,n}^-)} \right]^{\beta^2} \quad (\text{C.21})$$

we indeed get the $2n$ point function of the vertex operators. This shows that the contribution from the higher point function had to be considered to obtain the actual result.

C.2 $S(\rho_0||\rho_\beta)$ using OPEs (Irrelevant since we compute general case in next subsection)

Using (C.4), we have the following non-mixing OPE,

$$V_\beta(z_{k,n}^+) V_{-\beta}(z_{k+1,n}^-) \simeq \zeta_n^{-\beta^2} \left(1 - \beta \zeta_n J_k + \frac{\beta^2 \zeta_n^2}{2} : J_k^2 : + O(\zeta_n^3) \right), \quad k = 2, \dots, n-2 \quad (\text{C.22})$$

where we absorbed all the phases in the current $J_k \equiv J(e^{i\pi\theta_k})$, for $k = 2, \dots, n-2$. To compute the denominator of $\mathcal{F}_n$, we have to evaluate,

$$\text{tr}(\rho_0 \rho_\beta^{n-1}) = \langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) \prod_{k=1}^{n-2} \left[1 - \beta \zeta_n J_k + \frac{\beta^2 \zeta_n^2}{2} : J_k^2 : + O(\zeta_n^3) \right] \rangle \quad (\text{C.23})$$

Using the Ward Identity[8], we have,

$$\mathcal{J}_k \equiv \frac{\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) J(u) \rangle}{\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) \rangle} = \frac{-\beta}{u - z_{1,n}^-} + \frac{\beta}{u - z_{n-1,n}^+}. \quad (\text{C.24}) \quad \{\text{eq: normalization}\}$$

Then Wick's theorem [8] gives us,

$$\frac{\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) : J_k^2 : \rangle}{\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) \rangle} = \mathcal{J}_k^2, \quad (\text{C.25})$$

$$\frac{\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) J_k J_l \rangle}{\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) \rangle} = \mathcal{J}_k \mathcal{J}_l + \langle J_k J_l \rangle, \quad (k \neq l). \quad (\text{C.26})$$

Therefore, the correlation function evaluates to the following,

$$\langle V_\beta(z_{n-1,n}^+) V_{-\beta}(z_{1,n}^-) \rangle \left[1 - \underbrace{\beta \zeta_n \sum_{k=1}^{n-2} \mathcal{J}_k}_{\frac{\langle V_\beta V_{-\beta} J_k \rangle}{\langle V_\beta V_{-\beta} \rangle}} + \underbrace{\frac{\beta^2 \zeta_n^2}{2} \sum_{k=1}^{n-2} \mathcal{J}_k^2}_{\frac{\langle V_\beta V_{-\beta} : J_k^2 : \rangle}{\langle V_\beta V_{-\beta} \rangle}} + \underbrace{\beta^2 \zeta_n^2 \sum_{1 \leq k < l \leq n-2} (\mathcal{J}_k \mathcal{J}_l + \langle J_k J_l \rangle)}_{\frac{\langle V_\beta V_{-\beta} J_k J_l \rangle}{\langle V_\beta V_{-\beta} \rangle}} + \mathcal{O}(\zeta_n^3) \right].$$

C.2.1 3-point function

We evaluate the three-point function, $\mathcal{J}_k$. The individual terms in $(z_{k+1,n}^- - z_{k,n}^+) \mathcal{J}_k$ evaluates to,

$$\frac{z_{k+1,n}^- - z_{k,n}^+}{e^{i\pi\theta_k} - z_{1,n}^-} = \frac{2i \sin(\pi\epsilon/n) e^{i\pi(2k+1)/n}}{2i \sin(\pi(k - \epsilon/2)/n) e^{i\pi(k+1+\epsilon/2)/n}} = \sin(\pi\epsilon/n) (\cot(\pi(k - \epsilon/2)/n) + i) \quad (\text{C.27}) \quad \{\text{eq:Fn_after_w}\}$$

$$\frac{z_{k+1,n}^- - z_{k,n}^+}{e^{i\pi\theta_k} - z_{n-1,n}^+} = \frac{2i \sin(\pi\epsilon/n) e^{i\pi(2k+1)/n}}{2i \sin(\pi(k + 1 + \epsilon/2)/n) e^{i\pi(k-\epsilon/2)/n}} = \sin(\pi\epsilon/n) (\cot(\pi(k + 1 + \epsilon/2)/n) + i) \quad (\text{C.28}) \quad \{\text{eq: 3 point 1}\}$$

We can further expand this to $O(\epsilon^2)$,

$$\cot(\pi(k - \epsilon/2)/n) = \cot(\pi k/n) + \left(\frac{\pi\epsilon}{2n} \right) \csc^2(\pi k/n) + O(\epsilon^2) \quad (\text{C.29})$$

and similarly for the other cotangent. We thus have,

$$\begin{aligned} & \cot(\pi(k - \epsilon/2)/n) - \cot(\pi(k + 1 + \epsilon/2)/n) \\ &= \cot(\pi k/n) - \cot(\pi(k + 1)/n) + \frac{\pi\epsilon}{2n} (\csc^2(\pi k/n) + \csc^2(\pi(k + 1)/n)) + O(\epsilon^2) \end{aligned} \quad (\text{C.30})$$

We thus have,

$$\begin{aligned} -\beta \zeta_n \mathcal{J}_k &= \beta^2 \frac{\pi\epsilon}{n} (\cot(\pi k/n) - \cot(\pi(k + 1)/n)) + \frac{\beta^2}{2} \left(\frac{\pi\epsilon}{n} \right)^2 (\csc^2(\pi k/n) + \csc^2(\pi(k + 1)/n)) + O(\epsilon^3) \\ &= L_k^{(1)} + L_k^{(2)} + O(\epsilon^3) \end{aligned} \quad (\text{C.31})$$

where $L_k^{(1)}$ and $L_k^{(2)}$ are the terms of order ϵ and ϵ^2 respectively. Now we sum over k to get,

$$\sum_{k=1}^{n-2} L_k^{(1)} = 2\beta^2 \frac{\pi\epsilon}{n} \cot(\pi/n) \quad (\text{C.32}) \quad \{\text{eq: Lk1 and 1}\}$$

$$\sum_{k=1}^{n-2} L_k^{(2)} = \beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \left(\frac{8S_2(n, 1)}{n} - \csc^2(\pi/n) \right) \quad (\text{C.33}) \quad \{\text{eq: Lk1 and 1}\}$$

C.2.2 4-point function

Now at the second order, we have contributions from the connected and disconnected piece. The disconnected pieces combine,

$$\text{Disc} \equiv \frac{\beta^2 \zeta_n^2}{2} \sum_k \mathcal{J}_k^2 + \beta^2 \zeta_n^2 \sum_{k < l} \mathcal{J}_k \mathcal{J}_l = \frac{1}{2} \left(\beta \zeta_n \sum_{k=1}^{n-2} \mathcal{J}_k \right)^2 \quad (\text{C.34})$$

$$\implies \text{Disc} = \frac{1}{2} \left(\sum_{k=1}^{n-2} L_k^{(1)} \right)^2 = 2\beta^4 \left(\frac{\pi\epsilon}{n} \right)^2 \cot^2(\pi/n). \quad (\text{C.35})$$

The connected piece gives,

$$\mathcal{C} \equiv \beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \sum_{1 \leq k < l \leq n-2} \csc^2(\pi(l-k)/n) = 4\beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \left(S_2(n, 1) - \frac{4S_2(n, 1)}{n} + \frac{1}{4 \sin^2(\pi/n)} \right) \quad (\text{C.36})$$

We can add the $\sum_k L_k^{(2)}$ to the connected piece, to further simplify,

$$\begin{aligned} \mathcal{C} + \sum_k L_k^{(2)} &= \beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \left(\frac{8}{n} S_2(n, 1) - \csc^2(\pi/n) + 4S_2(n, 1) - \frac{16S_2(n, 1)}{n} + \csc^2(\pi/n) \right) \\ &= 4\beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \left(S_2(n, 1) - \frac{2S_2(n, 1)}{n} \right) \end{aligned} \quad (\text{C.37})$$

We can see how the divergent piece $\csc^2(\pi/n)$ gets cancelled exactly. Now, we sum over all the terms we have, to obtain,

$$\mathcal{G}_n \equiv 1 + \sum_k \left(L_k^{(1)} + L_k^{(2)} \right) + \text{Disc} + \mathcal{C} \quad (\text{C.38})$$

$$\equiv 1 + 2\beta^2 \frac{\pi\epsilon}{n} \cot(\pi/n) + 2\beta^4 \left(\frac{\pi\epsilon}{n} \right)^2 \cot^2(\pi/n) + 4\beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \left(S_2(n, 1) - \frac{2S_2(n, 1)}{n} \right) \quad (\text{C.39})$$

$$\implies \log \mathcal{G}_n = 2\beta^2 \frac{\pi\epsilon}{n} \cot(\pi/n) + 4\beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \left(S_2(n, 1) - \frac{2S_2(n, 1)}{n} \right) \quad (\text{C.40})$$

where in the last line we used,

$$\log(1 + A + A^2/2 + B) \sim A + B \quad (\text{C.41})$$

The full denominator takes the following form,

$$\mathcal{F}_n^{-1} = D(n)^{n-1} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1+\epsilon)/n)} \right)^{\beta^2} \mathcal{G}_n \quad (\text{C.42})$$

Now, if we take the log of $\sin(\pi(1+\epsilon)/n)$ and expand around $\log(\sin(\pi/n))$, we get,

$$\log(\sin(\pi(1+\epsilon)/n)) \simeq \log \sin(\pi/n) + (\pi\epsilon/n) \cot(\pi/n) + O(\epsilon^2) \quad (\text{C.43})$$

Then, the log of denominator gives,

$$-\log \mathcal{F}_n = (n-1) \log D(n) + \log \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1+\epsilon)/n)} \right)^{\beta^2} + \log \mathcal{G}_n \quad (\text{C.44})$$

$$\begin{aligned} &\simeq (n-1) \log D(n) + \beta^2 \log(\sin(\pi\epsilon/n)) - \beta^2 (\log(\sin(\pi/n)) + (\pi\epsilon/n) \cot(\pi/n)) \\ &+ 2\beta^2 \frac{\pi\epsilon}{n} \cot(\pi/n) + 4\beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \left(S_2(n, 1) - \frac{2S_2(n, 1)}{n} \right) \end{aligned} \quad (\text{C.45})$$

Notice, if we take the log of $\sin(\pi(1-\epsilon)/n)$ and expand around $\log(\sin(\pi/n))$ we get,

$$\log \sin(\pi(1-\epsilon)/n) \simeq \log \sin(\pi/n) - (\pi\epsilon/n) \cot(\pi/n) + O(\epsilon^2) \quad (\text{C.46})$$

Then, the log of the denominator simplifies to,

$$\begin{aligned} -\log \mathcal{F}_n &\simeq (n-1) \log D(n) + \beta^2 \log(\sin(\pi\epsilon/n)) - \beta^2 \log(\sin(\pi(1-\epsilon)/n)) \\ &+ 4\beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \left(S_2(n, 1) - \frac{2S_2(n, 1)}{n} \right) \end{aligned} \quad (\text{C.47})$$

Now, we finally have the relative entropy,

$$\begin{aligned} S(\rho_0 \parallel \rho_\beta) &= \partial_n \mathcal{F}_n|_{n=1} \\ &= -\partial_n \left(\beta^2 \log(\sin(\pi\epsilon/n)) - \beta^2 \log(\sin(\pi(1-\epsilon)/n)) + 4\beta^2 \left(\frac{\pi\epsilon}{n} \right)^2 \left(S_2(n, 1) - \frac{2S_2(n, 1)}{n} \right) \right) \Big|_{n=1} \\ &= \beta^2 \pi \epsilon \cot(\pi\epsilon) + \beta^2 \pi(1-\epsilon) \cot(\pi\epsilon) + \beta^2 \frac{\pi^2 \epsilon^2}{3} + \dots \\ &= \beta^2 \left(\frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} + \frac{\pi^2 \epsilon^2}{3} + \dots \right). \end{aligned} \quad (\text{C.48})$$

$$= \beta^2 \left(\frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} + \frac{\pi^2 \epsilon^2}{3} + \dots \right). \quad (\text{C.49})$$

Therefore, up to ϵ^2 our result using replica trick match with the exact result. The lesson learnt here is that we are supposed to include the contribution of currents from the non-mixing OPE channels as well.

C.3 $S(\rho_\alpha \parallel \rho_\beta)$ using OPEs

In the denominator, we have to compute $\text{tr}(\rho_\alpha \parallel \rho_\beta^{n-1})$. There are $(n-2)$ sites with $V_\beta V_{-\beta}$ OPEs. While the sites $e^{i\pi\theta_0}, e^{i\pi\theta_{n-1}}$ have mixing OPEs. The non-mixing OPE is given by,

$$V_{-\beta}(z_{k+1,n}^-) V_\beta(z_{k,n}^+) \simeq \zeta_n^{-\beta^2} \left(1 - \beta \zeta_n J_k + \frac{\beta^2 \zeta_n^2}{2} : J_k^2 : + O(\zeta_n^3) \right), \quad k = 1, \dots, n-2. \quad (\text{C.50})$$

For the mixing OPEs, we need to keep all the terms up to order ζ_n^2 . We need to evaluate $V_\alpha(z_{0,n}^+) V_{-\beta}(z_{1,n}^-)$ and $V_\beta(z_{n-1,n}^+) V_{-\alpha}(z_{0,n}^-)$. We thus need to evaluate the following,

$$V_\alpha(z) V_{-\beta}(w) =: e^{i\alpha\phi(z)} :: e^{-i\beta\phi(w)} : \quad (\text{C.51})$$

$$= e^{\langle i\alpha\phi(z)(-i\beta\phi(w)) \rangle} : e^{i(\alpha\phi(z) - \beta\phi(w))} : \quad (\text{C.52})$$

Thus, when we do the midpoint OPEs, i.e. by expanding the fields around the midpoints, we are left with the following mixing OPEs,

$$V_\alpha(z_{0,n}^+)V_{-\beta}(z_{1,n}^-) = \zeta_n^{-\alpha\beta}V_\delta(e^{i\pi\theta_0})\left(1 - \frac{s\zeta_n}{2}J_0 + \frac{s^2\zeta_n^2}{8}:J_0^2: + \frac{\delta\zeta_n^2}{8}\partial J_0 + O(\zeta_n^3)\right) \quad (\text{C.53})$$

$$V_\beta(z_{n-1,n}^+)V_{-\alpha}(z_{0,n}^-) = \zeta_n^{-\alpha\beta}V_{-\delta}(e^{i\pi\theta_{n-1}})\left(1 - \frac{s\zeta_n}{2}J_{n-1} + \frac{s^2\zeta_n^2}{8}:J_{n-1}^2: - \frac{\delta\zeta_n^2}{8}\partial J_{n-1} + O(\zeta_n^3)\right) \quad (\text{C.54})$$

where we defined,

$$s = \alpha + \beta, \quad \delta = \alpha - \beta. \quad (\text{C.55})$$

The denominator of $\mathcal{F}_n$ that we have to evaluate becomes,

$$\text{tr}(\rho_\alpha \rho_\beta^{n-1}) = \langle V_\delta(A)\mathcal{M}_A V_{-\delta}(B)\mathcal{M}_B \prod_{k=1}^{n-2} \left(1 - \beta\zeta_n J_k + \frac{\beta^2\zeta_n^2}{2}:J_k^2: + O(\zeta_n^3)\right) \rangle, \quad (\text{C.56})$$

$$A = e^{i\pi\theta_0}, \quad B = e^{i\pi\theta_{n-1}}, \quad (\text{C.57})$$

$$\mathcal{M}_A = 1 - \frac{s\zeta_n}{2}J_0 + \frac{s^2\zeta_n^2}{8}:J_0^2: + \frac{\delta\zeta_n^2}{8}\partial J_0 + O(\zeta_n^3), \quad (\text{C.58})$$

$$\mathcal{M}_B = 1 - \frac{s\zeta_n}{2}J_{n-1} + \frac{s^2\zeta_n^2}{8}:J_{n-1}^2: - \frac{\delta\zeta_n^2}{8}\partial J_{n-1} + O(\zeta_n^3) \quad (\text{C.59})$$

C.3.1 3-point function

We have 3-point contributions from terms such as,

$$V_\delta(A)J_0V_{-\delta}(B), \quad V_\delta(A)V_{-\delta}(B)J_{n-1}, \quad V_\delta(A)V_{-\delta}(B)J_k \quad k = 1, \dots, n-2, \quad (\text{C.60})$$

$$V_\delta\partial J_0V_{-\delta} \quad V_\delta\partial J_{n-1}V_{-\delta} \quad (\text{C.61})$$

three-point function from a current sitting at a site $k \neq \{0, n-1\}$, we have the following contribution⁵,

$$ie^{i\theta} \frac{\langle V_\delta(A)J(e^{i\pi\theta})V_{-\delta}(B) \rangle}{\langle V_\delta(A)V_{-\delta}(B) \rangle} = ie^{i\theta} \left(\frac{\delta}{e^{i\theta} - e^{i\pi/n}} - \frac{\delta}{e^{i\theta} - e^{-i\pi/n}} \right) \quad (\text{C.62})$$

$$= \frac{1}{2}\delta \left(\cot\left(\frac{\theta - \pi/n}{2}\right) - \cot\left(\frac{\theta + \pi/n}{2}\right) \right) \quad (\text{C.63})$$

$$\Rightarrow ie^{i\theta} \frac{\langle V_\delta(A)\partial J(e^{i\pi\theta})V_{-\delta}(B) \rangle}{\langle V_\delta(A)V_{-\delta}(B) \rangle} = -\frac{1}{4}\delta \left(\csc^2\left(\frac{\theta - \pi/n}{2}\right) - \csc^2\left(\frac{\theta + \pi/n}{2}\right) \right) \quad (\text{C.64})$$

⁵Current $J(e^{i\theta})$ comes with a phase $ie^{i\theta}$.

Using the above equations, the contributions from the currents at sitting at sites $k \neq \{0, n-1\}$ is given by,

$$\tilde{L}_k \equiv -\beta \zeta_n i e^{i\pi(2k+1)/n} \frac{\langle V_\delta(A) J_k V_{-\delta}(B) \rangle}{\langle V_\delta(A) V_{-\delta}(B) \rangle} \quad (\text{C.65})$$

$$= -\beta \zeta_n i e^{i\pi(2k+1)/n} \left(\frac{\delta}{e^{i\pi(2k+1)/n} - e^{i\pi/n}} - \frac{\delta}{e^{i\pi(2k+1)/n} - e^{i\pi(2n-1)/n}} \right) \quad (\text{C.66})$$

$$= -\beta \frac{\pi\epsilon}{n} \delta (\cot(\pi k/n) - \cot(\pi(k+1)/n)) \quad (\text{C.67})$$

$$\implies \sum_{k=1}^{n-2} \tilde{L}_k = -2\beta \frac{\pi\epsilon}{n} \delta \cot(\pi/n) \quad (\text{C.68})$$

The first two terms give us,

$$\tilde{L}_A \equiv -\frac{s\zeta_n i e^{i\pi/n}}{2} \frac{\langle V_\delta(A) J_0 V_{-\delta}(B) \rangle}{\langle V_\delta(A) V_{-\delta}(B) \rangle} = -s \frac{\pi\epsilon}{2n} \left(\frac{(-\delta)e^{i\pi/n}}{A-B} \right) = s\delta \frac{\pi\epsilon}{2n} (\cot(\pi/n) + i) \quad (\text{C.69})$$

Similarly,

$$\tilde{L}_B \equiv -\frac{s\zeta_n i e^{-i\pi/n}}{2} \frac{\langle V_\delta(A) J_{n-1} V_{-\delta}(B) \rangle}{\langle V_\delta(A) V_{-\delta}(B) \rangle} = s\delta \frac{\pi\epsilon}{2n} (\cot(\pi/n) - i) \quad (\text{C.70})$$

The total first order contribution is thus given by,

$$\tilde{L}_A + \tilde{L}_B + \sum_{k=1}^{n-2} \tilde{L}_k = (s - 2\beta) \delta \frac{\pi\epsilon}{n} \cot(\pi/n) = \delta^2 \frac{\pi\epsilon}{n} \cot(\pi/n) \quad (\text{C.71})$$

Hence, the denominator of $\mathcal{F}_n$ becomes,

$$D(n)^{n-1} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi/n)} \right)^{\delta^2} \left(1 + \tilde{L}_A + \tilde{L}_B + \sum_{k=1}^{n-2} \tilde{L}_k \right) \quad (\text{C.72})$$

Taking the logarithm of the above gives,

$$(n-1) \log(D(n)) + \delta^2 \left[(\log(\sin(\pi\epsilon)/n) - \log \sin(\pi/n)) + \frac{\pi\epsilon}{n} \cot(\pi/n) \right] \quad (\text{C.73})$$

$$\simeq (n-1) \log(D(n)) + \delta^2 \left(\log \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right) \right) \quad (\text{C.74})$$

The second line above is not an allowed manipulation, simply because it does not commute with $n \rightarrow 1$ limit. Now with this correction from three-point function, the relative entropy is given by,

$$S(\rho_\alpha \| \rho_\beta) = \partial_n \mathcal{F}_n|_{n=1} = \frac{\alpha^2 (\pi\epsilon)^2}{3} + \delta^2 \pi\epsilon \cot(\pi\epsilon) + \delta^2 \pi(1-\epsilon) \cot(\pi\epsilon) \quad (\text{C.75})$$

$$\begin{aligned} &= \frac{\alpha^2 (\pi\epsilon)^2}{3} + \delta^2 \pi \cot(\pi\epsilon) \\ &= \frac{\alpha^2 (\pi\epsilon)^2}{3} + (\alpha - \beta)^2 \left(\frac{1}{\epsilon} - \frac{\epsilon}{3} + O(\epsilon^3) \right) \end{aligned} \quad (\text{C.76})$$

However, the exact analytical relative entropy is,

$$S_{exact}(\rho_\alpha \parallel \rho_\beta) = (\alpha - \beta)^2 \left(1 - \pi(1 - \epsilon) \cot(\pi(1 - \epsilon)) \right) \quad (\text{C.77})$$

$$= (\alpha - \beta)^2 \left(\frac{1}{\epsilon} - \frac{\epsilon}{3} + \frac{\pi^2 \epsilon^2}{3} + O(\epsilon^3) \right) \quad (\text{C.78})$$

So, in order to correct the coefficient of ϵ^2 , we need to compute the correction due to 4-point function.

We can further compute the contribution from the ∂J terms,

$$W(A) \equiv \frac{\delta \zeta_n^2}{8} \langle V_\delta \partial J_0 V_{-\delta} \rangle = \frac{\delta^2}{8} \left(\frac{\pi \epsilon}{n} \right)^2 \csc^2(\pi/n) \quad (\text{C.79})$$

$$W(B) \equiv -\frac{\delta \zeta_n^2}{8} \langle V_\delta \partial J_{n-1} V_{-\delta} \rangle = \frac{\delta^2}{8} \left(\frac{\pi \epsilon}{n} \right)^2 \csc^2(\pi/n) \quad (\text{C.80})$$

C.3.2 4-point function

Now at the second order, we have contributions from the connected and disconnected piece. Since we already computed the contribution from the three-point function, which comes from the $(\beta \sum_{k=1}^{n-2} J_k + \frac{s}{2} J_0 + \frac{s}{2} J_{n-1})$, the disconnected piece can be directly written as,

$$\text{Disc} \equiv \frac{\zeta_n^2}{2} \left(\beta \sum_{k=1}^{n-2} J_k + \frac{s}{2} J_0 + \frac{s}{2} J_{n-1} \right)^2 \quad (\text{C.81})$$

This will exactly vanish when we expand the logarithm of the denominator, i.e.

$$\log \left(1 + \left(\tilde{L}_A + \tilde{L}_B + \sum_{k=1}^{n-2} \tilde{L}_k \right) + W(A) + W(B) + \text{Disc} + \mathcal{C} \right) \quad (\text{C.82})$$

$$\simeq \tilde{L}_A + \tilde{L}_B + \sum_{k=1}^{n-2} \tilde{L}_k + W(A) + W(B) + \mathcal{C} \quad (\text{C.83})$$

where $\mathcal{C}$ is the connected component. In the last line we used,

$$\log(1 + A + A^2/2 + B) \sim A + B \quad (\text{C.84})$$

The connected component $\mathcal{C}$ gives,

$$\mathcal{C} = \zeta_n^2 \left[s\beta \left(\frac{2S_2(n, 1)}{n} - 4 \csc^2(\pi/n) \right) + \frac{s^2}{4} \frac{1}{4} \csc^2(\pi/n) + \beta^2 \left(S_2(n, 1) - \frac{4S_2(n, 1)}{n} + \frac{1}{4} \csc^2(\pi/n) \right) \right] \quad (\text{C.85})$$

$$= \left(\frac{\pi \epsilon}{n} \right)^2 \left[\left(\frac{s^2}{4} + \beta^2 - s\beta \right) \csc^2(\pi/n) + \left(4\beta^2 + \frac{8\beta\delta}{n} \right) S_2(n, 1) \right] \quad (\text{C.86})$$

$$= \left(\frac{\pi \epsilon}{n} \right)^2 \frac{\delta^2}{4} \csc^2(\pi/n) + \left(\frac{\pi \epsilon}{n} \right)^2 \left(4\beta^2 + \frac{8\beta\delta}{n} \right) S_2(n, 1) \quad (\text{C.87})$$

We can now add the contribution of ∂J to the $\csc^2(\pi/n)$ in the above connected piece to obtain,

$$W(A) + W(B) + \left(\frac{\pi\epsilon}{n}\right)^2 \frac{\delta^2}{4} \csc^2(\pi/n) = \frac{\delta^2}{2} \left(\frac{\pi\epsilon}{n}\right)^2 \csc^2(\pi/n) \quad (\text{C.88})$$

This term is the second order correction to the modified two-point function. We can then write the denominator of $\mathcal{F}_n$,

$$D(n)^{n-1} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi/n)} \right)^{\delta^2} \left(1 + \tilde{L}_A + \tilde{L}_B + \sum_{k=1}^{n-2} \tilde{L}_k + W(A) + W(B) + \underbrace{\left(\frac{\pi\epsilon}{n} \right)^2 \frac{\delta^2}{4} \csc^2(\pi/n)}_{\langle V_\delta \partial J_0 V_{-\delta} \rangle + \langle V_\delta \partial J_{n-1} V_{-\delta} \rangle} \right) \quad (\text{C.89})$$

Taking the logarithm of the above gives,

$$(n-1) \log(D(n)) + \delta^2 \left[(\log(\sin(\pi\epsilon/n)) - \log \sin(\pi/n)) + \frac{\pi\epsilon}{n} \cot(\pi/n) + \frac{1}{2} \left(\frac{\pi\epsilon}{n} \right)^2 \csc^2(\pi/n) \right] \quad (\text{C.90})$$

$$\simeq (n-1) \log(D(n)) + \delta^2 \left(\log \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right) \right) \quad (\text{C.91})$$

The leftover $S_2(n, 1)$ piece correction from the connected piece would give us the coefficient of ϵ^2 in the relative entropy. Finally we obtain,

$$S(\rho_\alpha \| \rho_\beta) = \partial_n \mathcal{F}_n|_{n=1} = \frac{\alpha^2(\pi\epsilon)^2}{3} + \delta^2 \partial_n \left(\log \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right) \right) \Big|_{n=1} \quad (\text{C.92})$$

$$- (2\pi\epsilon)^2 (\beta^2 + 2\beta\delta) S'_2(1, 1) \quad (\text{C.93})$$

$$= \frac{\alpha^2(\pi\epsilon)^2}{3} + \delta^2 \pi \cot(\pi\epsilon) - (2\pi\epsilon)^2 (\beta^2 + 2\beta\delta) S'_2(1, 1) \quad (\text{C.94})$$

$$= \delta^2 \pi \cot(\pi\epsilon) + \frac{(\alpha - \beta)^2 (\pi\epsilon)^2}{3} \quad (\text{C.95})$$

$$\simeq (\alpha - \beta)^2 \left(\frac{1}{\epsilon} - \frac{\epsilon}{3} + \frac{\pi^2 \epsilon^2}{3} + O(\epsilon^3) \right) \quad (\text{C.96})$$

C.4 $S(\rho_{i\partial\phi} \| \rho_\beta)$ using OPEs

C.5 $S(\rho_\alpha \|_{\frac{1}{2}}(\rho_\alpha + \rho_\beta))$ using OPEs

We need to evaluate the following,

$$\text{tr} \left(\rho_\alpha \left(\frac{\rho_\alpha + \rho_\beta}{2} \right)^{n-1} \right). \quad (\text{C.97})$$

{appendix rho a

Without loss of generality, we keep β pairs at sites $j = 1, \dots, p$ and the α pairs at $j = 0, p+1, p+2, \dots, n-1$. Then the mixing OPEs are,

$$V_\beta(z_{p,n}^+)V_{-\alpha}(z_{p+1,n}^-) = \zeta_n^{-\alpha\beta}V_{-\delta}(e^{i\pi\theta_p})\left(1 - \frac{s\zeta_n}{2}J_p + \frac{s^2\zeta_n^2}{8}:J_p^2:-\frac{\delta\zeta_n^2}{8}\partial J_p + O(\zeta_n^3)\right) \quad (\text{C.98})$$

$$V_\alpha(z_{0,n}^+)V_{-\beta}(z_{1,n}^-) = \zeta_n^{-\alpha\beta}V_\delta(e^{i\pi\theta_0})\left(1 - \frac{s\zeta_n}{2}J_0 + \frac{s^2\zeta_n^2}{8}:J_0^2:+\frac{\delta\zeta_n^2}{8}\partial J_0 + O(\zeta_n^3)\right) \quad (\text{C.99})$$

The other sites have the usual non-mixing OPEs of $\alpha\alpha$ and $\beta\beta$ pairs.

C.5.1 3-point function

We can again define the analogous 3-point function contribution,

$$\hat{\mathcal{J}}_k \equiv ie^{i\pi(2k+1)/n} \frac{\langle V_\delta(e^{i\pi\theta_p})J_kV_{-\delta}(e^{i\pi\theta_0}) \rangle}{\langle V_\delta(e^{i\pi\theta_p})V_{-\delta}(e^{i\pi\theta_0}) \rangle} \quad (\text{C.100})$$

$$= \frac{\delta}{2} (\cot(\pi k/n) - \cot(\pi(k-p)/n)) \quad (\text{C.101})$$

Sum over $\beta\beta$ sites $j = 1, \dots, p-1$,

$$\hat{L}_p^{(\beta)} = -\zeta_n\beta\frac{\delta}{2}\sum_{j=1}^{p-1}(\cot(\pi k/n) - \cot(\pi(k-p)/n)) \quad (\text{C.102})$$

$$= -2\beta\delta\left(\frac{\pi\epsilon}{n}\right)\sum_{r=1}^{p-1}\cot(\pi r/n) \quad (\text{C.103})$$

Sum over $\alpha\alpha$ sites, $j = p+1, \dots, n-1$,

$$\hat{L}_p^{(\alpha)} = -\zeta_n\alpha\frac{\delta}{2}\sum_{k=p+1}^{n-1}(\cot(\pi k/n) - \cot(\pi(k-p)/n)) \quad (\text{C.104})$$

$$= \zeta_n\alpha\delta\left(\sum_{r=1}^{n-p-1}\cot(\pi r/n)\right) = 2\alpha\delta\left(\frac{\pi\epsilon}{n}\right)\left(\sum_{r=1}^p\cot(\pi r/n)\right) \quad (\text{C.105})$$

The contribution from the currents at the mixing endpoints,

$$\hat{L}_p^{(p)} \equiv -\frac{s\delta}{2}\left(\frac{\pi\epsilon}{n}\right)\cot(\pi p/n) \quad (\text{C.106})$$

$$\hat{L}_p^{(0)} \equiv -\frac{s\delta}{2}\left(\frac{\pi\epsilon}{n}\right)\cot(\pi p/n) \quad (\text{C.107})$$

Hence the net contribution from the three-point functions,

$$\hat{L}_p \equiv \hat{L}_p^{(\beta)} + \hat{L}_p^{(\alpha)} + \hat{L}_p^{(p)} + \hat{L}_p^{(0)} = \left(\frac{2\pi\epsilon}{n}\right)\delta^2\left(\sum_{r=1}^{p-1}\cot(\pi r/n) + \frac{1}{2}\cot(\pi p/n)\right) \quad (\text{C.108})$$

The second term precisely shifted the two-point function in previous calculation to give the correct divergent piece. For calculation purpose, let's define,

$$\hat{L}_p \equiv \hat{L}_p^{\text{sum}} + \hat{L}_p^{\text{div}}, \quad (\text{C.109}) \quad \{\text{eq: 3 point 1}\}$$

$$\hat{L}_p^{\text{sum}} = \left(\frac{2\pi\epsilon}{n}\right) \delta^2 \sum_{r=1}^{p-1} \cot(\pi r/n), \quad (\text{C.110}) \quad \{\text{eq: 3 point 1}\}$$

$$\hat{L}_p^{\text{div}} = \frac{\pi\epsilon}{n} \delta^2 \cot(\pi p/n) \quad (\text{C.111})$$

We thus have,

$$\mathcal{F}_n^{-1} = 2^{1-n} \left[1 + \sum_{p=1}^{n-1} (n-p) \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(p-\epsilon)/n)} \right)^{\delta^2} + \delta^2 \zeta_n \sum_{p=1}^{n-1} (n-p) \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{\delta^2} \sum_{r=1}^{p-1} \cot(\pi r/n) \right] \quad (\text{C.112})$$

We can further compute the contribution from the ∂J terms,

$$\hat{W}_0 \equiv \frac{\delta\zeta_n^2}{8} \langle V_\delta \partial J_0 V_{-\delta} \rangle = \frac{\delta^2}{8} \left(\frac{\pi\epsilon}{n} \right)^2 \csc^2(\pi p/n) \quad (\text{C.113})$$

$$\hat{W}_p \equiv \frac{\delta\zeta_n^2}{8} \langle V_\delta \partial J_p V_{-\delta} \rangle = \frac{\delta^2}{8} \left(\frac{\pi\epsilon}{n} \right)^2 \csc^2(\pi p/n) \quad (\text{C.114})$$

C.5.2 4-point function

Disconnected four point function would be again,

$$\text{Disc}_p \equiv \frac{\zeta_n^2}{2} \left(\beta \sum_{k=1}^{p-1} \mathcal{J}_k + \alpha \sum_{k=p+1}^{n-1} \mathcal{J}_k + \frac{s}{2} (\mathcal{J}_p + \mathcal{J}_0) \right)^2 = \frac{1}{2} \hat{L}_p^2 = \frac{1}{2} \left(\hat{L}_p^{\text{sum}} + \hat{L}_p^{\text{div}} \right)^2. \quad (\text{C.115})$$

The connected part is given by,

$$\begin{aligned} \mathcal{C}_p \equiv & \zeta_n^2 \left[\beta^2 \sum_{1 \leq k < l \leq p-1} \frac{1}{\left(2 \sin \frac{\pi(k-l)}{n}\right)^2} + \alpha^2 \sum_{p+1 \leq k < l \leq n-1} \frac{1}{\left(2 \sin \frac{\pi(k-l)}{n}\right)^2} + \alpha\beta \sum_{k=1}^{p-1} \sum_{l=p+1}^{n-1} \frac{1}{\left(2 \sin \frac{\pi(k-l)}{n}\right)^2} \right. \\ & + \frac{s\beta}{2} \sum_{k=1}^{p-1} \left(\frac{1}{\left(2 \sin \frac{\pi k}{n}\right)^2} + \frac{1}{\left(2 \sin \frac{\pi(p-k)}{n}\right)^2} \right) + \frac{s\alpha}{2} \sum_{k=p+1}^{n-1} \left(\frac{1}{\left(2 \sin \frac{\pi k}{n}\right)^2} + \frac{1}{\left(2 \sin \frac{\pi(k-p)}{n}\right)^2} \right) \\ & \left. + \frac{s^2}{4} \frac{1}{\left(2 \sin \frac{\pi p}{n}\right)^2} \right] \end{aligned} \quad (\text{C.116})$$

$$\begin{aligned} = & \zeta_n^2 \left[\frac{s^2}{4} S_2(n, 1) + \frac{s\delta(n-2p)}{2n} S_2(n, 1) + \frac{\delta^2}{4} \left[S_2(n, 1) - \frac{4S_2(n, 1)}{n} + \frac{1}{\left(2 \sin(\pi p/n)\right)^2} \right. \right. \\ & \left. \left. - \sum_{k=1}^{p-1} \sum_{l=p+1}^{n-1} \frac{2}{\left(2 \sin(\pi(k-l)/n)\right)^2} \right] \right] \end{aligned} \quad (\text{C.117})$$

$$\begin{aligned} = & \zeta_n^2 \left[\left(\frac{s^2}{4} + \frac{s\delta(n-2p)}{2n} + \frac{\delta^2}{4} \left(1 - \frac{4}{n} \right) \right) S_2(n, 1) + \frac{\delta^2}{4} \frac{1}{\left(2 \sin(\pi p/n)\right)^2} - \frac{\delta^2}{2} \sum_{k=1}^{p-1} \sum_{l=p+1}^{n-1} \frac{1}{\left(2 \sin(\pi(k-l)/n)\right)^2} \right] \end{aligned} \quad (\text{C.118})$$

As a sanity check, if we substitute $p = n - 1$ in the above equation, the last double sum vanishes. Then we do get back the connected component $\mathcal{C}$ we calculated earlier for the pure states, i.e. $\mathcal{C}_{n-1} = \mathcal{C}$.

To simplify the calculation in the later steps, we write the connected part as follows,

$$\mathcal{C}_p = \mathcal{C}_p^{\text{sum}} + \mathcal{C}_p^{\text{divergent}} \quad (\text{C.119})$$

$$\mathcal{C}_p^{\text{sum}} = \zeta_n^2 \left[\left(\frac{s^2}{4} + \frac{s\delta(n-2p)}{2n} + \frac{\delta^2}{4} \left(1 - \frac{4}{n} \right) \right) S_2(n, 1) - \frac{\delta^2}{2} \sum_{k=1}^{p-1} \sum_{l=p+1}^{n-1} \frac{1}{\left(2 \sin(\pi(k-l)/n)\right)^2} \right] \quad (\text{C.120})$$

$$\mathcal{C}_p^{\text{divergent}} = \zeta_n^2 \frac{\delta^2}{4} \frac{1}{\left(2 \sin(\pi p/n)\right)^2} \quad (\text{C.121})$$

Again, we can add the contribution due to the ∂J terms with the divergent piece from the full connected component, to obtain,

$$\hat{W}_0 + \hat{W}_p + \mathcal{C}_p^{\text{divergent}} = \frac{\delta^2}{2} \left(\frac{\pi\epsilon}{n} \right)^2 \csc^2(\pi p/n) \quad (\text{C.122})$$

We can then write the denominator,

$$\mathcal{F}_n^{-1} = 2^{1-n} \left[1 + \sum_{p=1}^{n-1} (n-p) D_n(\epsilon)^p \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{\delta^2} \times \right. \\ \left. \left(1 + \hat{L}_p + \hat{W}_0 + \hat{W}_p + \text{Disc}_p + \mathcal{C}_p^{\text{sum}} + \mathcal{C}_p^{\text{divergent}} \right) \right] \quad (\text{C.123})$$

$$\implies -\log \mathcal{F}_n = (1-n) \log 2 + \left[\sum_{p=1}^{n-1} (n-p) D_n(\epsilon)^p \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{\delta^2} \times \right. \\ \left. \left(1 + \hat{L}_p + \hat{W}_0 + \hat{W}_p + \text{Disc}_p + \mathcal{C}_p^{\text{sum}} + \mathcal{C}_p^{\text{divergent}} \right) \right] \quad (\text{C.124})$$

Clearly, the leading order contribution comes from the 2-point function of the primaries,

$$-\log \mathcal{F}_n \simeq (1-n) \log 2 + \left[\sum_{p=1}^{n-1} (n-p) \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{\delta^2} \right] \quad (\text{C.125})$$

$$\implies S(\rho_\alpha \| (\rho_\alpha + \rho_\beta)/2) = \log 2 - (\pi\epsilon)^{(\alpha-\beta)^2} \frac{\sqrt{\pi}\Gamma\left(1 + \frac{(\alpha-\beta)^2}{2}\right)}{4\Gamma\left(3/2 + \frac{(\alpha-\beta)^2}{2}\right)} + \dots \quad (\text{C.126})$$

However, keeping the higher order corrections gives us,

$$\mathcal{F}_n^{-1} \sim 2^{1-n} \left[1 + \sum_{p=1}^{n-1} (n-p) \left(1 + (\pi\epsilon)^2 \frac{ps\delta}{3} \left(1 - \frac{1}{n^2} \right) + O(\epsilon^4) \right) \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{\delta^2} \right. \\ \times \left(1 + \hat{L}_p + \hat{W}_0 + \hat{W}_p + \text{Disc}_p + \mathcal{C}_p^{\text{sum}} + \mathcal{C}_p^{\text{divergent}} \right) \\ \left. + \text{Broken words of the order } \epsilon^{2\delta^2} + \dots \right] \quad (\text{C.127})$$

where we expanded the normalization and conformal factor $D_n(\epsilon)^p$,

$$D_n(\epsilon)^p = \left(\frac{n \sin(\pi\epsilon/n)}{\sin(\pi\epsilon)} \right)^{2(\Delta_\alpha - \Delta_\beta)p} = \left(\frac{n \sin(\pi\epsilon/n)}{\sin(\pi\epsilon)} \right)^{s\delta p} \quad (\text{C.128})$$

$$\simeq 1 + (\pi\epsilon)^2 \frac{ps\delta}{3} \left(1 - \frac{1}{n^2} \right) + \dots \quad (\text{C.129})$$

We can decompose the paranthesis by,

$$\left(1 + \hat{L}_p + \hat{W}_0 + \hat{W}_p + \text{Disc}_p + \mathcal{C}_p^{\text{sum}} + \mathcal{C}_p^{\text{divergent}}\right) \quad (\text{C.130})$$

$$= \left[1 + \left(\hat{L}_p^{\text{div}} + \hat{L}_p^{\text{sum}}\right) + \hat{W}_0 + \hat{W}_p + \left(\frac{1}{2} \left(\hat{L}_p^{\text{div}}\right)^2 + \frac{1}{2} \left(\hat{L}_p^{\text{sum}}\right)^2 + \hat{L}_p^{\text{div}} \hat{L}_p^{\text{sum}}\right) + \mathcal{C}_p^{\text{sum}} + \mathcal{C}_p^{\text{divergent}}\right] \quad (\text{C.131})$$

$$\simeq \left(1 + \hat{L}_p^{\text{div}} + \hat{W}_0 + \hat{W}_p + \mathcal{C}_p^{\text{divergent}} + \frac{1}{2} \left(\hat{L}_p^{\text{div}}\right)^2\right) \left(1 + \hat{L}_p^{\text{sum}} + \mathcal{C}_p^{\text{sum}} + \frac{1}{2} \left(\hat{L}_p^{\text{sum}}\right)^2\right) \quad (\text{C.132})$$

The first factor shifts the two-point function by ϵ , since we have,

$$\left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)}\right)^{\delta^2} \left(1 + \hat{L}_p^{\text{div}} + \hat{W}_0 + \hat{W}_p + \mathcal{C}_p^{\text{divergent}} + \frac{1}{2} \left(\hat{L}_p^{\text{div}}\right)^2\right) \simeq \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(p-\epsilon)/n)}\right)^{\delta^2} + O(\epsilon^3). \quad (\text{C.133})$$

Taking the logarithm of the above, we get,

$$-\log \mathcal{F}_n \sim (1-n) \log 2 + \left[\sum_{p=1}^{n-1} (n-p) \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(p-\epsilon)/n)}\right)^{\delta^2} \left(1 + \hat{L}_p^{\text{sum}} + \frac{1}{2} \left(\hat{L}_p^{\text{sum}}\right)^2 + \mathcal{C}_p^{\text{sum}} + (\pi\epsilon)^2 \frac{ps\delta}{3} \left(1 - \frac{1}{n^2}\right)\right) \right] \quad (\text{C.134})$$

The lowest order contribution comes from the two-point function which is of the order ϵ^{δ^2} . However, unlike to the case of pure states, the corrections from three-point and four-point functions due to current is of the order ϵ^{δ^2+1} and ϵ^{δ^2+2} respectively. This is because in the mixture case, there are multiple words including the identity piece. However, there was just a single non-identity word in the pure states calculation.

C.6 $S\left(\frac{1}{2}(\rho_\alpha + \rho_\beta) \parallel \rho_\alpha\right)$ using OPEs

We need to evaluate the following,

$$\mathcal{F}_n = \frac{\text{tr} \left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right)^n \right)}{\text{tr} \left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right) \rho_\alpha^{n-1} \right)}. \quad (\text{C.135})$$

From the previous calculation, it is clear that the leading order contributions are from the identity pieces, and the subleading corrections are from the two-point functions of the primaries from the mixing OPEs. So, both for the numerator and the denominator, we will consider only contiguous words of α and β to obtain the relative entropy.

C.6.1 Numerator

The numerator becomes,

$$\text{tr} \left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right)^n \right) = 2^{-n} \left(1 + D_n(\epsilon)^n + \zeta_n^2 S_2(n, 1) (\alpha^2 + \beta^2 D_n(\epsilon)^n) + n \sum_{p=1}^{n-1} D_n(\epsilon)^p \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{\delta^2} + \dots \right) \quad (\text{C.136})$$

We are considering the uniform words $(\alpha, \dots, \alpha)$, $(\beta, \dots, \beta)$, which consist of the identity piece. We also consider the subleading terms which have a pair of mixing OPE, where we have a string of α connected to a string of β . These terms are always ϵ^{δ^2} suppressed from the leading order contribution (uniform words), as the later contains an identity element in it. We can thus write the numerator as follows,

$$\begin{aligned} \text{tr} \left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right)^n \right) &\simeq 2^{-n} \left(\underbrace{[1 + \zeta_n^2 \alpha^2 S_2(n, 1)]}_{(\alpha, \dots, \alpha)} + \underbrace{[1 + \zeta_n^2 \beta^2 S_2(n, 1)]}_{(\beta, \dots, \beta)} \right. \\ &\quad + \underbrace{n \sum_{p=1}^{n-1} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{\delta^2}}_{\substack{\text{Contiguous block of } p \text{ } \beta\text{'s connected to} \\ \text{contiguous block of } (n-p) \text{ } \alpha\text{'s:} \\ (\alpha, \dots, \alpha, \underbrace{\beta, \dots, \beta}_p, \alpha, \dots, \alpha)}} + \dots \Big). \end{aligned} \quad (\text{C.137})$$

We can now take the logarithm of the above,

$$\begin{aligned} \log \left(\text{tr} \left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right)^n \right) \right) &\simeq -n \log 2 + \log \left([1 + \zeta_n^2 \alpha^2 S_2(n, 1)] + [1 + \zeta_n^2 \beta^2 S_2(n, 1)] \right) \\ &\quad + \frac{n \sum_{p=1}^{n-1} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{\delta^2}}{\left([1 + \zeta_n^2 \alpha^2 S_2(n, 1)] + [1 + \zeta_n^2 \beta^2 S_2(n, 1)] \right)} + \dots \end{aligned} \quad (\text{C.138})$$

$$\begin{aligned} &\simeq -n \log 2 + \log \left([1 + \zeta_n^2 \alpha^2 S_2(n, 1)] + [1 + \zeta_n^2 \beta^2 S_2(n, 1)] \right) \\ &\quad + \frac{2\zeta_n^{\delta^2} S_2(n, \delta^2/2)}{\left([1 + \zeta_n^2 \alpha^2 S_2(n, 1)] + [1 + \zeta_n^2 \beta^2 S_2(n, 1)] \right)} \end{aligned} \quad (\text{C.139})$$

We can take the derivative at $n = 1$,

$$\left. \partial_n \left[\log \left(\text{tr} \left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right)^n \right) \right) \right] \right|_{n=1} \simeq -\log 2 + \frac{1}{2} \left((2\pi\epsilon)^2 (\alpha^2 + \beta^2) S_2'(1, 1) \right) + (2\pi\epsilon)^2 S_2'(1, \delta^2/2) + \dots \quad (\text{C.140})$$

C.6.2 Denominator

For the denominator, we just have two words,

$$\text{tr} \left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right) \rho_\alpha^{n-1} \right) = \frac{1}{2} \text{tr} (\rho_\alpha^n) + \frac{1}{2} \text{tr} (\rho_\beta \rho_\alpha^{n-1}) \quad (\text{C.141})$$

$$\simeq \frac{1}{2} \left(\underbrace{[1 + \zeta_n^2 \alpha^2 S_2(n, 1)]}_{(\alpha, \dots, \alpha)} + \underbrace{\left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi/n)} \right)^{\delta^2} \left(1 + \frac{\langle V_\delta V_{-\delta} J \rangle}{\langle V_\delta V_{-\delta} \rangle} + \frac{\langle V_\delta V_{-\delta} J J \rangle}{\langle V_\delta V_{-\delta} \rangle} + \dots \right)}_{(\beta, \alpha, \dots, \alpha)} + \dots \right) \quad (\text{C.142})$$

$$\simeq \frac{1}{2} \left(\underbrace{[1 + \zeta_n^2 \alpha^2 S_2(n, 1)]}_{(\alpha, \dots, \alpha)} + \underbrace{\left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right)^{\delta^2} \left(1 + \left(\frac{\pi\epsilon}{n} \right)^2 \left(4\alpha^2 - \frac{8\alpha\delta}{n} \right) S_2(n, 1) + \dots \right)}_{(\beta, \alpha, \dots, \alpha)} + \dots \right) \quad (\text{C.143})$$

The $\dots$ in both numerator and denominator refer to broken words and higher point functions. Notice that the second word in the denominator would give the divergence. It is the same term we had while we computed the relative entropy of pure states, which shifted the two-point function, and gave correct coefficient up to ϵ^2 terms of relative entropy.

Now we can take the logarithm of the denominator,

$$\log \left(\text{tr} \left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right) \rho_\alpha^{n-1} \right) \right) \simeq -\log 2 + \log \left([1 + \zeta_n^2 \alpha^2 S_2(n, 1)] + \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right)^{\delta^2} \left(1 + \left(\frac{\pi\epsilon}{n} \right)^2 \left(4\alpha^2 - \frac{8\alpha\delta}{n} \right) S_2(n, 1) \right) \right) \quad (\text{C.144})$$

Taking the derivative of the above at $n = 1$, we obtain,

$$\partial_n \left[\log \left(\text{tr} \left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right) \rho_\alpha^{n-1} \right) \right) \right] \Big|_{n=1} \simeq \frac{1}{2} (2\pi\epsilon)^2 \alpha^2 S_2'(1, 1) + \frac{1}{2} \partial_n \left[\left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right)^{\delta^2} \left(1 + \left(\frac{\pi\epsilon}{n} \right)^2 \left(4\alpha^2 - \frac{8\alpha\delta}{n} \right) S_2(n, 1) \right) \right] \Big|_{n=1} \quad (\text{C.145})$$

$$\simeq \frac{1}{2} (2\pi\epsilon)^2 \alpha^2 S_2'(1, 1) + \frac{\delta^2}{2} \partial_n \left[\log \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right) \right] \Big|_{n=1} + \frac{1}{2} (2\pi\epsilon)^2 (2\alpha\beta - \alpha^2) S_2'(1, 1) + \dots \quad (\text{C.146})$$

C.6.3 Relative entropy

Now combining both the numerator and the denominator, the contribution from the uniform word $(\alpha, \alpha, \dots, \alpha)$ gets cancelled. We are left with,

$$S\left(\left(\frac{\rho_\alpha + \rho_\beta}{2}\right) \parallel \rho_\alpha\right) = -\log 2 + \frac{(\alpha - \beta)^2}{2} \left(\pi \cot(\pi\epsilon) + \frac{(\pi\epsilon)^2}{3} \right) + (\pi\epsilon)^{(\alpha-\beta)^2} \frac{\sqrt{\pi}\Gamma\left(1 + \frac{(\alpha-\beta)^2}{2}\right)}{4\Gamma\left(3/2 + \frac{(\alpha-\beta)^2}{2}\right)} \\ + \text{Broken words } O(\epsilon^{2(\alpha-\beta)^2}) + \text{higher point functions } O(\epsilon^{(\alpha-\beta)^2+1}) + \dots \quad (\text{C.147})$$

$$\simeq \frac{(\alpha - \beta)^2}{2\epsilon} - \log 2 - \frac{\pi^2(\alpha - \beta)^2}{6}\epsilon + \frac{\pi^2(\alpha - \beta)^2}{6}\epsilon^2 + (\pi\epsilon)^{(\alpha-\beta)^2} \frac{\sqrt{\pi}\Gamma\left(1 + \frac{(\alpha-\beta)^2}{2}\right)}{4\Gamma\left(3/2 + \frac{(\alpha-\beta)^2}{2}\right)} + \dots \quad (\text{C.148})$$

$$= -\log 2 + \frac{(\alpha - \beta)^2}{2} \left(\frac{1}{\epsilon} - \frac{\pi^2}{3}\epsilon + \frac{\pi^2}{3}\epsilon^2 \right) + (\pi\epsilon)^{(\alpha-\beta)^2} \frac{\sqrt{\pi}\Gamma\left(1 + \frac{(\alpha-\beta)^2}{2}\right)}{4\Gamma\left(3/2 + \frac{(\alpha-\beta)^2}{2}\right)} + \dots \quad (\text{C.149})$$

D Identity relating Relative entropy of mixtures and pure states

Consider a mixed state, $\bar{\rho} = \sum_i p_i \rho_i$. For any mixed state σ , we then have,

$$\sum_i p_i S(\rho_i \parallel \sigma) = \sum_i p_i \text{tr}(\rho_i (\log \rho_i - \log \sigma)) \quad (\text{D.1})$$

$$= \sum_i p_i \text{tr}(\rho_i (\log \rho_i - \log \bar{\rho})) + \sum_i p_i \text{tr}(\rho_i (\log \bar{\rho} - \log \sigma)) \quad (\text{D.2})$$

$$= \sum_i p_i \text{tr}(\rho_i (\log \rho_i - \log \bar{\rho})) + \text{tr}\left(\sum_i (p_i \rho_i) (\log \bar{\rho} - \log \sigma)\right) \quad (\text{D.3})$$

$$= \sum_i p_i S(\rho_i \parallel \bar{\rho}) + S(\bar{\rho} \parallel \sigma). \quad (\text{D.4})$$

where in the second last line, we pushed the sum inside the trace, since trace is linear. Therefore,

$$\boxed{\sum_i p_i S(\rho_i \parallel \sigma) = \sum_i p_i S(\rho_i \parallel \bar{\rho}) + S(\bar{\rho} \parallel \sigma)} \quad (\text{D.5})$$

Note that the first term on the right-hand side is the Holevo information:

$$\sum_i p_i S(\rho_i \parallel \bar{\rho}) = S(\bar{\rho}) - \sum_i p_i S(\rho_i). \quad (\text{D.6})$$

We can use the fact that the Holevo information is independent of the state σ to obtain another useful relation. By writing the above relation for some σ_1 and σ_2 , and subtracting one from the other, we obtain,

$$S(\bar{\rho}\|\sigma_1) - S(\bar{\rho}\|\sigma_2) = \sum_i p_i [S(\rho_i\|\sigma_1) - S(\rho_i\|\sigma_2)] \quad (\text{D.7})$$

We can now further utilize the symmetry of relative entropy in the case of binary mixture of ρ_α and ρ_β , under the exchange of $\alpha \leftrightarrow \beta$, i.e.

$$S(\rho_\alpha\|p\rho_\alpha + (1-p)\rho_\beta) = S(\rho_\beta\|p\rho_\beta + (1-p)\rho_\alpha). \quad (\text{D.8})$$

Let $\bar{\rho}_p = p\rho_\alpha + (1-p)\rho_\beta$, we then have the following two relations,

$$pS(\rho_\alpha\|\sigma) + (1-p)S(\rho_\beta\|\sigma) = pS(\rho_\alpha\|\bar{\rho}_p) + (1-p)S(\rho_\beta\|\bar{\rho}_p) + S(\bar{\rho}_p\|\sigma) \quad (\text{D.9})$$

$$pS(\rho_\beta\|\sigma) + (1-p)S(\rho_\alpha\|\sigma) = pS(\rho_\beta\|\bar{\rho}_{1-p}) + (1-p)S(\rho_\alpha\|\bar{\rho}_{1-p}) + S(\bar{\rho}_{1-p}\|\sigma) \quad (\text{D.10})$$

Subtracting the second line from the first, the first two terms on the right-hand side gets cancelled by the virtue of symmetry, and we are left with,

$$(1-2p) \left[S(\rho_\beta\|\rho_\sigma) - S(\rho_\alpha\|\rho_\sigma) \right] = S(\bar{\rho}_p\|\sigma) - S(\bar{\rho}_{1-p}\|\sigma). \quad (\text{D.11})$$

D.1 Special case of averaged binary mixture

Now, consider the special case, $p_1 = p_2 = 1/2$, $\rho_1 = \rho_\alpha$, $\rho_2 = \rho_\beta$ and $\sigma = \rho_\alpha$. Plugging these in the general result we obtained above, we get,

$$\frac{1}{2}S(\rho_\alpha\|\rho_\alpha) + \frac{1}{2}S(\rho_\beta\|\rho_\alpha) = \frac{1}{2}S\left(\rho_\alpha\left\|\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\right.\right) + \frac{1}{2}S\left(\rho_\beta\left\|\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\right.\right) + S\left(\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\|\rho_\alpha\right) \quad (\text{D.12})$$

Since, the first term on the left side vanishes, we thus have,

$$\frac{1}{2}S(\rho_\beta\|\rho_\alpha) = \frac{1}{2}S\left(\rho_\alpha\left\|\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\right.\right) + \frac{1}{2}S\left(\rho_\beta\left\|\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\right.\right) + S\left(\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\|\rho_\alpha\right) \quad (\text{D.13})$$

Now, from our results, for both small and large interval up to the lowest order, for the averaged mixture, we have the symmetry,

$$S\left(\rho_\alpha\left\|\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\right.\right) = S\left(\rho_\beta\left\|\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\right.\right) + \dots \quad (\text{D.14})$$

Combining these results gives the following relation up to the leading order,

$$S\left(\rho_\alpha\left\|\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\right.\right) + S\left(\left(\frac{\rho_\alpha + \rho_\beta}{2}\right)\|\rho_\alpha\right) = \frac{1}{2}S(\rho_\beta\|\rho_\alpha) + \dots \quad (\text{D.15})$$

D.2 $x \rightarrow 1^-$ limit

up to the leading order, we have,

$$\begin{aligned}
S\left(\rho_\alpha \left\| \left(\frac{\rho_\alpha + \rho_\beta}{2} \right) \right.\right) + S\left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right) \left\| \rho_\alpha \right.\right) &= \left[\log 2 - (\pi\epsilon)^{(\alpha-\beta)^2} \frac{\sqrt{\pi}\Gamma\left(1 + \frac{(\alpha-\beta)^2}{2}\right)}{4\Gamma\left(3/2 + \frac{(\alpha-\beta)^2}{2}\right)} \right] \\
&+ \left[-\log 2 + \frac{(\alpha-\beta)^2}{2} \left(\frac{1}{\epsilon} - \frac{\pi^2}{3}\epsilon + \frac{\pi^2}{3}\epsilon^2 \right) \right. \\
&\quad \left. + (\pi\epsilon)^{(\alpha-\beta)^2} \frac{\sqrt{\pi}\Gamma\left(1 + \frac{(\alpha-\beta)^2}{2}\right)}{4\Gamma\left(3/2 + \frac{(\alpha-\beta)^2}{2}\right)} \right] + \dots
\end{aligned} \tag{D.16}$$

$$\begin{aligned}
&= \underbrace{\frac{(\alpha-\beta)^2}{2} \left(\frac{1}{\epsilon} - \frac{\pi^2}{3}\epsilon \right)}_{-\frac{1}{2}\langle \beta | \hat{K}_{\alpha,A} | \beta \rangle} + \underbrace{\frac{(\alpha-\beta)^2}{2} \left(\frac{\pi^2}{3}\epsilon^2 \right)}_{\frac{1}{2}S(\rho_{\beta,A} \| \rho_{0,A})} + \dots
\end{aligned} \tag{D.17}$$

$$= \frac{1}{2}S(\rho_\beta \| \rho_\alpha) + \dots \tag{D.18}$$

where in the second last line, we used the following relation,

$$S(\psi_{\bar{A}} \| \chi_{\bar{A}}) - S(\psi_A \| \chi_A) = -\langle \Psi | \hat{K}_{\chi,A} | \Psi \rangle. \tag{D.19}$$

where A corresponds to small interval, and $\bar{A}$ is the complement of A , i.e. large interval.

D.3 $x \rightarrow 0^+$ limit

$$\begin{aligned}
S\left(\rho_\alpha \left\| \left(\frac{\rho_\alpha + \rho_\beta}{2} \right) \right.\right) + S\left(\left(\frac{\rho_\alpha + \rho_\beta}{2} \right) \left\| \rho_\alpha \right.\right) &= \left[(\pi x)^2 \frac{\sqrt{\pi}\Gamma(2)}{4\Gamma(5/2)} (1 - 1/2)^2 (\alpha - \beta)^2 \right] \\
&+ \left[(\pi x)^2 \frac{\sqrt{\pi}\Gamma(2)}{4\Gamma(5/2)} (1 - 1/2)^2 (\alpha - \beta)^2 \right] + \dots
\end{aligned} \tag{D.20}$$

$$= \frac{(\alpha - \beta)^2}{2} \frac{(\pi x)^2}{3} + \dots \tag{D.21}$$

$$= \frac{1}{2}S(\rho_\beta \| \rho_\alpha) + \dots \tag{D.22}$$

This gives us a check that our results for both the large and small interval limits up to the leading order, do obey the identity of relative entropy given above.

E Large interval Relative Entropy

E.1 $S\left(\left(\frac{\rho_\Psi + \rho_\Phi}{2}\right) \parallel \rho_0\right)$ (Irrelevant since we compute general case in next subsection)

We want to calculate the relative entropy of the averaged binary mixture with respect to vacuum. We have,

$$\mathcal{F}_n = \frac{2^{-n} \left([1 + \sum_L (C_\Psi^L)^2 \zeta_n^2 S_2(n, \Delta_L)] + [1 + \sum_L (C_\Phi^L)^2 \zeta_n^2 S_2(n, \Delta_L)] + n \sum_{L'} (C_{\Psi\Phi}^{L'})^2 \sum_{p=1}^{n-1} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{2\Delta_{L'}} + \dots \right)}{2^{-1} \left(\left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right)^{2\Delta_\Psi} + \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right)^{2\Delta_\Phi} \right)} \quad (\text{E.1})$$

We can directly write the relative entropy by taking the log and its derivative at $n = 1$,

$$\begin{aligned} S\left(\left(\frac{\rho_\Psi + \rho_\Phi}{2}\right) \parallel \rho_0\right) &= -\log 2 + (\Delta_\Psi + \Delta_\Phi) \left(\frac{1}{\epsilon} - \frac{\pi^2}{3} \epsilon \right) + \sum_L \left(\frac{(C_\Psi^L)^2 + (C_\Phi^L)^2}{2} \right) \frac{\sqrt{\pi}\Gamma(1 + \Delta_L)}{4\Gamma(3/2 + \Delta_L)} (\pi\epsilon)^{2\Delta_L} \\ &\quad + \sum_{L'} (C_{\Psi\Phi}^{L'})^2 \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L'})}{4\Gamma(3/2 + \Delta_{L'})} (\pi\epsilon)^{2\Delta_{L'}} + \dots \quad (\text{E.2}) \\ &= -\log 2 + \frac{1}{2} (S(\rho_\Psi \parallel \rho_0) + S(\rho_\Phi \parallel \rho_0)) + \sum_{L'} (C_{\Psi\Phi}^{L'})^2 \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L'})}{4\Gamma(3/2 + \Delta_{L'})} (\pi\epsilon)^{2\Delta_{L'}} \quad (\text{E.3}) \end{aligned}$$

From the above result, it should be clear that the following relation holds true,

$$S\left(\left(\frac{\rho_\Psi + \rho_\Phi}{2}\right) \parallel \rho_0\right) + \log 2 - \sum_{L'} (C_{\Psi\Phi}^{L'})^2 \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L'})}{4\Gamma(3/2 + \Delta_{L'})} (\pi\epsilon)^{2\Delta_{L'}} \quad (\text{E.4})$$

$$= S\left(\left(\frac{\rho_\Psi + \rho_\Phi}{2}\right) \parallel \rho_0\right) + S\left(\rho_\Psi \parallel \left(\frac{\rho_\Psi + \rho_\Phi}{2}\right)\right) + \dots \quad (\text{E.5})$$

$$= S\left(\left(\frac{\rho_\Psi + \rho_\Phi}{2}\right) \parallel \rho_0\right) + \frac{1}{2} S\left(\rho_\Psi \parallel \left(\frac{\rho_\Psi + \rho_\Phi}{2}\right)\right) + \frac{1}{2} S\left(\rho_\Phi \parallel \left(\frac{\rho_\Psi + \rho_\Phi}{2}\right)\right) + \dots \quad (\text{E.6})$$

$$= \frac{1}{2} (S(\rho_\Psi \parallel \rho_0) + S(\rho_\Phi \parallel \rho_0)) + \dots \quad (\text{E.7})$$

where in the second line, we used the symmetry of the relative entropy. Thus our result for the relative entropy of the binary averaged mixture with respect to vacuum obeys the relation between relative entropy of mixtures and pure states.

For coherent states, we have,

$$S\left(\left(\frac{\rho_\alpha + \rho_\beta}{2}\right) \parallel \rho_0\right) = -\log 2 + \left(\frac{\alpha^2 + \beta^2}{2} \right) \left(\frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} + \frac{\pi^2 \epsilon^2}{3} \right) + (\pi\epsilon)^{(\alpha-\beta)^2} \frac{\sqrt{\pi}\Gamma\left(1 + \frac{(\alpha-\beta)^2}{2}\right)}{4\Gamma\left(3/2 + \frac{(\alpha-\beta)^2}{2}\right)} + \dots \quad (\text{E.8})$$

which matches with the result we obtained earlier.

E.2 $S(\bar{\rho}_{\mathbf{p}}\|\rho_0)$

Now consider a mixed state,

$$\bar{\rho}_{\mathbf{p}} = \sum_{i=1}^M p_i \rho_i \quad (\text{E.9})$$

$$\mathbf{p} = (p_1, p_2, \dots, p_M). \quad (\text{E.10})$$

The pure states ρ_i have scaling dimensions Δ_i . We consider the following OPEs,

$$\mathcal{O}_i \times \mathcal{O}_i \sim 1 + C_i^{L'} \mathcal{O}_{L_i} + \dots \quad (\text{E.11})$$

$$\mathcal{O}_i \times \mathcal{O}_j \sim C_{ij}^{L'} \mathcal{O}_{ij}^{L'} + \dots \quad (\text{E.12})$$

up to the normalization factors. Here $\mathcal{O}_{L_i}$ and $\mathcal{O}_{ij}^{L'}$ have scaling dimension Δ_{L_i} and $\Delta_{L'_{ij}}$. We then define,

$$D_{i,n}(\epsilon) = \left(\frac{n \sin(\pi\epsilon/n)}{\sin(\pi\epsilon)} \right)^{-2\Delta_{L_i}} \quad (\text{E.13})$$

$$\simeq 1 - \frac{\Delta_{L_i}(\pi\epsilon)^2}{3} \left(1 - \frac{1}{n^2} \right) + \dots, \quad D_{i,1}(\epsilon) = 1, \quad (\text{E.14})$$

$$D_{i|j,n}(\epsilon) = \frac{D_{i,n}(\epsilon)}{D_{j,n}(\epsilon)} = \left(\frac{n \sin(\pi\epsilon/n)}{\sin(\pi\epsilon)} \right)^{2(\Delta_{L_j} - \Delta_{L_i})} \quad (\text{E.15})$$

$$\simeq 1 - \frac{(\Delta_{L_j} - \Delta_{L_i})(\pi\epsilon)^2}{3} \left(1 - \frac{1}{n^2} \right) + \dots \quad (\text{E.16})$$

$$D_{i|j,1}(\epsilon) = 1, \quad (\text{E.17})$$

$$\partial_n D_{i,n}(\epsilon)|_{n=1} = \partial_n D_{i,n}(\epsilon)|_{n=1} = -2\Delta_{L_i} (1 - \pi\epsilon \cot(\pi\epsilon)) \quad (\text{E.18})$$

$$\simeq -2\Delta_{L_i} \left(\frac{\pi^2}{3} \epsilon^2 + O(\epsilon^4) \right). \quad (\text{E.19})$$

which carries the normalization, conformal factors and the scaling factors. Then, the numerator of $\mathcal{F}_n$ evaluates to the following,

$$\begin{aligned} & \sum_i (p_i D_{i,n}(\epsilon))^n \left[1 + \sum_{L_i} (C_i^{L_i})^2 \zeta_n^{2\Delta_{L_i}} S_2(n, \Delta_{L_i}) \right] \\ & + \sum_{1 \leq i < j \leq M} \sum_{L'_{ij}} n \left(C_{ij}^{L'_{ij}} \right)^2 \zeta_n^{2\Delta_{L'_{ij}}} \sum_{r=1}^{n-1} (p_i D_{i,n}(\epsilon))^{n-r} (p_j D_{j,n}(\epsilon))^r \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} + \dots \end{aligned} \quad (\text{E.20})$$

$$= \sum_i (p_i D_{i,n}(\epsilon))^n \left[1 + \sum_{L_i} (C_i^{L_i})^2 \zeta_n^{2\Delta_{L_i}} S_2(n, \Delta_{L_i}) \right] + \sum_{1 \leq i < j \leq M} \sum_{L'_{ij}} \left(C_{ij}^{L'_{ij}} \right)^2 \zeta_n^{2\Delta_{L'_{ij}}} \mu_{ij,n}^{(\mathbf{p})}(\epsilon) + \dots \quad (\text{E.21})$$

where we defined,

$$\mu_{ij,n}^{(\mathbf{p})}(\epsilon) = \sum_{r=1}^{n-1} n (p_i D_{i,n}(\epsilon))^{n-r} (p_j D_{j,n}(\epsilon))^r \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} \quad (\text{E.22})$$

Note that, since the ϵ dependence comes only from the factors of $D_{i,n}(\epsilon)$, $\mu_{ij,n}^{(\mathbf{p})}(\epsilon)$ would go as $1 + O(\epsilon^2) + \dots$.

The denominator is then given by,

$$\sum_i p_i D_{i,n}(\epsilon) \left(\frac{\sin(\pi \epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right)^{2\Delta_{L_i}} \quad (\text{E.23})$$

We can then take the logarithm and derivative of the above, to obtain,

$$S(\bar{\rho}_p \| \rho_0) = \sum_{i=1}^M p_i S(\rho_i \| \rho_0) - H(\mathbf{p}) + \sum_{1 \leq i < j \leq M} \sum_{L'_{ij}} \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \partial_n \mu_{ij,n}^{(\mathbf{p})}(\epsilon) \Big|_{n=1} + \dots \quad (\text{E.24})$$

$$H(p) = - \sum_i p_i \log p_i \quad (\text{E.25})$$

The contribution from $D_{i,n}(\epsilon)$ to the relative entropy would come only from $\mu_{ij,n}^{(\mathbf{p})}(\epsilon)$, which is the mixing OPE contribution. However, the contribution from $D_{i,n}(\epsilon)$ coming from identity and non-mixing OPEs vanishes due to cancellation from both numerator and denominator.

We can further reorganize the second summation in the numerator using conditional probabilities. For each pair of mixing OPEs, i, j , define

$$P_{ij} = p_i + p_j, \quad x_{ij} = \frac{p_i}{P_{ij}}. \quad (\text{E.26})$$

Then we can write,

$$p_i^{n-r} p_j^r = P_{ij}^n x_{ij}^{n-r} (1 - x_{ij})^r \quad (\text{E.27})$$

Then, the mixing OPE contribution is given by,

$$\sum_{1 \leq i < j \leq M} (p_i + p_j)^n \sum_{L'_{ij}} \sum_{r=1}^{n-1} n \left(C_{ij}^{L'} \right)^2 P_{ij}^n (x_{ij} D_{i,n}(\epsilon))^{n-r} ((1 - x_{ij}) D_{j,n}(\epsilon))^r \left(\frac{\sin(\pi \epsilon/n)}{\sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} \quad (\text{E.28})$$

Then, the relative entropy in terms of conditional probability is given by,

$$S(\bar{\rho}_p \| \rho_0) = \sum_{i=1}^M p_i S(\rho_i \| \rho_0) - H(\mathbf{p}) + \left[\sum_{i < j} \sum_{L'_{ij}} (p_i + p_j) \left(C_{ij}^{L'} \right)^2 \times \right. \\ \left. \partial_n \left(n P_{ij}^n \sum_{r=1}^{n-1} (x_{ij} D_{i,n}(\epsilon))^{n-r} ((1 - x_{ij}) D_{j,n}(\epsilon))^r \left(\frac{\sin(\pi \epsilon / n)}{\sin(\pi r / n)} \right)^{2\Delta_{L'_{ij}}} \right) \right]_{n=1} + \dots \quad (\text{E.29})$$

$$= \sum_{i=1}^M p_i S(\rho_i \| \rho_0) - H(\mathbf{p}) + \sum_{i < j} \sum_{L'_{ij}} (p_i + p_j) \left(C_{ij}^{L'} \right)^2 (2\pi \epsilon)^{2\Delta_{L'_{ij}}} \partial_n \eta_{ij,n}^{(\mathbf{p})}(\epsilon) \Big|_{n=1} + \dots \quad (\text{E.30})$$

$$\eta_{ij,n}^{(\mathbf{p})}(\epsilon) = \sum_{r=1}^{n-1} n (x_{ij} D_{i,n}(\epsilon))^{n-r} ((1 - x_{ij}) D_{j,n}(\epsilon))^r \left(\frac{1}{2 \sin(\pi r / n)} \right)^{2\Delta_{L'_{ij}}} \quad (\text{E.31})$$

Note that $\eta_{ij}^{(\mathbf{p})}$ is not a new replica function that we defined. It is related to $\mu_{ij}^{(\mathbf{p})}$,

$$\mu_{ij,n}^{(\mathbf{p})}(\epsilon) = P_{ij}^n \eta_{ij,n}^{(\mathbf{p})}(\epsilon). \quad (\text{E.32})$$

Then, as a function of ϵ it also behaves as, $1 + O(\epsilon^2) + \dots$.

As a consistency test, this result also reproduces our result for the relative entropy of averaged binary mixture with respect to vacuum. To see this, we can check the contribution from the mixing OPEs. We have $M = 2$ pure states, with $p_1 = p_2 = 1/2$. So the contribution from the mixing OPE, reduces to,

$$\sum_{L'} (C_{\Psi\Phi}^{L'})^2 (\pi \epsilon)^{2\Delta_{L'}} \partial_n \left(n 2^{-n} \sum_{r=1}^{n-1} D_{1,n}(\epsilon)^{n-r} D_{2,n}(\epsilon)^r \left(\frac{1}{2 \sin(\pi r / n)} \right)^{2\Delta_{L'_{ij}}} \right) \Big|_{n=1} \\ \simeq \sum_{L'} (C_{\Psi\Phi}^{L'})^2 (2\pi \epsilon)^{2\Delta_{L'}} S'_2(1, \Delta_{L'}) \quad (\text{E.33})$$

$$= \sum_{L'} (C_{\Psi\Phi}^{L'})^2 \frac{\sqrt{\pi} \Gamma(1 + \Delta_{L'})}{4 \Gamma(3/2 + \Delta_{L'})} (\pi \epsilon)^{2\Delta_{L'}} \quad (\text{E.34})$$

where in the second line we approximated $D_{i,n}(\epsilon) \simeq 1$. The Shannon term in the general result gives $-\log 2$, and thus we reproduce the result we obtained earlier.

E.3 $S(\rho_\alpha \| \bar{\rho}_{\mathbf{p}})$

Now we consider the case, where the first state is a pure state ρ_α and the reference state is a general mixture, $\bar{\rho}_{\mathbf{p}} = \sum_{i \neq \alpha} p_i \rho_i + p_\alpha \rho_\alpha$. The mixture contains ρ_α with a probability p_α . Then the numerator of $\mathcal{F}_n$ evaluates to,

$$D_{\alpha,n}(\epsilon)^n \left(1 + \sum_{L_\alpha} (C_\alpha^{L_\alpha})^2 \zeta_n^{2\Delta_{L_\alpha}} S_2(n, \Delta_{L_\alpha}) + \dots \right) \quad (\text{E.35})$$

while the denominator is given by,

$$\begin{aligned}
D_{\alpha,n}(\epsilon)^n p_{\alpha}^{n-1} & \left(1 + \sum_{L_{\alpha}} (C_{\alpha}^{L_{\alpha}})^2 \zeta_n^{2\Delta_{L_{\alpha}}} S_2(n, \Delta_{L_{\alpha}}) \right. \\
& \left. + \sum_{i \neq \alpha} \sum_{L'_{\alpha i}} (C_{\alpha i}^{L'})^2 \zeta_n^{2\Delta_{L'_{\alpha i}}} \sum_{r=1}^{n-1} \left(\frac{p_i D_{i,n}(\epsilon)}{p_{\alpha} D_{\alpha,n}(\epsilon)} \right)^r \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{\alpha i}}} + \dots \right)
\end{aligned} \tag{E.36}$$

Thus the relative entropy is then given by,

$$S(\rho_{\alpha} \| \bar{\rho}_{\mathbf{p}}) = -\log p_{\alpha} - \sum_{i \neq \alpha} \sum_{L'_{\alpha i}} (C_{\alpha i}^{L'})^2 (2\pi\epsilon)^{2\Delta_{L'_{\alpha i}}} \partial_n \theta_{i|\alpha,n}^{(\mathbf{p})}(\epsilon) \Big|_{n=1} + \dots \tag{E.37}$$

$$\theta_{i|j,n}^{(\mathbf{p})}(\epsilon) = \sum_{r=1}^{n-1} (n-r) \left(\frac{p_i D_{i,n}(\epsilon)}{p_j D_{j,n}(\epsilon)} \right)^r \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{\alpha i}}} \tag{E.38}$$

The analytic continuation of the derivative of $\theta_{i|j,n}^{(\mathbf{p})}(\epsilon)$ at $n = 1$ has been computed in the following Appendix F. This does indeed reproduce our previous result for the relative entropy of a pure state with respect to an averaged binary mixture (up to the leading-order term that we obtain by approximating $D_{i,n}(\epsilon) \simeq 1 + \dots$). We obtain $\log 2$ from the first term, and the mixing OPE of ρ_{α} with the other pure state would give the S'_2 term from θ_{α} term.

There is a relation between $\theta_{i|j}^{(\mathbf{p})}(n, \epsilon)$ and $\mu_{ij}^{(\mathbf{p})}$, which will be required in the following calculations. It can be seen as follows,

$$\partial_n \left((p_i D_{i,n}(\epsilon))^n \theta_{j|i,n}^{(\mathbf{p})}(\epsilon) \right) \Big|_{n=1} = \partial_n \left(p_i \theta_{j|i,n}^{(\mathbf{p})}(\epsilon) \right) \Big|_{n=1} = p_i \partial_n \theta_{j|i,n}^{(\mathbf{p})}(\epsilon) \Big|_{n=1} \tag{E.39}$$

Then, as a function of ϵ it also behaves as, $1 + O(\epsilon^2) + \dots$. Now consider the following,

$$\begin{aligned} (p_i D_{i,n}(\epsilon))^n \theta_{j|i,n}^{(\mathbf{p})}(\epsilon) + (p_j D_{j,n}(\epsilon))^n \theta_{i|j,n}^{(\mathbf{p})}(\epsilon) &= \sum_{r=1}^{n-1} (n-r) (p_i D_{i,n}(\epsilon))^{n-r} (p_j D_{j,n}(\epsilon))^r \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} \\ &\quad + \sum_{r=1}^{n-1} (n-r) (p_j D_{j,n}(\epsilon))^{n-r} (p_i D_{i,n}(\epsilon))^r \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} \end{aligned} \quad (\text{E.40})$$

$$\begin{aligned} &= \sum_{r=1}^{n-1} (n-r) (p_i D_{i,n}(\epsilon))^{n-r} (p_j D_{j,n}(\epsilon))^r \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} \\ &\quad + \sum_{r=1}^{n-1} r (p_j D_{j,n}(\epsilon))^r (p_i D_{i,n}(\epsilon))^{n-r} \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} \end{aligned} \quad (\text{E.41})$$

$$= n \sum_{r=1}^{n-1} (p_i D_{i,n}(\epsilon))^{n-r} (p_j D_{j,n}(\epsilon))^r \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} \quad (\text{E.42})$$

$$= \mu_{ij,n}^{(\mathbf{p})}(\epsilon) \quad (\text{E.43})$$

E.4 $S(\bar{\rho}_{\mathbf{p}} \parallel \rho_{\alpha})$

Here we consider the first state to be a general mixture, while the reference state is a pure state, that is also a part of the mixture. The numerator of $\mathcal{F}_n$ is given by,

$$\sum_i (p_i D_{i,n}(\epsilon))^n \left[1 + \sum_{L_i} (C_i^{L_i})^2 \zeta_n^2 S_2(n, \Delta_{L_i}) \right] + \sum_{1 \leq i < j \leq M} \sum_{L'_{ij}} n \left(C_{ij}^{L'} \right)^2 \zeta_n^{2\Delta_{L'_{ij}}} \mu_{ij,n}^{(\mathbf{p})}(\epsilon) + \dots \quad (\text{E.44})$$

while the denominator is given by,

$$p_{\alpha} D_{\alpha,n}(\epsilon)^n \left[1 + \sum_{L_i} (C_i^{L_i})^2 \zeta_n^2 S_2(n, \Delta_{L_i}) \right] + \sum_{i \neq \alpha} \sum_{L'_{\alpha i}} \left(C_{\alpha i}^{L'} \right)^2 \zeta_n^{2\Delta_{L'_{\alpha i}}} p_i D_{i,n}(\epsilon) D_{\alpha,n}(\epsilon)^{n-1} \left(\frac{1}{2 \sin(\pi/n)} \right)^{2\Delta_{L'_{\alpha i}}} + \dots \quad (\text{E.45})$$

Then, the relative entropy is given by,

$$\begin{aligned} S(\bar{\rho}_{\mathbf{p}} \parallel \rho_{\alpha}) &= -H(\mathbf{p}) + \sum_{i < j} \sum_{L'} \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \partial_n \mu_{ij}^{(\mathbf{p})}(n, \epsilon) \Big|_{n=1} - \sum_{i \neq \alpha} 2\Delta_{L_i} p_i \left(\frac{\pi^2 \epsilon^2}{3} + \dots \right) \\ &\quad + \underbrace{\sum_{i \neq \alpha} \sum_{L_i} p_i (C_i^{L_i})^2 (\pi\epsilon)^{2\Delta_{L_i}} \frac{\sqrt{\pi} \Gamma(\Delta_{L_i} + 1)}{4\Gamma(\Delta_{L_i} + 3/2)} - \sum_{i \neq \alpha} \sum_{L'_{\alpha i}} \left(C_{\alpha i}^{L'} \right)^2 (\pi\epsilon)^{2\Delta_{L'_{\alpha i}}} p_i \partial_n \left(\frac{D_{i,n}(\epsilon)/D_{\alpha,n}(\epsilon)}{\sin(\pi/n)} \right)^{\Delta_{L'_{\alpha i}}} \Big|_{n=1}}_{\stackrel{?}{=} \sum_i p_i S(\rho_i \parallel \rho_{\alpha})} + \dots \end{aligned} \quad (\text{E.46})$$

However, using the relation between the relative entropy of pure states and mixtures and the general result of relative entropy of pure pure state with respect to mixture, we obtain,

$$S(\bar{\rho}_{\mathbf{p}}\|\rho_{\alpha}) = \sum_i p_i S(\rho_i\|\rho_{\alpha}) - \sum_i p_i S(\rho_i\|\bar{\rho}_{\mathbf{p}}) \quad (\text{E.47})$$

$$= \sum_i p_i S(\rho_i\|\rho_{\alpha}) - \sum_i p_i \left[-\log p_i - \sum_{j \neq i} \sum_{L'_{ij}} \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \partial_n \theta_{j|i,n}^{(\mathbf{p})}(\epsilon) \Big|_{n=1} + \dots \right] \quad (\text{E.48})$$

$$= \sum_i p_i S(\rho_i\|\rho_{\alpha}) - H(\mathbf{p}) + \sum_{i < j} \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \partial_n \left(p_i^n \theta_{j|i,n}^{(\mathbf{p})}(\epsilon) + p_j^n \theta_{i|j,n}^{(\mathbf{p})}(\epsilon) \right) \Big|_{n=1} + \dots \quad (\text{E.49})$$

$$= \sum_i p_i S(\rho_i\|\rho_{\alpha}) - H(\mathbf{p}) + \sum_{i < j} \sum_{L'_{ij}} \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \partial_n \mu_{ij,n}^{(\mathbf{p})}(\epsilon) \Big|_{n=1} + \dots \quad (\text{E.50})$$

This suggests that the divergence term coming from the mixing OPE of a single word with $n - 1$ copies of the ρ_{α} and a single copy of ρ_i , can be expanded around ρ_i , rather than midpoints. That would precisely give us the pure state relative entropy divergence term. We are allowed to do this, as long as we keep only this leading order contribution. Note that, since the sub-leading order terms from the descendants would be also shifted, those contributions would be different from the mid-point OPE expansion sub-leading terms.

E.5 $S(\bar{\rho}_{\mathbf{p}}\|\bar{\rho}_{\mathbf{q}})$ with same support

Now consider two mixed states,

$$\bar{\rho}_{\mathbf{p}} = \sum_{i=1}^M p_i \rho_i, \quad \sum_{i=1}^M p_i = 1, \quad (\text{E.51})$$

$$\bar{\rho}_{\mathbf{q}} = \sum_{i=1}^M q_i \rho_i, \quad \sum_{i=1}^M q_i = 1. \quad (\text{E.52})$$

Then the numerator of $\mathcal{F}_n$ is given by,

$$\begin{aligned} & \sum_i (p_i D_{i,n}(\epsilon))^n \left[1 + \sum_{L_i} (C_i^{L_i})^2 \zeta_n^{2\Delta_{L_i}} S_2(n, \Delta_{L_i}) \right] \\ & + \sum_{1 \leq i < j \leq M} \sum_{L'_{ij}} \sum_{r=1}^{n-1} n \left(C_{ij}^{L'} \right)^2 (p_i D_{i,n}(\epsilon))^{n-r} (p_j D_{j,n}(\epsilon))^r \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi p/n)} \right)^{2\Delta_{L'_{ij}}} + \dots \end{aligned} \quad (\text{E.53})$$

$$= \sum_i (p_i D_{i,n}(\epsilon))^n \left[1 + \sum_{L_i} (C_i^{L_i})^2 \zeta_n^{2\Delta_{L_i}} S_2(n, \Delta_{L_i}) \right] + \sum_{1 \leq i < j \leq M} \sum_{L'_{ij}} \left(C_{ij}^{L'} \right)^2 \zeta_n^{2\Delta_{L'_{ij}}} \mu_{ij,n}^{(\mathbf{p})}(\epsilon) + \dots \quad (\text{E.54})$$

while the denominator is given by,

$$\left(\sum_i p_i q_i^{n-1} D_{i,n}(\epsilon)^n \left[1 + \sum_{L_i} (C_i^{L_i})^2 \zeta_n^2 S_2(n, \Delta_{L_i}) \right] + \sum_{i < j} \sum_{L'_{ij}} \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \tilde{\mu}_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) \right) \quad (\text{E.55})$$

where we defined,

$$\tilde{\mu}_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) = \sum_{r=1}^{n-1} D_{i,n}(\epsilon)^{n-r} D_{j,n}(\epsilon)^r \left[(n-r) p_i q_i^{n-r-1} q_j^r + r p_j q_j^{r-1} q_i^{n-r} \right] \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} \quad (\text{E.56})$$

Note that, it can be written in terms of the θ replica function we defined earlier,

$$\tilde{\mu}_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) = p_i q_i^{n-1} D_{i,n}(\epsilon)^n \theta_{j|i}^{(\mathbf{q})}(n, \epsilon) + p_j q_j^{n-1} D_{j,n}(\epsilon)^n \theta_{i|j}^{(\mathbf{q})}(n, \epsilon). \quad (\text{E.57})$$

Then, as a function of ϵ it also behaves as, $1 + O(\epsilon^2) + \dots$. Then the relative entropy is given by,

$$S(\bar{\rho}_{\mathbf{p}} \| \bar{\rho}_{\mathbf{q}}) = D_{KL}(\mathbf{p} \| \mathbf{q}) + \sum_{i < j} \sum_{L'_{ij}} \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \partial_n \left(\mu_{ij}^{(\mathbf{p})}(n, \epsilon) - \tilde{\mu}_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) \right) \Big|_{n=1} + \dots \quad (\text{E.58})$$

$$\begin{aligned} &= D_{KL}(\mathbf{p} \| \mathbf{q}) + \sum_{i < j} \sum_{L'_{ij}} \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \left(p_i \partial_n \left(\theta_{j|i}^{(\mathbf{p})}(n, \epsilon) - \theta_{j|i}^{(\mathbf{q})}(n, \epsilon) \right) \right. \\ &\quad \left. + p_j \partial_n \left(\theta_{i|j}^{(\mathbf{p})}(1, \epsilon) - \theta_{i|j}^{(\mathbf{q})}(1, \epsilon) \right) \right) \end{aligned} \quad (\text{E.59})$$

$$D_{KL}(\mathbf{p} \| \mathbf{q}) = \sum_i p_i \log(p_i/q_i) \quad (\text{E.60})$$

It can again be written in terms of conditional probabilities. For each pair i, j define,

$$P_{ij} = p_i + p_j, \quad Q_{ij} = q_i + q_j \quad (\text{E.61})$$

and the conditional probabilities,

$$x_{ij} = \frac{p_i}{p_i + p_j}, \quad y_{ij} = \frac{q_i}{q_i + q_j} \quad (\text{E.62})$$

Then for the denominator, we have,

$$(n-r) p_i q_i^{n-r-1} q_j^r + r p_j q_j^{r-1} q_i^{n-r} = P_{ij} Q_{ij}^{n-1} \left[(n-r) x_{ij} y_{ij}^{n-r-1} (1-y_{ij})^r + r (1-x_{ij}) (1-y_{ij})^{r-1} y_{ij}^{n-r} \right] \quad (\text{E.63})$$

Then we can define the following,

$$\begin{aligned} \tilde{\eta}_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) = \sum_{r=1}^{n-1} D_{i,n}(\epsilon)^{n-r} D_{j,n}(\epsilon)^r \left[(n-r) x_{ij} y_{ij}^{n-r-1} (1-y_{ij})^r \right. \\ \left. + r(1-x_{ij})(1-y_{ij})^{r-1} y_{ij}^{n-r} \right] \left(\frac{1}{2 \sin(\pi r/n)} \right)^{2\Delta_{L'_{ij}}} . \end{aligned} \quad (\text{E.64})$$

Similarly, $\tilde{\eta}$ is related to $\tilde{\mu}$,

$$\tilde{\mu}_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) = P_{ij} Q_{ij}^{n-1} \tilde{\eta}_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon). \quad (\text{E.65})$$

Then, as a function of ϵ it also behaves as, $1 + O(\epsilon^2) + \dots$. Finally, the relative entropy is given by,

$$S(\bar{\rho}_p \| \bar{\rho}_q) = D_{KL}(p \| q) + \sum_{i < j} \sum_{L'_{ij}} (p_i + p_j) \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \partial_n \left(\eta_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) - \tilde{\eta}_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) \right) \Big|_{n=1} + \dots \quad (\text{E.66})$$

E.6 $S(\bar{\rho}_p \| \bar{\rho}_q)$ with different support (Most general result, reproduces the rest of the cases)

Now we consider the relative entropy of two mixtures, however, here the first mixed state's support does not necessarily overlap with the support of the reference mixed state. Consider two mixtures,

$$\bar{\rho}_{\mathbf{p}} = \sum_{a=1}^M p_a \rho_{\Psi_a}, \quad \sum_{a=1}^M p_a = 1, \quad (\text{E.67})$$

$$\bar{\rho}_{\mathbf{q}} = \sum_{b=1}^N q_b \rho_{\Phi_b}, \quad \sum_{b=1}^N q_b = 1. \quad (\text{E.68})$$

Suppose C states are common to the two mixtures. Without any loss of generality, we consider,

$$\rho_{\Psi_c} = \rho_{\Phi_c} \equiv \rho_c, \quad c = 1, \dots, C. \quad (\text{E.69})$$

We can then introduce the label sets for the overlapping and non-overlapping states. Let the overlapping set be $\mathcal{C} = \{1, \dots, C\}$ for the common states. Then we can introduce, $\mathcal{A} = \{a : C < a \leq M\}$ for the states only appearing in $\bar{\rho}_{\mathbf{p}}$ and $\mathcal{B} = \{b : C < b \leq N\}$ for the states only appearing in $\bar{\rho}_{\mathbf{q}}$. Thus we can write,

$$\bar{\rho}_{\mathbf{p}} = \sum_{c \in \mathcal{C}} p_c \rho_c + \sum_{a \in \mathcal{A}} p_a \rho_{\Psi_a}, \quad (\text{E.70})$$

$$\bar{\rho}_{\mathbf{q}} = \sum_{c \in \mathcal{C}} q_c \rho_c + \sum_{b \in \mathcal{B}} q_b \rho_{\Phi_b}. \quad (\text{E.71})$$

Thus we want to evaluate the following,

$$\mathcal{F}_n = \frac{\text{tr} \left(\left(\sum_{c \in \mathcal{C}} p_c \rho_c + \sum_{a \in \mathcal{A}} p_a \rho_{\Psi_a} \right)^n \right)}{\text{tr} \left(\left(\sum_{c \in \mathcal{C}} p_c \rho_c + \sum_{a \in \mathcal{A}} p_a \rho_{\Psi_a} \right) \left(\sum_{c \in \mathcal{C}} q_c \rho_c + \sum_{b \in \mathcal{B}} q_b \rho_{\Phi_b} \right)^{n-1} \right)} \quad (\text{E.72})$$

The numerator evaluates to the following,

$$\sum_{a=1}^M (p_a D_{a,n}(\epsilon))^n \left(1 + \sum_{L_a} (C_a^{L_a})^2 \zeta_n^{2\Delta_{L_a}} S_2(n, \Delta_{L_a}) \right) + \sum_{1 < a < d \leq M} \sum_{L'_{ad}} \left(C_{ad}^{L'} \right)^2 \zeta_n^{2\Delta_{L'_{ad}}} \mu_{ad}^{(\mathbf{p})}(n, \epsilon) + \dots \quad (\text{E.73})$$

Then its contribution to the relative entropy is given by,

$$\begin{aligned} -H(\mathbf{p}) + \sum_{i \in (\mathcal{C} \cup \mathcal{A})} p_i (C_i^{L_i})^2 (\pi \epsilon)^{2\Delta_{L_i}} \frac{\sqrt{\pi} \Gamma(1 + \Delta_{L_i})}{4\Gamma(3/2 + \Delta_{L_i})} + \sum_{1 < a < d \leq M} \sum_{L'_{ad}} \left(C_{ad}^{L'} \right)^2 (2\pi \epsilon)^{2\Delta_{L'_{ad}}} \partial_n \mu_{ad}^{(\mathbf{p})}(n, \epsilon) \Big|_{n=1} \\ - \sum_i 2\Delta_{L_i} p_i \left(\frac{\pi^2 \epsilon^2}{3} + \dots \right) + \dots \end{aligned} \quad (\text{E.74})$$

where the ϵ^2 correction comes from the derivative of $D_{a,n}(\epsilon)$ at $n = 1$. The denominator is given by

$$\text{tr}(\bar{\rho}_{\mathbf{p}} \bar{\rho}_{\mathbf{q}}^{n-1}) = \sum_{a=1}^M p_a \text{tr}(\rho_{\Psi_a} \bar{\rho}_{\mathbf{q}}^{n-1}) \quad (\text{E.75})$$

which can be split into the common and non-common states contributions. For the common states ρ_c , we have the following contribution,

$$\begin{aligned} \sum_{c \in \mathcal{C}} p_c q_c^{n-1} D_{c,n}(\epsilon)^n \left(1 + \sum_{L_c} (C_c^{L_c})^2 \zeta_n^{2\Delta_{L_c}} S_2(n, \Delta_{L_c}) \right) + \sum_{c \in \mathcal{C}} \sum_{\substack{j \in \mathcal{C} \cup \mathcal{B} \\ j \neq c}} \sum_{L'_{cj}} p_c q_c^{n-1} D_{c,n}(\epsilon)^n \left(C_{cj}^{L'} \right)^2 \zeta_n^{2\Delta_{L'_{cj}}} \theta_{j|c}^{(\mathbf{q})}(n, \epsilon) + \\ = \sum_{c \in \mathcal{C}} p_c q_c^{n-1} D_{c,n}(\epsilon)^n \left(1 + \sum_{L_c} (C_c^{L_c})^2 \zeta_n^{2\Delta_{L_c}} S_2(n, \Delta_{L_c}) \right) + \sum_{\substack{c < d \\ c, d \in \mathcal{C}}} \sum_{L'_{cd}} \left(C_{cd}^{L'} \right)^2 \zeta_n^{2\Delta_{L'_{cd}}} \tilde{\mu}_{cd}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) \\ + \sum_{\substack{c \in \mathcal{C} \\ b \in \mathcal{B}}} \sum_{L'_{ab}} p_c q_c^{n-1} D_{c,n}(\epsilon)^n \left(C_{cb}^{L'} \right)^2 \zeta_n^{2\Delta_{L'_{cb}}} \theta_{b|c}^{(\mathbf{q})}(n, \epsilon) + \dots \end{aligned} \quad (\text{E.76})$$

For the non-common states, ρ_{Ψ_a} which do not occur in $\bar{\rho}_{\mathbf{q}}$, we will have only one word contribution, which would give the divergence since they have different support. For example, the contribution from the word ⁶

$$(\Psi_a, \Phi_b, \Phi_b, \dots, \Phi_b) = (p_a D_{a,n}(\epsilon)) (q_b D_{b,n}(\epsilon))^{n-1} \left(\frac{\sin(\pi \epsilon / n)}{\sin(\pi(1 - \epsilon) / n)} \right)^{2\Delta_{L'_{ab}}}. \quad (\text{E.78})$$

⁶Note that, since the contribution is only from one word, we shifted the primary from the midpoint OPE to $2 \sin(\pi(1 - \epsilon)/n)$ apart.

So, the complete denominator at this leading order contribution is given by,

$$\begin{aligned}
& \sum_{c \in \mathcal{C}} p_c q_c^{n-1} D_{c,n}(\epsilon)^n \left(1 + \sum_{L_c} (C_c^{L_c})^2 \zeta_n^{2\Delta_{L_c}} S_2(n, \Delta_{L_c}) \right) + \sum_{\substack{c < d \\ c, d \in \mathcal{C}}} \sum_{L'_{cd}} (C_{cd}^{L'})^2 \zeta_n^{2\Delta_{L'_{cd}}} \tilde{\mu}_{cd,n}^{(\mathbf{p}, \mathbf{q})}(\epsilon) \\
& + \sum_{\substack{c \in \mathcal{C} \\ b \in \mathcal{B}}} \sum_{L'_{ab}} p_c q_c^{n-1} D_{c,n}(\epsilon)^n (C_{cb}^{L'})^2 \zeta_n^{2\Delta_{L'_{cb}}} \theta_{b|c,n}^{(\mathbf{q})}(\epsilon) + \sum_{a \in \mathcal{A}} \sum_{b=1}^N (p_a D_{a,n}(\epsilon)) (q_b D_{b,n}(\epsilon))^{n-1} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right)^{2\Delta_{L'_{ab}}}
\end{aligned} \tag{E.79}$$

Now we can take the logarithm and the derivative at $n = 1$. However, notice that at $n = 1$ the denominator evaluates to

$$\sum_c p_c + \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a = P_{\mathcal{C}} + N P_{\mathcal{A}} \equiv Z, \tag{E.80}$$

$$P_{\mathcal{C}} = \sum_{c \in \mathcal{C}} p_c, \quad P_{\mathcal{A}} = \sum_{a \in \mathcal{A}} p_a = 1 - P_{\mathcal{C}}. \tag{E.81}$$

Then the contribution of the denominator to the relative entropy is given by,

$$\begin{aligned}
& \frac{1}{Z} \left[\sum_{c \in \mathcal{C}} p_c \left(\log q_c + \sum_{L_c} (C_c^{L_c})^2 (\pi\epsilon)^{2\Delta_{L_c}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_c})}{4\Gamma(3/2 + \Delta_{L_c})} \right) + \sum_{\substack{c < d \\ c, d \in \mathcal{C}}} \sum_{L'_{cd}} (C_{cd}^{L'})^2 (2\pi\epsilon)^{2\Delta_{L'_{cd}}} \partial_n \tilde{\mu}_{cd}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) \Big|_{n=1} \right. \\
& + \sum_{\substack{c \in \mathcal{C} \\ b \in \mathcal{B}}} \sum_{L'_{ab}} p_c (C_{cb}^{L'})^2 (2\pi\epsilon)^{2\Delta_{L'_{cb}}} \partial_n \theta_{b|c}^{(\mathbf{q})}(n, \epsilon) \Big|_{n=1} + \sum_{a \in \mathcal{A}} p_a \sum_{b=1}^N \left(\log q_b - 2\Delta_{L'_{ab}} \pi \cot(\pi\epsilon) \right) \\
& \left. + \sum_{c \in \mathcal{C}} (-2\Delta_{L_c}) p_c (1 - \pi\epsilon \cot(\pi\epsilon)) + N \sum_{a \in \mathcal{A}} (-2\Delta_{L_a}) p_a (1 - \pi\epsilon \cot(\pi\epsilon)) + \dots \right] \tag{E.82}
\end{aligned}$$

where the last line contribution comes from the derivative of $D_{a,n}(\epsilon)$ and $D_{c,n}(\epsilon)$ at $n = 1$. Combining the numerator and denominator contribution, we obtain,

$$\begin{aligned}
S(\bar{\rho}_{\mathbf{p}}|\bar{\rho}_{\mathbf{q}}) = & -H(\mathbf{p}) - \frac{1}{Z} \left(\sum_{c \in \mathcal{C}} p_c \log q_c + P_{\mathcal{A}} \sum_{b=1}^N \log q_b \right) + \left(1 - \frac{1}{Z} \right) \sum_{c \in \mathcal{C}} p_c (C_c^{L_c})^2 (\pi\epsilon)^{2\Delta_{L_c}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_c})}{4\Gamma(3/2 + \Delta_{L_c})} \\
& + \sum_{a \in \mathcal{A}} p_a (C_a^{L_a})^2 (\pi\epsilon)^{2\Delta_{L_a}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_a})}{4\Gamma(3/2 + \Delta_{L_a})} + \frac{1}{Z} \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a \left(2\Delta_{L'_{ab}} \pi \cot(\pi\epsilon) \right) \\
& + \mathcal{M}_{\mathbf{p}} - \frac{1}{Z} (\mathcal{M}_{\mathcal{CC}} + \mathcal{M}_{\mathcal{CB}}) - \frac{2(N-1)}{Z} \left(P_{\mathcal{A}} \sum_{c \in \mathcal{C}} p_c \Delta_{L_c} - P_{\mathcal{C}} \sum_{a \in \mathcal{A}} p_a \Delta_{L_a} \right) (1 - \pi\epsilon \cot(\pi\epsilon)) + \dots \\
Z = & P_{\mathcal{C}} + NP_{\mathcal{A}} \\
\mathcal{M}_{\mathbf{p}} = & \sum_{1 \leq a < d \leq M} \sum_{L'_{ad}} \left(C_{ad}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ad}}} \partial_n \mu_{ad,n}^{(\mathbf{p})}(\epsilon) \Big|_{n=1} \\
\mathcal{M}_{\mathcal{CC}} = & \sum_{\substack{c < d \\ c, d \in \mathcal{C}}} \sum_{L'_{cd}} \left(C_{cd}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{cd}}} \partial_n \tilde{\mu}_{cd,n}^{(\mathbf{p}, \mathbf{q})}(\epsilon) \Big|_{n=1} \\
\mathcal{M}_{\mathcal{CB}} = & \sum_{\substack{c \in \mathcal{C} \\ b \in \mathcal{B}}} \sum_{L'_{cb}} p_c \left(C_{cb}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{cb}}} \partial_n \theta_{b|c,n}^{(\mathbf{q})}(\epsilon) \Big|_{n=1}
\end{aligned} \tag{E.83}$$

This can be further simplified. Define p -averages,

$$\langle X \rangle_{\mathcal{C}} = \frac{1}{P_{\mathcal{C}}} \sum_{c \in \mathcal{C}} p_c X_c, \quad \langle X \rangle_{\mathcal{A}} = \frac{1}{P_{\mathcal{A}}} \sum_{a \in \mathcal{A}} p_a X_a. \tag{E.84}$$

Then we have,

$$P_{\mathcal{A}} \sum_{c \in \mathcal{C}} p_c \Delta_{L_c} - P_{\mathcal{C}} \sum_{a \in \mathcal{A}} p_a \Delta_{L_a} = P_{\mathcal{A}} P_{\mathcal{C}} (\langle \Delta_L \rangle_{\mathcal{C}} - \langle \Delta_L \rangle_{\mathcal{A}}) \tag{E.85}$$

Hence the contribution from the derivative of $D_{i,n}(\epsilon)$ simplifies to,

$$\begin{aligned}
& -\frac{2(N-1)}{Z} \left(P_{\mathcal{A}} \sum_{c \in \mathcal{C}} p_c \Delta_{L_c} - P_{\mathcal{C}} \sum_{a \in \mathcal{A}} p_a \Delta_{L_a} \right) (1 - \pi\epsilon \cot(\pi\epsilon)) \\
& = -\frac{2(N-1)P_{\mathcal{A}}P_{\mathcal{C}}}{Z} (\langle \Delta_L \rangle_{\mathcal{C}} - \langle \Delta_L \rangle_{\mathcal{A}}) (1 - \pi\epsilon \cot(\pi\epsilon)).
\end{aligned} \tag{E.86}$$

The divergent piece is simplified in the following way,

$$\frac{1}{Z} \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a \left(2\Delta_{L'_{ab}} \pi \cot(\pi\epsilon) \right) = \frac{1}{Z} \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a S(\rho_a \| \rho_b) - \frac{N}{Z} \sum_{a \in \mathcal{A}} p_a (C_a^{L_a})^2 (\pi\epsilon)^{2\Delta_{L_a}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_a})}{4\Gamma(3/2 + \Delta_{L_a})} \tag{E.87}$$

Therefore we have,

$$\sum_{a \in \mathcal{A}} p_a (C_a^{L_a})^2 (\pi\epsilon)^{2\Delta_{L_a}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_a})}{4\Gamma(3/2 + \Delta_{L_a})} + \frac{1}{Z} \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a \left(2\Delta_{L'_{ab}} \pi \cot(\pi\epsilon) \right) \quad (\text{E.88})$$

$$\begin{aligned} &= \frac{1}{Z} \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a S(\rho_a \| \rho_b) \\ &\quad + \left(1 - \frac{N}{Z} \right) \sum_{a \in \mathcal{A}} p_a (C_a^{L_a})^2 (\pi\epsilon)^{2\Delta_{L_a}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_a})}{4\Gamma(3/2 + \Delta_{L_a})} \end{aligned} \quad (\text{E.89})$$

$$\begin{aligned} &= \frac{1}{Z} \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a S(\rho_a \| \rho_b) \\ &\quad - \left(\frac{(N-1)P_C}{Z} \right) \sum_{a \in \mathcal{A}} p_a (C_a^{L_a})^2 (\pi\epsilon)^{2\Delta_{L_a}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_a})}{4\Gamma(3/2 + \Delta_{L_a})} \end{aligned} \quad (\text{E.90})$$

Simplifying the uniform words contribution from the common sector words, we get,

$$\left(1 - \frac{1}{Z} \right) \sum_{c \in \mathcal{C}} p_c (C_c^{L_c})^2 (\pi\epsilon)^{2\Delta_{L_c}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_c})}{4\Gamma(3/2 + \Delta_{L_c})} = \frac{N-1}{Z} P_{\mathcal{A}} \sum_{c \in \mathcal{C}} p_c (C_c^{L_c})^2 (\pi\epsilon)^{2\Delta_{L_c}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_c})}{4\Gamma(3/2 + \Delta_{L_c})} \quad (\text{E.91})$$

Adding all of the above contributions, we obtain,

$$\begin{aligned} &\left(1 - \frac{1}{Z} \right) \sum_{c \in \mathcal{C}} p_c (C_c^{L_c})^2 (\pi\epsilon)^{2\Delta_{L_c}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_c})}{4\Gamma(3/2 + \Delta_{L_c})} + \sum_{a \in \mathcal{A}} p_a (C_a^{L_a})^2 (\pi\epsilon)^{2\Delta_{L_a}} \frac{\sqrt{\pi}\Gamma(1 + \Delta_{L_a})}{4\Gamma(3/2 + \Delta_{L_a})} \\ &+ \frac{1}{Z} \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a \left(2\Delta_{L'_{ab}} \pi \cot(\pi\epsilon) \right) - \frac{2(N-1)}{Z} \left(P_{\mathcal{A}} \sum_{c \in \mathcal{C}} p_c \Delta_{L_c} - P_C \sum_{a \in \mathcal{A}} p_a \Delta_{L_a} \right) (1 - \pi\epsilon \cot(\pi\epsilon)) \end{aligned} \quad (\text{E.92})$$

$$\begin{aligned} &= \frac{1}{Z} \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a S(\rho_a \| \rho_b) + \frac{(N-1)P_{\mathcal{A}}P_C}{Z} \left[\left\langle (C^L)^2 (\pi\epsilon)^{2\Delta_L} \frac{\sqrt{\pi}\Gamma(1 + \Delta_L)}{4\Gamma(3/2 + \Delta_L)} \right\rangle_{\mathcal{C}} - \left\langle (C^L)^2 (\pi\epsilon)^{2\Delta_L} \frac{\sqrt{\pi}\Gamma(1 + \Delta_L)}{4\Gamma(3/2 + \Delta_L)} \right\rangle_{\mathcal{A}} \right. \\ &\quad \left. - 2(\langle \Delta_L \rangle_{\mathcal{C}} - \langle \Delta_L \rangle_{\mathcal{A}}) (1 - \pi\epsilon \cot(\pi\epsilon)) \right] \end{aligned} \quad (\text{E.93})$$

Finally, the relative entropy is given by,

$$\begin{aligned}
S(\bar{\rho}_{\mathbf{p}}|\bar{\rho}_{\mathbf{q}}) = & -H(\mathbf{p}) - \frac{1}{Z} \left(\sum_{c \in \mathcal{C}} p_c \log q_c + P_{\mathcal{A}} \sum_{b=1}^N \log q_b \right) + \frac{1}{Z} \sum_{a \in \mathcal{A}} \sum_{b=1}^N p_a S(\rho_a|\rho_b) \\
& + \frac{(N-1)P_{\mathcal{A}}P_{\mathcal{C}}}{Z} \left[\left\langle (C^L)^2 (\pi\epsilon)^{2\Delta_L} \frac{\sqrt{\pi}\Gamma(1+\Delta_L)}{4\Gamma(3/2+\Delta_L)} \right\rangle_{\mathcal{C}} - \left\langle (C^L)^2 (\pi\epsilon)^{2\Delta_L} \frac{\sqrt{\pi}\Gamma(1+\Delta_L)}{4\Gamma(3/2+\Delta_L)} \right\rangle_{\mathcal{A}} \right. \\
& \left. - 2(\langle \Delta_L \rangle_{\mathcal{C}} - \langle \Delta_L \rangle_{\mathcal{A}})(1 - \pi\epsilon \cot(\pi\epsilon)) \right] + \mathcal{M}_{\mathbf{p}} - \frac{1}{Z} (\mathcal{M}_{\mathcal{CC}} + \mathcal{M}_{\mathcal{CB}}) + \dots
\end{aligned} \tag{E.94}$$

Consistency tests:

All states are common: Then $\mathcal{A} = \phi = \mathcal{B}$, $C = M = N$. Then $P_{\mathcal{C}} = 1$, and $Z = 1$. Then the square bracket drops out. The contribution from $\mathcal{M}_{\mathcal{CB}}$ and the divergent piece vanishes. And we are left with,

$$\begin{aligned}
S(\bar{\rho}_{\mathbf{p}}|\bar{\rho}_{\mathbf{q}}) &= D_{KL}(\mathbf{p}|\mathbf{q}) + \mathcal{M}_{\mathbf{p}} - \mathcal{M}_{\mathcal{CC}} + \dots \tag{E.95} \\
&= D_{KL}(\mathbf{p}|\mathbf{q}) + \sum_{i < j} \sum_{L'_{ij}} \left(C_{ij}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ij}}} \partial_n \left(\mu_{ij}^{(\mathbf{p})}(n, \epsilon) - \tilde{\mu}_{ij}^{(\mathbf{p}, \mathbf{q})}(n, \epsilon) \right) \Big|_{n=1} + \dots \tag{E.96}
\end{aligned}$$

Mixture with respect to pure state: Consider the case, where the reference state is a part of the mixture. Then we have $N = 1$. Then $\log q_{\alpha} = 0$. The square bracket again drops out. We have $\mathcal{A} = \phi$. Lets label the reference pure state as ρ_{α} . Then, $Z = P_{\mathcal{C}} + P_{\mathcal{A}} = 1$ and $\mathcal{M}_{\mathcal{CC}} = \mathcal{M}_{\mathcal{CB}} = 0$. Thus, we are left with,

$$\begin{aligned}
S(\bar{\rho}_{\mathbf{p}}|\rho_{\alpha}) &= -H(\mathbf{p}) + \mathcal{M}_{\mathbf{p}} + \sum_i p_i S(\rho_i|\rho_{\alpha}) \tag{E.97} \\
&= -H(\mathbf{p}) + \sum_i p_i S(\rho_i|\rho_{\alpha}) + \sum_{1 \leq a < d \leq M} \sum_{L'_{ad}} \left(C_{ad}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ad}}} \partial_n \mu_{ad}^{(\mathbf{p})}(n, \epsilon) \Big|_{n=1} + \dots \tag{E.98}
\end{aligned}$$

Mixture with respect to vacuum: Then we have $N = 1$. Again, the square bracket vanishes. $\log q_{\alpha} = 0$. This gives us $C = 0, P_{\mathcal{A}} = 1$ and $Z = 1$. $\mathcal{M}_{\mathcal{CC}} = 0 = \mathcal{M}_{\mathcal{CB}}$. Then we are left with,

$$S(\bar{\rho}_{\mathbf{p}}|\rho_0) = \sum_a p_a S(\rho_a|\rho_0) - H(\mathbf{p}) + \sum_{1 \leq a < d \leq M} \sum_{L'_{ad}} \left(C_{ad}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ad}}} \partial_n \mu_{ad}^{(\mathbf{p})}(n, \epsilon) \Big|_{n=1} + \dots \tag{E.99}$$

Pure state with respect to Mixture: Here, $C = 1$. Since $\mathcal{A} = \phi$, entire thing in the square bracket vanishes. We also have $Z = 1$. Then we are left with,

$$S(\rho_{\alpha}|\bar{\rho}_{\mathbf{p}}) = -\log q_{\alpha} - \sum_{b \neq \alpha} \sum_{L'_{ab}} \left(C_{ab}^{L'} \right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ab}}} \partial_n \theta_{b|\alpha, n}^{(\mathbf{q})}(\epsilon) \Big|_{n=1} + \dots \tag{E.100}$$

Vacuum with respect to Mixture: Here, we have $\mathcal{C} = \phi$, $P_{\mathcal{A}} = 1$. This gives $Z = 1$. Square bracket vanishes again. We are left with,

$$S(\rho_0 \| \bar{\rho}_{\mathbf{q}}) = \frac{1}{N} \sum_i S(\rho_0 \| \rho_i) - \frac{1}{N} \sum_i \log q_i. \quad (\text{E.101})$$

Perturbation around vacuum reference state: We want to compute $S((1-t)\rho_0 + t\rho \| \rho_0)$ in the limit $t \rightarrow 0$. here, we have $N = 1$, so the square bracket vanishes. $\mathcal{B} = \phi$. And $Z = P_{\mathcal{C}} + NP_{\mathcal{A}} = t + (1-t) = 1$. Hence, we have,

$$S((1-t)\rho_0 + t\rho \| \rho_0) = (1-t) \log(1-t) + t \log t + tS(\rho \| \rho_0) + \sum_{L'_{ad}} \left(C_{ad}^{L'}\right)^2 (2\pi\epsilon)^{2\Delta_{L'_{ad}}} \partial_n \mu_{\rho\rho_0}^{(\mathbf{p})}(n, \epsilon) \Big|_{n=1} + \dots \quad (\text{E.102})$$

Note that, this is not Fisher information. Here, we first took the limit $x \rightarrow 1$, and then we consider $t \rightarrow 0$ limit, which is the opposite way of taking limits while computing Fisher Information.

F Analytic continuation of $\theta_{i|j,n}^{(\mathbf{p})}(\epsilon)$

Using the modular flow analytic continuation in [4], we extend that to λ -weighted replica sum. We compute the analytic continuation of the derivative of $\theta_{i|j,n}^{(\mathbf{p})}(\epsilon)$ at $n = 1$. {analytic cont.

Let us define,

$$S_n(\lambda, \Delta) \equiv \sum_{r=1}^{n-1} \frac{(n-r)\lambda^r}{(2 \sin(\pi r/n))^{2\Delta}}. \quad (\text{F.1})$$

Then we have,

$$\theta_{i|j,n}^{(\mathbf{p})} = S_n(\lambda_{i|j,n}(\epsilon), \Delta), \quad \lambda_{i|j,n}(\epsilon) = \frac{p_i D_{i,n}(\epsilon)}{p_j D_{j,n}(\epsilon)} \quad (\text{F.2})$$

It should be clear that, while computing the derivative of $\theta_{i|j,n}^{(\mathbf{p})}(\epsilon)$ at $n = 1$, the derivative of $\lambda_{i|j,n}(\epsilon)$ with respect to n vanishes. So, in the end result, we will be interested in the value of $\lambda_{i|j,n}(\epsilon)|_{n=1} = \frac{p_i}{p_j}$.

In order to evaluate $\partial_n S_n|_{n=1}$, define the following quantity,

$$A_n(\lambda, \Delta) = \sum_{r=1}^{n-1} \frac{\lambda^r}{(2 \sin(\pi r/n))^{2\Delta}}. \quad (\text{F.3})$$

Then, the replica sum can be written as

$$S_n(\lambda, \Delta) = (n - \lambda \partial_\lambda) A_n(\lambda, \Delta). \quad (\text{F.4})$$

From [4], we have the following result,

$$\lim_{n \rightarrow 1} \frac{1}{n-1} \frac{1}{2} \sum_{k \neq m} \langle \mathcal{O}_k \mathcal{O}_m \rangle = \int_{-\infty}^{\infty} \frac{ds}{8 \cosh^2(s/2)} \langle \mathcal{O}(\pi - is) \mathcal{O}(0) \rangle \quad (\text{F.5})$$

where s is the modular time and $\mathcal{O}_i$ are the primaries. When we set $\lambda = 1$, we obtain

$$\partial_n \left(\frac{n}{2} \sum_{r=1}^{n-1} \frac{1}{(2 \sin(\pi r/n))^{2\Delta}} \right) \Big|_{n=1} = \frac{2^{-2\Delta}}{8} \int_{-\infty}^{\infty} ds \frac{1}{\cosh^{2\Delta+2}(s/2)} \quad (\text{F.6})$$

$$= \frac{\sqrt{\pi} \Gamma(\Delta + 1)}{2^{2\Delta+2} \Gamma(\Delta + 3/2)} \quad (\text{F.7})$$

which corresponds to the result by Cardy, Tonni and collaborators in [3].

From [4], the distance between the two operators in the hyperbolic cylinder $S^1 \times H^{d-1}$ in d -dimensional CFT, which is $\pi - is$, is to be identified with $2\pi r/n$. In other words, $\sin(\pi r/n)$ becomes $\sin(\pi/2 - is/2) = \cosh(s/2)$. Then,

$$\lambda^r = \lambda^{\frac{n}{2\pi}(\pi - is)} = \lambda^{\frac{n}{2} - \frac{ins}{2\pi}}. \quad (\text{F.8})$$

Note that, the last equality is what we require, since we evaluate the function λ at $n = 1$ in the end. Then we have the following,

$$\lim_{n \rightarrow 1} \frac{A_n(\lambda, \Delta)}{n-1} = \partial_n A_n(\lambda, \Delta)|_{n=1} = \frac{2^{-2\Delta}}{4} \int_{-\infty}^{\infty} ds \frac{\lambda^{\frac{1}{2} - \frac{is}{2\pi}}}{\cosh^{2\Delta+2}(s/2)}. \quad (\text{F.9})$$

Thus, we want to consider the following,

$$\partial_n S_n(\lambda, \Delta)|_{n=1} = \partial_n \left(\sum_{r=1}^{n-1} \frac{(n-r)\lambda^r}{(2 \sin(\pi r/n))^{2\Delta}} \right) \Big|_{n=1} \quad (\text{F.10})$$

$$= \partial_n A_n(\lambda, \Delta)|_{n=1} - \partial_n \left(\sum_{r=1}^{n-1} \frac{r\lambda^r}{(2 \sin(\pi r/n))^{2\Delta}} \right) \Big|_{n=1} \quad (\text{F.11})$$

$$= \frac{2^{-2\Delta}}{4} \int_{-\infty}^{\infty} ds \left(\frac{1}{2} + \frac{is}{2\pi} \right) \frac{\lambda^{\frac{1}{2} - \frac{is}{2\pi}}}{\cosh^{2\Delta+2}(s/2)} \quad (\text{F.12})$$

$$(\text{F.13})$$

where in the last step we used the integral identity in modular time, by identifying $r = \frac{1}{2} - \frac{is}{2\pi}$. To evaluate this integral, define $q = \frac{\log \lambda}{2\pi}$. Then, we have,

$$\lambda^{1/2 - is/2\pi} = \sqrt{\lambda} e^{-iqs} \quad (\text{F.14})$$

$$\partial_n A_n(\lambda, \Delta) = \frac{2^{-2\Delta} \sqrt{\lambda}}{4} \int_{-\infty}^{\infty} ds \frac{e^{-iqs}}{\cosh^{2\Delta+2}(s/2)} \quad (\text{F.15})$$

$$= \frac{\sqrt{\lambda}}{\Gamma(2\Delta + 2)} \Gamma(\Delta + 1 + iq) \Gamma(\Delta + 1 - iq). \quad (\text{F.16})$$

We can then compute the derivative of the above with respect to λ to finally obtain,

$$\partial_n S_n(\lambda, \Delta)|_{n=1} = \frac{\sqrt{\lambda}}{\Gamma(2\Delta+2)} \left| \Gamma \left(\Delta + 1 + \frac{i \log \lambda}{2\pi} \right) \right|^2 \left[\frac{1}{2} + \frac{1}{\pi} \Im \psi \left(\Delta + 1 + \frac{i \log \lambda}{2\pi} \right) \right] \quad (\text{F.17})$$

As a sanity check, when $\lambda = 1$, we have,

$$\partial_n S_n(1, \Delta)|_{n=1} = \frac{\Gamma(\Delta+1)^2}{2\Gamma(2\Delta+2)} \quad (\text{F.18})$$

$$= 2^{-2\Delta} \frac{\sqrt{\pi} \Gamma(\Delta+1)}{4\Gamma(\Delta+3/2)} \quad (\text{F.19})$$

which agrees with our previous result. Thus we have,

$$\partial_n \theta_{i|j,n}^{(\mathbf{p})}(\epsilon)|_{n=1} = \frac{\sqrt{p_i/p_j}}{\Gamma(2\Delta_{L'_{ij}}+2)} \left| \Gamma \left(\Delta_{L'_{ij}} + 1 + \frac{i}{2\pi} \log \frac{p_i}{p_j} \right) \right|^2 \left[\frac{1}{2} + \frac{1}{\pi} \Im \psi \left(\Delta_{L'_{ij}} + 1 + \frac{i}{2\pi} \log \frac{p_i}{p_j} \right) \right] \quad (\text{F.20})$$

As a special case, if we have a binary mixture, such that $p_i = 1 - y$ and $p_j = y$, we obtain the analytic continuation of $\mu'(n, x, y; \Delta_{L'})$ at $n \rightarrow 1$. Thus we obtain,

$$\partial_n \mu(n, x, y; \Delta_{L'})|_{n=1} = \frac{\sqrt{\frac{1-y}{y}}}{\Gamma(2\Delta_{L'}+2)} \left| \Gamma \left(\Delta_{L'} + 1 + \frac{i \log \frac{1-y}{y}}{2\pi} \right) \right|^2 \left[\frac{1}{2} + \frac{1}{\pi} \Im \psi \left(\Delta_{L'} + 1 + \frac{i \log \frac{1-y}{y}}{2\pi} \right) \right] \quad (\text{F.21})$$

G Analytic continuation of three-point function $\mathcal{T}_{n,LMN}$

{Analytic cont.

In this section, we follow the method of modular flow used in [4] to analytically continue the contribution of three-point function $\langle \mathcal{O}_L \mathcal{O}_M \mathcal{O}_N \rangle$ to the relative entropy, where the three primaries $\mathcal{O}_L, \mathcal{O}_M, \mathcal{O}_N$ have scaling dimensions $\Delta_L, \Delta_M, \Delta_N$. This is the generalization of their result, since they considered the simple case where all the operators were the same, i.e. $\langle \mathcal{O}_L \mathcal{O}_L \mathcal{O}_L \rangle$.

Following Casini, Huerta, Myers map in [16], the three-point function on $\Sigma_1 = S^1 \times H^{d-1}$ is given by,

$$\begin{aligned} & \langle \mathcal{O}_L(\hat{\tau} - \tau_{s_1}) \mathcal{O}_M(\hat{\tau} - \tau_{s_2}) \mathcal{O}_N(\hat{\tau}) \rangle \\ &= \frac{2^{-(\Delta_L + \Delta_M + \Delta_N)} C_{LMN}}{(\cosh(\frac{s_1}{2}))^{\Delta_N + \Delta_L - \Delta_M} (\cosh(\frac{s_2}{2}))^{\Delta_M + \Delta_N - \Delta_L} \left[(i \sinh(\frac{s_2 - s_1}{2}))^2 \right]^{\frac{\Delta_L + \Delta_M - \Delta_N}{2}}} \end{aligned} \quad (\text{G.1})$$

Now the contribution of the three-point function to the relative entropy is given by the following [4],

$$\begin{aligned} \partial_n \mathcal{T}_n(\Delta_L, \Delta_M, \Delta_N)|_{n=1} &= \frac{(2\theta_0)^\Sigma C_L C_M C_N C_{LMN}}{16\pi 2^\Sigma} \\ &\times \int_{-\infty}^{\infty} ds_1 ds_2 \frac{i(s_2 - s_1)}{2\pi} \frac{i}{(\cosh \frac{s_1}{2})^{c+1} (\cosh \frac{s_2}{2})^{b+1} (\sinh \frac{s_2-s_1}{2}) \left[(i \sinh \frac{s_2-s_1}{2})^2 \right]^{a/2}} \end{aligned} \quad (\text{G.2})$$

Now consider the following auxiliary integral,

$$Q_{LM|N} = \frac{(2\pi x)^\Sigma C_L C_M C_N C_{LMN}}{16\pi 2^\Sigma} \mathcal{J}(a, b, c) \quad (\text{G.3})$$

$$\mathcal{J}(a, b, c) = \int_{-\infty}^{\infty} ds_1 ds_2 \frac{i}{(\cosh \frac{s_1}{2})^{c+1} (\cosh \frac{s_2}{2})^{b+1} (\sinh \frac{s_2-s_1}{2}) \left[(i \sinh \frac{s_2-s_1}{2})^2 \right]^{a/2}} \quad (\text{G.4})$$

$$a = \Delta_L + \Delta_M - \Delta_N \quad (\text{G.5})$$

$$b = \Delta_M + \Delta_N - \Delta_L \quad (\text{G.6})$$

$$c = \Delta_N + \Delta_L - \Delta_M \quad (\text{G.7})$$

$$\Sigma = \Delta_L + \Delta_M + \Delta_N \quad (\text{G.8})$$

Now, for general value of a , we can decouple the s_2 and s_1 integrals by inserting $\int du \delta(u - (s_2 - s_1))$ and then pass over to Fourier representation of delta,

$$\int_{-\infty}^{\infty} ds_1 ds_2 \frac{i}{(\cosh \frac{s_1}{2})^{c+1} (\cosh \frac{s_2}{2})^{b+1} \sinh \frac{s_2-s_1}{2} \left[(i \sinh \frac{s_2-s_1}{2})^2 \right]^{a/2}} \quad (\text{G.9})$$

$$= i \int \frac{dq}{2\pi} \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{b+1}} \right] \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{c+1}} \right] \left[\int du \frac{e^{iqu}}{\sinh \frac{u}{2} \left[(i \sinh \frac{u}{2})^2 \right]^{a/2}} \right] \quad (\text{G.10})$$

$$= i \int \frac{dq}{2\pi} \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{b+1}} \right] \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{c+1}} \right] \left[\int du \frac{e^{-\pi q} e^{iqu}}{(\cosh \frac{u}{2})^{a+1}} \right] \quad (\text{G.11})$$

where in the last line, we deformed the contour $u \rightarrow u + i\pi$. We now obtain the formula,

$$\begin{aligned} Q_{LM|N} &= \frac{(2\pi x)^\Sigma C_L C_M C_N C_{LMN}}{16\pi 2^\Sigma} \\ &\times \int_{-\infty}^{\infty} \frac{dq}{2\pi} e^{-\pi q} \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{a+1}} \right] \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{b+1}} \right] \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{c+1}} \right] \end{aligned} \quad (\text{G.12})$$

Note that, the above integral also gives the following,

$$\mathcal{J}(a, b, c) = \int_{-\infty}^{\infty} \frac{dq}{2\pi} e^{-\pi q} \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{a+1}} \right] \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{b+1}} \right] \left[\int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh(\frac{s}{2}))^{c+1}} \right]. \quad (\text{G.13})$$

The s integral is easy to evaluate by a change of variable,

$$\mathcal{G}_{\Delta}(q) \equiv \int_{-\infty}^{\infty} ds \frac{e^{iqs}}{(\cosh \frac{s}{2})^{\Delta+1}} = \frac{2^{1+\Delta} \Gamma(\frac{1+\Delta}{2} + iq) \Gamma(\frac{1+\Delta}{2} - iq)}{\Gamma(\Delta + 1)}. \quad (\text{G.14})$$

However, the resulting q integral in $Q_{LM|N}$ or $\mathcal{J}(a, b, c)$ have to be computed numerically. Note that, the convergence domain is,

$$a > -1, \quad b > -1, \quad c > -1. \quad (\text{G.15})$$

Whenever atleast one of them is equal to -1 , then the integral vanishes.

Now consider the case, when a is an even positive integer. This choice would get rid of the branch cuts. We are left with,

$$\mathcal{J}(a, b, c) = \int_{-\infty}^{\infty} ds_1 ds_2 \frac{i}{(\cosh \frac{s_1}{2})^{c+1} (\cosh \frac{s_2}{2})^{b+1} (\sinh \frac{s_2 - s_1}{2})^{a+1}} \quad (\text{G.16})$$

Note that, this is the generalization of the integral they had for $\Delta_L = \Delta_M = \Delta_N = \Delta$. To compute this, we exchange the integration variables $s_i = 2 \operatorname{arctanh}(2v_i)$. Then we have,

$$\tanh \frac{s_i}{2} = 2v_i, \quad (\text{G.17})$$

$$\cosh \frac{s_i}{2} = \frac{1}{\sqrt{1 - 4v_i^2}}, \quad (\text{G.18})$$

$$ds_i = \frac{4dv_i}{1 - 4v_i^2}, \quad (\text{G.19})$$

$$\sinh \frac{s_2 - s_1}{2} = \frac{2(v_2 - v_1)}{\sqrt{(1 - 4v_1^2)(1 - 4v_2^2)}}. \quad (\text{G.20})$$

Thus, we are now supposed to evaluate,

$$\mathcal{J}(a, b, c) = i^{a+1} 2^{3-a} \int_{-1/2}^{1/2} dv_1 \int_{-1/2}^{1/2} dv_2 \frac{(1 - 4v_1^2)^{\Delta_L} (1 - 4v_2^2)^{\Delta_M}}{(v_2 - v_1 + i0^+)^{a+1}} \quad (\text{G.21})$$

where we close the contour from above the pole. We now Fourier decouple $u = v_2 - v_1$. The singular Fourier integral in u is done by closing the contour from above the pole,

$$\int_{-\infty}^{\infty} du \frac{e^{ipu}}{(u + i0^+)^{a+1}} = -\frac{2\pi i^{a+1}}{\Gamma(a+1)} p^a \Theta(-p), \quad (\text{G.22})$$

where Θ is the usual Heaviside theta function. Define,

$$W_{\Delta}(p) \equiv \int_{-1/2}^{1/2} dv (1 - 4v^2)^{\Delta} e^{-ipv} \quad (\text{G.23})$$

$$= 2^{2\Delta} \sqrt{\pi} \Gamma(\Delta + 1) p^{-\Delta - \frac{1}{2}} J_{\Delta + \frac{1}{2}}\left(\frac{p}{2}\right) \quad (\text{G.24})$$

where $J_n(x)$ is Bessel's function of the first kind. Then we arrive at the following integral,

$$\mathcal{J}(a, b, c) = \frac{2^{3-a}}{\Gamma(a+1)} \int_0^{\infty} dp p^a W_{\Delta_L}(p) W_{\Delta_M}(p) \quad (\text{G.25})$$

which is clearly the generalization of their case where $\Delta_L = \Delta_M = \Delta$. We can now substitute the expression of $W_{\Delta}(p)$ in the above integral to obtain,

$$\mathcal{J}(a, b, c) = \frac{2^{3-a+2\Delta_L+2\Delta_M} \pi \Gamma(\Delta_L + 1) \Gamma(\Delta_M + 1)}{\Gamma(a+1)} \int_0^{\infty} dp p^{-\Delta_N-1} J_{\Delta_L + \frac{1}{2}}(p/2) J_{\Delta_M + \frac{1}{2}}(p/2). \quad (\text{G.26})$$

Now substitute $p = 2x$, to obtain,

$$\mathcal{J}(a, b, c) = \frac{2^{3+\Delta_L+\Delta_M} \pi \Gamma(\Delta_L + 1) \Gamma(\Delta_M + 1)}{\Gamma(a+1)} \int_0^{\infty} dx x^{-\Delta_N-1} J_{\Delta_L + \frac{1}{2}}(x) J_{\Delta_M + \frac{1}{2}}(x) \quad (\text{G.27})$$

Using the Infinite Integrals (Weber–Schafheitlin Type),

$$\int_0^{\infty} dx x^{-\lambda} J_{\mu}(x) J_{\nu}(x) = \frac{\Gamma(\lambda) \Gamma\left(\frac{\mu+\nu-\lambda+1}{2}\right)}{2^{\lambda} \Gamma\left(\frac{-\mu+\nu+\lambda+1}{2}\right) \Gamma\left(\frac{\mu+\nu+\lambda+1}{2}\right) \Gamma\left(\frac{\mu-\nu+\lambda+1}{2}\right)} \quad (\text{G.28})$$

we substitute, $\lambda = \Delta_N + 1$, $\mu = \Delta_L + 1/2$ and $\nu = \Delta_M + 1/2$ to obtain,

$$\int_0^{\infty} dx x^{-\Delta_N-1} J_{\Delta_L + \frac{1}{2}}(x) J_{\Delta_M + \frac{1}{2}}(x) = \frac{\Gamma(\Delta_N + 1) \Gamma\left(\frac{a+1}{2}\right)}{2^{\Delta_N+1} \Gamma\left(1 + \frac{b}{2}\right) \Gamma\left(1 + \frac{c}{2}\right) \Gamma\left(\frac{\Sigma+3}{2}\right)}. \quad (\text{G.29})$$

Combining these results gives,

$$\mathcal{J}(a, b, c) = 2^{a+2} \pi \frac{\Gamma(\Delta_L + 1) \Gamma(\Delta_M + 1) \Gamma(\Delta_N + 1) \Gamma\left(\frac{a+1}{2}\right)}{\Gamma(a+1) \Gamma\left(1 + \frac{b}{2}\right) \Gamma\left(1 + \frac{c}{2}\right) \Gamma\left(\frac{\Sigma+3}{2}\right)}. \quad (\text{G.30})$$

Now using the Gamma duplication formula, we get,

$$\mathcal{J}(a, b, c) = 4\pi^{3/2} \frac{\Gamma(\Delta_L + 1) \Gamma(\Delta_M + 1) \Gamma(\Delta_N + 1)}{\Gamma\left(1 + \frac{a}{2}\right) \Gamma\left(1 + \frac{b}{2}\right) \Gamma\left(1 + \frac{c}{2}\right) \Gamma\left(\frac{\Sigma+3}{2}\right)}. \quad (\text{G.31})$$

which is symmetric with $\{L, M, N\}$. This also says that $Q_{LM|N}$ is equal to the Q 's with the different permutation of $\{L, M, N\}$. This gives,

$$Q_{LM|N} = \frac{\sqrt{\pi}}{4} C_L C_M C_N C_{LMN} (\pi x)^{\Sigma} \frac{\Gamma(\Delta_L + 1) \Gamma(\Delta_M + 1) \Gamma(\Delta_N + 1)}{\Gamma\left(1 + \frac{a}{2}\right) \Gamma\left(1 + \frac{b}{2}\right) \Gamma\left(1 + \frac{c}{2}\right) \Gamma\left(\frac{\Sigma+3}{2}\right)}. \quad (\text{G.32})$$

As a sanity check, we substitute $\Delta_L = \Delta_M = \Delta_N = \Delta$. Then we have $a = b = c = \Delta = \Sigma/3$. Then the above equation gives,

$$\mathcal{J}(\Delta, \Delta, \Delta) = 4\pi^{3/2} \frac{\Gamma(\Delta + 1)^3}{\Gamma(1 + \frac{\Delta}{2})^3 \Gamma(\frac{3+3\Delta}{2})} \quad (\text{G.33})$$

Using the duplication formula, it simplifies to,

$$\mathcal{J}(\Delta, \Delta, \Delta) = 2^{3\Delta+2} \frac{\Gamma(\frac{1+\Delta}{2})^3}{\Gamma(\frac{3+3\Delta}{2})} \quad (\text{G.34})$$

We then obtain,

$$Q_{LL|L} = \frac{(2\pi x)^\Sigma (C_L)^3 C_{LLL}}{16\pi 2^\Sigma} \mathcal{J}(\Delta, \Delta, \Delta) \quad (\text{G.35})$$

$$= (2\pi x)^{3\Delta} (C_L)^3 C_{LLL} \frac{\Gamma(\frac{1+\Delta}{2})^3}{4\pi \Gamma(\frac{3+3\Delta}{2})} \quad (\text{G.36})$$

which agrees with the result of [4].

Now we move on to compute the actual integral that gives the contribution of the three-point function,

$$\mathcal{I}_{LM|N} \equiv \int_{-\infty}^{\infty} ds_1 ds_2 \frac{i(s_2 - s_1)}{2\pi} \frac{i}{(\cosh \frac{s_1}{2})^{c+1} (\cosh \frac{s_2}{2})^{b+1} (\sinh \frac{s_2 - s_1}{2}) \left[(i \sinh \frac{s_2 - s_1}{2})^2 \right]^{a/2}} \quad (\text{G.37})$$

$$= i \int \frac{dq}{2\pi} \left[\int ds_1 \frac{e^{iqs_1}}{(\cosh \frac{s_1}{2})^{c+1}} \right] \left[\int ds_2 \frac{e^{iqs_2}}{(\cosh \frac{s_2}{2})^{b+1}} \right] \left[\int du \frac{\frac{iu}{2\pi} e^{iqu}}{\sinh \frac{u}{2} \left[(i \sinh \frac{u}{2})^2 \right]^{a/2}} \right] \quad (\text{G.38})$$

$$= i \int \frac{dq}{2\pi} \left[\int ds_1 \frac{e^{iqs_1}}{(\cosh \frac{s_1}{2})^{c+1}} \right] \left[\int ds_2 \frac{e^{iqs_2}}{(\cosh \frac{s_2}{2})^{b+1}} \right] \frac{\partial_q}{2\pi} \left[\int du \frac{e^{iqu}}{\sinh \frac{u}{2} \left[(i \sinh \frac{u}{2})^2 \right]^{a/2}} \right] \quad (\text{G.39})$$

$$= \int \frac{dq}{2\pi} \left[\int ds_1 \frac{e^{iqs_1}}{(\cosh \frac{s_1}{2})^{c+1}} \right] \left[\int ds_2 \frac{e^{iqs_2}}{(\cosh \frac{s_2}{2})^{b+1}} \right] \frac{\partial_q}{2\pi} \left[\int du \frac{e^{-\pi q} e^{iqu}}{(\cosh \frac{u}{2})^{a+1}} \right] \quad (\text{G.40})$$

where in the third line we have replaced the $\frac{iu}{2\pi}$ factor by a $\frac{1}{2\pi} \partial_q$ and in the last line we have again deformed the u contour as $u \rightarrow u + i\pi$. We want to express the above integral in terms of $\mathcal{J}(a, b, c)$. By adding the cyclic permutations of $\{L, M, N\}$, one obtains,

$$\mathcal{I}_{LM|N} + \mathcal{I}_{MN|L} + \mathcal{I}_{NL|M} = \frac{1}{2\pi} \int \frac{dq}{2\pi} e^{-\pi q} \left[\partial_q (\mathcal{G}_a \mathcal{G}_b \mathcal{G}_c) - 3\pi \mathcal{G}_a \mathcal{G}_b \mathcal{G}_c \right] \quad (\text{G.41})$$

Now by integration by parts, the boundary term vanishes, and we obtain,

$$\mathcal{I}_{LM|N} + \mathcal{I}_{MN|L} + \mathcal{I}_{NL|M} = - \int \frac{dq}{2\pi} e^{-\pi q} \mathcal{G}_a \mathcal{G}_b \mathcal{G}_c \quad (\text{G.42})$$

$$= -\mathcal{J}(a, b, c). \quad (\text{G.43})$$

Now, we can sum over different permutations of $\{L, M, N\}$. Since,

$$\sum_{L,M,N}^{\text{ord}} \mathcal{I}_{LM|N} = \sum_{L,M,N}^{\text{ord}} \mathcal{I}_{MN|L} = \sum_{L,M,N}^{\text{ord}} \mathcal{I}_{NL|M} \quad (\text{G.44})$$

then adding all of the above gives

$$3 \sum_{L,M,N}^{\text{ord}} \mathcal{I}_{LM|N} = - \sum_{L,M,N}^{\text{ord}} \mathcal{J}(a, b, c). \quad (\text{G.45})$$

Thus the contribution to relative entropy is

$$\sum_{L,M,N}^{\text{ord}} \partial_n \mathcal{T}_n(\Delta_L, \Delta_M, \Delta_N)|_{n=1} = -\frac{1}{3} \sum_{L,M,N}^{\text{ord}} Q_{LM|N} \quad (\text{G.46})$$

$$= -\frac{\sqrt{\pi}}{12} \sum_{L,M,N}^{\text{ord}} C_L C_M C_N C_{LMN}(\pi x)^\Sigma \frac{\Gamma(\Delta_L + 1) \Gamma(\Delta_M + 1) \Gamma(\Delta_N + 1)}{\Gamma\left(1 + \frac{a}{2}\right) \Gamma\left(1 + \frac{b}{2}\right) \Gamma\left(1 + \frac{c}{2}\right) \Gamma\left(\frac{\Sigma+3}{2}\right)} \quad (\text{G.47})$$

$$= -\frac{\sqrt{\pi}}{12} \sum_{L,M,N}^{\text{ord}} C_L C_M C_N C_{LMN}(\pi x)^{\Delta_L + \Delta_M + \Delta_N} \times \frac{\Gamma(\Delta_L + 1) \Gamma(\Delta_M + 1) \Gamma(\Delta_N + 1)}{\Gamma\left(1 + \frac{\Delta_L + \Delta_M - \Delta_N}{2}\right) \Gamma\left(1 + \frac{\Delta_M + \Delta_N - \Delta_L}{2}\right) \Gamma\left(1 + \frac{\Delta_N + \Delta_L - \Delta_M}{2}\right) \Gamma\left(\frac{\Delta_L + \Delta_M + \Delta_N + 3}{2}\right)} \quad (\text{G.48})$$

For the simplicity of notation, we define

$$f_{LMN} \equiv -\frac{\sqrt{\pi}}{12} \frac{\Gamma(\Delta_L + 1) \Gamma(\Delta_M + 1) \Gamma(\Delta_N + 1)}{\Gamma\left(1 + \frac{\Delta_L + \Delta_M - \Delta_N}{2}\right) \Gamma\left(1 + \frac{\Delta_M + \Delta_N - \Delta_L}{2}\right) \Gamma\left(1 + \frac{\Delta_N + \Delta_L - \Delta_M}{2}\right) \Gamma\left(\frac{\Delta_L + \Delta_M + \Delta_N + 3}{2}\right)} \quad (\text{G.49})$$

Consider the simple case, $\Delta_L = \Delta_M = \Delta_N$, then we obtain,

$$\partial_n \mathcal{T}_n(\Delta_L, \Delta_L, \Delta_L)|_{n=1} = -\frac{\sqrt{\pi}}{12} C_L^3 C_{LLL}(\pi x)^{3\Delta} \frac{\Gamma(\Delta_L + 1)^3}{\Gamma(1 + \Delta_L/2)^3 \Gamma((3 + 3\Delta_L)/2)} \quad (\text{G.50})$$

$$\equiv C_L^3 C_{LLL} f_{\Delta_L, \Delta_M, \Delta_N}(\pi x)^{3\Delta_L} \quad (\text{G.51})$$

$$= -C_L^3 C_{LLL} \frac{2^{3\Delta_L} \Gamma\left(\frac{1+\Delta_L}{2}\right)^3}{12\pi \Gamma\left(\frac{3+3\Delta_L}{2}\right)} (\pi x)^{3\Delta_L} \quad (\text{G.52})$$

which agrees with the result of [4].

If we choose one of the operators to be vacuum, the above result won't reproduce the result of two-point function we had earlier. This is because, to start with we assumed that the operators $\mathcal{O}_L, \mathcal{O}_M, \mathcal{O}_N$ are non-vacuum operators. We can then use the result of two-point function in [4], which matches with the replica trick result from [3], and denote,

$$f_{LL} \equiv \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)}. \quad (\text{G.53})$$

Note that, if we substitute $\mathcal{O}_N$ as the vacuum operator ($I \equiv \mathcal{O}_0$), then

$$f_{LL0} = -\frac{\sqrt{\pi}}{12} \frac{\Gamma(\Delta_L + 1)}{\Gamma(\Delta_L + 3/2)} \quad (\text{G.54})$$

$$= -f_{LL}/3. \quad (\text{G.55})$$

H Analytic continuation of three-point function $\mu_G^{\Psi|\Phi}$

{Appendix: an

In this section, we follow the method of modular flow used in [4] to analytically continue the contribution of three-point function $\langle \mathcal{O}_{L'} \mathcal{O}_{L'} \mathcal{O}_L \rangle$ to the relative entropy in $x \rightarrow 1^-$ limit of a pure state with respect to binary mixture. Let the dimension of $\mathcal{O}_{L'}$ be $\Delta_{L'}$ and the dimension of $\mathcal{O}_L$ be Δ_L . When we compute $S(\rho_\Psi \| (y\rho_\Psi + (1-y)\rho_\Phi))$, the three-point function that contributes to it from the denominator of $\mathcal{F}_n$ takes the following form,

$$\mu_G^{\Psi|\Phi}(n, x, y; \Delta_{L'}, \Delta_L) \equiv \mu_G = \sum_{p=1}^{n-1} \frac{q^p (n-p) D_n(\epsilon)^p}{(\sin(\pi p/n))^{2\Delta_{L'} - \Delta_L}} (C_\Phi^L G_p(n) + C_\Psi^L G_{n-p}(n)) \quad (\text{H.1}) \quad \{\text{eq: definition}\}$$

$$q = y^{-1} - 1 \quad (\text{H.2}) \quad \{\text{eq: definition}\}$$

$$G_p(n) = \sum_{m=1}^{p-1} \frac{1}{(\sin(\pi m/n) \sin(\pi(p-m)/n))^{\Delta_L}} \quad (\text{H.3})$$

We want to compute the analytic continuation of the derivative of μ_G as $n \rightarrow 1$. Let us define,

$$\lambda_n(\epsilon) \equiv q D_n(\epsilon). \quad (\text{H.4})$$

It is clear that, at $n = 1$,

$$D_1(\epsilon) = 1, \quad \lambda_1 = q = \frac{1-y}{y}. \quad (\text{H.5})$$

We now define the weighted three-point replica sum in terms of the new variables (call $\lambda \equiv \lambda_n(\epsilon)$),

$$A_n(\lambda; \Delta_{L'}, \Delta_L) \equiv \sum_{p=1}^{n-1} \lambda^p \frac{G_p(n)}{(\sin \frac{\pi p}{n})^{2\Delta_{L'} - \Delta_L}} \quad (\text{H.6})$$

$$= \sum_{p=1}^{n-1} \sum_{m=1}^{p-1} \frac{\lambda^p}{(\sin \frac{\pi m}{n})^{\Delta_L} \left(\sin \frac{\pi(p-m)}{n}\right)^{\Delta_L} (\sin \frac{\pi p}{n})^{2\Delta_{L'} - \Delta_L}} \quad (\text{H.7})$$

Then, in order to write μ_G in terms of $A_n(\lambda; \Delta_{L'}, \Delta_L)$, the first term is straight forward,

$$\sum_{p=1}^{n-1} \frac{q^p D_n(\epsilon)^p (n-p) G_p(n)}{\left(\sin \frac{\pi p}{n}\right)^{2\Delta_{L'} - \Delta_L}} = (n - \lambda \partial_\lambda) A_n(\lambda; \Delta_{L'}, \Delta_L). \quad (\text{H.8})$$

For the second term, make the change $r = n - p$. Then we obtain,

$$\sum_{p=1}^{n-1} \frac{q^p D_n(\epsilon)^p (n-p) G_{n-p}(n)}{\left(\sin \frac{\pi p}{n}\right)^{2\Delta_{L'} - \Delta_L}} = \lambda^n \sum_{p=1}^{n-1} r \lambda^{-r} \frac{G_r(n)}{\left(\sin \frac{\pi p}{n}\right)^{2\Delta_{L'} - \Delta_L}} \quad (\text{H.9})$$

Let $z = \lambda^{-1}$, then we finally have,

$$\mu_G = C_\Phi^L (n - \lambda \partial_\lambda) A_n(\lambda; \Delta_{L'}, \Delta_L) + C_\Psi^L \lambda^n (z \partial_z A_n(z))|_{z=\lambda^{-1}} \quad (\text{H.10})$$

$$\implies \partial_n \mu_G|_{n=1} = C_\Phi^L (1 - \lambda \partial_\lambda) \partial_n A_n(\lambda; \Delta_{L'}, \Delta_L)|_{n=1} + C_\Psi^L \lambda [z \partial_z (\partial_n A_n(\lambda; \Delta_{L'}, \Delta_L)|_{n=1})]_{z=\lambda^{-1}} \quad (\text{H.11})$$

From our previous result of the analytic continuation of the weighted two-point function in Appendix F, and that of the three-point function G, we can analogously define,

$$\nu \equiv \frac{\log \lambda}{2\pi} \quad (\text{H.12})$$

$$\implies \lambda^{\frac{1}{2} - \frac{is_2}{2\pi}} = \sqrt{\lambda} e^{-i\nu s_2}. \quad (\text{H.13})$$

Then the integral we need to compute is the following,

$$\begin{aligned} \mathcal{I}_\lambda(\Delta_L, \Delta_{L'}) &= \sqrt{\lambda} \int_{-\infty}^{\infty} ds_1 \int_{-\infty}^{\infty} ds_2 e^{-i\nu s_2} \frac{i(s_2 - s_1)}{2\pi} \\ &\quad \times \frac{i}{\left(\cosh \frac{s_1}{2}\right)^{\Delta_L+1} \left(\cosh \frac{s_2}{2}\right)^{2\Delta_{L'} - \Delta_L+1} \left(\sinh \frac{s_2 - s_1}{2}\right) \left[\left(i \sinh \frac{s_2 - s_1}{2}\right)^2 \right]^{\Delta_L/2}} \end{aligned} \quad (\text{H.14})$$

Also, we have the following relation,

$$\partial_n A_n(\lambda; \Delta_{L'}, \Delta_L)|_{n=1} = \frac{3}{16\pi} \mathcal{I}_\lambda(\Delta_L, \Delta_{L'}). \quad (\text{H.15}) \quad \{\text{eq: relation}\}$$

Similarly, we can introduce the auxiliary weighted integral,

$$\mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) = \sqrt{\lambda} \int_{-\infty}^{\infty} ds_1 \int_{-\infty}^{\infty} ds_2 \frac{i e^{-i\nu s_2}}{\left(\cosh \frac{s_1}{2}\right)^{\Delta_L+1} \left(\cosh \frac{s_2}{2}\right)^{2\Delta_{L'} - \Delta_L+1} \left(\sinh \frac{s_2 - s_1}{2}\right) \left[\left(i \sinh \frac{s_2 - s_1}{2}\right)^2 \right]^{\Delta_L/2}} \quad (\text{H.16})$$

Note that, for $\lambda = 1$, this reduces to $\mathcal{J}(\Delta_L, 2\Delta_{L'} - \Delta_L, \Delta_L)$ specialization of our auxiliary integral defined in the previous Appendix G. Now we can Fourier decouple $u = s_2 - s_1$. Define,

$$\hat{G}_\Delta(\omega) = \int_{-\infty}^{\infty} ds \frac{e^{iqs}}{\left(\cosh \frac{s}{2}\right)^{\Delta+1}} = \frac{2^{1+\Delta} \Gamma\left(\frac{1+\Delta}{2} + iq\right) \Gamma\left(\frac{1+\Delta}{2} - iq\right)}{\Gamma(\Delta+1)}. \quad (\text{H.17})$$

After Fourier decoupling of $u = s_2 - s_1$ and deformation of contour $u \rightarrow u + i\pi$, we obtain,

$$\mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) = \sqrt{\lambda} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-\pi\omega} \hat{G}_{\Delta_L}(\omega)^2 \hat{G}_{2\Delta_{L'} - \Delta_L}(\omega + \nu) \quad (\text{H.18})$$

Note that, this is the generalization of the previous three-point unweighted integral. We can then directly write down the actual integral,

$$\mathcal{I}_\lambda(\Delta_L, \Delta_{L'}) = \sqrt{\lambda} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \hat{G}_{\Delta_L}(\omega) \hat{G}_{2\Delta_{L'} - \Delta_L}(\omega + \nu) \frac{1}{2\pi} \partial_\omega \left[e^{-\pi\omega} \hat{G}_{\Delta_L}(\omega) \right] \quad (\text{H.19})$$

We want to write the above integral in terms of $\mathcal{J}_\lambda(\Delta_L, \Delta_{L'})$. We define,

$$A \equiv \sqrt{\lambda} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-\pi\omega} \hat{G}_{\Delta_L}(\omega) \hat{G}'_{\Delta_L}(\omega) \hat{G}_{2\Delta_{L'} - \Delta_L}(\omega + \nu) \quad (\text{H.20})$$

$$K \equiv \sqrt{\lambda} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-\pi\omega} \hat{G}_{\Delta_L}(\omega)^2 \hat{G}'_{2\Delta_{L'} - \Delta_L}(\omega + \nu). \quad (\text{H.21})$$

Then integrating the total derivative, we obtain,

$$-\pi \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) + 2A + K = 0 \quad (\text{H.22})$$

Since we defined, $\lambda = e^{2\pi\nu}$, we then have,

$$\partial_\nu \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) = \pi \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) + K. \quad (\text{H.23})$$

Eliminating K , we get,

$$-\pi \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) + 2A + \partial_\nu \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) - \pi \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) = 0, \quad (\text{H.24})$$

$$\implies A = \pi \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) - \frac{1}{2} \partial_\nu \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}). \quad (\text{H.25})$$

But we have,

$$\mathcal{I}_\lambda(\Delta_L, \Delta_{L'}) = \frac{1}{2\pi} (A - \pi \mathcal{J}_\lambda(\Delta_L, \Delta_{L'})). \quad (\text{H.26})$$

Therefore,

$$\mathcal{I}_\lambda(\Delta_L, \Delta_{L'}) = -\frac{1}{4\pi} \partial_\nu \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) \quad (\text{H.27}) \quad \{\text{eq: relation}\}$$

$$= -\frac{1}{2} \lambda \partial_\lambda \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}). \quad (\text{H.28}) \quad \{\text{eq: relation}\}$$

Now we consider the case, when $\Delta_L \in 2\mathbb{Z}_{\geq 0}$. This gets rid of branch cuts. By exchanging the integration variables as done in previous appendix, we thus end up with the following integral,

$$\mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) = i^{\Delta_L+1} 2^{3-\Delta_L} \sqrt{\lambda} \int_{-1/2}^{1/2} dv_1 \int_{-1/2}^{1/2} dv_2 \frac{(1-4v_1^2)^{\Delta_L} (1-4v_2^2)^{\Delta_{L'}}}{(v_2 - v_1 + i0^+)^{\Delta_L+1}} \left(\frac{1+2v_2}{1-2v_2} \right)^{-i\nu}. \quad (\text{H.29})$$

Here the extra factor arises from the phase, i.e.

$$e^{-i\nu s_2} = \left(\frac{1+2v_2}{1-2v_2} \right)^{-i\nu}. \quad (\text{H.30})$$

We map the integration region to $[0, 1]$ by the following change of variables,

$$x = v_1 + 1/2, \quad t = v_2 + 1/2. \quad (\text{H.31})$$

Then we have,

$$\mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) = i^{\Delta_L+1} 2^{3+\Delta_L+2\Delta_{L'}} \sqrt{\lambda} \int_0^1 dx \int_0^1 dt \frac{x^{\Delta_L} (1-x)^{\Delta_L} t^{\Delta_{L'}-i\nu} (1-t)^{\Delta_{L'}+i\nu}}{(t-x+i0^+)^{\Delta_L+1}}. \quad (\text{H.32})$$

For integer d we have the following integral,

$$Q_d(z) = \frac{1}{2^{d+1}} \int_{-1}^1 du, \frac{(1-u^2)^d}{(z-u)^{d+1}}. \quad (\text{H.33})$$

Set $z = 2t - 1 + i0^+$, and $u = 2x - 1$. Then we have,

$$Q_d(2t - 1 + i0^+) = \frac{1}{2^{d+1}} \int_0^1 2dx \frac{(4x(1-x))^d}{(2(t-x)) + i0^+)^d} \quad (\text{H.34})$$

$$= \frac{1}{2} \int_0^1 dx \frac{x^d (1-x)^d}{(t-x+i0^+)^{d+1}}. \quad (\text{H.35})$$

Using this, we obtain,

$$\mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) = i^{\Delta_L+1} 2^{3+\Delta_L+2\Delta_{L'}} \sqrt{\lambda} \int_0^1 dt t^{\Delta_{L'}-i\nu} (1-t)^{\Delta_{L'}+i\nu} 2Q_{\Delta_L}(2t-1+i0^+). \quad (\text{H.36})$$

We also have the following relation between the Legendre function of the first and second kind,

$$Q_d(x+i0^+) = Q_d(x) - \frac{i\pi}{2} P_d(x) \quad (\text{H.37})$$

$$\frac{2}{\pi} \sin(\pi d) Q_d(x) = \cos(\pi d) P_d(x) - P_d(-x). \quad (\text{H.38})$$

Therefore,

$$2Q_d(x+i0^+) = \frac{\pi}{\sin(\pi d)} [e^{-i\pi d} P_d(x) - P_d(-x)] \quad (\text{H.39})$$

Substituting $x = 2t - 1$, we get,

$$2Q_d(2t-1+i0^+) = \frac{\pi}{\sin(\pi d)} [e^{-i\pi d} P_d(2t-1) - P_d(1-2t)] \quad (\text{H.40})$$

For the simplicity of notation, let's define,

$$a = \Delta_{L'} + 1 - i\nu, \quad b = \Delta_{L'} + 1 + i\nu, \quad \mathcal{C}_{\Delta_L, \Delta_{L'}} = i^{\Delta_L+1} 2^{3+\Delta_L+2\Delta_{L'}}. \quad (\text{H.41})$$

Thus we end up with,

$$\mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) = \mathcal{C}_{\Delta_L, \Delta_{L'}} \sqrt{\lambda} \frac{\pi}{\sin(\pi\Delta_L)} \left[e^{-i\pi\Delta_L} \int_0^1 dt t^{a-1} (1-t)^{b-1} P_{\Delta_L}(2t-1) - \int_0^1 dt t^{a-1} (1-t)^{b-1} P_{\Delta_L}(1-2t) \right] \quad (\text{H.42}) \quad \{\text{eq: Legendre}\}$$

For any general Δ_L , we have the hypergeometric representation of the Legendre function,

$$P_d(x) = {}_2F_1 \left(-d, d+1; 1; \frac{1-x}{2} \right) \quad (\text{H.43})$$

and the identity,

$$\int_0^1 dt t^{A-1} (1-t)^{B-1} {}_2F_1(\alpha, \beta; \gamma; t) = B(a, b) {}_3F_2 \left(\alpha, \beta, A; \gamma, A+B; 1 \right) \quad (\text{H.44})$$

Using the hypergeometric representation and the identity above, we get,

$$\begin{aligned} \mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) &= i^{\Delta_L+1} 2^{3+\Delta_L+2\Delta_{L'}} \sqrt{\lambda} \frac{\pi B(\Delta_{L'} + 1 - i\nu, \Delta_{L'} + 1 + i\nu)}{\sin(\pi\Delta_L)} \\ &\times \left[e^{-i\pi\Delta_L} {}_3F_2 \left(-\Delta_L, \Delta_L + 1, \Delta_{L'} + 1 + i\nu; 1, 2\Delta_{L'} + 2; 1 \right) - {}_3F_2 \left(-\Delta_L, \Delta_L + 1, \Delta_{L'} + 1 - i\nu; 1, 2\Delta_{L'} + 2; 1 \right) \right] \end{aligned} \quad (\text{H.45})$$

Define,

$$\mathcal{H}_{\Delta_L, \Delta_{L'}}(\nu) = \frac{\pi B(a, b)}{\sin(\pi\Delta_L)} \left[e^{-i\pi\Delta_L} {}_3F_2 \left(-\Delta_L, \Delta_L + 1, b; 1, 2\Delta_{L'} + 2; 1 \right) - {}_3F_2 \left(-\Delta_L, \Delta_L + 1, a; 1, 2\Delta_{L'} + 2; 1 \right) \right] \quad (\text{H.46}) \quad \{\text{eq: H_DeltaL}\}$$

Then we have,

$$\mathcal{J}_\lambda(\Delta_L, \Delta_{L'}) = i^{\Delta_L+1} 2^{3+\Delta_L+2\Delta_{L'}} e^{\pi\nu} \mathcal{H}_{\Delta_L, \Delta_{L'}}(\nu), \quad (\text{H.47})$$

$$\implies \mathcal{I}_\lambda(\Delta_L, \Delta_{L'}) = -\frac{i^{\Delta_L+1} 2^{3+\Delta_L+2\Delta_{L'}} e^{\pi\nu}}{4\pi} \left(\pi \mathcal{H}_{\Delta_L, \Delta_{L'}}(\nu) + \mathcal{H}'_{\Delta_L, \Delta_{L'}}(\nu) \right) \quad (\text{H.48})$$

We can now compute the derivative of μ_G with respect to n as $n \rightarrow 1$. We have,

$$\mathcal{I} \equiv \mathcal{I}_\lambda(\Delta_L, \Delta_{L'}), \quad \mathcal{H} \equiv \mathcal{H}_{\Delta_L, \Delta_{L'}}(\nu) \quad (\text{H.49})$$

$$\implies (1 - \lambda \partial_\lambda) \mathcal{I} = \frac{C_{\Delta_L, \Delta_{L'}} e^{\pi\nu}}{8\pi^2} \left[-2\pi (\pi \mathcal{H} + \mathcal{H}') + \pi^2 \mathcal{H} + 2\pi \mathcal{H}' + \mathcal{H}'' \right] \quad (\text{H.50})$$

and

$$\lambda(z\partial_z\mathcal{I})|_{z=\lambda^{-1}} = -\frac{C_{\Delta_L,\Delta_{L'}}e^{\pi\nu}}{8}\left[\mathcal{H}(-\nu) + \frac{2}{\pi}\mathcal{H}'(-\nu) + \frac{1}{\pi^2}\mathcal{H}''(-\nu)\right] \quad (\text{H.51})$$

Using the above two, we obtain,

$$\begin{aligned} \partial_n \mu_G^{\Psi|\Phi}(n, x, y; \Delta_{L'}, \Delta_L)|_{n=1} = & \frac{3i^{\Delta_L+1}2^{3+\Delta_L+2\Delta_{L'}}e^{\pi\nu}}{128\pi} \left[C_\Phi \left(-\mathcal{H}_{\Delta_L,\Delta_{L'}}(\nu) + \frac{\mathcal{H}''_{\Delta_L,\Delta_{L'}}(\nu)}{\pi^2} \right) \right. \\ & \left. - C_\Psi \left(\mathcal{H}_{\Delta_L,\Delta_{L'}}(-\nu) + \frac{2}{\pi}\mathcal{H}'_{\Delta_L,\Delta_{L'}}(-\nu) + \frac{\mathcal{H}''_{\Delta_L,\Delta_{L'}}(-\nu)}{\pi^2} \right) \right] \end{aligned} \quad (\text{H.52}) \quad \{\text{eq: analytic}\}$$

I Analytic continuation sub-leading term for coherent states

{appendix: su

We want to compute the sub-leading contribution of relative entropy of mixture of coherent states. There are two contributions, as given in (C.109). We compute those contributions separately in this appendix.

I.1 Analytic continuation of the contribution from $\hat{L}_p^{\text{sum}}$

We consider $\Delta_L = \Delta_{i\partial\phi} = 1$. Note that the analytic continuation of three point function (H.52) is not defined at $\Delta_L = 1$. If we work with $\mathcal{J}_\lambda(\Delta_L, \Delta_{L'})$, it can be shown that the square bracket in (H.42) vanishes for $\Delta_L \in \mathbb{Z}_{\geq 0}$. For $\Delta_L \in \mathbb{Z}$, we have,

$$P_{\Delta_L}(-x) = (-1)^{\Delta_L} P_{\Delta_L}(x). \quad (\text{I.1})$$

Thus, the square bracket in (H.42) evaluates to,

$$\begin{aligned} & \left[e^{-i\pi\Delta_L} \int_0^1 dt t^{a-1} (1-t)^{b-1} P_{\Delta_L}(2t-1) - \int_0^1 dt t^{a-1} (1-t)^{b-1} P_{\Delta_L}(1-2t) \right] \\ & \propto \left[(-1)^{\Delta_L} \int_0^1 dt t^{a-1} (1-t)^{b-1} P_{\Delta_L}(2t-1) \right. \\ & \quad \left. - (-1)^{\Delta_L} \int_0^1 dt t^{a-1} (1-t)^{b-1} P_{\Delta_L}(2t-1) \right] \end{aligned} \quad (\text{I.2})$$

$$= 0. \quad (\text{I.3})$$

Hence, we need to take limit properly to obtain the analytic continuation at integer Δ_L . For simplicity, denote,

$$F_z(\Delta) \equiv {}_3F_2 \left(\begin{matrix} -\Delta, \Delta+1, z \\ 1, 2M \end{matrix}; 1 \right) \quad (\text{I.4})$$

$$M \equiv \Delta_{L'} + 1 \quad (\text{I.5})$$

$$N(\Delta) \equiv e^{-i\pi\Delta} F_b(\Delta) - F_a(\Delta). \quad (\text{I.6})$$

Then,

$$\mathcal{J}_\lambda(\Delta, \Delta_{L'}) = i^{\Delta+1} 2^{3+\Delta+2\Delta_{L'}} \sqrt{\lambda} B(a, b) \frac{\pi N(\Delta)}{\sin(\pi \Delta)}. \quad (\text{I.7})$$

We use l'Hopital derivative at $\Delta = 1$,

$$\mathcal{J}_\lambda(1, \Delta_{L'}) = 2^{4+2\Delta_{L'}} \sqrt{\lambda} B(a, b) N'(1), \quad (\text{I.8})$$

$$= 2^{4+2\Delta_{L'}} \sqrt{\lambda} B(a, b) (i\pi F_b(1) - F'_b(1) - F'_a(1)). \quad (\text{I.9})$$

We know,

$$F_b(1) = -\frac{i\nu}{M}. \quad (\text{I.10})$$

We need to evaluate the derivatives of the general Hypergeometric functions. We use the integral representation,

$$P_\Delta(1-2t) = {}_2F_1(-\Delta, \Delta+1; 1; t). \quad (\text{I.11})$$

Then we have,

$$F_z(\Delta) = \frac{1}{B(z, 2M-z)} \int_0^1 dt t^{z-1} (1-t)^{2M-z-1} P_\Delta(1-2t). \quad (\text{I.12})$$

Therefore,

$$F'_z(1) = \frac{1}{B(z, 2M-z)} \int_0^1 dt t^{z-1} (1-t)^{2M-z-1} \partial_\Delta P_\Delta(1-2t) \Big|_{\Delta=1} \quad (\text{I.13})$$

$$= \frac{1}{B(z, 2M-z)} \int_0^1 dt t^{z-1} (1-t)^{2M-z-1} [(1-2t) \log(1-t) - 2t] \quad (\text{I.14})$$

where in the second line we plugged in the derivative of P_Δ at $\Delta = 1$. Now to evaluate the Beta-function integrals, we use the following integral,

$$\frac{1}{B(z, 2M-z)} \int_0^1 dt t^z (1-t)^{2M-z-1} \log(1-t) = \frac{z}{2M} [\psi(2M-z) - \psi(2M)], \quad (\text{I.15})$$

$$\frac{1}{B(z, 2M-z)} \int_0^1 dt t^z (1-t)^{2M-z-1} = \frac{z}{2M}. \quad (\text{I.16})$$

Hence,

$$F'_z(1) = \psi(2M-z) - \psi(2M) - \frac{z}{M} [\psi(2M-z) - \psi(2M+1)] - \frac{z}{M}. \quad (\text{I.17})$$

Therefore,

$$N'(1) = 2 - \frac{1}{M} + \frac{\pi\nu}{M} + \frac{i\nu}{M} [\psi(a) - \psi(b)]. \quad (\text{I.18})$$

Finally, we obtain,

$$\begin{aligned} \mathcal{J}_\lambda(1, \Delta_{L'}) &= 2^{4+2\Delta_{L'}} \sqrt{\lambda} B(\Delta_{L'} + 1 - i\nu, \Delta_{L'} + 1 + i\nu) \\ &\times \left[2 - \frac{1}{\Delta_{L'} + 1} + \frac{\pi\nu}{\Delta_{L'} + 1} + \frac{i\nu}{\Delta_{L'} + 1} (\psi(\Delta_{L'} + 1 - i\nu) - \psi(\Delta_{L'} + 1 + i\nu)) \right] \end{aligned} \quad (\text{I.19})$$

This gives the regular continuation at Δ_L . Now we can proceed to evaluate the contribution of the three-point function. We first note that

$$\partial_\nu \log B(\Delta_{L'} + 1 - i\nu, \Delta_{L'} + 1 + i\nu) = -i\psi(\Delta_{L'} + 1 - i\nu) + i\psi(\Delta_{L'} + 1 + i\nu), \quad (\text{I.20})$$

and therefore,

$$\partial_\nu B(\Delta_{L'} + 1 - i\nu, \Delta_{L'} + 1 + i\nu) \Big|_{\nu=0} = 0, \quad (\text{I.21})$$

$$\partial_\nu^2 B(\Delta_{L'} + 1 - i\nu, \Delta_{L'} + 1 + i\nu) \Big|_{\nu=0} = -2B(\Delta_{L'} + 1, \Delta_{L'} + 1) \psi^{(1)}(\Delta_{L'} + 1), \quad (\text{I.22})$$

where $\psi^{(1)}$ denotes the trigamma function. For the remaining factor in the square bracket, we use

$$\psi(\Delta_{L'} + 1 - i\nu) - \psi(\Delta_{L'} + 1 + i\nu) = -2i\nu \psi^{(1)}(\Delta_{L'} + 1) + O(\nu^3). \quad (\text{I.23})$$

Hence,

$$\begin{aligned} &\left[2 - \frac{1}{\Delta_{L'} + 1} + \frac{\pi\nu}{\Delta_{L'} + 1} + \frac{i\nu}{\Delta_{L'} + 1} (\psi(\Delta_{L'} + 1 - i\nu) - \psi(\Delta_{L'} + 1 + i\nu)) \right] \Big|_{\nu=0} \\ &= 2 - \frac{1}{\Delta_{L'} + 1}, \end{aligned} \quad (\text{I.24})$$

$$\begin{aligned} &\partial_\nu \left[2 - \frac{1}{\Delta_{L'} + 1} + \frac{\pi\nu}{\Delta_{L'} + 1} + \frac{i\nu}{\Delta_{L'} + 1} (\psi(\Delta_{L'} + 1 - i\nu) - \psi(\Delta_{L'} + 1 + i\nu)) \right] \Big|_{\nu=0} \\ &= \frac{\pi}{\Delta_{L'} + 1}, \end{aligned} \quad (\text{I.25})$$

$$\begin{aligned} &\partial_\nu^2 \left[2 - \frac{1}{\Delta_{L'} + 1} + \frac{\pi\nu}{\Delta_{L'} + 1} + \frac{i\nu}{\Delta_{L'} + 1} (\psi(\Delta_{L'} + 1 - i\nu) - \psi(\Delta_{L'} + 1 + i\nu)) \right] \Big|_{\nu=0} \\ &= \frac{4}{\Delta_{L'} + 1} \psi^{(1)}(\Delta_{L'} + 1). \end{aligned} \quad (\text{I.26})$$

Combining these terms, the first derivative of $\mathcal{J}_\lambda$ at $\nu = 0$ is

$$\partial_\nu \mathcal{J}_\lambda(1, \Delta_{L'}) \Big|_{\nu=0} = 2^{5+2\Delta_{L'}} \pi B(\Delta_{L'} + 1, \Delta_{L'} + 1), \quad (\text{I.27}) \quad \{\text{eq: del nu J}\}$$

while the second derivative is

$$\begin{aligned} \partial_\nu^2 \mathcal{J}_\lambda(1, \Delta_{L'}) \Big|_{\nu=0} &= 2^{4+2\Delta_{L'}} B(\Delta_{L'} + 1, \Delta_{L'} + 1) \left[\pi^2 \left(2 + \frac{1}{\Delta_{L'} + 1} \right) \right. \\ &\quad \left. + \left(-4 + \frac{6}{\Delta_{L'} + 1} \right) \psi^{(1)}(\Delta_{L'} + 1) \right]. \end{aligned} \quad (\text{I.28})$$

From (H.27), for the equal mixture, we obtain

$$(1 - \lambda \partial_\lambda) \mathcal{I}_\lambda(1, \Delta_{L'}) \Big|_{\lambda=1} = -\frac{1}{4\pi} \partial_\nu \mathcal{J}_\lambda(1, \Delta_{L'}) \Big|_{\nu=0} + \frac{1}{8\pi^2} \partial_\nu^2 \mathcal{J}_\lambda(1, \Delta_{L'}) \Big|_{\nu=0} \quad (\text{I.29})$$

$$= -\frac{2^{2\Delta_{L'}+1} B(\Delta_{L'} + 1, \Delta_{L'} + 1)}{\Delta_{L'} + 1} \left[2\Delta_{L'} + 1 + \frac{2(2\Delta_{L'} - 1)}{\pi^2} \psi^{(1)}(\Delta_{L'} + 1) \right]. \quad (\text{I.30})$$

Using (H.15), we can now apply the above analytic continuation to the $\hat{L}_p^{\text{sum}}$ contribution in Appendix C.5. For $\Delta_L = 1$, the function $G_p(n)$ becomes

$$G_p(n) = \sum_{r=1}^{p-1} \frac{1}{\sin(\pi r/n) \sin(\pi(p-r)/n)}. \quad (\text{I.31})$$

Using

$$\cot A + \cot B = \frac{\sin(A+B)}{\sin A \sin B}, \quad (\text{I.32})$$

and relabelling $r \rightarrow p-r$ in the second term, we obtain

$$G_p(n) = \frac{2}{\sin(\pi p/n)} \sum_{r=1}^{p-1} \cot\left(\frac{\pi r}{n}\right), \quad (\text{I.33}) \quad \{\text{eq: } G_p(n) \text{ id}\}$$

or equivalently,

$$2 \sum_{r=1}^{p-1} \cot\left(\frac{\pi r}{n}\right) = \sin\left(\frac{\pi p}{n}\right) G_p(n). \quad (\text{I.34})$$

Then,

$$\delta^2 \zeta_n \sum_{p=1}^{n-1} (n-p) \lambda^p \left(\frac{\sin(\pi \epsilon/n)}{\sin(\pi p/n)} \right)^{2\Delta_{L'}} \sum_{r=1}^{p-1} \cot\left(\frac{\pi r}{n}\right) = \frac{\delta^2 \zeta_n}{2} \sin^{2\Delta_{L'}} \left(\frac{\pi \epsilon}{n} \right) (n - \lambda \partial_\lambda) A_n(\lambda; \Delta_{L'}, 1). \quad (\text{I.35})$$

For the averaged binary mixture, $\lambda_n(\epsilon) = D_n(\epsilon)$ and $\lambda|_{n=1} = 1$. Since $A_1(\lambda; \Delta_{L'}, 1) = 0$ for arbitrary λ , the derivatives of λ_n and of the prefactor do not contribute at $n = 1$. Hence,

$$\begin{aligned} \left[\log 2 - S \left(\rho_\alpha \left\| \frac{\rho_\alpha + \rho_\beta}{2} \right\| \right) \right] \Big|_{\hat{L}^{\text{sum}}} &= \delta^2 \sin^{2\Delta_{L'}+1}(\pi \epsilon) (1 - \lambda \partial_\lambda) \partial_n A_n(\lambda; \Delta_{L'}, 1) \Big|_{\lambda=1}^{n=1} \\ &= -\frac{3\delta^2 2^{2\Delta_{L'}-3} B(\Delta_{L'} + 1, \Delta_{L'} + 1)}{\pi(\Delta_{L'} + 1)} \\ &\quad \times \left[2\Delta_{L'} + 1 + \frac{2(2\Delta_{L'} - 1)}{\pi^2} \psi^{(1)}(\Delta_{L'} + 1) \right] \sin^{2\Delta_{L'}+1}(\pi \epsilon). \end{aligned} \quad (\text{I.36})$$

In the $\epsilon \rightarrow 0$ limit, this becomes

$$\left[\log 2 - S \left(\rho_\alpha \left\| \frac{\rho_\alpha + \rho_\beta}{2} \right\| \right) \right] \Big|_{\hat{L}^{\text{sum}}} = - \frac{3\delta^2 2^{2\Delta_{L'}-3} B(\Delta_{L'} + 1, \Delta_{L'} + 1)}{\pi(\Delta_{L'} + 1)} \\ \times \left[2\Delta_{L'} + 1 + \frac{2(2\Delta_{L'} - 1)}{\pi^2} \psi^{(1)}(\Delta_{L'} + 1) \right] (\pi\epsilon)^{2\Delta_{L'}+1} + \dots \quad (\text{I.37})$$

Note that, the above result is symmetric under $\alpha \leftrightarrow \beta$. This was a result particular to coherent states. It need not be true for any general CFT. We next consider the contribution from $\hat{L}_p^{\text{div}}$.

I.2 Analytic continuation of the divergent term contribution

We now consider the contribution from $\hat{L}_p^{\text{div}}$. Using (I.33) with $p \rightarrow n - p$, together with

$$\sum_{r=1}^{n-1} \cot \left(\frac{\pi r}{n} \right) = 0, \quad (\text{I.38})$$

we obtain

$$G_{n-p}(n) - G_p(n) = \frac{2 \cot(\pi p/n)}{\sin(\pi p/n)}. \quad (\text{I.39})$$

The contribution from $\hat{L}_p^{\text{div}}$ is therefore

$$-\log \mathcal{F}_n|_{\hat{L}^{\text{div}}} = \delta^2 \frac{\pi\epsilon}{n} \sin^{2\Delta_{L'}} \left(\frac{\pi\epsilon}{n} \right) \sum_{p=1}^{n-1} (n-p) \lambda^p \frac{\cot(\pi p/n)}{\sin^{2\Delta_{L'}}(\pi p/n)}, \quad (\text{I.40})$$

where $2\Delta_{L'} = \delta^2$. On the other hand, from the definition of A_n ,

$$\lambda^n z \partial_z A_n(z; \Delta_{L'}, 1)|_{z=\lambda^{-1}} - (n - \lambda \partial_\lambda) A_n(\lambda; \Delta_{L'}, 1) \\ = \sum_{p=1}^{n-1} (n-p) \lambda^p \frac{G_{n-p}(n) - G_p(n)}{\sin^{2\Delta_{L'}-1}(\pi p/n)} \\ = 2 \sum_{p=1}^{n-1} (n-p) \lambda^p \frac{\cot(\pi p/n)}{\sin^{2\Delta_{L'}}(\pi p/n)}. \quad (\text{I.41})$$

Hence, the divergent contribution can be expressed entirely in terms of the replica sum A_n as

$$-\log \mathcal{F}_n|_{\hat{L}^{\text{div}}} = \frac{\delta^2}{2} \frac{\pi\epsilon}{n} \sin^{2\Delta_{L'}} \left(\frac{\pi\epsilon}{n} \right) \left[\lambda^n z \partial_z A_n(z; \Delta_{L'}, 1)|_{z=\lambda^{-1}} - (n - \lambda \partial_\lambda) A_n(\lambda; \Delta_{L'}, 1) \right]. \quad (\text{I.42})$$

For the averaged binary mixture, $\lambda|_{n=1} = 1$. Since $A_1(\lambda; \Delta_{L'}, 1) = 0$ for arbitrary λ , we obtain

$$\left[\log 2 - S \left(\rho_\alpha \left\| \frac{\rho_\alpha + \rho_\beta}{2} \right\| \right) \right] \Big|_{\hat{L}^{\text{div}}} = \frac{\delta^2}{2} (\pi\epsilon) \sin^{2\Delta_{L'}}(\pi\epsilon) \left[\lambda \partial_\lambda - (1 - \lambda \partial_\lambda) \right] \partial_n A_n(\lambda; \Delta_{L'}, 1) \Big|_{\lambda=1}. \quad (\text{I.43})$$

We will also need $\partial_n A_n$ itself at $\lambda = 1$. From (H.15), (H.27) and (I.27), we have

$$\begin{aligned} \partial_n A_n(1; \Delta_{L'}, 1) \Big|_{n=1} &= -\frac{3}{64\pi^2} \partial_\nu \mathcal{J}_\lambda(1, \Delta_{L'}) \Big|_{\nu=0} \\ &= -\frac{3(2^{2\Delta_{L'}-1})}{\pi} B(\Delta_{L'} + 1, \Delta_{L'} + 1). \end{aligned} \quad (\text{I.44})$$

Combining this with the result for $(1 - \lambda \partial_\lambda) \partial_n A_n$ obtained above, we find

$$\begin{aligned} \left[\log 2 - S \left(\rho_\alpha \left\| \frac{\rho_\alpha + \rho_\beta}{2} \right\| \right) \right] \Big|_{\hat{L}^{\text{div}}} &= -\frac{3\delta^2 2^{2\Delta_{L'}-3} B(\Delta_{L'} + 1, \Delta_{L'} + 1)}{\pi(\Delta_{L'} + 1)} \\ &\quad \times \left[1 - \frac{2(2\Delta_{L'} - 1)}{\pi^2} \psi^{(1)}(\Delta_{L'} + 1) \right] (\pi\epsilon)^{2\Delta_{L'}+1} + \dots \end{aligned} \quad (\text{I.45})$$

I.3 Sub-leading contribution of relative entropy of coherent states

We now apply the analytic continuations above to the averaged mixture of coherent states,

$$S \left(\rho_\alpha \left\| \frac{\rho_\alpha + \rho_\beta}{2} \right\| \right), \quad \delta = \alpha - \beta. \quad (\text{I.46})$$

For the current contribution,

$$\Delta_L = 1, \quad 2\Delta_{L'} = \delta^2. \quad (\text{I.47})$$

Using the result obtained above, the contribution from $\hat{L}_p^{\text{sum}}$ is

$$\begin{aligned} \left[\log 2 - S \left(\rho_\alpha \left\| \frac{\rho_\alpha + \rho_\beta}{2} \right\| \right) \right] \Big|_{\hat{L}^{\text{sum}}} &= -\frac{3\delta^2 2^{\delta^2-3} B \left(1 + \frac{\delta^2}{2}, 1 + \frac{\delta^2}{2} \right)}{\pi \left(1 + \frac{\delta^2}{2} \right)} \\ &\quad \times \left[\delta^2 + 1 + \frac{2(\delta^2 - 1)}{\pi^2} \psi^{(1)} \left(1 + \frac{\delta^2}{2} \right) \right] (\pi\epsilon)^{\delta^2+1} + \dots \end{aligned} \quad (\text{I.48})$$

Similarly, the contribution from $\hat{L}_p^{\text{div}}$ is

$$\begin{aligned} \left[\log 2 - S \left(\rho_\alpha \left\| \frac{\rho_\alpha + \rho_\beta}{2} \right\| \right) \right] \Big|_{\hat{L}^{\text{div}}} &= -\frac{3\delta^2 2^{\delta^2-3} B \left(1 + \frac{\delta^2}{2}, 1 + \frac{\delta^2}{2} \right)}{\pi \left(1 + \frac{\delta^2}{2} \right)} \\ &\quad \times \left[1 - \frac{2(\delta^2 - 1)}{\pi^2} \psi^{(1)} \left(1 + \frac{\delta^2}{2} \right) \right] (\pi\epsilon)^{\delta^2+1} + \dots \end{aligned} \quad (\text{I.49})$$

Adding the two contributions, the terms proportional to the trigamma function cancel, and we obtain

$$\begin{aligned} \left[\log 2 - S \left(\rho_\alpha \left\| \frac{\rho_\alpha + \rho_\beta}{2} \right\| \right) \right] \Big|_{\text{sub.}} &= -\frac{3\delta^2 2^{\delta^2-2}}{\pi} B \left(1 + \frac{\delta^2}{2}, 1 + \frac{\delta^2}{2} \right) (\pi\epsilon)^{\delta^2+1} + \dots \\ &= -\frac{3\delta^2}{2\pi} \frac{\sqrt{\pi} \Gamma \left(1 + \frac{\delta^2}{2} \right)}{4\Gamma \left(\frac{3}{2} + \frac{\delta^2}{2} \right)} (\pi\epsilon)^{\delta^2+1} + \dots \end{aligned} \quad (\text{I.50})$$

Including the leading two-point contribution, we finally obtain

$$S \left(\rho_\alpha \left\| \frac{\rho_\alpha + \rho_\beta}{2} \right\| \right) = \log 2 - \frac{\sqrt{\pi} \Gamma \left(1 + \frac{\delta^2}{2} \right)}{4\Gamma \left(\frac{3}{2} + \frac{\delta^2}{2} \right)} (\pi\epsilon)^{\delta^2} + \frac{3\delta^2}{2\pi} \frac{\sqrt{\pi} \Gamma \left(1 + \frac{\delta^2}{2} \right)}{4\Gamma \left(\frac{3}{2} + \frac{\delta^2}{2} \right)} (\pi\epsilon)^{\delta^2+1} + \dots \quad (\text{I.51})$$

Thus, up to the first sub-leading order, the result depends only on $\delta^2 = (\alpha - \beta)^2$ and is therefore symmetric under $\alpha \leftrightarrow \beta$. For $\delta^2 > 1$, the broken-word contribution starts at $O(\epsilon^{2\delta^2})$ and does not affect the first sub-leading term. For $\delta^2 = 1$, it contributes at the same order as the terms above and must be included separately.

Symmetry of coherent states relative entropy: For a generic CFT, the three-point function contribution derived in the previous appendix need not be symmetric under the exchange $\Psi \leftrightarrow \Phi$, since it consist of independent structures,

$$C_\Phi^L G_p(n) + C_\Psi^L G_{n-p}(n) \quad (\text{I.52})$$

whose analytic continuation are in general different. However, for coherent states, the current algebra has an additional simplification. Writing,

$$\delta = \alpha - \beta, \quad s = \alpha + \beta, \quad (\text{I.53})$$

the Ward identity gives,

$$\frac{\langle V_\delta V_{-\delta} J \rangle}{\langle V_\delta V_{-\delta} \rangle} \propto \delta, \quad (\text{I.54})$$

while the ordinary $\alpha\alpha$ and $\beta\beta$ OPEs carry current coefficients proportional to α and β respectively. In addition, the two mixing OPEs contain current descendants proportional to s . Combining all current insertions, we find,

$$\begin{aligned} \hat{L}_p &= 2\delta^2 \frac{\pi\epsilon}{n} \sum_{r=1}^{p-1} \cot \left(\frac{\pi r}{n} \right) + \delta^2 \frac{\pi\epsilon}{n} \cot \left(\frac{\pi p}{n} \right) \\ &= \frac{\delta^2}{2} \frac{\pi\epsilon}{n} \sin \left(\frac{\pi p}{n} \right) \left[G_p(n) + G_{n-p}(n) \right] \end{aligned} \quad (\text{I.55})$$

$$\propto \delta^2 \quad (\text{I.56})$$

Thus, the two replica structures appear with the same effective coefficient and the dependence

J Contribution of descendants to Relative entropy

{Descendant con

J.1 Level 2 descendant

The two-point function can be first taylor expanded, to obtain,

$$S(\rho_\Psi \parallel \rho_\Phi) \Big|_{\langle \mathcal{O}_L \mathcal{O}_L \rangle} = \sum_L (C_\Psi^L - C_\Phi^L)^2 f_{LL} \sin^{2\Delta_L}(\pi x) \quad (\text{J.1})$$

$$= \sum_L (C_\Psi^L - C_\Phi^L)^2 f_{LL}(\pi x)^{2\Delta_L} \left[1 - \frac{\Delta_L}{3}(\pi x)^2 + \dots \right]. \quad (\text{J.2})$$

Next, we consider the standard endpoint OPE [8],

$$\begin{aligned} \Psi(z, \bar{z}) \Psi^*(0, 0) = C_\Psi^L z^{h_L - 2h_\Psi} \bar{z}^{\bar{h}_L - 2\bar{h}_\Psi} \left[\mathcal{O}_L + \beta_1 z L_{-1} \mathcal{O}_L + \bar{\beta}_1 \bar{z} \bar{L}_{-1} \mathcal{O}_L + z^2 (\beta_{11} L_{-1}^2 + \beta_2 L_{-2}) \mathcal{O}_L \right. \\ \left. + \bar{z}^2 (\bar{\beta}_{11} \bar{L}_{-1}^2 + \bar{\beta}_2 \bar{L}_{-2}) \mathcal{O}_L + z \bar{z} \beta_1 \bar{\beta}_1 L_{-1} \bar{L}_{-1} \mathcal{O}_L + \dots \right] \end{aligned} \quad (\text{J.3})$$

For our case, where we have non-mixing OPE, we have,

$$\beta_1 = \bar{\beta}_1 = \frac{1}{2}, \quad (\text{J.4})$$

$$2(2h_L + 1)\beta_{11} + 3\beta_2 = \frac{h_L + 1}{2}, \quad (\text{J.5})$$

$$6h_L\beta_{11} + \left(4h_L + \frac{c}{2}\right)\beta_2 = h_L + h_\Psi \quad (\text{J.6})$$

Now, we translate the operators to the midpoint, to obtain the midpoint OPEs. Note that, again the level-one term disappears at the midpoint. This can be seen, when we expand the first primary $\mathcal{O}_L$ and the level-one descendant,

$$\mathcal{O}_L(-l/2) \simeq \mathcal{O}_L - \frac{l}{2} L_{-1} \mathcal{O}_L + \dots, \quad (\text{J.7})$$

$$L_{-1} \mathcal{O}_L \left(-\frac{l}{2} \right) \simeq L_{-1} \mathcal{O}_L + \dots \quad (\text{J.8})$$

where $\dots$ refer to anti-holomorphic and level-two descendants. Hence, the coefficient of $L_{-1} \mathcal{O}_L$, and the anti-holomorphic part is given by,

$$-\frac{1}{2} + \beta_1 = 0 \quad (\text{J.9})$$

which agrees with the result of the next subsection, that the odd level descendants vanish in midpoint OPEs. We can now translate the level-two descendants,

$$\left[\left(\beta_{11} - \frac{1}{8} \right) L_{-1}^2 + \beta_2 L_{-2} \right] \quad (\text{J.10})$$

where the $-\frac{1}{8}$ comes from translating the primary.

For the $L_{-1}\bar{L}_{-1}$ descendant, we have contribution from translations of primary as well as level-one descendants. They all add up to zero. Hence we are left with the following midpoint OPE at the second-level order,

$$\Psi(l/2)\Psi^*(-l/2) \simeq C_\Psi^L \left[\mathcal{O}_L + l^2 \left(\left(\beta_{11} - \frac{1}{8} \right) L_{-1}^2 + \beta_2 L_{-2} \right) \mathcal{O}_L + \bar{l}^2 \left(\left(\bar{\beta}_{11} - \frac{1}{8} \right) \bar{L}_{-1}^2 + \bar{\beta}_2 \bar{L}_{-2} \right) \mathcal{O}_L + \dots \right] \quad (\text{J.11})$$

Now we can compute the contribution to the relative entropy from the level-two descendant terms, such as $\langle \mathcal{O}_L L_{-1}^2 \mathcal{O}_L \rangle$ and $\langle \mathcal{O}_L L_{-2} \mathcal{O}_L \rangle$. The first one is easy to compute,

$$\langle \mathcal{O}_L(z, \bar{z}) L_{-1}^2 \mathcal{O}_L(0, 0) \rangle = \partial^2 \langle \mathcal{O}_L \mathcal{O}_L \rangle \quad (\text{J.12})$$

$$= \frac{2h_L(2h_L + 1)}{z^{2h_L+2} \bar{z}^{2h_L}} \quad (\text{J.13})$$

The other level-two descendant can be computed as follows,

$$L_{-2} \mathcal{O}_L(0) \oint_0 \frac{dw}{2\pi i} \frac{T(w) \mathcal{O}_L(0)}{w}. \quad (\text{J.14})$$

Using the conformal Ward identity,

$$T(w) \mathcal{O}_L(z, \bar{z}) \mathcal{O}_L(0, 0) = \left[\frac{h_L}{(w-z)^2} + \frac{1}{(w-z)} \partial_z + \frac{h_L}{w^2} + \frac{1}{w} \partial_0 \right] \langle \mathcal{O}_L \mathcal{O}_L \rangle \quad (\text{J.15})$$

Now we multiply $1/w$ in the above equation, and take the residue at $w = 0$. The non-vanishing contributions come from the operators at z . Therefore, we obtain,

$$\langle \mathcal{O}_L(z, \bar{z}) L_{-2} \mathcal{O}_L \rangle = \frac{3h_L}{z^{2h_L+2} \bar{z}^{2h_L}} \quad (\text{J.16})$$

Now adding the contributions from the level-two descendants, with their OPE coefficients, we obtain,

$$2 \left[\left(\beta_{11} - \frac{1}{8} \right) \langle \mathcal{O}_L L_{-1}^2 \mathcal{O}_L \rangle + \beta_2 \langle \mathcal{O}_L L_{-2} \mathcal{O}_L \rangle \right] = \frac{\frac{h_L}{2}}{z^{2h_L+2} \bar{z}^{2h_L}}. \quad (\text{J.17})$$

The factor of 2 is due to the two different orientation, i.e. $\langle \mathcal{O}_L(z) L_{-1}^2 \mathcal{O}(w) \rangle$ and $\langle \mathcal{O}_L(w) L_{-1}^2 \mathcal{O}(z) \rangle$. Thus, the contribution to the relative entropy from the descendants is given by,

$$S(\rho_\Psi \| \rho_\Phi) \Big|_{[\mathcal{O}_L \mathcal{O}_L]_{\text{level-2}}} = \sum_L \frac{\Delta_L}{2} (C_\Psi^L - C_\Phi^L)^2 \frac{\sqrt{\pi} \Gamma(\Delta_L + 2)}{4\Gamma(\Delta_L + 5/2)} (\pi x)^{2\Delta_L+2} + \dots \quad (\text{J.18})$$

$$= \sum_L \frac{\Delta_L(\Delta_L + 1)}{2\Delta_L + 3} f_{\Delta_L, \Delta_L} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L+2} + \dots \quad (\text{J.19})$$

Adding the subleading contribution from the Taylor expansion of the two-point function, we finally obtain the contribution of the terms of the order $x^{2\Delta_L+2}$,

$$S(\rho_\Psi \| \rho_\Phi) \Big|_{[\mathcal{O}_L \mathcal{O}_L]_{\text{level-2}}} = \frac{\Delta_L^2}{3(2\Delta_L + 3)} f_{\Delta_L, \Delta_L} (C_\Psi^L - C_\Phi^L)^2 (\pi x)^{2\Delta_L+2} + \dots \quad (\text{J.20})$$

J.2 Odd level descendants

In this subsection, using the method used in [8], we show that the odd level descendants do not contribute to the relative entropy. This is an artefact of using the midpoint OPEs. The idea would be to compare the OPE coefficients β ($\tilde{\beta}$ in our case, which are the coefficients obtained after shifting operators to the midpoint) with the coefficients determined using conformal symmetry. Consider the three-point function,

$$\langle \Psi(l/2) \Psi^*(-l/2) \mathcal{O}_L(w) \rangle = C_\Psi^L l^{h_L-2h_\Psi} \left(w^2 - \frac{l^2}{4} \right)^{-h_L} \quad (\text{J.21})$$

$$\simeq C_\Psi^L l^{h_L-2h_\Psi} w^{-2h_L} \left[1 + \frac{h_L}{4} \frac{l^2}{w^2} + \frac{h_L(h_L+1)}{32} \frac{l^4}{w^4} + \dots \right] \quad (\text{J.22}) \quad \{\text{eq: 3 point}\}$$

Now compare this the midpoint OPE,

$$\begin{aligned} \Psi(l/2) \Psi^*(-l/2) = C_\Psi^L l^{h_L-2h_\Psi} & \left[\mathcal{O}_L(0) + l \tilde{\beta}_1 L_{-1} \mathcal{O}_L(0) + l^2 \left(\tilde{\beta}_2 L_{-2} + \tilde{\beta}_{11} L_{-1}^2 \right) \mathcal{O}_L(0) \right. \\ & \left. + l^3 \left(\tilde{\beta}_{111} L_{-1}^3 + \tilde{\beta}_{21} L_{-2} L_{-1} + \tilde{\beta}_3 L_{-3} \right) \mathcal{O}_L(0) + \dots \right] \quad (\text{J.23}) \end{aligned}$$

Now, taking the expectation value of the above OPE with $\mathcal{O}_L(w)$, we obtain,

$$\begin{aligned} \langle \Psi(l/2) \Psi^*(-l/2) \mathcal{O}_L(w) \rangle = C_\Psi^L l^{h_L-2h_\Psi} & \left[\langle \mathcal{O}_L(0) \mathcal{O}_L(w) \rangle + l \langle \tilde{\beta}_1 L_{-1} \mathcal{O}_L(0) \mathcal{O}_L(w) \rangle \right. \\ & + l^2 \langle \left(\tilde{\beta}_2 L_{-2} + \tilde{\beta}_{11} L_{-1}^2 \right) \mathcal{O}_L(0) \mathcal{O}_L(w) \rangle \\ & \left. + l^3 \langle \left(\tilde{\beta}_{111} L_{-1}^3 + \tilde{\beta}_{21} L_{-2} L_{-1} + \tilde{\beta}_3 L_{-3} \right) \mathcal{O}_L(0) \mathcal{O}_L(w) \rangle + \dots \right] \quad (\text{J.24}) \end{aligned}$$

However, from the earlier three-point function computation in (J.22), we only had even powers of l . This suggests that the net contribution of every odd level descendant must vanish.

K Large Interval for mixture of coherent states (Irrelevant, since this is just a check using exact 2n point function)

K.1 Exact analytical calculation

Leading and sub-leading terms For chiral vertex operators, the exact 2n-point correlation function is given by,

$$\left\langle \prod_{l=1}^{2n} V_{q_l}(z_l) \right\rangle = \prod_{l < r} (z_l - z_r)^{q_l q_r}. \quad (\text{K.1})$$

Then, for the replica insertions, we have a word, $(a_0, a_1, \dots, a_{n-1})$ for which the 2n-point function is given by,

$$\left| \left\langle \prod_{j=0}^{n-1} V_{a_j}(z_j^+) V_{-a_j}(z_j^-) \right\rangle \right| = \prod_{j=0}^{n-1} \left[2 \sin \left(\frac{\pi x}{n} \right) \right]^{-a_j^2} \prod_{0 \leq j < k \leq n-1} \left[\frac{\left(2 \sin \frac{\pi(k-j)}{n} \right)^2}{\left| 2 \sin \frac{\pi(k-j-x)}{n} \right| \left| 2 \sin \frac{\pi(k-j+x)}{n} \right|} \right]^{a_j a_k} \quad (\text{K.2})$$

We can also write the above expression as,

$$\begin{aligned} \left| \left\langle \prod_{j=0}^{n-1} V_{a_j}(z_j^+) V_{-a_j}(z_j^-) \right\rangle \right| &= \prod_{j=0}^{n-1} \left[2 \sin \left(\frac{\pi x}{n} \right) \right]^{-a_j^2} \prod_{0 \leq j < k \leq n-1} \left[2 \sin \left(\frac{\pi(k-j)}{n} \right) \right]^{2a_j a_k} \\ &\times \prod_{\substack{0 \leq j, k \leq n-1 \\ j \neq k}} \left| 2 \sin \left(\frac{\pi(k-j-1+\epsilon)}{n} \right) \right|^{-a_j a_k}. \end{aligned} \quad (\text{K.3})$$

K.1.1 Ratio of the two words

Define the following,

$$\mathcal{P}_r(x) \equiv \frac{(2 \sin \frac{\pi r}{n})^2}{|2 \sin \frac{\pi(r-x)}{n}| |2 \sin \frac{\pi(r+x)}{n}|} \quad (\text{K.4})$$

Then the contiguous word can be written as,

$$\begin{aligned} |\mathcal{C}_c(n, x)| &= \left[2 \sin \left(\frac{\pi x}{n} \right) \right]^{-[2\beta^2 + (n-2)\alpha^2]} \mathcal{P}_1(n, x)^{\beta^2} \prod_{k=2}^{n-1} \mathcal{P}_k(n, x)^{\alpha\beta} \prod_{k=2}^{n-1} \mathcal{P}_{k-1}(n, x)^{\alpha\beta} \\ &\times \prod_{2 \leq j < k \leq n-1} \mathcal{P}_{k-j}(n, x)^{\alpha^2}. \end{aligned} \quad (\text{K.5})$$

Similarly, the broken word is given by,

$$\begin{aligned} |\mathcal{C}_b^{(m)}(n, x)| &= \left[2 \sin \left(\frac{\pi x}{n} \right) \right]^{-[2\beta^2 + (n-2)\alpha^2]} \mathcal{P}_m(n, x)^{\beta^2} \prod_{\substack{k=1 \\ k \neq m}}^{n-1} \mathcal{P}_k(n, x)^{\alpha\beta} \prod_{j=1}^{m-1} \mathcal{P}_{m-j}(n, x)^{\alpha\beta} \\ &\times \prod_{k=m+1}^{n-1} \mathcal{P}_{k-m}(n, x)^{\alpha\beta} \prod_{\substack{0 \leq j < k \leq n-1 \\ j, k \neq 0, m}} \mathcal{P}_{k-j}(n, x)^{\alpha^2}. \end{aligned} \quad (\text{K.6})$$

The first term in both the correlators cancel exactly. However, rather than comparing the large product separately, we compare the individual pairs in the two words. All pairs not involving sites 1 or m are exactly the same in both correlators and cancel. For the remaining pairs,

Pair (0,1) Ratio is given by

$$\frac{\mathcal{P}_1(x)^{\alpha\beta}}{\mathcal{P}_1(x)^{\beta^2}} = \mathcal{P}_1(x)^{\alpha\beta - \beta^2} \quad (\text{K.7})$$

Pair (0,m) Ratio is given by

$$\frac{\mathcal{P}_m(x)^{\beta^2}}{\mathcal{P}_m(x)^{\alpha\beta}} = \mathcal{P}_m(x)^{\beta^2 - \alpha\beta} \quad (\text{K.8})$$

Pair (1,m) Ratio is given by

$$\frac{\mathcal{P}_{m-1}(x)^{\alpha\beta}}{\mathcal{P}_{m-1}(x)^{\alpha\beta}} = 1 \quad (\text{K.9})$$

Pair involving 1 or m and the remaining α sites Let

$$S = \{2, \dots, n-1\} \setminus \{m\} \quad (\text{K.10})$$

The remaining ratio is,

$$\prod_{k \in S} \left(\frac{\mathcal{P}_{|m-k|}(x)}{\mathcal{P}_{|1-k|}(x)} \right)^{\alpha\beta - \alpha^2} \quad (\text{K.11})$$

Therefore we have,

$$\frac{\mathcal{C}_b^{(m)}(n, x)}{\mathcal{C}_c(n, x)} = \mathcal{P}_1(x)^{\alpha\beta - \beta^2} \mathcal{P}_m(x)^{\beta^2 - \alpha\beta} \prod_{k=2, k \neq m}^{n-1} \left(\frac{\mathcal{P}_{|m-k|}(x)}{\mathcal{P}_{|1-k|}(x)} \right)^{\alpha\beta - \alpha^2} \quad (\text{K.12})$$

We compute the numerator and denominator of the product in the right hand side separately. The numerator is given by,

$$\prod_{k=2, k \neq m}^{n-1} \mathcal{P}_{|m-k|}(x) = \left(\prod_{r=1}^{m-2} \mathcal{P}_r \right) \left(\prod_{r=1}^{n-1-m} \mathcal{P}_r \right) \quad (\text{K.13})$$

Similarly the denominator is given by,

$$\prod_{k=2, k \neq m}^{n-1} \mathcal{P}_{|1-k|} = \frac{\prod_{r=1}^{n-2} \mathcal{P}_r}{\mathcal{P}_{m-1}} \quad (\text{K.14})$$

$$= \frac{\mathcal{P}_{m-1} \left(\prod_{r=1}^{m-2} \mathcal{P}_r \right) \left(\prod_{r=m}^{n-2} \mathcal{P}_r \right)}{\mathcal{P}_{m-1}} \quad (\text{K.15})$$

$$= \left(\prod_{r=1}^{m-2} \mathcal{P}_r \right) \left(\prod_{r=m}^{n-2} \mathcal{P}_r \right) \quad (\text{K.16})$$

$$= \left(\prod_{r=1}^{m-2} \mathcal{P}_r \right) \left(\prod_{s=2}^{n-m} \mathcal{P}_s \right) \quad (\text{K.17})$$

where in the last step we used the symmetry $\mathcal{P}_{n-r} = \mathcal{P}_r$. Consequently,

$$\prod_{k=2, k \neq m}^{n-1} \left(\frac{\mathcal{P}_{|m-k|}(x)}{\mathcal{P}_{|1-k|}(x)} \right) = \frac{\mathcal{P}_1(x)}{\mathcal{P}_m(x)} \quad (\text{K.18})$$

Therefore,

$$\left| \frac{\mathcal{C}_b^{(m)}(n, x)}{\mathcal{C}_c(n, x)} \right| = \mathcal{P}_1(x)^{\alpha\beta-\beta^2} \mathcal{P}_m(x)^{\beta^2-\alpha\beta} \left(\frac{\mathcal{P}_1(x)}{\mathcal{P}_m(x)} \right)^{\alpha\beta-\alpha^2} \quad (\text{K.19})$$

$$= \left(\frac{\mathcal{P}_m(x)}{\mathcal{P}_1(x)} \right)^{(\alpha-\beta)^2} \quad (\text{K.20})$$

$$= \left(\frac{(2 \sin \frac{\pi m}{n})^2 |2 \sin \frac{\pi \epsilon}{n}| |2 \sin \frac{\pi(2-\epsilon)}{n}|}{(2 \sin \frac{\pi}{n})^2 |2 \sin \frac{\pi(m-1+\epsilon)}{n}| |2 \sin \frac{\pi(m+1-\epsilon)}{n}|} \right)^{(\alpha-\beta)^2} \quad (\text{K.21})$$

$$\underset{m \neq \{1, n-1\}}{\simeq} \left(2 \sin \frac{\pi \epsilon}{n} \right)^{(\alpha-\beta)^2} \left(\frac{(2 \sin \frac{\pi m}{n})^2 (2 \sin \frac{2\pi}{n})}{(2 \sin \frac{\pi}{n})^2 \left(2 \sin \frac{\pi(m-1)}{n} \right) \left(2 \sin \frac{\pi(m+1)}{n} \right)} \right)^{(\alpha-\beta)^2}, \quad (\text{K.22})$$

K.2 Using OPEs

Leading and sub-leading terms From the denominator of (3.93), we assumed that the leading order contribution comes from the terms which have a string of $\Phi\Phi^*$ and a string of $\Psi\Psi^*$. That gave us mixing OPEs at two sites, while non-mixing OPEs at the remaining $n-2$ sites. For simplicity, consider the averaged mixture, i.e. $y = 1/2$ case. For $p = 2$ case, which corresponds to the insertions of two $\Phi\Phi^*$ and $(n-2)$ $\Psi\Psi^*$ operators, we will compare two terms in the denominator. One would correspond to a single string of $\Phi\Phi^*$ connected to a string of $\Psi\Psi^*$. It will have 2 mixing OPE and $n-2$ non-mixing OPEs that include $n-3$ $\Psi\Psi^*$ and one $\Phi\Phi^*$ OPE. The other term would correspond to the case where the $\Phi\Phi^*$ insertions are separated, and hence there are 4 mixing OPEs, while $n-4$ $\Psi\Psi^*$ non-mixing OPEs. Particularly, we focus on chiral Vertex operators. So up to the leading order, we have the following OPEs,

$$V_\alpha V_{-\alpha} = \zeta_n^{-\alpha^2} (1 + \alpha \zeta_n J + \dots) \quad (\text{K.23})$$

$$V_\beta V_{-\beta} = \zeta_n^{-\beta^2} (1 + \beta \zeta_n J + \dots) \quad (\text{K.24})$$

$$V_\alpha V_{-\beta} = \zeta_n^{-\alpha\beta} (V_{\alpha-\beta} + \dots) \quad (\text{K.25})$$

$$V_\beta V_{-\alpha} = \zeta_n^{-\alpha\beta} (V_{\beta-\alpha} + \dots) \quad (\text{K.26})$$

K.2.1 Contiguous $p = 2$ word

We insert $V_\beta V_{-\beta}$ at z_0 and z_1 . Thus, we have the following OPEs,

$$V_\beta(z_{1,n}^+) V_{-\alpha}(z_{2,n}^-) = \zeta_n^{-\alpha\beta} \left(V_{\beta-\alpha}(e^{i\pi\theta_1}) + O(\zeta_n) \right) \quad (\text{K.27})$$

$$V_\alpha(z_{n-1,n}^+) V_{-\beta}(z_{0,n}^-) = \zeta_n^{-\alpha\beta} \left(V_{\alpha-\beta}(e^{i\pi\theta_{n-1}}) + O(\zeta_n) \right) \quad (\text{K.28})$$

$$V_\alpha(z_{j,n}^+) V_{-\alpha}(z_{j,n}^-) = \zeta_n^{-\alpha^2} (1 + O(\zeta_n)), \quad j = 2, \dots, n-2. \quad (\text{K.29})$$

The above $2n$ point function is thus given by,

$$\zeta_n^{-(n-3)\alpha^2-\beta^2-2\alpha\beta} \langle V_{\beta-\alpha}(e^{i\pi\theta_1}) V_{\alpha-\beta}(e^{i\pi\theta_{n-1}}) \rangle (1 + O(\zeta_n)) \quad (\text{K.30})$$

$$= \zeta_n^{-(n-3)\alpha^2-\beta^2-2\alpha\beta} (2 \sin(2\pi/n))^{-(\alpha-\beta)^2} (1 + O(\zeta_n)) \quad (\text{K.31})$$

K.2.2 Broken $p = 2$ word

Now for the broken word, we insert $V_\beta V_{-\beta}$ at z_0 and z_m , where $1 < m < n - 1$. Thus, we have the following OPEs,

$$V_\beta(z_{0,n}^+) V_{-\alpha}(z_{1,n}^-) = \zeta_n^{-\alpha\beta} \left(V_{\beta-\alpha}(e^{i\pi\theta_0}) + O(\zeta_n) \right) \quad (\text{K.32})$$

$$V_\alpha(z_{m-1,n}^+) V_{-\beta}(z_{m,n}^+) = \zeta_n^{-\alpha\beta} \left(V_{\alpha-\beta}(e^{i\pi\theta_{m-1}}) + O(\zeta_n) \right) \quad (\text{K.33})$$

$$V_\beta(z_{m,n}^+) V_{-\alpha}(z_{m+1,n}^-) = \zeta_n^{-\alpha\beta} \left(V_{\beta-\alpha}(e^{i\pi\theta_m}) + O(\zeta_n) \right) \quad (\text{K.34})$$

$$V_\alpha(z_{n-1,n}^+) V_{-\beta}(z_{0,n}^-) = \zeta_n^{-\alpha\beta} \left(V_{\alpha-\beta}(e^{i\pi\theta_{n-1}}) + O(\zeta_n) \right) \quad (\text{K.35})$$

Hence, we have the following $2n$ point function,

$$\zeta_n^{-(n-4)\alpha^2-4\alpha\beta} \langle V_{\alpha-\beta}(e^{i\pi\theta_{n-1}}) V_{\beta-\alpha}(e^{i\pi\theta_0}) V_{\alpha-\beta}(e^{i\pi\theta_{m-1}}) V_{\beta-\alpha}(e^{i\pi\theta_m}) \rangle (1 + O(\zeta_n)) \quad (\text{K.36})$$

$$= \zeta_n^{-(n-4)\alpha^2-4\alpha\beta} \left(\frac{(e^{i\pi\theta_{n-1}} - e^{i\pi\theta_{m-1}})(e^{i\pi\theta_0} - e^{i\pi\theta_m})}{(e^{i\pi\theta_{n-1}} - e^{i\pi\theta_0})(e^{i\pi\theta_{n-1}} - e^{i\pi\theta_m})(e^{i\pi\theta_0} - e^{i\pi\theta_{m-1}})(e^{i\pi\theta_{m-1}} - e^{i\pi\theta_m})} \right)^{(\alpha-\beta)^2} (1 + O(\zeta_n)) \quad (\text{K.37})$$

$$= \zeta_n^{-(n-4)\alpha^2-4\alpha\beta} \left(\frac{(2 \sin(\pi m/n))^2}{(2 \sin(\pi/n))^2 (2 \sin(\pi(m-1)/n)) (2 \sin(\pi(m+1)/n))} \right)^{(\alpha-\beta)^2} (1 + O(\zeta_n)) \quad (\text{K.38})$$

Now we can compare the contiguous and broken correlators,

$$\frac{\zeta_n^{-(n-4)\alpha^2-4\alpha\beta} \langle V_{\alpha-\beta}(e^{i\pi\theta_{n-1}}) V_{\beta-\alpha}(e^{i\pi\theta_0}) V_{\alpha-\beta}(e^{i\pi\theta_{m-1}}) V_{\beta-\alpha}(e^{i\pi\theta_m}) \rangle (1 + O(\zeta_n))}{\zeta_n^{-(n-3)\alpha^2-\beta^2-2\alpha\beta} (2 \sin(2\pi/n))^{-(\alpha-\beta)^2} (1 + O(\zeta_n))} \quad (\text{K.39})$$

$$= \zeta_n^{(\alpha-\beta)^2} \left(\frac{(2 \sin(2\pi/n)) (2 \sin(\pi m/n))^2}{(2 \sin(\pi/n))^2 (2 \sin(\pi(m-1)/n)) (2 \sin(\pi(m+1)/n))} \right)^{(\alpha-\beta)^2} (1 + O(\zeta_n)) \quad (\text{K.40})$$

Since $\zeta_n = 2 \sin(\pi\epsilon/n) \sim 2\pi\epsilon/n$, the ratio tends to zero. Similarly, for $p > 1$, it can be shown that the term having the least number of mixing OPEs would be the leading term. Hence, in our calculations the dominant contribution came from the terms which had a contiguous string of $\Psi\Psi^*$ connected to a contiguous string of $\Phi\Phi^*$. These calculations can be extended for any general 2d CFT, since we just require the order of the terms.

L Numerical methods

{app:num-methods}

This appendix specifies the numerical calculations used in Sec. 4. The calculations evaluate finite-chain reduced density matrices directly.

L.1 States, subsystems, and sampling

{app:num-samp1}

We use periodic spin-1/2 chains with N sites and unit lattice spacing, so their circumference is $R = N$ and the interval length l counts sites. We take contiguous subsystems $A = \{0, \dots, l-1\}$, $B = A^c$. All active mixture fits and figures use $l = 6, 7, 8, 9, 10$. The Ising states are $0, \sigma$, and ϵ ; the boson states are $00, ss$, and $3s, s$. Each of the three unordered pairs is evaluated in both directions. We use natural logarithms and the incoherent mixture $\bar{\rho}_C = (\rho_C^\Psi + \rho_C^\Phi)/2$, for $C = A, B$. For both equal and unequal weights, the complete system-size grid is

$$\mathcal{G}_m(c) = \{(N, l) : N \in 4\mathbb{N}, 6 \leq l \leq 10, l \in \mathbb{N}, N \leq N_{\max}^{(m)}, 0 < l/N \leq c\}, \quad (\text{L.1}) \quad \{\text{eq:num-extend}\}$$

where $N_{\max}^{(\text{Ising})} = 8192$ and $N_{\max}^{(\text{boson})} = 16384$. The evaluated grids use $c = 0.1$ for small intervals and $c = 0.05$ for large complements. At each fixed source weight they contain 183 168 and 181 980 directed observations, respectively, or 365 148 in total. All required geometry, state-pair, and direction records are present, with no duplicate records. Chain sizes are uniformly spaced by four; the fractions l/N are not uniformly spaced. All displayed mixture fits use $c = 0.02$ and $c = 0.005$, respectively. For Sec. 4.4, replace $\bar{\rho}_C$ by $\rho_{C,y} = y\rho_C^\Psi + (1-y)\rho_C^\Phi$ and repeat the complete grid at $y_j = (78 + j)/195$, $j = 0, \dots, 39$. These forty probabilities yield 7 326 720 small-interval and 7 279 200 large-interval observations, in addition to the separately retained half-weight reference. Each direction is indexed by its source weight; the reverse direction to the same mixture has weight $1 - y$.

For each group, observations are selected individually by their fraction l/R : a chain need not contribute all five lengths inside the fitting window. Repeated fractions arising from different (N, l) remain distinct observations. Table. 6 gives the windows and counts. The interval figures show only fitted observations; lengths share markers and colors, while residual colors identify models. In the probability figures, each marker is an independently fitted leading coefficient, and the marker style identifies the source direction.

Regime	CFT	Upper cutoff	Points	Sizes	Min. training
Small x	Ising	0.02	9744	1974	9739
Small x	Boson	0.02	19984	4022	19979
Small ϵ	Ising	0.005	8245	1749	8240
Small ϵ	Boson	0.005	18485	3797	18480

Table 6. Complete step-4 sampling per ordered pair and source weight for all three fitting models. The point and size counts apply to $y = 1/2$ and to each of the forty unequal weights. The lower endpoint is zero and is open. “Min. training” gives the smallest number of observations left after omitting a complete chain size in the equal-mixture sensitivity checks. These are deterministic lattice data, not independent random samples.

{tab:num-window}

L.2 Exact Ising primary states and interval matrices

{app:num-ising}

We use the critical periodic spin Hamiltonian

$$H = - \sum_{j=0}^{N-1} X_j X_{j+1} - \sum_{j=0}^{N-1} Z_j, \quad X_N = X_0, \quad P = \prod_j Z_j. \quad (\text{L.2}) \quad \{\text{eq:num-ising-1}\}$$

The Jordan–Wigner Majoranas $\gamma_{2j} = (\prod_{k<j} Z_k) X_j$ and $\gamma_{2j+1} = (\prod_{k<j} Z_k) Y_j$ reduce this Hamiltonian to a quadratic form [17, 18]. Even spin parity gives antiperiodic fermions. Their vacuum represents 0, and occupying the two modes at momenta $\pm\pi/N$ represents ε . Odd spin parity gives periodic fermions; their lowest state with the zero-mode occupation fixed to $P = -1$ represents σ . All three states have zero spin momentum. Their exact energies are

$$E_0 = -2 \csc \frac{\pi}{2N}, \quad E_\sigma = -2 \cot \frac{\pi}{2N}, \quad E_\varepsilon = E_0 + 8 \sin \frac{\pi}{2N}. \quad (\text{L.3}) \quad \{\text{eq:num-ising-2}\}$$

With velocity two, these gaps approach the scaling dimensions $0, 1/8, 1$, respectively.

For reproducibility, the real antisymmetric covariance is $G_{rs} = \langle i\gamma_r \gamma_s \rangle$ for $r \neq s$ and $G_{rr} = 0$. Set $L_M = 2N$, $k_m = 2\pi(m+t)/L_M$, and $w_m = -\text{sgn}(\sin k_m)$. The twist is $t = 1/2$ for $(0, \varepsilon)$ and $t = 0$ for σ . Reverse w_m at $m = 0, N-1, N, L_M-1$ for ε . For σ , set $w_0 = w_N = 0$ and include the parity-fixing term in

$$(G_\Psi)_{rs} = \frac{i}{L_M} \sum_{m=0}^{L_M-1} w_m e^{ik_m(r-s)} + \begin{cases} [(-1)^r - (-1)^s]/L_M, & \Psi = \sigma, \\ 0, & \Psi = 0, \varepsilon. \end{cases} \quad (\text{L.4}) \quad \{\text{eq:num-covari-1}\}$$

This exact Fourier construction avoids a full Hamiltonian eigensolve. We check antisymmetry, purity $G_\Psi^2 = -\mathbb{I}$, and the energies in Eq. (L.3).

Restrict G_Ψ to the first $2l$ Majoranas. If an orthogonal change of basis gives blocks $\begin{pmatrix} 0 & g_j \\ -g_j & 0 \end{pmatrix}$, and b_r are the correspondingly rotated Majoranas, then

$$\rho_A^\Psi = 2^{-l} \prod_{j=0}^{l-1} (\mathbb{I} + g_j i b_{2j} b_{2j+1}). \quad (\text{L.5}) \quad \{\text{eq:num-ising-3}\}$$

All Jordan–Wigner strings lie inside this interval, so this is also its spin density matrix. We form the 2^l -dimensional component matrices and then their mixture. Although each component is Gaussian, the mixture generally is not; its entropy is evaluated from the actual matrix, not an averaged covariance.

L.3 Compact-boson states and exact environment contraction

{app:num-boson}

The compactification radius of our compact free boson CFT example is $R_c = \sqrt{2}$. In the vertex notation of the main text, $(\alpha, \bar{\alpha}) = (n_L, n_R)/\sqrt{2}$ and $(h, \bar{h}) = (\alpha^2, \bar{\alpha}^2)/2$. The integer charge labels $(n_L, n_R) = (0, 0), (1, 1), (3, 1)$ correspond to $|00\rangle, |ss\rangle, |3s, s\rangle$, respectively. On

sites $z_j = e^{2\pi i j/N}$, the auxiliary vertex-block construction with charges at zero and W gives a Jastrow polynomial [19]. For an occupied set O its normalized amplitude is

$$\begin{aligned}\psi_{n_L, n_R}(O) &= \frac{1}{\sqrt{\mathcal{Z}_{n_L, n_R}}} \prod_{j \in O} z_j^{n_L+1} (1 - z_j/W)^{n_R} \prod_{i < j \in O} (z_j - z_i)^2, \\ |O| &= \frac{N - n_L - n_R}{2}, \quad W = 10^8.\end{aligned}\tag{L.6} \quad \{\text{eq:num-boson-v}\}$$

$\mathcal{Z}_{n_L, n_R}$ is the sum of squared amplitudes before normalization. The three states have occupations $N/2, N/2 - 1, N/2 - 2$ and are exactly orthogonal. The complex phase of this amplitude is retained in every off-diagonal matrix element. The finite endpoint W is part of the supplied lattice-state convention; it approximates the strict insertion at infinity.

To compute both density matrices and cross-state transitions, let $U, V \subset A$ be occupied sets and let $Y \subset B$ be their common environment. For states $\Psi_1 = \Psi, \Psi_2 = \Phi$, with occupations M_i and unnormalized polynomials P_i , we need

$$(\mathbb{T}_A^{12})_{UV} = \frac{1}{\sqrt{\mathcal{Z}_1 \mathcal{Z}_2}} \sum_{\substack{Y \subset B \\ |Y|=M_1-|U|=M_2-|V|}} P_1(U \cup Y) P_2(V \cup Y)^*.\tag{L.7} \quad \{\text{eq:num-boson-v}\}$$

It vanishes if the two required environment occupations differ.

We contract this sum through confluent Vandermonde identities and Pfaffians, building on the moment method for the Haldane–Shastry state [20]. Explicitly, set $m = M_1 - |U| = M_2 - |V|$, $\nu_j = \mathbf{1}_{j \in U} + \mathbf{1}_{j \in V}$, $d = M_1 + M_2$, $g_i(z) = (1 - z/W)^{n_{R,i}}$, and

$$\begin{aligned}L_i(U) &= \prod_{j \in U} z_j^{n_{L,i}+1} g_i(z_j) \prod_{j < k \in U} (z_j - z_k)^2, \\ w(z) &= z^{n_{L,1} - n_{L,2} - 2M_2 + 2} g_1(z) \overline{g_2(z)}, \\ \mathcal{E}_{UV} &= \sum_{\substack{Y \subset B \\ |Y|=m}} \prod_{i < j \in Y} (z_i - z_j)^4 \prod_{i \in Y} \left[w(z_i) \prod_{j \in A} (z_j - z_i)^{2\nu_j} \right],\end{aligned}$$

so that $(\mathbb{T}_A^{12})_{UV} = L_1(U) \overline{L_2(V)} \prod_{j \in V} z_j^{-2m} \mathcal{E}_{UV} / \sqrt{\mathcal{Z}_1 \mathcal{Z}_2}$. Let $f(z) = (1, z, \dots, z^{d-1})^T$ and define

$$\mathbb{M}_B = \sum_{j \in B} w(z_j) [f(z_j) f'(z_j)^T - f'(z_j) f(z_j)^T].\tag{L.8} \quad \{\text{eq:num-moment}\}$$

At each site of A , a matrix E_{UV} contains no column if $\nu_j = 0$, $f(z_j)$ if $\nu_j = 1$, and $f(z_j), f'(z_j)$ if $\nu_j = 2$, in site order. With $q = \sum_j \nu_j$ and $\Delta_{\text{ext}} = \prod_{i < j \in A} (z_j - z_i)^{\nu_i \nu_j}$,

$$\mathcal{E}_{UV} = \frac{(-1)^{q(q-1)/2}}{\Delta_{\text{ext}}} \text{Pf} \begin{pmatrix} \mathbb{M}_B & E_{UV} \\ -E_{UV}^T & 0 \end{pmatrix}.\tag{L.9} \quad \{\text{eq:num-border}\}$$

Here $d + q$ is even, including for transitions with odd d . The formula defines the contraction without requiring an inverse of a possibly singular cross-moment matrix. Numerical evaluation uses sparse full-circle moments, a rank- $2l$ subtraction of the interval sites. This extends the norm contraction to off-diagonal reduced matrices and transitions.

L.4 Large intervals from a small Gram matrix

{app:num-gram}

A global pure state has Schmidt rank at most 2^l across the cut, even though B has $N - l$ sites. The mixture on B therefore has rank at most 2^{l+1} . Its calculation needs the transition $\mathsf{T}_A^{\Psi\Phi} = \text{tr}_B |\Psi\rangle\langle\Phi|$, as well as ρ_A^Ψ, ρ_A^Φ ; the individual reduced states alone do not specify their Schmidt-support overlap on B .

Writing $|\Psi\rangle = \sum_{u,v} X_\Psi(v,u) |u\rangle_A |v\rangle_B$ and $Z = (X_\Psi, X_\Phi)/\sqrt{2}$ gives $ZZ^\dagger = \bar{\rho}_B$ and

$$\mathcal{G} = Z^\dagger Z = \frac{1}{2} \begin{pmatrix} (\rho_A^\Psi)^T & (\mathsf{T}_A^{\Psi\Phi})^* \\ (\mathsf{T}_A^{\Psi\Phi})^T & (\rho_A^\Phi)^T \end{pmatrix}. \quad (\text{L.10}) \quad \{\text{eq:num-gram}\}$$

The nonzero eigenvalues λ_k of $\mathcal{G}$ equal those of $\bar{\rho}_B$. For its normalized eigenvectors v_k and the first-block projector P_Ψ , the directed entropy is

$$S(\rho_B^\Psi \| \bar{\rho}_B) = \text{tr}(\rho_A^\Psi \log \rho_A^\Psi) - 2 \sum_{\lambda_k > 0} \lambda_k (v_k^\dagger P_\Psi v_k) \log \lambda_k. \quad (\text{L.11}) \quad \{\text{eq:num-gram-ent}\}$$

Purity supplies the first term, and the second follows from $X_\Psi = \sqrt{2} Z E_\Psi$, where E_Ψ embeds the first block. The reversed direction uses P_Φ and ρ_A^Φ . We diagonalize common parity sectors on B for Ising and common occupation sectors for the boson. A constant phase on $\mathsf{T}_A^{\Psi\Phi}$ is immaterial, but its configuration-dependent phases must be retained.

For Ising, direct Gaussian transition formulas between the orthogonal primaries would divide by zero overlap. We instead choose a local Hermitian unitary $U_A = \gamma_0 = X_0$ for pairs containing σ , or $U_A = i\gamma_0\gamma_1 = -Z_0$ for $(0, \varepsilon)$, and set $|\Phi'\rangle = U_A |\Phi\rangle$. This gives a nonzero anchor overlap on the sampled grid. Gaussian product identities [21, 22] give

$$\begin{aligned} K &= \mathbb{I} - G_\Psi G_{\Phi'}, & o &= 2^{-N/2} |\det K|^{1/4}, \\ K Z_A &= (\mathbb{I} - i G_\Psi) E_A, & G_{*,A} &= i [E_A^T (\mathbb{I} - i G_{\Phi'}) Z_A - \mathbb{I}_{2l}], \end{aligned} \quad (\text{L.12}) \quad \{\text{eq:num-anchor}\}$$

where E_A selects the first $2l$ coordinates. Fourier application of G_Ψ and a pivoted real LU factorization of K avoid forming a full complex inverse; its log determinant also gives $\log o$. For a complex antisymmetric H on the interval modes, Wick reconstruction is

$$\mathcal{R}(H) = 2^{-l} \sum_{\substack{S \subset \{0, \dots, 2l-1\} \\ |S| \text{ even}}} i^{|S|/2} \text{Pf}(H_S) \prod_{r \in S}^{\rightarrow} \gamma_r. \quad (\text{L.13}) \quad \{\text{eq:num-wick}\}$$

The product is ordered by increasing index and the empty Pfaffian is one. Restoring the anchor yields $\mathsf{T}_A^{\Psi\Phi} = o \mathcal{R}(G_{*,A}) U_A$, up to a single irrelevant phase. Independent small-chain wavefunction checks validate this use of the Gaussian identities. The boson transitions are obtained directly from Eq. (L.7).

L.5 Entropy evaluation and control of cancellation

{app:num-precis}

For ordinary fractions, diagonalizing the component and mixture matrices gives $S(\rho \| \bar{\rho}) = \text{tr} \rho \log \rho - \text{tr} \rho \log \bar{\rho}$. The cross trace uses the weights of ρ in the eigenbasis of $\bar{\rho}$; the matrices

need not commute. The primary check uses the same spectral trace with $\bar{\rho}$ replaced by ρ_A^Φ and a target-logarithm floor of 10^{-15} .

At very small fractions, subtracting nearly equal entropy traces, or subtracting $S(\rho_B^\Psi \|\bar{\rho}_B)$ from $\log 2$, loses relative precision. We evaluate the remainder directly using matrix-function divided differences [23]. Let $f(t) = t \log t$, $D = \text{diag}(d_i) > 0$, and $H = D + E = V \text{diag}(\lambda_k) V^\dagger$. Then

$$\begin{aligned} \mathcal{R}_i &= [f(H) - f(D)]_{ii} - f'(d_i) E_{ii} \\ &= \sum_k |(V^\dagger E)_{ki}|^2 \frac{\lambda_k \log(\lambda_k/d_i) - \lambda_k + d_i}{(\lambda_k - d_i)^2}. \end{aligned} \quad (\text{L.14}) \quad \{\text{eq:num-stable}\}$$

The identity follows by expanding in the eigenvectors of H and using $(\lambda_k - d_i) V_{ik} = (EV)_{ik}$. At $\lambda = d$ the quotient is $1/(2d)$. With $u = (\lambda - d)/d$, its stable form is

$$\frac{1}{d} \frac{(1+u) \log(1+u) - u}{u^2} = \frac{1}{d} \sum_{n=0}^{\infty} \frac{(-u)^n}{(n+1)(n+2)}. \quad (\text{L.15}) \quad \{\text{eq:num-scalar}\}$$

Ten terms are used for $|u| < 0.02$, where the omitted tail is below double-precision accuracy; otherwise the closed expression is used.

For a small interval, take D to be the diagonalized mixture and H the source density matrix in that basis. Equal traces give $S(\rho_A^\Psi \|\bar{\rho}_A) = \sum_i \mathcal{R}_i$. For the complement, take the unweighted Gram matrix

$$H = \begin{pmatrix} \rho_A^\Psi & \mathsf{T}_A^{\Psi\Phi} \\ (\mathsf{T}_A^{\Psi\Phi})^\dagger & \rho_A^\Phi \end{pmatrix} = 2\mathcal{G}^*. \quad (\text{L.16}) \quad \{\text{eq:num-unweig}\}$$

Diagonalize its two diagonal blocks separately and let D contain their eigenvalues. The perturbation E then has only off-diagonal blocks. Equation (L.11) gives

$$\begin{aligned} \log 2 - S(\rho_B^\Psi \|\bar{\rho}_B) &= \text{tr} P_\Psi [f(H) - f(D)] = \sum_i (P_\Psi)_{ii} \mathcal{R}_i, \\ \log 2 - S(\rho_B^\Phi \|\bar{\rho}_B) &= \sum_i (P_\Phi)_{ii} \mathcal{R}_i. \end{aligned} \quad (\text{L.17}) \quad \{\text{eq:num-stable}\}$$

This permits a deficit much smaller than machine precision relative to $\log 2$ to be resolved on its own scale.

For unequal weights, use the weighted Gram matrix

$$H_y = \begin{pmatrix} y\rho_A^\Psi & \sqrt{y(1-y)} \mathsf{T}_A^{\Psi\Phi} \\ \sqrt{y(1-y)} (\mathsf{T}_A^{\Psi\Phi})^\dagger & (1-y)\rho_A^\Phi \end{pmatrix}. \quad (\text{L.18}) \quad \{\text{eq:num-weight}\}$$

After diagonalizing its two diagonal blocks, let D_y contain their eigenvalues and evaluate the same remainders $\mathcal{R}_i(H_y, D_y)$ in Eq. (L.14). The two deficits to this fixed mixture are

$$\begin{aligned} -\log y - S(\rho_B^\Psi \|\rho_{B,y}) &= \frac{1}{y} \sum_{i \in \Psi} \mathcal{R}_i(H_y, D_y), \\ -\log(1-y) - S(\rho_B^\Phi \|\rho_{B,y}) &= \frac{1}{1-y} \sum_{i \in \Phi} \mathcal{R}_i(H_y, D_y). \end{aligned} \quad (\text{L.19}) \quad \{\text{eq:num-weight}\}$$

Thus the large-interval probability data are obtained directly as stable deficits, without subtracting nearly equal entropies. For small intervals the diagonalized target is simply $\rho_{A,y}$, and the same trace remainder gives the relative entropy.

The identities are exact on positive support. In the double-precision implementation, reference eigenvalues below $\tau = 10^{-15}$ are discarded within each symmetry sector. This support truncation is a numerical approximation. The returned equal-mixture production checkpoints record 816 558 passing numerical checks. The unequal-weight run records another 12 142 275 passing checks across all 6116 model-size jobs, including density-matrix constraints, retained-support sensitivity, Gram positivity, and reproduction of the saved half-weight values. All returned checkpoints pass the completeness audit, and every reported unequal-weight fit has been replayed from the merged data. These checks concern the finite-chain calculations, not the lattice-to-CFT discrepancy. In particular, agreement of two trace-log evaluations in absolute error alone cannot establish the relative accuracy of a very small deficit.

L.6 Fitting models, residuals, and window dependence

Use $z = x$ for a small interval and $z = \epsilon = 1 - x$ for a large interval. Let v_i denote the corresponding measured value of $S(\rho_A^\Psi \|\bar{\rho}_A)$ or $\log 2 - S(\rho_B^\Psi \|\bar{\rho}_B)$. At unequal weight these become $S(\rho_A^\Psi \|\rho_{A,y})$ and $-\log y - S(\rho_B^\Psi \|\rho_{B,y})$, respectively; each weight and source direction is fitted independently on the same geometry grid. For each ordered pair (Ψ, Φ) , we compare

$$M_{\text{free}}(z) = C_{\text{free}} z^r, \quad M_1(z) = c_0 z^{r_0}, \quad M_2(z) = c_0 z^{r_0} + c_1 z^{r_1}. \quad (\text{L.20})$$

The free fit estimates $C_{\text{free}} > 0$ and r without theoretical constraints; its fitted exponent is denoted $\hat{r}$. M_1 and M_2 impose the powers in Table. 7 and fit one and two amplitudes, respectively; the M_2 amplitudes have no sign constraints. In the equal-mixture interval figures, the CFT curve includes the analytical leading term and each complete known next term: the small-Ising sixth-order terms and the large-boson sixth- and third-order terms listed in Table 7. In the other channels it includes the leading term only. The probability figures instead compare the fitted leading coefficient directly with $c_0^{\text{CFT}}(y)$. For unequal weights, the present next-coefficient comparison is restricted to the known small-Ising $0 \leftrightarrow \varepsilon$ sixth-order terms.

Every fit minimizes unweighted squared residuals in the observable, $\sum_{i:z_i \in W} [v_i - M(z_i)]^2$, on the window in Table. 6. For fixed powers, scale the columns as $X_{ij} = (z_i/z_\star)^{r_j}$, where $z_\star$ is the upper cutoff, solve by singular-value decomposition, and convert the fitted amplitudes back by dividing by $z_\star^{r_j}$. For a trial free exponent the optimal scaled amplitude is

$$A(r) = \frac{\sum_i v_i (z_i/z_\star)^r}{\sum_i (z_i/z_\star)^{2r}}, \quad C_{\text{free}} = A(r) z_\star^{-r}. \quad (\text{L.21})$$

A scalar search of the profiled objective provides an independent check on the two-parameter optimization [24]. The signed amplitude discrepancies are

$$\eta_j(M)[\%] = 100 \left(\frac{c_j(M)}{c_j^{\text{CFT}}} - 1 \right), \quad j = 0, 1. \quad (\text{L.22})$$

CFT	Pair (both directions)	Small interval		Large interval	
		r_0	r_1	$r_0 = 2\Delta_{L'}$	r_1
Ising	$(0, \sigma)$	2	4	1/4	1/2
Ising	$(0, \varepsilon)$	4	6	2	4
Ising	(σ, ε)	2	4	1/4	1/2
Boson	$(00, ss)$	2	4	1	2
Boson	$(00, (3s, s))$	2	4	5	6
Boson	$(ss, (3s, s))$	2	4	2	3

Table 7. Powers used in M_1, M_2 . At $y = 1/2$, complete next coefficients are available for the small-interval Ising $0 \leftrightarrow \varepsilon$ pair and the large-interval boson $00 \leftrightarrow 3s, s$ and $ss \leftrightarrow 3s, s$ pairs. The other next amplitudes remain numerical estimates without a complete analytical comparison. The unequal-weight next-coefficient tests use the small-Ising $0 \leftrightarrow \varepsilon$ pair.

{tab:num-powers}

Here η_0 is reported for both M_1 and M_2 , and η_1 only for M_2 where a complete nonzero c_1^{CFT} is available. All discrepancies are computed from the unrounded fitted amplitudes. For large intervals, both fitted and analytical coefficients refer to the deficit $\log 2 - S$ at $y = 1/2$, and to $-\log y - S$ at unequal source weight y . The pointwise figure residual and the free-power discrepancy are

$$R_i(M)[\%] = 100 \frac{v_i - M(z_i)}{v_i}, \quad \eta_r[\%] = 100 \left(\frac{\hat{r}}{r_0} - 1 \right). \quad (\text{L.23}) \quad \{\text{eq:num-point-}$$

Residual ordinates are linear. Omitting all observations at one chain size and refitting tests the sensitivity of the coefficients; it does not give a statistical confidence interval for these deterministic calculations. In particular, a decrease in the training residual is guaranteed by the nesting of M_1 in M_2 and is not itself evidence of more accurate CFT coefficients.

The common windows in Table 6 provide a fixed comparison across both models and every direction. They are not windows of uniformly one-percent amplitude accuracy. For $y = 1/2$, an exhaustive scan of the observed large-interval endpoints, including zero-anchored intervals, gives 85 059 733 selections with at least forty observations and all five lengths for each CFT. None simultaneously has $|\eta_0(M_2)| \leq 1\%$ in every direction and $|\eta_1(M_2)| \leq 10\%$ wherever the next coefficient is known, even before imposing $|\eta_r| \leq 1\%$. This is a result for the present finite grid and unweighted fitting models, not a statement about the continuum limit. Small l/R and large l are separate requirements; increasing N at fixed $l = 6, \dots, 10$ does not remove finite-length effects.

M Old text (Irrelevant)

M.1 Relative entropy of mixture of Coherent states

The calculation below will not reproduce the coherent states result, even if we successfully do the analytical continuation. The reason is that here we are not considering the sub-leading terms in the mixing channels, which will contribute at the same order that of the three point

function considered here $\langle V_{\alpha-\beta} V_{\beta-\alpha} J \rangle$. Even further, we have assumed that the primaries in the mixing OPEs, $V_{\alpha-\beta}$ and $V_{\beta-\alpha}$, are $2 \sin(p\pi(1-\epsilon)/n)$ separated, instead of $2 \sin(\pi p/n)$.

We consider a mixture of coherent states. We will evaluate the relative entropy $S(\rho_\alpha \| y\rho_\alpha + (1-y)\rho_\beta)$, where the ρ_α, ρ_β are reduced density matrices of some excited states created by Vertex operators V_α, V_β . As discussed in the previous section, the numerator of $\mathcal{F}_n$ evaluates to $1 + \zeta_n^{2\Delta_L} f(n)$, however for the denominator, we get the divergent piece from the two-point function of the vertex operators $V_{\alpha-\beta}$ and $V_{\beta-\alpha}$. This gets multiplied with the two-point function of the currents J , with the appropriate OPE coefficient that sits in front of J . The denominator takes the following form,

$$\sum_{p=1}^{n-1} y^{n-1-p} (1-y)^p D_n(\epsilon)^p (n-p) (2 \sin(p\pi(1-\epsilon)/n))^{-2(\alpha-\beta)^2} [1 + \zeta_n^2 g_p(n) + \dots] \quad (\text{M.1})$$

$$g(p) = \frac{1}{2} \sum_{k \neq k'=0}^{n-1} c_k c_{k'} (2 \sin(\pi(k-k')/n))^{-2} \quad (\text{M.2})$$

$$c_k = \begin{cases} \beta, & k = 1, 2, \dots, p \\ \alpha, & k = 0, p+1, p+2, \dots, n-1 \end{cases} \quad (\text{M.3})$$

We can introduce an indicator function,

$$\chi_k^{(p)} = \begin{cases} 1 & k \in \{1, 2, \dots, p\} \\ 0 & k \in \{0, p+1, p+2, \dots, n-1\} \end{cases} \quad (\text{M.4})$$

to write

$$c_k = \alpha + (\beta - \alpha) \chi_k^{(p)} \quad (\text{M.5})$$

Then $g_p(n)$ simplifies to,

$$g_p(n) = \frac{1}{2} \sum_{k \neq k'=0}^{n-1} c_k c_{k'} (2 \sin(\pi(k-k')/n))^{-2} \quad (\text{M.6})$$

$$= \alpha^2 S_2(n) + \frac{\alpha(\beta - \alpha)}{2} \sum_{k \neq k'}^{n-1} \frac{\chi_k + \chi_{k'}}{(2 \sin(\pi(k-k')/n))^2} + \frac{1}{2} (\beta - \alpha)^2 \sum_{k \neq k'=0}^{n-1} \frac{\chi_k \chi_{k'}}{(2 \sin(\pi(k-k')/n))^2} \quad (\text{M.7})$$

The second sum can be simplified to,

$$\frac{\alpha(\beta - \alpha)}{2} \sum_{k \neq k'}^{n-1} \frac{\chi_k + \chi_{k'}}{(2 \sin(\pi(k-k')/n))^2} = \alpha(\beta - \alpha) \sum_{k \neq k'=0}^{n-1} \frac{\chi_k}{(2 \sin(\pi(k-k')/n))^2} \quad (\text{M.8})$$

$$= \alpha(\beta - \alpha) \sum_{k=1}^p \sum_{k' \neq k} \frac{1}{(2 \sin(\pi(k-k')/n))^2} \quad (\text{M.9})$$

$$= \alpha(\beta - \alpha) p \frac{2S_2(n)}{n} \quad (\text{M.10})$$

where in the last step, the sum over k' is $\frac{2S_2(n)}{n}$. Consider the third sum,

$$\frac{1}{2}(\beta - \alpha)^2 \sum_{k \neq k'=0}^{n-1} \frac{\chi_k \chi_{k'}}{(2 \sin(\pi(k - k')/n))^2} = \frac{1}{2}(\beta - \alpha)^2 \sum_{k \neq k'=1}^p \frac{\chi_k \chi_{k'}}{(2 \sin(\pi(k - k')/n))^2} \quad (\text{M.11})$$

$$= (\beta - \alpha)^2 \sum_{m=1}^{p-1} \frac{p - m}{(2 \sin(\pi m/n))^2} \quad (\text{M.12})$$

Now we can write the full $\mathcal{F}_n$,

$$\mathcal{F}_n = \frac{1 + \zeta_n^2 f(n) + \dots}{y^{n-1} \left(1 + \zeta_n^2 f(n) + \sum_{p=1}^{n-1} (y^{-1} - 1)^p D_n(\epsilon)^p (n - p) \frac{\zeta_n^{2(\alpha-\beta)^2}}{(2 \sin(p\pi(1-\epsilon)/n))^2} (1 + \zeta_n^2 g_p(n) + \dots) \right)} \quad (\text{M.13})$$

$$g(p) = \left(\alpha^2 + \frac{2\alpha(\beta - \alpha)p}{n} \right) S_2(n) + (\beta - \alpha)^2 H_p(n) \quad (\text{M.14})$$

$$H_p(n) = \sum_{m=1}^{p-1} \frac{(p - m)}{(2 \sin(\pi m/n))^2}. \quad (\text{M.15})$$

To simplify the notation, we define,

$$q = y^{-1} - 1, \quad \lambda_p = D_n(\epsilon)^p \left(\frac{\zeta_n}{\sin(p\pi(1 - \epsilon)/n)} \right)^{(\alpha-\beta)^2} \quad (\text{M.16})$$

$$\mu_0(n, y) = \sum_{p=1}^{n-1} q^p (n - p) \lambda_p \quad (\text{M.17})$$

$$\mu_1(n, y) = \sum_{p=1}^{n-1} q^p p (n - p) \lambda_p \quad (\text{M.18})$$

$$\mu_H(n, y) = \sum_{p=1}^{n-1} q^p (n - p) \lambda_p H_p(n) \quad (\text{M.19})$$

In terms of these sums, we get,

$$\mathcal{F}_n = \frac{1 + \zeta_n^2 f(n) + \dots}{y^{n-1} \left(1 + \zeta_n^2 f(n) + \mu_0(n, y) + \zeta_n^2 \left(\alpha^2 S_2(n) \mu_0(n, y) + \frac{2\alpha(\beta-\alpha)S_2(n)}{n} \mu_1(n, y) + (\beta - \alpha)^2 \mu_H(n, y) \right) \right)} \quad (\text{M.20})$$

$$\partial_n \log \mathcal{F}_n = -\log y - \mu'_0(1, y) - (\pi\epsilon)^2 (\beta - \alpha)^2 \partial_n \left(\sum_{p=1}^{n-1} q^p (n - p) \lambda_p H_p \right) \Big|_{n=1} \quad (\text{M.21})$$

where in the last step, we used the fact $S_2(n) = 0$, when the derivative hits μ_0, μ_1 and is evaluated at $n = 1$.

The limit $y \rightarrow 1$ is trivial, since $q \rightarrow 0$, and hence $\mu_0 = \mu_1 = \mu_H = 0$. $\mathcal{F}_n = 1$, and hence the relative entropy vanishes as expected.

For the other limit, $y \rightarrow 0$ limit, we pull the y^{n-1} back into the paranthesis in the denominator. Then in the sums μ_0, μ_1, μ_H , only $p = n - 1$ contributes. Then we have the simplification,

$$\lambda_{n-1} = D_n(\epsilon)^{n-1} \quad (\text{M.22})$$

$$\mu_0(n, y) = \lambda_{n-1} \quad (\text{M.23})$$

$$\mu_1 = (n-1)\lambda_{n-1} \quad (\text{M.24})$$

$$\mu_H = \lambda_{n-1} \sum_{p=1}^{n-2} \frac{n-1-m}{(2 \sin(\pi m/n))^2} = \lambda_{n-1} \sum_{m=1}^{n-1} \frac{n-1-m}{(2 \sin(\pi m/n))^2} \quad (\text{M.25})$$

$$= \lambda_{n-1} \frac{n-2}{n} S_2(n) \quad (\text{M.26})$$

In this limit, we have

$$\mathcal{F}_n = \frac{1 + \zeta_n^2 f(n) + \dots}{\lambda_{n-1} \left(1 + \zeta_n^2 \left(\alpha^2 S_2(n) + \frac{2\alpha(\beta-\alpha)(n-1)}{n} S_2(n) + (\beta-\alpha)^2 \left(\frac{n-2}{n} \right) S_2(n) \right) \right)} \quad (\text{M.27})$$

Taking the log and derivative, the λ_{n-1} piece gives the divergent piece, the $\alpha^2 S_2(n)$ in the denominator cancels with the numerator, while the second term proportional to $\frac{n-1}{n}$ does not contribute to relative entropy when we take the limit $n \rightarrow 1$. So we are left with,

$$S(\rho_\alpha \| \rho_\beta) = (\alpha - \beta)^2 \left(\frac{1}{\epsilon} - \frac{\epsilon}{3} \right) + (\alpha - \beta)^2 \frac{(\pi\epsilon)^2}{3} + \dots \quad (\text{M.28})$$

Similarly, for the non-chiral vertex operators, we obtain,

$$S(\rho_\alpha \| \rho_\beta) = 2(\alpha - \beta)^2 \left(\frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} \right) + 2(\alpha - \beta)^2 \frac{(\pi\epsilon)^2}{3} + \dots \quad (\text{M.29})$$

M.2 Renyi entropy of Vertex operators

{Renyi entropy}

In [25], they define

$$F_\Psi^n(x) = \frac{\text{tr}(\rho_\Psi^n)}{\text{tr}(\rho_0^n)}, \quad (\text{M.30})$$

where Ψ is some excited state. In the $x \rightarrow 0$ limit, they show for any excited state generated by the operator Ψ ,

$$F_\Psi^{(n)}(x) \sim 1 + \frac{h + \bar{h}}{3} \left(\frac{1}{n} - n \right) (\pi x)^2 + O(x^{2\Delta_L}). \quad (\text{M.31})$$

assuming that $\Delta_L > 1$. This piece comes from the $D_n(x)$, without considering the contribution from any two-point function of primaries from OPEs. This can be shown by considering,

$$F_\Psi^{(n)}(x) = \frac{\text{tr}(\rho_\Psi^n)}{\text{tr}(\rho_0^n)} \quad (\text{M.32})$$

$$\simeq D_n(x)^{-n} \quad (\text{M.33})$$

$$= \left(\frac{\sin(\pi x)}{n \sin(\pi x/n)} \right)^{2n\Delta_\Psi} \quad (\text{M.34})$$

$$\simeq \left(\frac{\pi x - \frac{(\pi x)^3}{6} + \dots}{n \left(\frac{\pi x}{n} - \frac{(\pi x)^3}{6n^3} + \dots \right)} \right)^{2n\Delta_\Psi} \quad (\text{M.35})$$

$$= \left(\frac{1 - \frac{(\pi x)^2}{6}}{1 - \frac{(\pi x)^2}{6n^2}} \right)^{2n\Delta_\Psi} \quad (\text{M.36})$$

$$\simeq 1 + 2n\Delta_\Psi \frac{(\pi x)^2}{6} \left(\frac{1}{n^2} - 1 \right) + \dots \quad (\text{M.37})$$

$$= 1 + \frac{h + \bar{h}}{3} \left(\frac{1}{n} - n \right) (\pi x)^2 + \dots \quad (\text{M.38})$$

When $\Delta_L = 1$, we have to take the higher order terms into account which will contribute as it is $O(x^2)$. This is the case with massless free bosons. In particular, using the 2n-point function of the coherent states, they show that

$$F_{\Psi[r,s]}^{(n)}(x) = 1, \quad \forall n, r, s. \quad (\text{M.39})$$

where $\Psi[r, s] = e^{i(\alpha_+\phi + \alpha_-\phi)}$ is the vertex operator. This shows that the Renyi entropy of any order n for coherent states is the same as that of the ground state. Now, using the replica trick, we can show that for vertex operators, this contribution coming from current $J = i\partial\phi$ exactly cancels the piece coming from $D_n(x)$. The contribution from the two-point functions is proportional to $S_2(n, 1)$. We compute it below,

$$S_2(n, \Delta) = \frac{n}{2} \sum_{m \in \mathbb{Z}} [ng_\Delta(nm) - g_\Delta(m)] \quad (\text{M.40})$$

We take $\Delta \sim 1 + \varepsilon$ in order to deal with singularities of Gamma functions (Note that, this ε has nothing to do with the primary in Ising CFT). Then we have,

$$g_{1+\varepsilon}(m) = 2^2 \Gamma(-1 - 2\varepsilon) \frac{\sin(\pi(1 + \varepsilon))}{\pi} \frac{\Gamma(m+1)}{\Gamma(m)} \quad (\text{M.41})$$

$$\simeq 4 \left(\frac{1}{2\varepsilon} \right) \left(\frac{-\pi\varepsilon}{\pi} \right) m \quad (\text{M.42})$$

$$= -2m. \quad (\text{M.43})$$

Similarly, we have

$$g_{1+\varepsilon}(nm) = -2nm. \quad (\text{M.44})$$

We can then write the sum of all two-point functions to be,

$$S_2(n, 1) = 2^{-2} \frac{n}{2} \sum_{m \in \mathbb{Z}} [n(2nm) + 2m] \quad (\text{M.45})$$

$$= -2^{-2} n(n^2 - 1) 2 \sum_{m \in \mathbb{Z}} m \quad (\text{M.46})$$

$$= -2^{-2} n(n^2 - 1) \zeta(-1) \quad (\text{M.47})$$

$$= \frac{n(n^2 - 1)}{24} \quad (\text{M.48})$$

Now adding the contribution from both holomorphic and anti-holomorphic currents, we get the total contribution to be,

$$2(C_\Psi^L)^2 \left(\frac{2\pi x}{n} \right)^2 \frac{n(n^2 - 1)}{24} = (C_\Psi^L)^2 \frac{(\pi x)^2}{3} \left(n - \frac{1}{n} \right) \quad (\text{M.49})$$

For coherent states, $C_\Psi^L = \alpha$. We finally get,

$$F_\Psi^{(n)} = 1 + \frac{\alpha^2}{3} \left(\frac{1}{n} - n \right) (\pi x)^2 + \frac{\alpha^2}{3} \left(n - \frac{1}{n} \right) (\pi x)^2 + \dots \quad (\text{M.50})$$

$$= 1 + O(x^3) \quad (\text{M.51})$$

where the higher order terms are a result from the higher point functions. We essentially showed the cancellation of the terms of the order of $O(x^2)$ explicitly. This could be further extended by computing higher order terms in the 2n-point function.

M.3 $S_2(1/2, \Delta)$ computation for Logarithmic Negativity

We compute $S_2(1/2, \Delta)$ which is required to measure logarithmic negativity [26],[27],[28]. We can see that from the definition of Logarithmic negativity,

$$\mathcal{E} = \lim_{n \rightarrow 1/2} 2 \log \text{Tr} \rho_\Psi^n = S_{1/2}(\rho_\Psi) \quad (\text{M.52})$$

where $S_n(\rho_\Psi)$ is the n^{th} order Renyi entropy of reduced density matrix of some excited state ρ_Ψ . Since, $\text{Tr}(\rho_\Psi^n)$ involves $S_2(n, \Delta)$ term from the two-point functions, the negativity is related to $S_2(1/2, \Delta)$. We follow Cardy, Calabrese and others [3] method to analytically continue the sum $S_2(n, \Delta)$ to $n = 1/2$.

We want to analytically continue,

$$S_2(n, \Delta) = 2^{-2\Delta} n \left(\frac{n-1}{2} g_\Delta(0) + \sum_{k=1}^{\infty} (n g_\Delta(nk) - g_\Delta(k)) \right). \quad (\text{M.53})$$

Using the Gamma function identity,

$$\Gamma(p)\Gamma(q) = 2\Gamma(p+q) \int_0^{\pi/2} (\cos \phi)^{2p-1} (\sin \phi)^{2q-1} d\phi, \quad (\text{M.54})$$

we can write the integral representation of $g_\Delta(k)$,

$$g_\Delta(k) = \frac{2^{2\Delta} \sec(\pi\Delta)}{\pi} \sin(\pi(\Delta - k)) \int_0^{\pi/2} d\phi \left(\frac{\sin(2\phi)}{2} \right)^{2\Delta-1} \tan(2\phi)^{2k}. \quad (\text{M.55})$$

We thus have,

$$g_\Delta(k) = C_\Delta \int_0^{\pi/2} d\phi A_\Delta(\phi) \sin(\pi(\Delta - k)) v^{2k}, \quad (\text{M.56})$$

$$C_\Delta \equiv \frac{2^{2\Delta} \sec(\pi\Delta)}{\pi}, \quad A_\Delta(\phi) = (\sin \phi \cos \phi)^{2\Delta-1}, \quad v \equiv \tan(\phi). \quad (\text{M.57})$$

We can then write the sum,

$$S_2(n, \Delta) = 2^{-2\Delta} C_\Delta \int_0^{\pi/2} d\phi A_\Delta(\phi) n \left(\frac{n-1}{2} \sin(\pi\Delta) + nI_n(v) - I_1(v) \right) \quad (\text{M.58})$$

$$I_n(v) \equiv \sum_{k=1}^{\infty} \sin(\pi(\Delta - nk)) v^{2nk}. \quad (\text{M.59})$$

We want to now evaluate the geometric sum $I_n(v)$. Let $r = v^{2n}$. Then,

$$I_n(v) = \sum_{k=1}^{\infty} \sin(\pi(\Delta - nk)) r^k \quad (\text{M.60})$$

$$= \sin(\pi\Delta) \sum_{k=1}^{\infty} r^k \cos(\pi nk) - \cos(\pi\Delta) \sum_{k=1}^{\infty} r^k \sin(\pi nk). \quad (\text{M.61})$$

Using the standard geometric sum,

$$\sum_{k=1}^{\infty} r^k e^{ik\theta} = \frac{re^{i\theta}}{1 - re^{i\theta}}, \quad |r| < 1, \quad (\text{M.62})$$

the real and imaginary parts of which are required for our calculation.

$$\sum_{k=1}^{\infty} r^k \cos(k\theta) = \frac{r \cos(\theta) - r^2}{1 - 2r \cos(\theta) + r^2}, \quad (\text{M.63})$$

$$\sum_{k=1}^{\infty} r^k \sin(k\theta) = \frac{r \sin(\theta)}{1 - 2r \cos(\theta) + r^2}. \quad (\text{M.64})$$

Then we have,

$$I_n(v) = \frac{\sin(\pi\Delta) (r \cos(\pi n) - r^2)}{1 - 2r \cos(\pi n) + r^2} - \frac{\cos(\pi\Delta) r \sin(\pi n)}{1 - 2r \cos(\pi n) + r^2} \quad (\text{M.65})$$

$$= \frac{v^{2n} (\sin(\pi(\Delta - n)) - \sin(\pi\Delta) v^{2n})}{1 - 2 \cos(\pi n) v^{2n} + v^{4n}} \quad (\text{M.66})$$

Now we can compute,

$$I_1(v) = -\sin(\pi\Delta) \left(\frac{v^2}{1+v^2} \right) \quad (\text{M.67})$$

$$I_{1/2}(v) = -\frac{\cos(\pi\Delta)v + \sin(\pi\Delta)v^2}{1+v^2}. \quad (\text{M.68})$$

This gives us the desired sum,

$$S_2(1/2/\Delta) = 2^{-2\Delta} C_\Delta \int_0^{\pi/2} d\phi \frac{A_\Delta(\phi)}{2} \left(-\frac{1}{4} \sin(\pi\Delta) - \frac{1}{2} \cos(\pi\Delta) \frac{v}{1+v^2} + \frac{1}{2} \sin(\pi\Delta) \frac{v^2}{1+v^2} \right) \quad (\text{M.69})$$

$$= 2^{-2\Delta} C_\Delta \int_0^{\pi/2} d\phi \frac{A_\Delta(\phi)}{2} \left(-\frac{1}{4} \sin(\pi\Delta) - \frac{1}{2} \cos(\pi\Delta) \sin(\phi) \cos(\phi) + \frac{1}{2} \sin(\pi\Delta) \sin^2(\phi) \right). \quad (\text{M.70})$$

Since the integrand is symmetric under $\phi \leftrightarrow \frac{\pi}{2} - \phi$, the third term cancels the first term. We are thus left with,

$$S_2(1/2, \Delta) = -\frac{2^{-2\Delta} C_\Delta \cos(\pi\Delta)}{4} \int_0^{\pi/2} d\phi A_\Delta(\phi) \sin(\phi) \cos(\phi), \quad (\text{M.71})$$

$$= -\frac{1}{4\pi} \int_0^{\pi/2} (\sin \phi \cos \phi)^{2\Delta} d\phi, \quad (\text{M.72})$$

$$= -\frac{1}{8\pi} B(\Delta + 1/2, \Delta + 1/2), \quad (\text{M.73})$$

where in the last step we used the integral representation of Beta function,

$$\int_0^{\pi/2} d\phi \sin^{p-1} \phi \cos^{q-1} \phi = \frac{1}{2} B(\Delta + 1/2, \Delta + 1/2). \quad (\text{M.74})$$

Finally using the Gamma duplication formula, we obtain,

$$S_2(1/2, \Delta) = -\frac{2^{-2\Delta-3} \Gamma(\Delta + 1/2)}{\sqrt{\pi} \Gamma(\Delta + 1)}. \quad (\text{M.75})$$

Consistency test 1

For coherent states, we have $\Delta = 1$, which gives $S_2(1/2, 1) = -1/64$. This does agree with the following identity for $\Delta = 1$,

$$\sum_{j=1}^{n-1} \csc^2(\pi j/n) = \frac{n^2 - 1}{3}. \quad (\text{M.76})$$

For $n = 1/2$, the identity along with the $(n/2)$ factor in the definition of $S_2(n, \Delta)$ gives $(2^{-2}) \left(\frac{1/2}{2} \right) \left(\frac{(1/2)^2 - 1}{3} \right) = -1/64$, in accordance to our result.

Consistency test 2

For coherent states, $F_\alpha^n(x) = \frac{\text{tr}(\rho_\Psi^n)}{\text{tr}(\rho_0^n)} = 1$ for any excited state created by the chiral Vertex operator V_α . We showed in Appendix M.2 that this relation indeed holds true up to second order. The contribution from the currents cancelled the contribution from the Jacobian factors and normalizations. From the definition of Logarithmic Negativity, we have

$$\Delta\mathcal{E}_\Psi = \mathcal{E}_\Psi - \mathcal{E}_0 = 2 \log F_\Psi^{1/2}(x), \quad (\text{M.77})$$

For coherent states, we then have,

$$F_\alpha^{n=1/2}(x) = 1 + \frac{\alpha^2}{6} \left(\frac{1}{n} - n \right) (\pi x)^2 \Big|_{n=1/2} + \alpha^2 \left(\frac{2\pi x}{n} \right)^2 S_2(n, 1) \Big|_{n=1/2} + O(x^3) \quad (\text{M.78})$$

$$= 1 + O(x^3), \quad (\text{M.79})$$

where the two terms in the first line cancel after we plugged in the value of $S_2(1/2, 1) = -1/64$. Equivalently, this shows that there is no change in Logarithmic negativity, i.e. $\Delta\mathcal{E}_\alpha = 0$ for coherent states.

Similarly, we can compute $\Delta\mathcal{E}_{i\partial\phi}$, $\Delta\mathcal{E}_\sigma$ and $\Delta\mathcal{E}_\varepsilon$. In [9–11], it was shown that

$$F_\varepsilon^n(x) = F_{i\partial\phi}^n(x) \quad (\text{M.80})$$

$$F_\sigma^n(x) = F_\alpha^n(x) = 1. \quad (\text{M.81})$$

This directly gives us $\Delta\mathcal{E}_\sigma = 0$. For ε and $J = i\partial\phi$,

$$F_{i\partial\phi}^n(x) = F_\varepsilon^n(x) = \left(\frac{2 \sin(\pi x)}{n} \right)^{2n} \left(\frac{\Gamma\left(\frac{1+n+n \csc(\pi x)}{2}\right)}{\Gamma\left(\frac{1-n+n \csc(\pi x)}{2}\right)} \right)^2 \quad (\text{M.82})$$

We thus obtain,

$$\Delta\mathcal{E}_{i\partial\phi} = \Delta\mathcal{E}_\varepsilon = 2 \log(4 \sin(\pi x)) + 4 \log \frac{\Gamma\left(\frac{3+\csc(\pi x)}{4}\right)}{\Gamma\left(\frac{1+\csc(\pi x)}{4}\right)} \quad (\text{M.83})$$

$$\underset{x \rightarrow 0}{\simeq} (\pi x)^2 - \frac{17}{6} (\pi x)^4 + O(x^6) \quad (\text{M.84})$$

We have,

$$F_\Psi^{(n)}(x) = \frac{\text{tr}(\rho_\Psi^n)}{\text{tr}(\rho_0^n)} \quad (\text{M.85})$$

$$\simeq D_n(x)^{-n} \quad (\text{M.86})$$

$$= \left(\frac{\sin(\pi x)}{n \sin(\pi x/n)} \right)^{2n\Delta_\Psi} \quad (\text{M.87})$$

$$\simeq \left(\frac{\pi x - \frac{(\pi x)^3}{6} + \frac{(\pi x)^5}{120} + \dots}{n \left(\frac{\pi x}{n} - \frac{(\pi x)^3}{6n^3} + \frac{(\pi x)^5}{120n^5} + \dots \right)} \right)^{2n\Delta_\Psi} \quad (\text{M.88})$$

$$\simeq 1 + 2n\Delta_\Psi a_n (\pi x)^2 + \left(2nb_n \Delta_\Psi + \frac{2n\Delta_\Psi (2n\Delta_\Psi - 1)}{2} a_n^2 \right) (\pi x)^4 + O(x^6) \quad (\text{M.89})$$

where we defined,

$$a_n = \frac{1}{6} \left(\frac{1}{n^2} - 1 \right), \quad b_n = \frac{3n^4 - 10n^2 + 7}{360n^4}. \quad (\text{M.90})$$

This gives,

$$\log F_\Psi^n(x) = \frac{\Delta_\Psi}{3} \left(\frac{1}{n} - n \right) (\pi x)^2 + \frac{\Delta_\Psi}{90} \left(\frac{1}{n^3} - n \right) (\pi x)^4 + O(x^6). \quad (\text{M.91})$$

This is the identity and conformal factor contribution up to the fourth order. For $n = 1/2$ and $\Delta = 1$, we have,

$$\Delta \mathcal{E}_{i\partial\phi} \Big|_{D_n(x)^{-n}} = \Delta \mathcal{E}_\varepsilon \Big|_{D_n(x)^{-n}} = 2 \log F_{i\partial\phi}^{1/2}(x) = 2 \log F_\varepsilon^{1/2}(x) = (\pi x)^2 + \frac{1}{6}(\pi x)^4 + O(x^6) \quad (\text{M.92})$$

Now, we can add the contribution from the two-point function of the stress tensor $\langle TT \rangle$,

$$\Delta \mathcal{E}_{i\partial\phi} \Big|_{TT} = 2 \log F_{i\partial\phi}^{1/2}(x) \Big|_{TT} \equiv 2 \left(C_{(i\partial\phi)(i\partial\phi)}^T \right)^2 \left(\frac{1}{2} \right) \left(\frac{2\pi x}{n} \right)^4 S_2(n, 2) \Big|_{n=1/2} = -3(\pi x)^4 \quad (\text{M.93})$$

$$\Delta \mathcal{E}_\varepsilon \Big|_{TT} = 2 \log F_\varepsilon^{1/2}(x) \Big|_{TT} \equiv 4 \left(C_{\varepsilon\varepsilon}^T \right)^2 \left(\frac{1}{4} \right) \left(\frac{2\pi x}{n} \right)^4 S_2(n, 2) \Big|_{n=1/2} = -3(\pi x)^4 \quad (\text{M.94})$$

where we used $S_2(1/2, 2) = -\frac{3}{1024}$. Notice, for ε we have $c = 1/2$, but we have contribution from both holomorphic and anti-holomorphic part. While for $i\partial\phi$, we have $c = 1$ and just the holomorphic contribution. So the net contribution from stress tensor is the same for both the cases. Adding both the contribution to the identity piece, we finally obtain,

$$\Delta \mathcal{E}_{i\partial\phi} = \Delta \mathcal{E}_\varepsilon = (\pi x)^2 - \frac{17}{6}(\pi x)^4 + O(x^6) \quad (\text{M.95})$$

in accordance with the exact result up to the fourth order.

M.4 Symmetry Resolved Relative Entropy

We extend our calculation of relative entropy of pure states to the symmetry resolution of relative entropy of the pure states. In the $x \rightarrow 0$ limit, we end up using OPE of close-by operators. The exact result of symmetry resolution of relative entropy of pure states for various examples has been carried out in [29].

The symmetry-resolved Renyi Relative entropy is given by,

$$S_n(\rho_\alpha(q) \parallel \rho_\beta(q)) = -\log \frac{p^\alpha(q)}{p^\beta(q)} + \frac{1}{1-n} \log \frac{p_n^{\alpha|\beta}(q)}{p_n^\alpha(q)} + S_n(\rho_\alpha \parallel \rho_\beta), \quad (\text{M.96})$$

where the generalized probabilities are defined as,

$$p_n^\alpha(q) = \frac{\text{tr}(\rho_\alpha^n \Pi_q)}{\text{tr}(\rho_\alpha^n)}, \quad p_n^{\alpha|\beta}(q) = \frac{\text{tr}(\rho_\alpha \rho_\beta^{n-1} \Pi_q)}{\text{tr}(\rho_\alpha \rho_\beta^{n-1})}, \quad (\text{M.97})$$

$$p^\alpha(q) = p_1^\alpha(q), \quad (\text{M.98})$$

$$p^\beta(q) = p_q^\beta(q). \quad (\text{M.99})$$

At $n \rightarrow 1$, we have the Symmetry Resolved Relative Entropy,

$$S(\rho_\alpha(q) \parallel \rho_\beta(q)) = -\log \frac{p^\alpha(q)}{p^\beta(q)} - \frac{1}{p^\alpha(q)} \partial_n [p_n^{\alpha|\beta}(q) - p_n^\alpha(q)] \Big|_{n=1} + S(\rho_\alpha \parallel \rho_\beta). \quad (\text{M.100})$$

We want to compute the first two terms of the right hand side. We define the charged moments, as the Fourier transformations of the generalized probabilities,

$$p_n^\alpha(q) \equiv \int \frac{d\mu}{2\pi} e^{-i\mu q} p_n^\alpha(q). \quad (\text{M.101})$$

Then, we have the charged moments of the excited states,

$$p_n^\alpha(\mu) = \frac{\text{tr}(\rho_\alpha^n e^{i\mu Q_A})}{\text{tr}(\rho_\alpha^n)}, \quad p_n^{\alpha|\beta}(\mu) = \frac{\text{tr}(\rho_\alpha \rho_\beta^{n-1} e^{i\mu Q_A})}{\text{tr}(\rho_\alpha \rho_\beta^{n-1})}. \quad (\text{M.102})$$

The operator $e^{i\mu Q_A}$ can be represented by two flux fields at the endpoints of the intervals. After uniformization mapping, the interval endpoints are sent to $0, \infty$.

M.4.1 Coherent states

For coherent states, the flux insertions are implemented by the vertex operator,

$$\mathcal{V} = V_{\frac{\mu}{2\pi}} = e^{\frac{i\mu}{2\pi} \phi} \quad (\text{M.103})$$

whose conformal weights are given by $(h_\mu, \bar{h}_\mu) = \left(\frac{1}{2} \left(\frac{\mu}{2\pi}\right)^2, \frac{1}{2} \left(\frac{\mu}{2\pi}\right)^2\right)$. So now we have a $2n+2$ point function,

$$\left\langle V_{\frac{\mu}{2\pi}}(\infty) V_{-\frac{\mu}{2\pi}}(0) \prod_{k=0}^{n-1} V_{\gamma_k}(z_{k,n}^+) V_{-\gamma_k}(z_{k,n}^-) \right\rangle, \quad (\text{M.104})$$

where γ_k depends on the replica word. For ρ_α^n , $\gamma_k = \alpha$ for all values of k , while for $\rho_\alpha \rho_\beta^{n-1}$, $\gamma_0 = \alpha$ and $\gamma_k = \beta$ for $k \neq 0$. After we do the OPE of the close by operators, we will end up with three and 4 point functions, such as,

$$\frac{\langle V_{\frac{\mu}{2\pi}}(\infty) V_{-\frac{\mu}{2\pi}}(0) J(z) \rangle}{\langle V_{\frac{\mu}{2\pi}}(\infty) V_{-\frac{\mu}{2\pi}}(0) \rangle} = -\frac{\mu}{2\pi z}, \quad (\text{M.105})$$

$$\frac{\langle V_{\frac{\mu}{2\pi}}(\infty) V_{-\frac{\mu}{2\pi}}(0) : J^2 : (z) \rangle}{\langle V_{\frac{\mu}{2\pi}}(\infty) V_{-\frac{\mu}{2\pi}}(0) \rangle} = \frac{\mu^2}{4\pi^2 z^2}, \quad (\text{M.106})$$

$$\frac{\langle V_{\frac{\mu}{2\pi}}(\infty) V_{-\frac{\mu}{2\pi}}(0) J(z) J(w) \rangle}{\langle V_{\frac{\mu}{2\pi}}(\infty) V_{-\frac{\mu}{2\pi}}(0) \rangle} = \frac{1}{(z-w)^2} + \frac{\mu^2}{4\pi^2 zw}. \quad (\text{M.107})$$

These are fixed by current Ward identities. Note that the first term in the four point function is the ordinary vacuum current two-point function, which gives the old relative entropy answer, while the second term of the four point function gives the flux-background term.

M.4.2 Three-point function

The three-point function contribution is of the first order in ζ_n , and is given by the sum of the $U(1)$ charges of each of the operator insertions on the circle,

$$R_n^{(1)} = \sum_k (-i\gamma_k \zeta_n z_k) \left(-\frac{\mu}{2\pi z_k} \right) \quad (\text{M.108})$$

$$= i\frac{\mu}{2\pi} \zeta_n \sum_k \gamma_k. \quad (\text{M.109})$$

Similarly the $:J^2:$ term gives,

$$R_{n,;J^2}^{(2)} = -\frac{\mu^2 \zeta_n^2}{8\pi^2} \sum_k \gamma_k^2 \quad (\text{M.110})$$

M.4.3 Four-point function

The four point function in the charged background gives,

$$R_{n,JJ}^{(2)} = -\frac{\mu^2 \zeta_n^2}{4\pi^2} \sum_{k<j} \gamma_k \gamma_j, \quad (\text{M.111})$$

$$= -\frac{\mu^2 \zeta_n^2}{8\pi^2} \left(\sum_k \gamma_k \right)^2. \quad (\text{M.112})$$

Combining the three-point and four-point function, we obtain,

$$R_n(\{\gamma_k\}; \mu) = R_n^{(1)} + R_{n,;J^2}^{(2)} + R_{n,JJ}^{(2)} \quad (\text{M.113})$$

$$= 1 + i\frac{\mu}{2\pi} \zeta_n \sum_k \gamma_k - \frac{1}{2} \left(\frac{\mu}{2\pi} \right)^2 \zeta_n^2 \left(\sum_k \gamma_k \right)^2. \quad (\text{M.114})$$

M.4.4 Computation of $\log \frac{p^\alpha(q)}{p^\beta(q)}$

For ρ_α^n , we have,

$$p_n^\alpha(\mu) = p_n^0(\mu) \left[1 + i\mu\alpha x - \frac{1}{2}\mu^2\alpha^2 x^2 + O(x^3) \right] \quad (\text{M.115})$$

while for $\rho_\alpha \rho_\beta^{n-1}$ we have,

$$p_n^{\alpha|\beta}(\mu) = \partial_n^0(\mu) \left[1 + i\mu \frac{\alpha + (n-1)\beta}{n} x - \frac{1}{2}\mu^2 \left(\frac{\alpha + (n-1)\beta}{n} \right)^2 x^2 + O(x^3) \right] \quad (\text{M.116})$$

After Fourier transform, we obtain the generalized probabilities,

$$p_n^\alpha(q) = p_n^0(q) - \alpha x \partial_q p_n^0(q) + \frac{\alpha^2 x^2}{2} \partial_q^2 p_n^0(q) + O(x^3), \quad (\text{M.117})$$

$$p_n^{\alpha|\beta}(q) = p_n^0(q) - \frac{\alpha + (n-1)\beta}{n} \partial_q p_n^0(q) + \frac{\left(\frac{\alpha + (n-1)\beta}{n}\right)^2}{2} \partial_q^2 p_n^0(q) + O(x^3). \quad (\text{M.118})$$

Similarly, at $n = 1$,

$$p^\alpha(q) = p_1^0(q) - \alpha x p_1^{0'}(q) + \frac{\alpha^2 x^2}{2} p_1^{0''}(q) + O(x^3), \quad (\text{M.119})$$

$$p^\beta(q) = p_1^0(q) - \beta x p_1^{0'}(q) + \frac{\beta^2 x^2}{2} p_1^{0''}(q) + O(x^3), \quad (\text{M.120})$$

Hence, by expanding the logarithm in small x , we have

$$\log p^\alpha(q) = \log p_n^0(q) - \alpha x \frac{p_n^{0'}}{p_n^0} + \frac{\alpha^2 x^2}{2} \left(\frac{p_n^{0''}}{p_n^0} - \left(\frac{p_n^{0'}}{p_n^0} \right)^2 \right) + O(x^3), \quad (\text{M.121})$$

$$\log p^\beta(q) = \log p_n^0(q) - \beta x \frac{p_n^{0'}}{p_n^0} + \frac{\beta^2 x^2}{2} \left(\frac{p_n^{0''}}{p_n^0} - \left(\frac{p_n^{0'}}{p_n^0} \right)^2 \right) + O(x^3). \quad (\text{M.122})$$

For simplicity of notations, we introduce,

$$\delta \equiv \alpha - \beta, \quad a_n \equiv \frac{\alpha + (n-1)\beta}{n}, \quad (\text{M.123})$$

$$L_n(q) \equiv \log p_n^0(q), \quad (\text{M.124})$$

$$\implies L'_n = \frac{p_n^{0'}}{p_n^0}, \quad L''_n = \frac{p_n^{0''}}{p_n^0} - \left(\frac{p_n^{0'}}{p_n^0} \right)^2. \quad (\text{M.125})$$

Then we have,

$$-\log \frac{p^\alpha(q)}{p^\beta(q)} = \delta x L'_1 - \frac{\alpha^2 - \beta^2}{2} x^2 L''_1 + O(x^3). \quad (\text{M.126})$$

M.4.5 Computation of $\frac{1}{p^\alpha(q)} \partial_n [p_n^{\alpha|\beta}(q) - p_n^\alpha(q)] \big|_{n=1}$

We have,

$$p_n^{\alpha|\beta}(q) - p_n^\alpha(q) = -(a_n - \alpha) x p_n^{0'} + \frac{a_n^2 - \alpha^2}{2} x^2 p_n^{0''} + O(x^3), \quad (\text{M.127})$$

$$\implies \partial_n [p_n^{\alpha|\beta}(q) - p_n^\alpha(q)] \big|_{n=1} = \delta x p_1^{0'} - \alpha \delta x^2 p_1^{0''} + O(x^3). \quad (\text{M.128})$$

The expansion of the inverse of the generalized probability is given by,

$$-\frac{1}{p^\alpha(q)} = -\frac{1}{p_1^0} \left[1 + \alpha x \frac{p_1^{0'}}{p_1^0} + O(x^2) \right] \quad (\text{M.129})$$

Combining the above two expressions, we obtain,

$$-\frac{1}{p^\alpha(q)}\partial_n[p_n^{\alpha|\beta}(q) - p_n^\alpha(q)]|_{n=1} = -\delta x L'_1 + \alpha \delta x^2 L''_1 + O(x^3). \quad (\text{M.130})$$

Combining these results gives the Symmetry Resolved Relative Entropy,

$$S(\rho_\alpha(q)\|\rho_\beta(q)) = (\alpha - \beta)^2 x^2 \left[\frac{\pi^2}{3} + \frac{1}{2} \partial_q^2 \log p_1^0(q) \right] + O(x^3). \quad (\text{M.131})$$

M.4.6 Coherent states exact result comparison

Since for compact free boson, the vacuum charged moment is Gaussian,

$$p_n^0(\mu) = \exp \left[-\frac{1}{2} \langle (\Delta q)^2 \rangle_n^0 \mu^2 + i\mu \bar{q} \right], \quad (\text{M.132})$$

therefore,

$$p_n^0(q) = \frac{1}{\sqrt{2\pi \langle (\Delta q)^2 \rangle_n^0}} \exp \left[-\frac{(q - \bar{q})^2}{2 \langle (\Delta q)^2 \rangle_n^0} \right]. \quad (\text{M.133})$$

Then, we have,

$$\partial_q^2 \log p_1^0(q) = -\frac{1}{\langle (\Delta q)^2 \rangle_1^0}. \quad (\text{M.134})$$

The Symmetry Resolved Relative Entropy of coherent states is thus given by,

$$S(\rho_\alpha(q)\|\rho_\beta(q)) = \frac{(\alpha - \beta)^2}{3} (\pi x)^2 - \frac{(\alpha - \beta)^2 x^2}{2 \langle (\Delta q)^2 \rangle_1^0} + O(x^3), \quad (\text{M.135})$$

which agrees with the small x expansion given in [29].

M.4.7 Mixtures

Check, if the above result can be generalized to the mixture case,

$$S(\bar{\rho}_{\mathbf{p}}(q)\|\bar{\rho}_{\mathbf{q}}(q)) \stackrel{?}{=} \frac{(\langle \alpha \rangle_{\mathbf{p}} - \langle \beta \rangle_{\mathbf{q}})^2}{3} (\pi x)^2 - \frac{(\langle \alpha \rangle_{\mathbf{p}} - \langle \beta \rangle_{\mathbf{q}})^2 x^2}{2 \langle (\Delta q)^2 \rangle_1^0} + O(x^3), \quad (\text{M.136})$$

where

$$\bar{\rho}_{\mathbf{p}} = \sum_{a=1}^M p_a \rho_{\alpha_a}, \quad \sum_{a=1}^M p_a = 1, \quad (\text{M.137})$$

$$\bar{\rho}_{\mathbf{q}} = \sum_{b=1}^M q_b \rho_{\beta_b}, \quad \sum_{b=1}^M q_b = 1, \quad (\text{M.138})$$

$$\langle \alpha \rangle_{\mathbf{p}} = \sum_{a=1}^M p_a \alpha_a, \quad \langle \beta \rangle_{\mathbf{q}} = \sum_{b=1}^M q_b \beta_b. \quad (\text{M.139})$$

M.5 Is there an OPE expansion for $S(\rho_0||\rho_\Phi)$?

where the first factor comes the normalization of the density matrices, and we have used $\sin(\pi x) = \sin(\pi(1-x))$. We also used the following OPE for the Φ and Φ^* with identity close to it,

$$\begin{aligned}\Phi(z_{1,n}^-)I(z_{0,n}^+) &\sim \Phi(z_{0,n}^+) + \dots \\ \Phi^*(z_{n-1,n}^+)I(z_{0,n}^-) &\sim \Phi^*(z_{0,n}^-) + \dots\end{aligned}\tag{M.140}$$

We compute the two and three-point functions above

$$\langle \Phi^*(z_{0,n}^+) \Phi(z_{0,n}^-) \rangle = \left(2 \sin \left(\frac{\pi(1-\epsilon)}{n} \right) \right)^{-2\Delta_\Phi} e^{-3\pi i s_\Phi} \tag{M.141}$$

$$\langle \Phi(z_{n-1,n}^+) \Phi^*(z_{1,n}^-) \rangle = \left(2 \sin \left(\frac{\pi(1+\epsilon)}{n} \right) \right)^{-2\Delta_\Phi} e^{-3\pi i s_\Phi} \tag{M.142}$$

$$\begin{aligned}\langle \Phi(z_{0,n}^-) \Phi^*(z_{0,n}^+) \mathcal{O}_L(e^{i\pi(2k+1)/n}) \rangle &= C_\Phi^L \left(2 \sin \left(\frac{\pi(1-\epsilon)}{n} \right) \right)^{-2\Delta_\Phi + \Delta_L} \left(2 \sin \left(\frac{\pi(k+1-\epsilon/2)}{n} \right) \right)^{-\Delta_L} \times \\ &\quad \left(2 \sin \left(\frac{\pi(k+\epsilon/2)}{n} \right) \right)^{-\Delta_L} e^{-3\pi i s_\Phi} e^{-i\pi s_L((2k+1)/n+1/2)} \\ &= C_\Phi^L \left(2 \sin \left(\frac{\pi(1-\epsilon)}{n} \right) \right)^{-2\Delta_\Phi + \Delta_L} \times \\ &\quad (2 \cos(\pi(1-\epsilon)/n) - 2 \cos(\pi(2k+1)/n))^{-\Delta_L} e^{-3\pi i s_\Phi} e^{-i\pi s_L((2k+1)/n+1/2)}\end{aligned}\tag{M.143}$$

$$\begin{aligned}\langle \Phi(z_{n-1,n}^+) \Phi^*(z_{1,n}^-) \mathcal{O}_L(e^{i\pi(2k+1)/n}) \rangle &= e^{3\pi i s_\Phi/2} e^{-i\pi(s_L+s_\Phi)((2k+1)/(2n)+1/2)} \times \\ &\quad (C_\Phi^L)^2 \left(2 \sin \left(\frac{\pi(1+\epsilon)}{n} \right) \right)^{\Delta_L} \times \left(2 \sin \left(\frac{\pi(2k+1)}{2n} \right) \right)^{-2\Delta_L}\end{aligned}\tag{M.144}$$

Therefore, in the small ϵ expansion we have

$$\begin{aligned}\langle \tilde{\mathcal{O}}_{0|\Phi}^{(n-1)} \rangle &= \left(\frac{n \sin(\pi\epsilon/n)}{\sin(\pi\epsilon)} \right)^{-2\Delta_\Phi(n-1)} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1-\epsilon)/n)} \right)^{2\Delta_\Phi} \times \\ &\quad \left(1 + C_\Phi^L (\sin(\pi\epsilon/n) \sin(\pi(1-\epsilon)/n))^{\Delta_L} \times \right. \\ &\quad \left. \sum_{k=1}^{n-2} \left(\frac{1}{\sin(\pi(k+1-\epsilon/2)/n) \sin(\pi(k+\epsilon/2)/n)} \right)^{-\Delta_L} + \dots \right).\end{aligned}\tag{M.145}$$

$$\begin{aligned} \langle \tilde{\mathcal{O}}_{0|\Phi}^{(n-1)} \rangle &= \left(\frac{n \sin(\pi\epsilon/n)}{\sin(\pi\epsilon)} \right)^{-2\Delta_\Phi(n-1)} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1+\epsilon)/n)} \right)^{2\Delta_\Phi} \times \\ &\quad \left(1 + (C_\Phi^L)^2 e^{4\pi i s_\Phi} \frac{(\sin(\pi(1+\epsilon)/n))^{2\Delta_\Phi+\Delta_L}}{(\sin(\pi\epsilon/n))^{2\Delta_\Phi-\Delta_L}} \times \sum_{k=1}^{n-2} \frac{e^{i\pi(s_L-s_\Phi)((2k+1)/(2n))}}{\left(\sin\left(\frac{\pi(2k+1)}{2n}\right)\right)^{2\Delta_L}} + \dots \right) \end{aligned} \quad (\text{M.146})$$

$$\begin{aligned} &= \left(\frac{n \sin(\pi\epsilon/n)}{\sin(\pi\epsilon)} \right)^{-2\Delta_\Phi(n-1)} \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1+\epsilon)/n)} \right)^{2\Delta_\Phi} \times \\ &\quad \left(1 + (C_\Phi^L)^2 e^{4\pi i s_\Phi} \left(\frac{\sin(\pi(1+\epsilon)/n)}{\sin(\pi\epsilon/n)} \right)^{2\Delta_\Phi} (\sin(\pi(1+\epsilon)/n) \sin(\pi\epsilon/n))^{\Delta_L} \sum_{k=1}^{n-2} \frac{1}{\left(\sin\left(\frac{\pi(2k+1)}{2n}\right)\right)^{2\Delta_L}} + \dots \right) \end{aligned} \quad (\text{M.147})$$

If we did not have any phase term in the sum in the last line of the above equation, then we end up with

$$\sum_{k=0}^{n-1} \frac{1}{\sin^2(\pi(2k+1)/(2n))^\Delta} = \frac{2}{2n} \sum_{k=1}^{2n-1} \frac{1}{\left(\sin^2\left(\frac{\pi j}{2n}\right)\right)^\Delta} - \frac{2}{n} S_2(n, \Delta) \quad (\text{M.148})$$

Here we wrote the sum over the odd sites to be the sum over all $2n-1$ sites, and subtract the sum over even sites, which is $2S_2(n)/n$. Note that we have to subtract the terms for $k=1, n-1$ as well. Thus we can simplify the sum,

$$\sum_{k=1}^{n-2} \frac{1}{\left(\sin\left(\frac{\pi(2k+1)}{2n}\right)\right)^{2\Delta_L}} = 2^{2\Delta_L} \left(\frac{S_2(2n, \Delta_L)}{n} - \frac{2S_2(n, \Delta_L)}{n} - \frac{2}{(\sin(\pi/(2n)))^{2\Delta_L}} \right) \quad (\text{M.149})$$

The correction to the divergent piece comes when we take the derivative of log of the things inside parentheses. We have the divergent piece in the denominator as the common factor. When the derivative hits the summation, we get,

$$S'_2(2) - S_2(2) - 2S'_2(1) \quad (\text{M.150})$$

While the derivative hitting the term outside the sum gives,

$$\left(\left(\frac{\sin(\pi(1+\epsilon)/n)}{\sin(\pi\epsilon/n)} \right)^{2\Delta_\Phi} (\sin(\pi(1+\epsilon)/n) \sin(\pi\epsilon/n))^{\Delta_L} \right)' \Big|_{n=1} \quad (\text{M.151})$$

$$= (-1)^{2\Delta_\Phi+\Delta_L} (\pi\epsilon)^{2\Delta_L} \left((2\Delta_\Phi + \Delta_L) \left(-\frac{1}{\epsilon} + \frac{\pi^2\epsilon}{3} + \dots \right) - 2\Delta_L \left(1 - \frac{\pi^2\epsilon^2}{3} \dots \right) \right) \quad (\text{M.152})$$

So this piece is divergent, but is suppressed from the leading divergent piece. Similarly, the other piece gives,

$$S'_2(2, \Delta_L) - 1 - 2 \frac{\sqrt{\pi}\Gamma(\Delta_L+1)}{4\Gamma(\Delta_L+3/2)} \quad (\text{M.153})$$

where we used $2^{2\Delta_L} S_2(2) = 1$. We compute $S'_2(2)$ in the Appendix [M.6](#),

$$S'_2(2) = \frac{2^{2\Delta} \sin(\pi\Delta)}{\pi} (I(\Delta + 2, -2\Delta, 1) + I(\Delta, -2\Delta, 1) + 2\partial_a I(a, -2\Delta, 1)|_{a=\Delta}) \quad (\text{M.154})$$

$$I(a, b, c) = \int_0^1 dt t^{a-1} (1-t)^{b-1} (1+t)^{-c} = B(a, b) {}_2F_1(a, c; a+b; -1) \quad (\text{M.155})$$

$$\partial_a I(a, b, c) = \int_0^1 dt t^{a-1} (1-t)^{b-1} (1+t)^{-c} \ln t = \partial_a (B(a, b) {}_2F_1(a, c; a+b; -1)) \quad (\text{M.156})$$

As long as $S'_2(2)$ is finite, these corrections are suppressed. Finally, we obtain the relative entropy,

$$\begin{aligned} S(\rho_0 \| \rho_\Phi) &= -2\Delta_\Phi \left(\frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} + \dots \right) \\ &\quad + (-1)^{2\Delta_\Phi + \Delta_L} (C_\Phi^L)^2 (\pi\epsilon)^{2\Delta_L} \left((2\Delta_\Phi + \Delta_L) \left(\frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} + \dots \right) + 2\Delta_L \left(1 - \frac{\pi^2 \epsilon^2}{3} \right) \right. \\ &\quad \left. - \left(S'_2(2) + 1 - 2 \frac{\sqrt{\pi} \Gamma(\Delta + 1)}{4\Gamma(\Delta + 3/2)} \right) \right) \end{aligned} \quad (\text{M.157})$$

We need to take the logarithm followed by a ∂_n at $n \rightarrow 1$. It is straightforward to see

$$\begin{aligned} (\partial_n \log (D_n(\epsilon)^{n-1}))_{n \rightarrow 1} &= 0 \\ \left(\partial_n \log \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1+\epsilon)/n)} \right) \right)_{n \rightarrow 1} &= \pi \cot(\pi\epsilon) = \frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} + O(\epsilon^3) \\ \frac{\sin(\pi\epsilon/n)}{\sin(\pi(1+\epsilon)/n)} &= \left(\frac{\pi\epsilon}{n \sin(\pi/n)} \right) \left(1 - \frac{\cot(\pi/n) \pi\epsilon}{n} + O(\epsilon^2) \right) \end{aligned} \quad (\text{M.158})$$

Therefore,

$$\begin{aligned} S(\rho_0 \| \rho_\beta) &= 4h_\beta \left(\frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} \right) - (C_\beta^L)^2 \partial_n \mathcal{T}(n)|_{n \rightarrow 1} + O(\epsilon^3) \\ \mathcal{T}(n) &= \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(1+\epsilon)/n)} \right)^{\Delta_L} \sum_{k=0}^{n-2} (2 \cos(\pi(1+\epsilon)/n) - 2 \cos(2\pi k/n))^{-\Delta_L} \\ &= \left(\frac{\pi\epsilon}{4n} \right)^{\Delta_L} T(n) (1 - O(\epsilon)) \\ T(n) &= \sum_{k=0}^{n-2} \left(\sin\left(\frac{\pi}{n}\right) \sin\left(\frac{(2k+1)}{2n}\right) \sin\left(\frac{(2k-1)}{2n}\right) \right)^{-\Delta_L} \end{aligned} \quad (\text{M.159})$$

$$\begin{aligned}
S(\rho_0\|\rho_\Phi) &= 2\Delta_\Phi \left(\frac{1}{\epsilon} - \sum_{k=1}^{\infty} \frac{2^{2k}|B_{2k}|}{(2k)!} \epsilon^{2k-1} \right) - C_\Phi^L \partial_n \mathcal{T}(n)|_{n \rightarrow 1} \\
\mathcal{T}(n) &= (\sin(\pi\epsilon/n) \sin(\pi(1-\epsilon)/n))^{\Delta_L} \sum_{k=1}^{n-2} \left(\frac{1}{\sin(\pi(k+1-\epsilon/2)/n) \sin(\pi(k+\epsilon/2)/n)} \right)^{-\Delta_L} \\
&= (\sin(\pi\epsilon/n) \sin(\pi(1-\epsilon)/n))^{\Delta_L} T(n) (1 - O(\epsilon)) \\
T(n) &= \sum_{k=1}^{n-2} \left(\frac{1}{\sin(\pi(k+1-\epsilon/2)/n) \sin(\pi(k+\epsilon/2)/n)} \right)^{-\Delta_L}
\end{aligned}$$

Therefore,

$$S(\rho_0\|\rho_\beta) = 4h_\beta \left(\frac{1}{\epsilon} - \frac{\pi^2 \epsilon}{3} \right) - (C_\beta^L)^2 (\pi\epsilon)^{\Delta_L} \mathcal{T}'(1) + \dots \quad (\text{M.160})$$

$$S(\rho_0\|\rho_\beta) = 2\Delta_\Phi \left(\frac{1}{\epsilon} - \sum_{k=1}^K \frac{2^{2k}|B_{2k}|}{(2k)!} \pi^{2k} \epsilon^{2k-1} \right) - C_\Phi^L \mathcal{T}'(1) + \dots$$

where we have used the fact that at $n = 1$ there should be no contribution coming from OPE expansions, so $T(1) = 0$.

Again, with the use of the OPE in eq. [2.105], we obtain

$$S(\rho_0\|\rho_\beta) = (1 - \delta_{\Psi\Phi}) \left(2\Delta_\Phi \left(\frac{1}{\epsilon} - \frac{\epsilon}{3} \right) - C_\Phi \mathcal{T}'(1) + \dots \right) \quad (\text{M.161})$$

This makes it clear that the divergence of $S(\rho_0\|\rho_\beta)$ comes from the $1/(1-x)$ term in the first term with residue that is universally fixed to be $4h_\beta$. The comparison with the relative entropy $S(\rho_0\|\rho_\beta)$ of the coherent states suggests that the term $T(n)$ does not contribute to the relative entropy, i.e. its analytic continuation vanishes at this order.

M.6 $S'_2(2)$

{Analytic cont.

We want to compute the derivative of $S_2(n)$ at $n = 2$. Thus we require,

$$\partial_n S_n|_{n=2} = \frac{1}{2} \sum_{m \geq 1} (g(2m) + 2mg'(2m)) + \frac{1}{2} \sum_{m \geq 0} (g(-2m) - 2mg'(-2m)) \quad (\text{M.162})$$

$$= \frac{1}{2} \sum_{m \geq 1} (g(2m) + 2mg'(2m)) + \frac{1}{2} \sum_{m \geq 0} (g(2m) + 2mg'(2m)) \quad (\text{M.163})$$

where in the last line, we used the reflection property ($g(-m) = g(m)$) at integer m . Then using the integral representation of $g_\Delta(m)$ and $g'_\Delta(m)$, we get,

$$\partial_n S_n|_{n=2} = \frac{A_\Delta}{2} \left[\int_0^1 dt (1-t)^{-2\Delta} \left(\sum_{m \geq 1} t^{2m+\Delta-1} (1+2m \ln t) + \sum_{m \geq 0} t^{2m+\Delta-1} (1+2m \ln t) \right) \right] \quad (\text{M.164})$$

We use the following geometric sums,

$$\sum_{m \geq 1} t^{2m} = \frac{t^2}{1 - t^2}, \quad (\text{M.165})$$

$$\sum_{m \geq 1} 2mt^{2m} = \frac{2t}{(1 - t^2)^2} \quad (\text{M.166})$$

to obtain,

$$\begin{aligned} \partial_n S_n|_{n=2} = \frac{A_\Delta}{2} & \left(\int_0^1 dt t^{\Delta+1} (1-t)^{-2\Delta-1} (1+t)^{-1} + \int_0^1 dt t^{\Delta-1} (1-t)^{-2\Delta-1} (1+t)^{-1} \right. \\ & \left. + 2 \int_0^1 dt t^\Delta (1-t)^{-2\Delta-1} (1+t)^{-1} \ln t \right) \end{aligned} \quad (\text{M.167})$$

We use the following identities,

$$\int_0^1 dt t^{a-1} (1-t)^{b-1} (1+t)^{-c} = B(a, b) {}_2F_1(a, c; a+b; -1) \quad (\text{M.168})$$

$$\int_0^1 dt t^{a-1} (1-t)^{b-1} (1+t)^{-c} \ln t = \partial_z (B(z, b) {}_2F_1(z, c; z+b; -1))|_{z=a} \quad (\text{M.169})$$

Denote

$$I(a, b, c) = B(a, b) {}_2F_1(a, c; a+b; -1). \quad (\text{M.170})$$

$$(\text{M.171})$$

Combining these results gives,

$$\partial_n S_2(n)|_{n=2} = \frac{A_\Delta}{2} (I(\Delta+2, -2\Delta, 1) + I(\Delta, -2\Delta, 1) + 2\partial_a I(a, -2\Delta, 1)|_{a=\Delta+1}). \quad (\text{M.172})$$

M.7 Fisher Information

$S(\rho_0 \| y\rho_0 + (1-y)\rho_\Phi)$.

The denominator starts from 1, coming from $p = 0$. For $p = 1$, we just have a sum over two-point functions.

$$y^{n-1} \left(1 + \frac{(n-1)(y^{-1} - 1)D_n(\epsilon)\zeta_n^{2\Delta_\Phi}}{(2\sin(\pi(1-\epsilon)/n))^{2\Delta_\Phi}} + \dots \right) \quad (\text{M.173})$$

For $p = 2$, we can have either have a string of $\Phi\Phi^*$ of length $p = 2$, or a pair of $\Phi\Phi^*$ which do not share common site. The contribution from $p = 2$ is,

$$((y^{-1} - 1)D_n(\epsilon))^2 \left((n-2) \left(\frac{\sin(\pi\epsilon/n)}{\sin(\pi(2-\epsilon)/n)} \right)^{2\Delta_\Phi} + O(\epsilon^{4\Delta_\Phi}) \right) \quad (\text{M.174})$$

where the higher order terms come from the non-stringy contribution, which is clearly sub-leading term. Ignoring the non-stringy contribution, we have,

$$\mathcal{F}_n = \frac{1}{y^{n-1} \left(1 + \zeta_n^{2\Delta_\Phi} \sum_{p=1}^{n-1} (n-p) \frac{((y^{-1}-1)(D_n(\epsilon))^p}{(2 \sin(\pi(p-\epsilon)/n))^{2\Delta_\Phi}} + \dots \right)} \quad (\text{M.175})$$

If we want to consider $y \rightarrow 1$ limit and keep the term up to the order of $(1-y)^2$ to relate it to quantum Fisher information, we thus have the following in the limit $y \rightarrow 1$,

$$\begin{aligned} S(\rho_0 \| (\rho_0 + (1-y)(\rho_\Phi - \rho_0))) &= -\log y - \partial_n \left((n-1)(y^{-1}-1)D_n(\epsilon) \left(\frac{\zeta_n}{2 \sin(\pi(1-\epsilon)/n)} \right)^{2\Delta_\Phi} \right. \\ &\quad \left. + (n-2) ((y^{-1}-1)(D_n(\epsilon))^2 \left(\frac{\zeta_n}{2 \sin(\pi(2-\epsilon)/n)} \right)^{2\Delta_\Phi} + \dots \right) \Big|_{n=1} \end{aligned} \quad (\text{M.176})$$

Notice, when the derivative hits $(n-1)$, in the limit $y \rightarrow 1$, the contribution from $(y^{-1}-1) \simeq (1-y)$ gets cancelled from $\log y$ expansion. Hence, we are left with terms of order $(1-y)^2$ and higher. Therefore, in the limit $y \rightarrow 1$, for coherent states up to the smallest order we have,

$$\begin{aligned} S(\rho_0 \| (\rho_0 + (1-y)(\rho_\beta - \rho_0))) &\simeq (1-y)^2/2 - \partial_n \left[(n-2) ((y^{-1}-1)D_n(\epsilon))^2 \left(\frac{\zeta_n}{2 \sin(\pi(2-\epsilon)/n)} \right)^{2\Delta_\Phi} \times \right. \\ &\quad \left(1 + 2\beta^2 \left(\frac{\pi\epsilon}{n} \right) \cot(\pi/n) + \frac{1}{2} \left(2\beta^2 \left(\frac{\pi\epsilon}{n} \right) \cot(\pi/n) \right)^2 \right. \\ &\quad \left. \left. + \left(\frac{\pi\epsilon}{n} \right)^2 \beta^2 \csc^2(\pi/n) + \dots \right) \right] \Big|_{n=1} \end{aligned} \quad (\text{M.177})$$

$$\propto (1-y)^2 \quad (\text{M.178})$$

where in the first line, the first term comes from the expansion of $-\log y$. We also added the corrections due to the three-point and four point functions in the first line.

$$S(\rho_0 \| (\rho_0 + (1-y)(\rho_\Phi - \rho_0))) = (-1)^{2\Delta_\Phi+1} 2^{-2\Delta_\Phi} (1-y)^2 + (1-y)^2 \partial_n \left(D_n(\epsilon)^2 \left(\frac{\zeta_n}{2 \sin(2\pi(1-\epsilon)/n)} \right)^{2\Delta_\Phi} \right) \quad (\text{M.179})$$

where the first term is a result of the derivative hitting $(n-2)$, which gives $(-1)^{2\Delta_\Phi}$ from

$\sin(2\pi(1-\epsilon)/n)^{2\Delta_\Phi}$. We then have,

$$S(\rho_0 \| (\rho_0 + (1-y)(\rho_\Phi - \rho_0))) = (-1)^{2\Delta_\Phi+1} 2^{-2\Delta_\Phi} (1-y)^2 \\ + (1-y)^2 \partial_n \left(\left(\frac{n \sin(\pi\epsilon/n)}{\sin(\pi\epsilon)} \right)^{-4\Delta_\Phi} \times \left(\frac{\zeta_n}{2 \sin(2\pi(1-\epsilon)/n)} \right)^{2\Delta_\Phi} \right) \quad (\text{M.180})$$

$$= (-1)^{2\Delta_\Phi+1} 2^{-2\Delta_\Phi} (1-y)^2 \\ + (1-y)^2 (\sin(\pi\epsilon))^{4\Delta_\Phi} \times \partial_n \left(\frac{1}{n^2 \sin(\pi\epsilon/n) \sin(2\pi(1-\epsilon)/n)} \right)^{2\Delta_\Phi} \quad (\text{M.181})$$

$$= (-1)^{2\Delta_\Phi+1} 2^{-2\Delta_\Phi} (1-y)^2 + 2\Delta_\Phi (1-y)^2 (\pi\epsilon)^{4\Delta_\Phi} \left(-2 + \frac{\pi\epsilon}{n^2} \cot\left(\frac{\pi\epsilon}{n}\right) \right. \\ \left. + \frac{2\pi(1-\epsilon)}{n^2} \cot\left(\frac{2\pi(1-\epsilon)}{n}\right) \right) \Big|_{n=1} \quad (\text{M.182})$$

$$= (-1)^{2\Delta_\Phi+1} 2^{-2\Delta_\Phi} (1-y)^2 + 2\Delta_\Phi (1-y)^2 (\pi\epsilon)^{4\Delta_\Phi} \left(-2 + \pi\epsilon \left(\frac{1}{\pi\epsilon} - \frac{\pi\epsilon}{3} + \dots \right) \right. \\ \left. - 2\pi(1-\epsilon) \left(\frac{1}{2\pi\epsilon} - \frac{2\pi\epsilon}{3} + \dots \right) \right) \quad (\text{M.183})$$

$$= (-1)^{2\Delta_\Phi+1} 2^{-2\Delta_\Phi} (1-y)^2 + 2\Delta_\Phi (1-y)^2 (\pi\epsilon)^{4\Delta_\Phi} \left(-\frac{1}{\epsilon} + \frac{4\pi^2\epsilon}{3} - \frac{5\pi^2\epsilon^2}{3} + \dots \right) \quad (\text{M.184})$$

If we consider the following,

$$S(\rho_\Psi \| (\rho_\Psi + (1-y)(\rho_0 - \rho_\Psi))) \quad (\text{M.185})$$

$$= \partial_n \log \left[\frac{1 + \zeta_n^{2\Delta_L} f(n) + \dots}{y^{n-1} \left(1 + \zeta_n^{2\Delta_L} f(n) + (n-2) ((y^{-1} - 1) D_n(\epsilon))^2 \left(\frac{\zeta_n}{2 \sin(2\pi(1+\epsilon)/n)} \right)^{2\Delta_\Phi} + \dots \right)} \right] \quad (\text{M.186})$$

Here we did not include the contribution from $p = 1$ since it does not contribute to the relative entropy, because of the $n - 1$ factor sitting in front of it. Also notice that the numerator gets

cancelled exactly from the $p = 0$ term in the denominator. This gives us,

$$S(\rho_\Psi \| (\rho_\Psi + (1-y)(\rho_0 - \rho_\Psi))) = -2^{-2\Delta_\Phi}(1-y)^2 + 2\Delta_\Phi(1-y)^2(\pi\epsilon)^{4\Delta_\Phi} \left(-2 + \frac{\pi\epsilon}{n^2} \cot\left(\frac{\pi\epsilon}{n}\right) + \frac{2\pi(1+\epsilon)}{n} \cot\left(\frac{2\pi(1+\epsilon)}{n}\right) \right) \Big|_{n=1} \quad (\text{M.187})$$

$$= -2^{-2\Delta_\Phi}(1-y)^2 + 2\Delta_\Phi(1-y)^2(\pi\epsilon)^{4\Delta_\Phi} \left(-2 + \pi\epsilon \left(\frac{1}{\pi\epsilon} - \frac{\pi\epsilon}{3} + \dots \right) + 2\pi(1+\epsilon) \left(\frac{1}{2\pi\epsilon} - \frac{2\pi\epsilon}{3} + \dots \right) \right) \quad (\text{M.188})$$

$$= -2^{-2\Delta_\Phi}(1-y)^2 + 2\Delta_\Phi(1-y)^2(\pi\epsilon)^{4\Delta_\Phi} \left(\frac{1}{\epsilon} - \frac{4\pi^2\epsilon}{3} - \frac{5\pi^2\epsilon^2}{3} + \dots \right) \quad (\text{M.189})$$

Here, we get positive divergence, in contrast to the previous case.

M.8 Pole-matching analytic continuation of $g(p; x)$ and $h(n-p, p; x)$

Here, we reformulate the analytic continuation of the building blocks $g(p; x)$ and $h(n-p, p; x)$ using the method used in the appendix of [2]. The idea is to this: To find the analytic cotinuation of a meromorphic function $f_1(z)$, you find another entire function with the same poles and zeros $f_2(z)$ such that $f_1(z)/f_2(z)$ is bounded. Then, by Liouville's theorem it must be a constant C . This means that $Cf_2(z)$ is the analytic continuation of $f_1(z)$.

The basic building block and the definitions

Introduce

$$f_{n,m}(x) := \frac{\sin^2(\pi m/n)}{\sin(\pi(m+x)/n) \sin(\pi(m-x)/n)}, \quad m \in \mathbb{Z}. \quad (\text{M.190})$$

For integer $n \geq 2$ and $p \in \{1, \dots, n-1\}$, the functions appearing in the correlator are

$$g(p; x) = \prod_{m=1}^{p-1} f_{n,m}(x)^{p-m}, \quad (\text{M.191}) \quad \{\text{eq:g-def-poles}\}$$

$$h(q, p; x) = \prod_{l=1}^p \prod_{k=1}^q f_{n,k+l-1}(x) = \prod_{r=1}^{n-1} f_{n,r}(x)^{w_r(p,q)}, \quad q = n-p, \quad (\text{M.192}) \quad \{\text{eq:h-def-poles}\}$$

where $w_r(p, q)$ counts the number of pairs (k, l) such that $k + l - 1 = r$.

Poles of $g(p; x)$ as a function of x

From (M.191), all x -dependence sits in the denominator sines. For fixed m , the factor $\sin(\pi(m \pm x)/n)$ vanishes iff $m \pm x = \ell n$ for some $\ell \in \mathbb{Z}$, i.e.

$$x = \pm m + \ell n. \quad (\text{M.193})$$

Hence the pole set of $g(p; x)$ is

$$\text{Pole}(g) = \left\{ x = \pm m + \ell n \mid m = 1, \dots, p-1, \ell \in \mathbb{Z} \right\}. \quad (\text{M.194})$$

Moreover, each vanishing sine gives a simple pole, and the factor $f_{n,m}(x)$ is raised to power $(p-m)$. Therefore the *pole order* of $g(p; x)$ at $x = \pm m + \ell n$ equals $(p-m)$:

$$\text{ord}_{x=\pm m+\ell n} g(p; x) = p-m. \quad (\text{M.195})$$

Finally, $g(p; x)$ has no zeros as a function of x , since the numerator $\sin^2(\pi m/n)$ is independent of x .

A multiple-sine candidate and its divisor

We now introduce a candidate $g_{\text{MS}}(p; x)$ built from multiple sine functions. Let $S_2(z|1, n)$ and $S_3(z|1, 1, n)$ denote the double and triple sine functions (in a standard normalization), characterized by the shift relations

$$\frac{S_2(z+1|1, n)}{S_2(z|1, n)} = 2 \sin\left(\frac{\pi z}{n}\right), \quad \frac{S_3(z+1|1, 1, n)}{S_3(z|1, 1, n)} = \frac{1}{S_2(z|1, n)}. \quad (\text{M.196}) \quad \{\text{eq:S2S3-shift}\}$$

Define the triangular-product package

$$T_p(z) := \frac{S_3(z+2|1, 1, n)^p}{S_3(z+p+1|1, 1, n) S_3(z+1|1, 1, n)^{p-1}}, \quad (\text{M.197}) \quad \{\text{eq:T-def}\}$$

and the candidate analytic continuation

$$g_{\text{MS}}(p; x) := \frac{T_p(0)^2}{T_p(x) T_p(-x)}. \quad (\text{M.198}) \quad \{\text{eq:gMS-def}\}$$

Claim (telescoping to the finite sine product). By iterating the shift relations (M.196), one checks that $T_p(z)$ is proportional to the triangular sine product

$$T_p(z) \propto \prod_{m=1}^{p-1} \sin\left(\frac{\pi(m+z)}{n}\right)^{p-m}, \quad (\text{M.199}) \quad \{\text{eq:T-telescope}\}$$

where the proportionality factor is independent of z (it may depend on n, p). Consequently, (M.198) implies that for integer n , $g_{\text{MS}}(p; x)$ reproduces the same pole lattice and the same pole orders as $g(p; x)$.

Zeros/poles of $g_{\text{MS}}(p; x)$. From (M.199), $T_p(z)$ has zeros whenever $z = \ell n - m$ for $m = 1, \dots, p-1$, with zero order $(p-m)$. Therefore $g_{\text{MS}}(p; x)$ has poles whenever $T_p(x) = 0$ or $T_p(-x) = 0$, i.e.

$$x = \ell n - m \quad \text{or} \quad x = -(\ell n - m) = m - \ell n \quad \Longleftrightarrow \quad x = \pm m + \ell n, \quad (\text{M.200})$$

and the pole order at each such point equals $(p-m)$. In particular,

$$\text{Pole}(g_{\text{MS}}) = \text{Pole}(g), \quad \text{ord}_{x=\pm m+\ell n} g_{\text{MS}}(p; x) = p-m. \quad (\text{M.201})$$

Like g , the function g_{MS} has no zeros as a function of x .

The ratio has no poles and is bounded

Define the ratio

$$R(x) := \frac{g_{\text{MS}}(p; x)}{g(p; x)}. \quad (\text{M.202})$$

Since g_{MS} and g have the same poles with the same orders and no zeros, all singularities cancel:

$$\boxed{R(x) \text{ is entire in } x.} \quad (\text{M.203})$$

Moreover, both g and g_{MS} are even and n -periodic in x , hence so is R :

$$R(-x) = R(x), \quad R(x+n) = R(x). \quad (\text{M.204}) \quad \{\text{eq:R-periodic}\}$$

To argue boundedness, it suffices (by periodicity) to bound R on a single vertical strip $\{x = u + iv : u \in [0, n]\}$. On such a strip, the original $g(p; x)$ obeys an exponential-type estimate as $|v| \rightarrow \infty$: for each fixed m ,

$$\left| \sin\left(\frac{\pi(m \pm (u + iv))}{n}\right) \right| \sim \frac{1}{2} e^{\pi|v|/n}, \quad |v| \rightarrow \infty, \quad (\text{M.205}) \quad \{\text{eq:R-periodic}\}$$

so

$$|g(p; u + iv)| \sim C \exp\left[-\frac{2\pi|v|}{n} \sum_{m=1}^{p-1} (p-m)\right] = C \exp\left[-\frac{\pi p(p-1)}{n} |v|\right], \quad |v| \rightarrow \infty, \quad (\text{M.206}) \quad \{\text{eq:g-asymp}\}$$

with C independent of v (uniformly for u in compact sets away from poles).

On the other hand, the multiple sine functions S_2, S_3 have well-known large-imaginary asymptotics of the form

$$\log S_r(u + iv | \boldsymbol{\omega}) = \pm \frac{\pi}{r! \omega_1 \cdots \omega_r} |v|^r + \text{lower powers in } |v| + O(1), \quad |v| \rightarrow \infty, \quad (\text{M.207}) \quad \{\text{eq:g-asymp}\}$$

so that in ratios such as (M.197) the leading polynomial growth cancels, and $T_p(z)$ has exponential type matching the finite sine product in (M.199). Consequently $g_{\text{MS}}(p; u + iv)$ has the same exponential type as $g(p; u + iv)$, and hence their ratio $R(u + iv)$ remains bounded as $|v| \rightarrow \infty$ uniformly in $u \in [0, n]$. Together with continuity on the compact region $\{u \in [0, n], |v| \leq V\}$, we conclude:

$$\boxed{R(x) \text{ is bounded on } \mathbb{C}.} \quad (\text{M.208})$$

Since R is entire and bounded, Liouville's theorem implies R is constant. Finally, $g(p; 0) = 1$ by definition and $g_{\text{MS}}(p; 0) = 1$ by (M.198), so $R(0) = 1$ and thus

$$\boxed{g_{\text{MS}}(p; x) = g(p; x) \quad \text{for all } x \text{ (at integer } n\text{).}} \quad (\text{M.209})$$

This is precisely the “same poles + bounded ratio” strategy.

Corollary: $h(n-p, p; x)$ from a full-product identity

Using (M.192) with $q = n - p$ and the explicit multiplicities

$$w_r(p, n-p) = \begin{cases} r, & 1 \leq r \leq p, \\ p, & p < r \leq n-p, \\ n-r, & n-p < r \leq n-1, \end{cases} \quad (\text{M.210})$$

one may rewrite $h(n-p, p; x)$ as

$$h(n-p, p; x) = \left(\prod_{r=1}^{n-1} f_{n,r}(x) \right)^p g(p; x)^{-2}. \quad (\text{M.211}) \quad \{\text{eq:h-via-full}\}$$

The appendix identity of [2] proves

$$\prod_{r=1}^{n-1} f_{n,r}(x) = \left(\frac{n \sin(\pi x/n)}{\sin(\pi x)} \right)^2, \quad (\text{M.212}) \quad \{\text{eq:h-via-full}\}$$

again by the same poles/bounded ratio argument. Substituting into (M.211) yields

$$\boxed{h(n-p, p; x) = \left(\frac{n \sin(\pi x/n)}{\sin(\pi x)} \right)^{2p} g(p; x)^{-2}}, \quad (\text{M.213})$$

so the analytic continuation of h follows once that of g is fixed by the pole-matching method.

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